

Edited by

PATRIZIA ROMANO, MAURIZIO CIANI and ANGELA CAPECE

CRAFT BEER

New Frontiers in Brewing Science



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Edited by

Patrizia Romano

Department of Economy, University Marcatorum, Rome, Italy

Maurizio Ciani

Department of Life and Environmental Sciences (DiSVA), Polytechnic
University of Marche, Ancona, Italy

Angela Capece

Department of Agricultural, Forest, Food, and Environmental Sciences (DAFE),
University of Basilicata, Potenza, Italy



ACADEMIC PRESS

An imprint of Elsevier

Academic Press is an imprint of Elsevier
125 London Wall, London EC2Y 5AS, United Kingdom
50 Hampshire Street, 5th Floor, Cambridge, MA 02139, United States

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ISBN: 978-0-443-16015-8

For information on all Academic Press publications visit our website at
<https://www.elsevier.com/books-and-journals>

Publisher: Mara Conner
Acquisitions Editor: Kelsey Connors
Editorial Project Manager: Emerald Li
Production Project Manager: Sruthi Sathesh
Cover Designer: Christian Bilbow

Typeset by TNQ Technologies



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Contributors

Antonietta Baiano

University of Foggia, Department of Agricultural Sciences, Food, Natural Resources, and Engineering, Foggia, Italy

Matthew L. Bochman

Indiana University, Molecular & Cellular Biochemistry Department, Bloomington, IN, United States

Julieta Burini

Centro de Referencia en Levaduras y Tecnología Cervecera (CRELTEC), Instituto Andino Patagónico de Tecnologías Biológicas y Geoambientales (IPATEC, CONICET - UNComahue), Bariloche, Argentina

Laura Canonico

Department of Life and Environmental Sciences (DiSVA), Polytechnic University of Marche, Ancona, Italy

Angela Capece

Department of Agricultural, Forest, Food, and Environmental Sciences (DAFE), University of Basilicata, Potenza, Italy

Maurizio Ciani

Department of Life and Environmental Sciences (DiSVA), Polytechnic University of Marche, Ancona, Italy

Francesca Comitini

Department of Life and Environmental Sciences (DiSVA), Polytechnic University of Marche, Ancona, Italy

Ray Daniels

Cicerone Certification Program, Chicago, IL, United States; Siebel Institute of Technology, Chicago, IL, United States

Christian Garavaglia

University of Milano-Bicocca, Department of Economics, Management and Statistics (DEMS), Milan, Italy; Bocconi University, Invernizzi Center for Research on Innovation, Organization, Strategy and Entrepreneurship (ICRIOS), Milan, Italy

Dolores González de Llano

Institute of Food Science and Research (CIAL), CSIC—UAM, C/ Nicolás Cabrera 9, Madrid, Spain

Nancy Hoalst-Pullen

Kennesaw State University, Professor of Geography and Geospatial Sciences, Department of Geography and Anthropology, Kennesaw, GA, United States

Edyta Kordialik-Bogacka

Lodz University of Technology, Institute of Fermentation Technology and Microbiology, Faculty of Biotechnology and Food Sciences, Lodz, Poland

Diego Libkind

Centro de Referencia en Levaduras y Tecnología Cervecera (CRELTEC), Instituto Andino Patagónico de Tecnologías Biológicas y Geoambientales (IPATEC, CONICET - UNComahue), Bariloche, Argentina

M. Victoria Moreno-Arribas

Institute of Food Science and Research (CIAL), CSIC—UAM, C/ Nicolás Cabrera 9, Madrid, Spain

Renan E. Araujo Piraine

University of São Paulo, Ribeirão Preto, Brazil

Nils Rettberg

Versuchs- und Lehranstalt für Brauerei in Berlin (VLB) e.V., Research Institute for Raw Materials and Beverage Analysis, Berlin, Germany

Magaly Rodríguez-Saavedra

Institute of Food Science and Research (CIAL), CSIC—UAM, C/ Nicolás Cabrera 9, Madrid, Spain

Patrizia Romano

Universitas Mercatorum, Department of Economy, Rome, Italy

Emanuele Zannini

Università di Roma La Sapienza, Department of Environmental Biology, Rome, Italy; University College Cork, School of Food and Nutritional Sciences, Cork, Ireland

Preface

This book provides a comprehensive picture of the main aspects of craft beer, an innovative product that is attracting growing interest from researchers and consumers worldwide. This increased interest indicates that the potential market niche of this product goes well beyond the home brewer movement. The craft beer industry was born in the United State, and this symbolically indicated the start of the international revolution in the brewing industry. Modern consumers are interested in finding authentic, unique, fresh, diverse, and high-quality beers. Craft brewers have answered this demand by reinterpreting historic beer styles obtained with ancient raw materials and developing new ones. Craft beer is typically produced at local microbreweries which create a strong connection with the local region where the beer is produced. Microbreweries filled the market niches left by standardized industrial beers and are identified with the terms “craft brewery” and “artisanal brewery” to indicate breweries that brew different types of beer on a small scale, distinguishing them from larger breweries mass-producing beers. According to the Brewers Association, a key feature of craft beer is innovation, which means that craft brewers interpret historic beer styles with unique twists and develop new styles beyond mainstream beer.

Craft beer is typically made from traditional ingredients, such as malted barley and hops, while local and sometimes nontraditional ingredients are added for distinctiveness. The typical properties of craft beers also derive from the production process, which comprises operations such as refermentation in bottles, or the absence of treatments such as pasteurization and microfiltration that are normally used in industrial brewing. Because of its modalities of production, craft beer is more susceptible to spoilage and chemical hazards. For this reason, the implementation of a HACCP plan with practical preventive procedures for manufacturing craft beer can ensure product safety.

The craft beer market encompasses both innovative and traditional beers that have gained consumer appreciation because of their distinctive and unexpected sensorial characteristics.

Indeed, market demands increasingly directed toward beers (with the absence of pasteurization and microfiltration treatments) have pushed research toward biotechnological studies in which fermenting yeast is the center of the experiments. Yeast in craft beers became the maximum expression of bio-flavor, able to give a characterizing impact to each product. In this regard, *Saccharomyces* and non-*Saccharomyces* yeasts may play a relevant role in the diversification and enhance distinctive sensorial characteristics. Over time, the awareness has grown that the choice of yeast to use in beer production is fundamental to obtaining a product with distinctive characteristics that consumers appreciate.

The modern consumer is also very aware of sustainability aspects and beer is one of the most resource intensive as brewing beer uses significant amounts of water and energy, has a massive carbon footprint, and produces substantial air, water, and solid wastes. In response to this problem, brewers have adopted diverse and wide-ranging economic, environmental, and social sustainability strategies and practices that promote water conservation, reduce waste and carbon emissions, improve energy efficiency and job conditions, and establish a sustainable workplace culture.

This book is aimed to collect the scientific knowledge available until now on craft beer, by including different issues concerning the history and legal definition, description of raw materials, and aspects related to the production process of the main beer styles, with emphasis on traditional and innovative microbial resources involved in the fermentative process, other than issues related to organoleptic quality, critical points and nutritional concerns, but also an issue related to the emerging problems, such as the sustainability of the production process.

This book is addressed to different readers, which will be students of university courses in food-related topics, researchers interested in fermented beverages, and specialists in quality management systems and marketing of beer and food companies.

Definition, history, and market of craft beer

1

Christian Garavaglia^{1,2}

¹University of Milano-Bicocca, Department of Economics, Management and Statistics (DEMS), Milan, Italy; ²Bocconi University, Invernizzi Center for Research on Innovation, Organization, Strategy and Entrepreneurship (ICRIOS), Milan, Italy

1. Introduction

Craft breweries have transformed the international beer industry. Following a century of breweries' consolidation, domination of a few global multinationals, and the homogenization of beer, a few small craft breweries entered the market and revolutionized the industry (Garavaglia & Swinnen, 2018a). After an initial slow start and diffusion, the phenomenon of craft brewing gradually expanded, gaining the interest of an increasing number of consumers. A rising number of craft breweries entered the market that were small in size, with differentiated products, experimenting with new ingredients in production and most were initially (and many are still currently) local in scope. The macroproducers of beer seemed to show indifference to this market change in the beginning, or perhaps they had strategically decided to observe the experimentation of the new phenomenon and take time to react. Eventually, the growth and impact of craft breweries stimulated strategic reactions among the macroproducers, which subsequently began to produce new types of beers, acquire craft breweries, and consolidate their positions in distribution channels. After the advent of craft brewing, the beer industry would never be the same.

The purpose of this chapter is to investigate the revolution of craft brewing. In particular, we endeavor to define craft beer, provide a brief history, and discuss the main issues of the current craft beer market. The chapter is structured as follows. First, we discuss the difficulties of outlining common definitions of craft beer and craft brewing. Given the heterogeneity of various countries, a generally acknowledged definition of craft beer has not yet emerged; however, it is crucial to establish some universal criteria for identifying what craft beer is. Second, we present the history of craft brewing, distinguishing five building blocks of the birth and diffusion of craft beer around the world. We assert that the tradition, history, and culture of beer in the traditional beer nations of the United Kingdom (UK), Belgium, and Germany represent the roots of craft brewing and craft beer. The burgeoning craft beer industry in the United States (US) signified the start of the worldwide craft beer revolution in the brewing industry. The processes of legitimation and social contagion contributed to disseminating craft brewing around the globe. Third, we analyze the relevant features and competitive issues of the craft

beer market, discussing how the co-existence of small craft producers and macrobreweries establishes an interesting and vivid market environment.

2. Defining craft beer

Considering the heterogeneity of various countries and different traditions in beer brewing, and given the various national legislation, no consensus has been reached concerning a general definition for the terms craft beer and craft brewery. However, it is crucial to define craft beer and craft breweries because government legislators may introduce policies tailored to such producers, and because consumers who positively value craft beer are willing to pay higher prices (Elzinga et al., 2018; Fertő et al., 2018; Garavaglia & Swinnen, 2018b); therefore, firms need to know the factors that consumers identify with craft beer.

Generally speaking, the term craft beer refers to beer production that has been conducted in the past few decades in opposition to the mass-produced beer of traditional, large-scale incumbent breweries. Some other terms have often been used to indicate this phenomenon as synonymous with craft breweries, including microbreweries, artisanal breweries, or local breweries.

Craft brewing emerged in all industrialized countries in the world at different times in the past decades. Diverse criteria have been used to define craft breweries and craft beer, each of which has limitations. The most acknowledged and accepted aspect that is needed to accurately define a beer as craft is ownership that is independent from other national or multinational beer producers (Garavaglia & Swinnen, 2018b). The motivation behind this criterion is the origin of the term craft brewery. As the number of breweries declined, a few very large industrial breweries began to dominate the market, while smaller firms exited the market or were acquired. While the smaller breweries that were acquired could retain their official names, they were under the control of the larger producers that acquired them, with no independence to manage the brewing process. The interpretation of this criterion in accordance with the craft perspective is that the breweries must be independent of incumbent mass producers and free from third-producer control. Recently, in some countries (e.g., Italy) craft breweries have emphasized the importance of independence by formalizing special logos that only “real” independent craft breweries can use.

This criterion has some limits. First, if a craft brewery is acquired by a multinational, it may lose its independence but continues to produce the same type of beer (Garavaglia and Swinnen (2018b) referred to these as ex-craft breweries). Second, the logic of the independence requirement could also lead to considering firms like the multinational Guinness as craft. Of course, this is not the case of a craft brewery; therefore, additional criteria are needed to define a brewery as craft.

We maintain that another crucial consideration is size. Craft breweries are usually small in scale. However, notably, breweries’ production scale can be directly related to the size of the country in which they operate. Some potentially embarrassing comparisons could then emerge in which a craft brewery in the US could be much larger than leading multinational brewing facilities in many smaller countries.

Being small often relates to being local in scope. In their definition of craft beer, the American Brewers Association specifies that craft brewers tend to be fully involved in their communities. Craft breweries engage in local activities such as beer festivals, local promotions, and connections with local agricultural activities (Garavaglia, 2020; Schnell, 2013; Schnell & Reese, 2003). Garavaglia (2020)

identified three types of relationships between craft beer and space, which include using local dialect in the name for a beer, using geographical places in the name for a firm/brand, and using local ingredients in beer production. All craft breweries started as small and locally oriented plants. This criterion also has obvious limitations, the most obvious of which is that multinationals are local for the consumers that live near a production plant, but we would not consider the mass-produced product craft beer. In addition, some small independent craft breweries are very export-oriented and have only loose ties with local communities.

In some countries, the definition of craft beer has been linked to the production process or use of specific ingredients. For example, craft beer is not subject to pasteurization or even microfiltration in Italy. Pasteurization is a heat treatment process during the packaging phase to kill pathogens and keep the beer sterile. Microfiltration removes yeast particles from the fermenting mixture. These treatments increase beers' shelf life and guarantee product consistency. However, these processes can also flatten some organoleptic fragrance characteristics of beer and reduce their flavor and aroma, contributing to loss of the distinctive beer taste. This criterion might be relevant in Italy, but it is not considered in other countries.

Given these considerations, we claim that the factor of age is an important criterion in the definition of craft beer. Craft brewing is a relatively recent phenomenon. The first craft beer producers in most countries started brewing in the 1980s and 1990s (Garavaglia and Swinnen, 2018a). However, this aspect opens up a discussion concerning the circumstances in traditional beer nations. It is important to note that traditional craft beer production has been conducted in Europe for centuries, particularly in countries with a long history of beer brewing such as Germany, Belgium, and the UK. This means that some old breweries have always been independent and small in scale, for example, the case of the family breweries in Belgium. However, when referring to contemporary craft beer, we consider recent breweries founded in the past few decades in contrast to mass production by large-scale incumbent breweries. This means that while old, small, independent breweries might be craft in style, they are not the focus of our analysis of the modern phenomenon of craft brewing. We develop this discussion in Section 3.

The consumer perspective can also help to define craft beer. Garavaglia and Mussini (2020) analyzed Italian consumers' preferences, concluding that consumers consider craft beer to be produced by an independent (family owned) firm in a small-scale plant and nonpasteurized.

In conclusion, considering the idiosyncrasies of each country, we posit that craft breweries could be defined as *small* (in relative terms with respect to their country), *independent*, and *young* breweries that have entered the beer market in the past few decades and have stood out with product differentiation in contrast to the standard homogeneous multinationals' beers.

3. The origin of craft beer

The beer industry experienced a century of brewery consolidation, which culminated in the domination of a few global multinationals and the standardization of beer. The number of breweries declined in all industrialized nations during much of the 20th century. Only when new craft breweries entered the market did the number of breweries increase again. Therefore, we assert that a good indication of the take-off of the craft beer market is a growing number of breweries, and a turnaround in the number of breweries in a country indicates the beginning of craft brewing. Fig. 1.1 illustrates this dynamic for some selected countries: It reports the number of breweries per capita (million inhabitants).

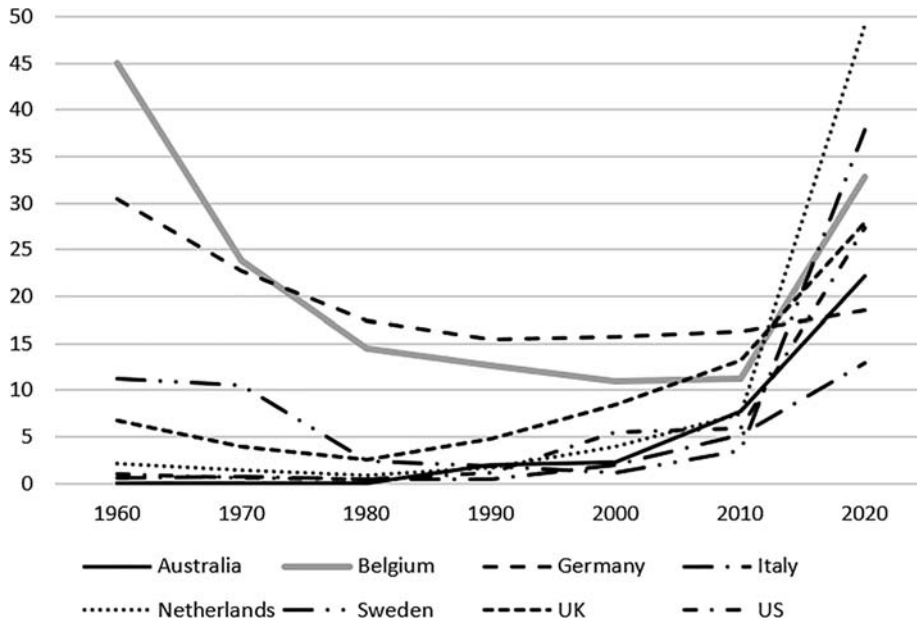


FIGURE 1.1

Number of breweries per capita (million inhabitants) in selected countries, 1960–2020.

Sources: Garavaglia, C., & Swinnen, J. (2018b). *Economics of the craft beer revolution: A comparative international perspective*. In: Garavaglia, C., & Swinnen, J. (Eds.) *Economic perspectives on craft beer: A revolution in the global beer industry*. Palgrave Macmillan, <https://brewersofeurope.eu/eu-beer-trends/european-union/>, <https://brewsnews.com.au/carbon/assets/0007e6/000003/States-of-Brewing-Report.pdf>; Häggqvist, H. (2023), *Drivers of market concentration in the Swedish brewery sector, 1910–1990*, Paper presented at the 48th economic and business history society conference, Porto, Portugal (Häggqvist, 2023).

Fig. 1.1 clearly demonstrates the continuous decline in the number of breweries for most of the 20th century due to consolidation and bankruptcies in the mainstream beer industry, and a subsequent growth in the number of breweries in more recent years, which is attributable to the entry of craft breweries. Data also reveal significant differences between countries in the emergence and growth of craft breweries. In countries where post-World War II consolidation of breweries was stronger, the turnaround in the number of breweries occurred earlier, indicating that craft brewing emerged earlier. That was the case in, for example, the US and the Netherlands, where the total number of breweries was at its lowest point around 1980. In those countries, the decline in macrobreweries was more than compensated for by the growth in craft breweries from the 1980s onwards. For example, the number of craft brewers (37) in the US exceeded the number of macrobrewers (34) for the first time in 1985. Since then, the number of brewers has grown to more than 9600, the vast majority of which are craft-type breweries.

It is more complex to identify the beginning of craft brewing in traditional beer countries such as Belgium, the UK, and Germany. In these countries, the small-scale production of local beer in establishments that are often owned by families has always been present. However, these countries also

exhibit a clear time period when new, mostly smaller, breweries entered the market producing new specialty beers.

Considering that the craft beer revolution began with the revival of traditional ways of brewing beer and producing traditional old beer styles, revitalizing what historically had been produced in some traditional beer nations, we identify the main traits of beer brewing and facts in the traditional nations that have been a source of inspiration, contagion, and imitation for subsequent pioneering craft breweries around the world.

In this study, we construct a proposed taxonomy to characterize the birth and history of craft beer in five (not necessarily sequential) steps that defined the main events that encouraged the emergence of craft brewing around the world.

3.1 UK: The rebellion of the campaign for real-ale association (CAMRA)

The United Kingdom has historically boasted a great number of small local breweries. Many of these breweries produced traditional styles of beer such as porters and ales for relatively small local markets. Brewery consolidation in the UK led to a dramatic decrease in these small breweries that resulted in industry concentration and the standardization of available beers. In fact, during the 1960s and 1970s, only six breweries, known as the Big Six (Allied, Bass, Courage, Scottish and Newcastle, Watneys, and Whitbread), controlled and monopolized the market in entire regions of the UK through direct ownership of hundreds of pubs that were only authorized to sell their beers. These six large breweries produced a standardized beer that was homogeneous in organoleptic characteristics and easier to store and serve. In this period, a growing number of consumers became dissatisfied with the decreased variety of beers from industry concentration and product homogenization, and this sentiment nurtured the creation of the *Campaign for Real-Ale* association (CAMRA) in 1971.¹

CAMRA is a movement of beer enthusiasts who supported the revival of “real-ale,” that is, beers brewed using traditional procedures, nonpasteurized and nonfiltered, and served hand-pumped (Bamforth & Cabras, 2016). CAMRA actions amplified consumers’ knowledge of traditional ales and gradually expanded the association’s activities from demand for more beer variety to preservation of brewing traditions and historical values. CAMRA soon passed from being perceived as a reactionary organization fighting for the preservation of real-ale, to a sort of lobbying group promoting the appreciation of British beer. The contribution of CAMRA to the renaissance of traditional real British ale in the UK is widely acknowledged. Gourvish & Wilson (1994) asserted that CAMRA contributed to preserving a niche for traditional beer brewing in the beer industry, notwithstanding the comparative difficulties of transportation and service for this kind of beer. The birth of CAMRA begat the birth of a decisive impulse for founding new breweries to revive the historical brewing tradition in the United Kingdom. The first craft breweries were attracted by the opportunity of growing market demand for real ales (Mason & McNally, 1997). Regional and local CAMRA branches also stimulated local beer festivals to support local breweries (Cabras, 2018).

The success of CAMRA in the UK inspired similar organizations in other countries. For example, in the Netherlands a small group of Dutch beer enthusiasts who regularly visited England contacted all the Dutch members of CAMRA to set up a Dutch craft beer consumers’ organization, called the PINT association, whose goal was “to make beer culture important again in the Netherlands” by supporting

¹<https://camra.org.uk>.

the production and consumption of traditional beers. PINT activities contributed to the emergence of a community of enthusiastic craft beer consumers and also encouraged hobby brewers.

3.2 Belgium: The “craft beer nation forever”

Although the number of breweries in Belgium suffered the same dramatic decline as other industrialized nations, recording a decline in the number of breweries from more than 3000 in 1900 to around 150 in 1980, a large variety of different types of beer survived in Belgium to a greater extent than in the other countries.

The survival of a large variety of beer styles in Belgium was possible because of small independent breweries that often focused their production on traditional Belgian beer styles. Abbey and Trappist beers and many other specialty beers were traditionally brewed under top fermentation with a relatively high alcohol percentage and represented the distinctive way of brewing beer in Belgium. In addition, Gueuze and Sour beers have always been considered unique and highly differentiated products that have been significantly produced only in Belgium. For these reasons, we contend that Belgium has been a “craft beer nation forever” (Swinnen & Briski, 2017) that inspired many craft brewers around the world.

3.3 Germany: Traditional small breweries above all

Germany is another beer nation that has had a key role in the history of craft brewing. Even with increased industry consolidation in the past decades, the degree of concentration in Germany remains low compared with other countries. Germany has always had a far greater number of breweries than any other nation, most of which are small and local producers. Therefore, the German beer market has been characterized by a huge brand variety. There are various reasons that the German beer market is so fragmented into many smaller local markets. Among others, a key role has been played by German consumers’ preferences, which are local in nature (Adams, 2006). This particularly occurs in southern states like Baden-Württemberg and Bavaria. German consumers tend to purchase locally produced beers, which sustains the survival of many small firms that are limited to local markets. This proclivity also created strong barriers to larger national and multinational producers’ market penetration (Scherer et al., 1975). Although these traditional German breweries differ from common definitions of craft breweries, they do present similar characteristics as they are small in scale, typically owned by traditional families, independent from multinationals or other large breweries and embedded in the local community.

The high number of small firms and resulting vast brand variety are the main cause of the weaker and later diffusion of the craft beer trend in Germany. Furthermore, the German case, with its peculiar structure of small-scale beer production, often with links to family owned taverns where local beer can be consumed have been an iconic symbol for other craft brewers around the world.

3.4 US: The symbol of the craft beer revolution

The key moment for craft brewing in the US was Fritz Maytag’s purchase of the Anchor Brewing Company in 1965, which had been founded in San Francisco by a German brewer in 1896. The brewery characterized itself as producing a characterful product, but after WWII and into the 60s the

beer frequently became contaminated, turning it sour (Ogle, 2005). Fritz Maytag bought 51% of the brewery for a few thousand dollars in 1965, saving it from bankruptcy and revitalizing the business, later purchasing it outright in 1969.² In 1971, Maytag began to produce beer using traditional European brewing practices that differed from other domestic light-bodied lager beers of the leading firms, Anheuser-Busch, Schlitz, Pabst, and Falstaff.

In 1976, Jack McAuliffe, Suzy Stern, and Jane Zimmerman founded the New Albion Brewing Company in Sonoma, California. In particular, McAuliffe's contribution was crucial for the development of craft brewing in the US. He gained experience with the US Navy in Scotland, where he was inspired by European beers, and began producing beer as a homebrewer upon his return (Johnson, 1993). He came into contact with Fritz Maytag, who motivated him to start a new entrepreneurial endeavor of producing craft beer.

In 1981, Ken Grossman and Paul Camusi founded the Sierra Nevada Brewing Company in Chico, California, after being inspired by visits to Maytag's and McAuliffe's brewing facilities and Grossman's homebrewing experience, which was influenced by Fred Eckhardt's book (Eckhardt, 1969) about brewing at home (Elzinga et al., 2018).

Jim Koch was another influential entrepreneur. After his experience in the Boston Consulting Group, he entered the craft beer market as a contract brewer, founding the Boston Beer Company in 1984 (Elzinga et al., 2018).

In addition to these pioneering craft brewers, three other craft beer promoters and two chemists merit particular mention because of their influence on the development and diffusion of craft beer (Elzinga et al., 2018). Beer writers such as Michael Jackson, Fred Eckhardt, and Charles Papazian contributed heavily to communicating about the growing variety of tastes and types of beers, like in Jackson's books and stimulating homebrewing thanks to Eckhardt's and Papazian's books. Finally, Professor Michael Lewis who was in brewing science at the University of California Davis and the consultant Joseph Owades were important technical support and consultants for McAuliffe and Maytag.

The origin of craft brewing in the US has a remarkable importance because it symbolically signified the beginning of the international revolution in craft beer in the brewing industry.

3.5 Global connections

The brewing traditions of the historical beer nations, the United Kingdom, Belgium, and Germany described in Sections 3.1.–3.3 represented a clear inspiration for early brewers in the US and around the world (Garavaglia & Swinnen, 2018b). Most of the craft beers that were initially produced in the US have European brewing origins (Bostwick, 2014), as described in Section 3.4. In addition, many pioneering entrepreneurs in craft brewing in various countries around the globe were inspired, influenced, and fascinated by the small, independent, traditional brewers in the UK, Belgium, and Germany (Garavaglia & Swinnen, 2018b), establishing an ideal connection between contemporary craft breweries and old, small, independent breweries in the historical European beer nations.

For their part, the pioneering entrepreneurs in craft brewing became a source of inspiration and information that fostered the diffusion of craft beer around the world. The diffusion of the first craft

²<https://web.archive.org/web/20090527200950/http://www.fundinguniverse.com/company-histories/Anchor-Brewing-Company-Company-History.html>.

breweries gradually legitimized this fresh new way of producing beer, increasing the degree of institutional and social acceptance of this new type of business in the beer market. Sociological and economic explanations have demonstrated how legitimation increases as new business typologies diffuse, helping to overcome the difficulties related to newness (Carroll, 1997). Gradually, the new type of business represented by craft brewing became a natural way of doing business in the beer market for potential new entrepreneurs. Furthermore, as the legitimation effect increases, opportunities associated with the new business forms emerge because of the greater ease of attracting capital and customers, increased possibilities for identifying proper suppliers and employees and fewer institutional barriers.

Moreover, entrepreneurial activities are influenced by social contagion. Given the uncertainty and risk related to the decision to start a new business, entrepreneurs observe the experiences of similar firms to assuage their uncertainties about what to do, how to do it and future profitability. This practice often causes entrepreneurs to believe and act in similar ways (Anglin et al., 2018; Angst et al., 2010). Social contagion was also facilitated by the growth of international travel, which allowed people to experience the local beer and tavern traditions in beer nations and discover varieties of beer and tastes beyond the prevalent lager. Increased rates of commercial integration and exports also contributed to spreading the knowledge about beer varieties.

As a consequence, craft brewing gradually disseminated in all industrialized countries, creating a distinctive and innovative environment and establishing a deep sentiment that has been at the core of the craft beer success.

4. The craft beer market

One of the primary effective aspects of the craft beer movement is the rediscovery of the concept of “craft” that the producers contributed to developing. The notion of craft evokes notions of sentiment, passion, and artistry. Beer craftsmanship is a means through which a new generation of young beer enthusiasts began to relate to their own work with embodied and affective engagement, revealing meaning and personal identity (Sennett, 2008). The notion of craft served as a means to address the personification of skills and emotions such as passion and satisfaction at work in the brewery (Thurnell-Read, 2014). In craft production, the goods are seen as manifestations of the maker’s identity. A large proportion of the first craft brewing entrepreneurs in many countries were young people who were “in love with” the idea of producing something new and different. Craft brewers expressed themselves in their products, demonstrating passion and creativity. They were producing something that they personally enjoyed.

The advent of craft brewing was part of a shift in thinking around the world that brought a new direction for countering mass consumption as consumers began using products to express their identities and consumer–product relationships began to reflect crucial life themes and identity concerns (Fournier, 1998). An exemplary expression of this was the birth of Slow Food. The Slow Food movement was initiated in Italy in 1986 and became an international association in 1989. This association clearly represents consumers’ changing attitudes and demonstrates the existence of a community of consumers who want to become part of a broader process beyond mere consumption, in which consumers are asked to “take care of producers that are marginalized by global, standardized commodity circuits” (Sassatelli & Davolio, 2010).

Early craft beer producers were first consumers in search of something new. The natural consequence, following decades of brewery consolidation and homogenization of the product that led to the prevalence of extremely similar types of lager beer (according to their organoleptic characteristics), was these early producers creating something different, with more distinctive flavors than the lager produced by mass producers, awakening and encountering the preferences of a new wave of consumers.

The increase in the variety of the beer supplied by craft breweries primarily occurred with an orientation to local markets. Very often, early craft breweries invoked geography and specific places in their brand names, highlighting their connections to these places and local communities. The use of beer names and images that reflect the place where the beer is produced to establish local identities and attachment to places has been employed by many craft breweries in various countries. Schnell & Reese (2003) and Schnell (2013) analyzed the practices of US craft breweries that employ historical figures, local characters, landmarks, historical and climatic events, nostalgic images of yesteryear, historical lifeways and images of nature in their names, and the artwork on their labels, consciously engaging in a process of neolocalism. Garavaglia (2020) discussed how the current Italian beer scene has numerous examples of beer linked to local identities through the use of dialect in the beer's name, the name of geographical locations in firm/brand names and using local ingredients in beer production. Locally rooted names, images, and ingredients create a sense of belonging to a unique place for the people who live there and also a chance to share one's pride for a locale's distinctiveness with other people. With craft beer, consumers are not only able to choose a specific brand but an idea that is the connection to a place and the identities of people who employ their craft knowledge in production (Khermouch, 1995).

Year by year, craft brewing has progressively attracted consumers' attention, which has contributed to the substantial growth of the craft beer market since its advent. Data about the number of craft breweries reveal how this phenomenon gained remarkable increasing success on all continents (Garavaglia & Swinnen, 2018a).

Fig. 1.1 shows the growth in the number of breweries in more recent years, which is attributable to the entry of craft breweries, as reported in Table 1.1. This recent remarkable increase in the number of craft breweries is explained by the ongoing success of craft beer among consumers, which represented a business opportunity for many investors. Specifically, relatively low barriers of entry in the craft beer market and the dissemination of public knowledge and training courses on producing beer have encouraged increased number of investors and passionate individuals to start a new craft brewery (Bronnenberg et al., 2022; Zang et al., 2015).

Furthermore, the value of the craft beer market registered significant growth rates. For example, in 10 years, retail dollar sales of craft beer rose from 19.3% of the US beer market to 24.7%³ and it is expected to continue growing.⁴

Along with craft breweries, other actors have key roles in the beer market, including public authorities, macrobreweries and distributors.

In the beginning, public authorities were unprepared to manage the emergence of craft breweries in most cases because of a lack of legislation and the newness of the phenomenon. Garavaglia & Swinnen

³See estimates provided by the American Brewers Association <https://www.brewersassociation.org/press-releases/craft-brewer-volume-share-of-u-s-beer-market-reaches-double-digits-in-2014/> and <https://www.brewersassociation.org/statistics-and-data/national-beer-stats/>.

⁴<https://www.marketdataforecast.com/market-reports/craft-beer-market>.

Table 1.1 The number of craft breweries in some selected countries, 1985–2022.

Country	1985	1990	1995	2000	2005	2010	2015	2020	2022
Australia	3	34	26	43	93	172	358		
Austria						101	114	118	130
Canada						277	610	1123	
Czech Republic						65	202	506	
Denmark	0	1				157	161 ^a	190	230
Finland						22	41	92	111
France						322	690	2100	2300
Germany						646	723	908	865
Hungary					90	50	64	75	70
Italy	0	1	7	59	127	313	685	756	870
Japan			82	288	237	206			677
Norway						21	91	129	
Poland						20	100		
Portugal	0	0	0	0	0	0	60	115	105
Slovakia						14	45	75	75
Spain	0	1	3	7		46	409	261	251
Sweden						34	214	387	435
UK						778	1828	1815	1779
USA	37	249	998	1469	1591	1756	4803	8921	9675

^aIndicates that the number of microbreweries of 2014 was used.

Sources: Garavaglia, C., & Swinnen, J. (2018b), *Economics of the craft beer revolution: A comparative international perspective*, In: Garavaglia, C., & Swinnen, J. (Eds.) *Economic perspectives on craft beer: A revolution in the global beer industry*. Palgrave Macmillan, <https://www.brewersassociation.org/statistics-and-data/national-beer-stats/>, <https://mainichi.jp/english/articles/20230407/p2a/00m/0op/039000c>, <https://www.statista.com/statistics/1171283/japan-number-microbreweries/>; Thomson, S. (2023). *Distinguishing the Canadian craft beer market: A comparative analysis of western and eastern regions and implications for brewing ingredient companies*. Preprints.

(2018b) reported how restrictive regulations that were tailored to and designed for mass producers hampered the activities of the first craft brewers in Italy, Spain, Colombia, China, and others. Public authorities' legislative development was crucial for the initial progress of the craft beer scene, as well as legalization for homebrewing, which was the first experience of many entrepreneurs who later decided to exploit this experience professionally and found a craft brewery (Bottero, 2005; Carroll & Swaminathan, 2000; Kadile & Biraglia, 2016). Public authorities also established associated taxation systems. In many industries, including the beer industry, small and medium-sized enterprises (SMEs) seek support through lower tax rates tailored to SMEs. Some countries offer reduced excise taxes for smaller brewers, establishing lower tax rates below a threshold of production to support craft breweries.

Considering that craft brewing was basically born in opposition to the homogenization of mass-produced beers, product competition is the most symbolic match between macro- and craft breweries. Craft breweries have pursued a product hyper-differentiation strategy (Clemons et al., 2006)

since their beginning. Many craft breweries have enlarged their portfolio of beers, frequently introducing a new beer or a different style. Following this route, craft breweries recognized the increased importance of differentiation in a highly competitive market, diminishing the importance of price as the principal determinant of consumers' product selection. In contrast, macroproducers have always benefitted from lower price as an attractive factor. Furthermore, the consistency of the macrobreweries' beers opposes craft beers' variability in quality and shelf life. In addition, macrobreweries in many countries (e.g., Italy) have stepped into the craft beer segment, starting to produce beers with crafty images or varieties that they never directly produced before such as IPA and unfiltered beers. Macrobreweries have also started to highlight the intrinsic characteristics of their beers in advertising, emphasizing ingredients and qualities such as the use of national cereals and hops in production processes, reinforcing the image of domestic producers rather than multinational companies.

Another strategy pursued by macroproducers to enter the craft beer market is mergers and acquisitions (M&A). After macroproducers' initial disinterest, in response to the growth and innovations linked to craft breweries, they began to acquire them. However, being acquired meant losing the prerequisite of independence for craft beer consumers, ultimately losing the nature of the craft brewery. As noted in [Section 2](#), ownership independence is central to the craft beer philosophy and the prerequisite of independence is probably the only criterion that is acknowledged across countries to define a craft brewery. As a consequence, craft beer consumers and publicans have harshly criticized such acquisitions, considering them to be a betrayal of craft beer origins and ceased purchasing these beers or selling them on their taps, respectively.

Distribution channels are one of the main competitive differences between craft and macrobreweries. Macrobreweries have always competed fiercely to strengthen their positions in the distribution system by bargaining for shelf space in supermarkets and beer distributors. Craft beers have begun to appear in off-trade channels such as hypermarkets, supermarkets, and mini markets; however, its penetration in this channel is relatively smaller than macrobreweries' beers. In addition, in the on-trade channel, macrobreweries often leverage control over bars and retailers, using exclusive purchase contracts to compete with craft beer at the bars they control and push their beer portfolio. However, the number of independent bars (or those that do not own the taps but have agreed with a distributor to manage some of their taps freely) has grown in recent years. These changes were a key factor in the growth of the craft beer market because of the sales and promotion and diffusion of the craft beer culture.

Beer consumption has increased around the world in recent years, opening opportunities for craft and macrobreweries. The beer market is extremely vivid and dynamic thanks to this ongoing competition.

5. Conclusions

In this chapter, we analyzed the craft beer phenomenon. We contend that this study contributes to providing a clear definition of craft breweries, which can be identified by the advent of *small*, *independent*, and *young* producers in the beer market that have been founded in recent decades and differentiate themselves with the production of novel beers in opposition to the mass-produced homogeneous beer made by old, large-scale incumbent breweries. We proposed a five-step taxonomy to identify the genesis and history of craft producers. Finally, we discussed the main relevant competition

issues in the market, describing how the contemporary market is partitioned into co-existing large multinationals and small craft firms. Even if these two different types of breweries tend to compete more with those of similar size (Carroll & Swaminathan, 2000; Hannan & Freeman, 1989), we discussed the competitive interactions of craft and macrobreweries in partially overlapping actions and strategies. The degree of competition between large and craft breweries has amplified in the past few years as multinationals have stepped into the craft beer market competing through product differentiation, M&A strategies and distribution tactics. These aspects have established vivid dynamics in the industry that should be considered in industry evolution studies, explicitly delineating renewed competition in a “David and Goliath” fashion (Garavaglia, 2022). We contend that this is an important topic for the future of the beer industry from a strategic perspective.

In conclusion, this investigation could be of interest and inspiration for further research for scholars who are involved in business history to disentangle the birth of a new market, for scholars of entrepreneurship, for scholars who study industrial dynamics and industry evolution, for scholars who are interested in the sociological aspects of the birth and diffusion of new organizational forms of production and for scholars of strategy who analyze the competitive environment populated by different types of firms.

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Raw materials in craft beers

2

Edyta Kordialik-Bogacka

Lodz University of Technology, Institute of Fermentation Technology and Microbiology, Faculty of Biotechnology and Food Sciences, Lodz, Poland

1. Introduction

Modern consumers are interested in finding authentic, unique, fresh, diverse, and high-quality beers. Craft brewers have answered this demand by reinterpreting historic beer styles obtained with ancient raw materials and developing new beer styles, applying premium ingredients along with nontraditional ones. Assorted blends of malted or raw grains from various cereals, combinations of traditional and new hops, fruits, vegetables, herbs, spices, and other natural ingredients are used by craft beer makers to extend and diversify their ranges with new aromas and flavors. Craft brewers also add value by creating a strong connection with the local region where the beer is produced. They often use locally sourced ingredients to support their communities and create a distinctive image. Specific soil features and environmental factors have an impact on the brewing properties of indigenous ingredients, such as cereals, malts, hops, microbes, and adjuncts (Bowen & Miller, 2023). This enables craft brewers to emphasize regional attributes in their promotional campaigns. The notion of *terroir*, long associated with the wine industry, has also become popular in craft brewing and can increase recognition for regional craft beers.

To create a beer with certain desired characteristics, it is necessary to select appropriate raw materials. Water, malted barley, hops, and yeasts are the traditional raw materials used in brewing. When producing wheat beer, barley malt is at least partially replaced with wheat malt. However, various cereals are nowadays used to make beer, including rye and sorghum as malted and unmalted grains, rice, corn, oats, and pseudocereals such as buckwheat, quinoa, and amaranth—particularly for manufacturing gluten-free beer (Benucci et al., 2020).

Cereals are a source of carbohydrates, mainly starch, which has to be broken down into mono-, di-, and trisaccharides before fermentation, in a process called mashing. During fermentation, these sugars are converted by yeast into alcohol and carbon dioxide. Cereals possess almost no enzymes. Therefore, prior to being used for brewing cereals are normally malted to induce the synthesis of enzymes required to hydrolyze starch into fermentable sugars. Cereal ingredients also have an impact on beer body, color, flavor, and foam stability.

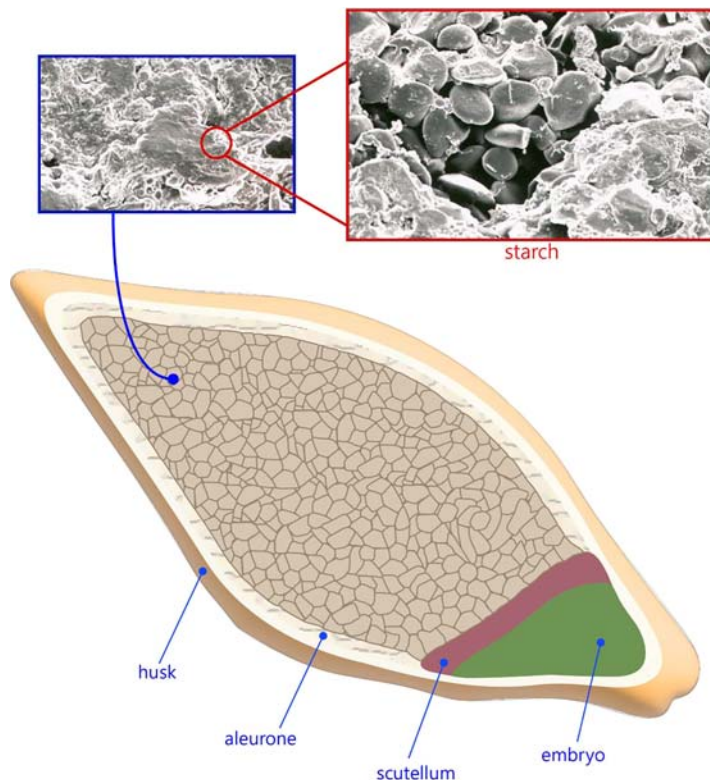
2. Barley

Barley originates from the Middle East. It ranks fourth worldwide in terms of both acreage sown and production volume, with 151.62 million metric tons harvested annually (Statista, 2024). Barley (*Hordeum vulgare*) belongs to botanical family of Gramineae grasses. Its cultivars can be planted in spring or fall. Fall-planted (winter barleys) are grown in regions where the winters are relatively mild. Both the variety of barley and local vegetation conditions influence grain quality (Herb et al., 2017). Only barley grains from certain cultivars and of suitable quality can be used in malting and brewing.

The main components of barley are carbohydrates, primary starch (around 60%), β -glucans (2%–10%), and arabinoxylans as well as proteins (8%–13%), free lipids (2%–3%), minerals (1.5%–2.5%), vitamins, and phenolic compounds (Fox, 2009). The barley grain is composed of three main parts: a large endosperm (80% of the cereal grain), the embryo or germ, and the outer layers (husk, pericarp, testa, and aleurone layers) (Briggs et al., 2004) (Fig. 2.1). The endosperm consists of large cells composed of large and small starch granules in a matrix of proteins and gums. The cell walls are mainly made of β -glucans (hemicellulose) and pentosans (gummy polysaccharides) with some protein (Meussdoerffer & Zarnkow, 2009).

FIGURE 2.1

Barley kernel.



There are two types of barley: two-row and six-row, depending on the arrangement of the seeds on the ear. Two-row barley is predominant in Europe, while six-row is more widespread in the US. Two-row kernels are larger and more uniformly plump than six-row kernels (Rani & Bhardwaj, 2021). They contain more starch and less protein. They also have a thinner husk, which makes them more desirable for brewers. Two-row barley gives a slightly higher brewhouse yield and is generally preferred. On the contrary, the activity of diastatic enzymes in six-row barley is a little higher. When unmalted grains (adjuncts) are used as nonenzymatic starch sources, the higher diastatic power of six-row barley can be advantageous in achieving full conversion of the mash.

The widespread use of barley in malting and brewing is attributable to several key characteristics, including high starch content, low content of lipids, and the presence of a tightly adhering husk. The husk protects the embryo from damage during grain handling and malting and provides a filter bed during lautering (Fox, 2009; Heisel, 2006). However, the amount of husk in barley should not be too high, since it contains phenol compounds that give astringent flavors to beer. Moreover, the thicker husk, the lower the malt extract. Low malt extract is also associated with high barley protein, low kernel size, and plumpness (Heisel, 2006).

3. Malting process

Malt is grain that has been germinated under controlled conditions up to a specific endpoint and then kiln-dried. The purpose of malting is to develop enzymes which break down the cell walls and the matrix surrounding the starch granules, partially converting storage substances (starch, protein) into simpler, soluble materials. These materials form the extract of wort. β -glucans are split into glucose, and high-molecular proteins are split into polypeptides, peptides, and amino acids. Starch is split into dextrins and maltose. Barley grains contain enzymes that are responsible for catalyzing the degradation of storage materials such as starches and proteins. However, most of these enzymes are inactive in the dormant barley grain. During germination, some enzymes are activated (such as β -amylase) and other new enzymes are formed (e.g., α -amylase).

The process of breaking down the cell walls and the protein and gum matrix that encases the starch granules softens the kernels, giving them a mealy texture. This transformation is referred to as modification. The next steps involved in the malting process are steeping in water, germination, and kilning, which is the drying phase. Kiln-drying green malt denatures many of the enzymes, but some remain, including those necessary for converting starch into fermentable sugars. The enzymatic activity developed in malt must be maintained until the brewer is prepared to use the malted grain in the brewing process. A large variety of malt types are available, differing in color and flavor. Pale (base) malts are the main source of enzymes. Darker malts (amber, crystal, chocolate, and black) are applied to supplement the pale malt and produce various beer styles.

Although malting may seem easy to perform, it is in reality a complex process. There are specific conditions that have to be met and many parameters need to be controlled to produce malt with the desired characteristics. This requires from maltsters a certain level of expertise, since the characteristics of barley and other grains change depending on the variety, soil, or weather conditions.

Plump and clean kernels with high germinative energy and adequate levels of protein are required for malting. At the start of the malting process, the grain is steeped under controlled conditions to induce biochemical and physical reactions. The growth of rootlets and the acrospire occurs alongside

the generation of enzymes within the grain. The grains are immersed in water until they increase their water content from less than 15% to 40%–43% in the case of pale grains or 44%–47% for dark malts (Palmer, 2018; Rani & Bhardwaj, 2021). After several hours, the water is drained, exposing the grains to air. Following an air rest, the grains are put back into water for another steeping cycle. Air rest periods are crucial for supplying oxygen and removing carbon dioxide, the end product of respiration. With higher water uptake, the intensity of grain respiration increases, which increases demand for oxygen. Typically, this cycle of submerging the grains in water and then allowing them to rest in the air is repeated twice.

The grain is then transferred to germination compartments. Air is supplied continuously, and the appropriate temperature and moisture levels are maintained to allow for the synthesis of enzymes. Too high temperature will result in lower enzyme formation and an excessive loss of extract, due to intensive respiration and rootlet growth. Enzyme formation can also drop due to insufficient air or water. The germinating barley is turned frequently to ensure the same conditions (temperature, humidity, oxygen access) for each grain in the bed and prevent the rootlets from becoming matted. With the aid of the enzymes, the cell walls are degraded completely and the proteins in the endosperm are modified, making the grain friable. This can be observed toward the end of germination, by rubbing the grain between finger and thumb. Unfortunately, modification proceeds at different rates at different points in the endosperm. The regions near the embryo degrade first, while the regions at the tip of the grain are the last to be degraded. This can result in differences in the degree of modification, as well as leading to the presence of unmodified material in the grain. Breaking down the cell walls and the matrix of proteins and gums that surround the starch granules results in the starch becoming more accessible for enzymatic action. This exposure is crucial for the subsequent stages in the brewhouse, where the enzymes will further degrade the starches into fermentable sugars, which are essential for the fermentation process.

As more enzymes are formed and grain modification progresses, storage substances in the grain are converted into rootlet/acrospire tissue and are utilized to provide energy for metabolic processes. This provokes malting losses and the reduction in the final malt yield. Therefore, it is essential to optimize the combination of germination parameters based on the type of malt being produced. In the case of pale malt, germination requires 3–4 days at about 14–18°C (Kleinwächter et al., 2012; Narziß et al., 2023). Colored malts require longer time and slightly higher temperature.

The barley is dried to suppress growth, make the grains stable during storage, and confer the appropriate malty flavor and color. However, care must be taken to protect enzymes from denaturation and prevent gelatinization of the starch. Gelatinization is caused when the structure of the starch granules is disrupted. If the temperature is raised too rapidly when the moisture content of the grains is too high, it can result in the formation of steely, glass-like grains. Therefore, at the beginning of kilning the temperature should be kept between 50–65°C. To obtain low-color malts, when the moisture decreases to about 12%, the temperature during kilning should be increased to about 82°C (Narziß et al., 2023; Rani & Bhardwaj, 2021).

During drying, the water content in the grain is reduced from over 40% to less than 5%. Kilning involves circulating large volumes of air at elevated temperatures through the bed of germinated grains. When pale malt is obtained, the maltsters try to avoid the formation of Maillard reaction products and their intermediates (nonenzymatic browning), since they make the beer darker. They are also precursors to stale off-flavors occurring during storage. Maillard compounds are produced during

malt kilning or roasting at elevated temperature, in complex and diverse reactions between reducing sugars and the amino groups found in amino acids, as well as in peptides and proteins.

Light-colored malts such as pale ale malt, pilsner malt, and malted wheat are dried at temperatures of up to 80–95°C. Munich malt is typically cured at 100–105°C for 5–6 h (Narziß et al., 2023). Aromatic malt is typically cured at higher temperature, up to 115°C. Light malts typically form the foundation or primary component of the mixture of grains (known as the grain bill) used in brewing. They facilitate enzymatic hydrolysis during mashing and provide a high saccharification yield.

Before the malt is transferred to storage silos for aging, the rootlets are removed. The whole process of malting takes about 1 week. The malt must be aged in silos for a few weeks prior to brewing (Narziß et al., 2023).

4. Special malts

Small amounts of special malts are used in grist to give specific aromas, flavors, and colors. Special malts are produced by selecting suitable barley types and varying the malting conditions. The production of colored malts requires barley with higher protein content (Narziß et al., 2023). This is because, during germination, a more intense process of hydrolysis is needed to break down complex molecules to simple sugars and amino acids. These simpler compounds then serve as the substrates in Maillard reactions. Some specialty malts, such as Vienna and Munich malts, can still be used as base malts. Vienna malt can be used as a base for golden ales or Vienna lagers, while Munich malt can be used for dark Dunkel lagers. The color of Munich malt ranges from 15 to 25 EBC (European Brewery Convention) units, which puts it in the darker spectrum of base malts used in brewing (Kreis, 2009). Vienna malt is lighter, since both its level of modification and kilning temperature are lower. Like pale malt, Munich and Vienna malts possess enzymatic activity. The enzymatic activity and extract yields of colored malts decrease with more intense heat treatment.

For caramel malts (Carapils, CaraMunich, Crystal), higher temperatures are employed during the last 2 days of germination. Caramel malts are kilned at temperatures ranging from 80°C to 180°C (Kreis, 2009). They produce malty beer and are available in different colors. Crystal malt is named for the crystalline structure of its endosperm. It is made by drying the liquefied endosperm in a kiln or special roasting drum. Crystal malt is produced from high-nitrogen barley that is over-modified and roasted in high moisture without ventilation. The temperature is raised to 150°C, which leads to denaturation of the enzymes. Crystal malts do not require mashing, since during roasting the starch is degraded to sugars. Crystal malts can only be steeped in hot water. Their golden to deep red color has high coloring potential (50–320 EBC units) (Wang et al., 2022).

Roasted malts (Chocolate, Black) are obtained from well-modified, prekilned pale malt, which is conditioned (5%–10% increase in moisture) then heated to 180–220°C in a roasting drum (Kreis, 2009). Roasted malts have strong coloring potential and give a burnt flavor. They can only replace about 1%–2% of the basic malt, since they can give the beer a sharp bitterness (Narziß et al., 2023). To retain their aromas, colored malts should be used when they are as fresh as possible. Storage time should be minimized to retain their aromas (Briggs et al., 2004).

There are many types of special malts, with a wide range of colors. In addition, maltsters now offer organic malts, malts made from heritage barley varieties, and traditional barley varieties grown in a defined area (terroir malts), as well as blends of malts tailored to brewers' requirements (Evans et al.,

2024; Stewart et al., 2023). However, it is simpler for brewers to make beer from malt produced from one barley variety (not a mixture of different varieties) and from homogenous malt batches rather than mixtures of under- and over-modified malts.

5. Other malted grains

Apart from barley, other grains, both cereals and pseudocereals, can also be malted and used for brewing. Wheat malts are the most commonly applied. Wheat malt is used for wheat beers, including Weissbier and Witbier. It can also replace barley malt in other types of beer. Wheat malts produce wort with higher extract content compared with barley malts, since wheat and so wheat malt does not possess a husk. However, when wheat malts constitute high percentage of the grist, it can cause a problem with lautering.

6. Malt quality

There are many parameters that describe the properties and consequently the processability of malt and lastly beer chemical and sensory characteristics as well as its colloidal and flavor stability. The quality of malt may be evaluated by assessing the appearance of the grains, their mechanical, physical, and chemical properties as well as the resulting wort. Protocols for assessing worts have been established by the European Brewery Convention (EBC), the American Society of Brewing Chemists (ASBC), and the Institute of Brewing (IoB). Different malts, such as pale malt and caramel or roasted malts, have specific standards and characteristics that define them. Brewers' requirements for malt are determined by the style of beer produced, the use of other sugar sources in brewing, or the production facilities available. Malt is a critical factor in the productions of good beer, in particular when craft brewers use a high proportion of unmalted materials, without exogenous enzymes. The enzymatic activity of malt must ensure all the starch present in the mash is hydrolyzed.

Other parameters are important for the assessment of pale and colored malt, not all of which are equally important. In the case of pale malt, numerous attributes are considered, whereas for colored malts solely color, moisture, and extract are typically measured. Brewers should determine which malt attributes are crucial to achieve a specific beer profile. Based on these requirements, they should specify the desired malt quality, in terms of a range of results for certain malt parameters.

The key parameters listed in almost all malt specifications are: sieving (kernel size fraction or grading >2.5 mm); foreign matter content (screenings <2.2 mm); moisture, extract content; extract difference (fine/coarse grind extract difference or F/C difference); wort viscosity; wort color; total protein or total nitrogen (TN) content; soluble protein or soluble nitrogen (SN) content, and Kolbach index (KI). Additional parameters that can be specified include friability, wholly and partly unmodified grains, hectoliter (volumetric) weight, 1000 kernel weight, hot water extract (HWE), saccharification time/rate, diastatic power (DP), α -amylase activity, β -glucanase activity, wort pH, boiled wort color, the content of free amino nitrogen (FAN), β -glucan content, and dimethyl sulfide precursors (DMS-precursors, DMS-P), wort fermentability, N-Nitrosodimethylamine (NDMA) content, and barley variety. However, many attributes give mutually subsidiary information on malt characteristics. There is, for example, a positive correlation between the amount of glassy corns, the wort viscosity, and the β -glucan content in wort and mash filterability. There is also a positive correlation between

saccharification time and enzymatic activity in malt, in particular α -amylase activity. Therefore, to evaluate malt quality, it is not necessary to examine an endless list of malt attributes. It is more necessary to understand their relevancy and interrelationships.

Generally, the malt used for brewing should be clean, with a pleasant, clean smell and flavor, uncontaminated with mold, free of foreign matter, and without broken kernels. The malt should not contain mycotoxins, pesticides, heavy metals, or NDMA, since nitrosamines are probable human carcinogens. Quality control methods for malt can be found in specialized publications, such as *Analytica EBC*, *MEBAK*, and *Analytica ASBC*.

The use of low-quality malt (frequently under-modified and/or inhomogenous) can lead to many problems, such as failure to recover the assumed extract in the brewhouse, slow wort separation, cloudy wort, excessive break formation in the boil, prolonged brewing, low fermentability (owing to too few fermentable sugars and/or FAN), inconsistent fermentation, short filter runs, excessive consumption of filter aids, and finally haze and off-flavors in beer. All these issues increase costs and disrupt production. The Malt Certificate of Analysis (MCA) issued by maltsters provides valuable guidelines for avoiding processing issues. Therefore, carefully considering essential malt parameters, as detailed below, is crucial for successful brewing.

6.1 Sieving test

The sieving test evaluates the uniformity in kernel size within a batch of malt. It specifically measures the percentage of plump kernels, defined as those with a diameter exceeding 2.5 mm, present in the lot. Plumper grains typically yield more extract; therefore, higher percentages in the sieving test results suggest a better extract yield.

6.2 Moisture

The moisture content (MC) in malt should not exceed 5% (Narziß et al., 2023). If the grains have a higher moisture level, brewers end up purchasing more water and less dry matter, which is not desirable. Excess moisture can also lead to mold growth on the grains, potentially spoiling the beer's flavor and producing mycotoxins that are harmful to human health. Moreover, malt contaminated with *Fusarium* mold can cause uncontrolled foaming in the beer, a phenomenon known as gushing.

6.3 Extract of malt

The malt extract (ME) value determines its potential for producing wort solubles in a standardized mashing program. Different brewing organizations, such as the European Brewery Convention (EBC), the American Society of Brewing Chemists (ASBC), and the Institute of Brewing and Distilling (IoB), each have their own protocols for mashing, measuring the extract, and expressing the results. Despite the differences in detailed methodologies among these organizations, the process utilized to determine the malt extract value is collectively referred to as the “standard mashing procedure.” This procedure provides a consistent basis for assessing the quality and efficiency of malt in the brewing process.

The main soluble compounds in wort are fermentable sugars, including maltose but also small dextrins, nitrogenous materials, tannins, vitamins, and trace elements. The wort produced in a standard mashing procedure is analyzed to measure saccharification time, speed of filtration, pH, color, and

viscosity, SN content, and FAN content. The extract value should be as high as possible. For example, for pilsner malt, it should preferably not be less than 80% d.w (Narziß et al., 2023). The malt extract results are expressed as the percentage of extract in dry matter, after excluding the malt's water content or are given on an "as sampled" or wet basis. A low extract yield indicates under- or over-modified malt, although it can also result from small kernels. Smaller kernels contain less starch and more husk.

6.4 Hot water extract of malt

Hot water extract (HWE) is also a measure of the malt's potential for producing wort solubles. However, it is measured using a constant temperature infusion mashing process with coarsely ground grains. This procedure can also be applied for ancillary analyses, including of saccharification time/rate, color, pH, viscosity of wort, fermentability, SN content, and FAN content.

6.5 Extract difference

Extract difference (F/C difference) shows the difference in extract content in wort when finely ground or coarsely ground malt are used. The extract difference should be low.

6.6 Total nitrogen or total protein

The total nitrogen (TN) level in malt is often presented as total protein content (TPC). This value is calculated by multiplying the value of TN by 6.25. The TN value indicates the quantity of nitrogenous compounds, such as proteins, polypeptides, peptides, and amino acids in grains. The amounts of nitrogenous compounds in barley and malt obtained from a particular barley are similar, since the barley determines the level of nitrogenous materials in the malt. Brewers prefer not to use malt with high protein content, since the higher the amount of protein the lower the amount of starch in the grain, with a lower extract yield as a result. Therefore, TPC should be in the range of 10%–11% d.w (Narziß et al., 2023).

6.7 Soluble nitrogen or soluble protein

The majority of proteins in barley are insoluble. They need to be degraded during malting to then be dissolved during mashing. The SN content describes the proportion of nitrogenous materials from malt that become soluble and can be measured in the wort. The amounts of nitrogenous materials in wort should be sufficient to ensure suitable beer foam stability and palate fullness, but without causing colloidal instability in the beer. The SN content in malt should usually be in the range of 0.64%–0.70% d.w (Narziß et al., 2023).

6.8 Kolbach index

The Kolbach index (KI) is the ratio of the quantity of the soluble protein to the total protein. It indicates the degree of grain modification. The KI value should be in the range of 35–45 KI, preferably from 38 to 42 KI since over-modification leads to lower beer foam stability (Narziß et al., 2023).

6.9 Free amino nitrogen

Free amino nitrogen (FAN) composes a proportion of the SN. The FAN content of malt gives an estimate of the content of amino acids, ammonia, and terminal nitrogen groups of peptides and proteins. These compounds are necessary for yeast to grow at the start of fermentation, as they are the sources of nitrogen in wort. However, excessive FAN in wort is also undesirable, since it often causes poor beer foam stability, defects in flavor due to excessive amounts of fermentation by-products, and increases in color, as well as leading to watery and thin-bodied beer.

6.10 Color

The color specified in malt specifications relates to the color of wort produced under laboratory conditions using a standardized mashing method. This serves as an approximate indicator of the sweet wort's color. Malt color is quantified in European Brewery Convention (EBC) units or Standard Reference Method (SRM) units. To approximately convert a color value from EBC to SRM, you divide the EBC value by roughly 2 (1.97).

6.11 Wort viscosity

High wort viscosity essentially results from poor malt modification and high β -glucan content. High β -glucan can cause problems with wort separation, extract recovery, and beer filtration, often requiring large quantities of filter aids. Corn, rice, and syrups used for brewing mitigate problems induced by the high β -glucan content in malt.

6.12 Diastatic power

Diastatic power (DP) measures the combined activity of α - and β -amylase in malt. It is expressed in Windish Kolbach units (WK units) or Lintner degrees ($^{\circ}$ Lintner), where one Lintner degree equals the sum of the value of DP in WK units plus 16, all divided by 3.5.

6.13 Dimethyl sulfide precursors (DMS-precursors, DMS-P)

Dimethyl sulfide (DMS) is formed from its precursor, S-methyl-methionine, during thermal treatment of malt, mash, and wort. It imparts an undesirable cooked sweetcorn flavor to beer. This problem does not concern colored malt, since intense thermal treatment removes DMS-precursors. The amount of DMS-precursors should usually be lower than 5–7 ppm (Narziß et al., 2023).

6.14 Friability of malt

Friability or malt fragility relates to the degree of malt modification. Higher values, above 90%, reflect well-modified, friable, easy-to-mill grains. Friability is measured using a friabilimeter, in which the ground kernels are pressed against a sieve. The material that falls through the mesh is weighed. The distribution of ground malt particles can be further analyzed, providing data on the percentages of partially glassy grains and completely glassy (ungerminated) grains. Friability is a good indicator of lautering performance.

6.15 Glassiness or mealiness

The glassiness of a malted grain—or, conversely, its mealiness—characterizes the degree of endosperm modification. Electron microscopy can be used to examine the endosperm structure, although traditionally it is assessed by cutting the grains transversally. Malt with high glassiness can lead to challenges during processing and lower extract yields in the brewhouse.

7. Adjuncts: sources of extract other than malt

Innovations in craft brewing are often related to the rediscovery of ancient grains or the use of new blends of grains, often in unmalted forms. While barley malt remains the primary source of sugars for beer fermentation, it may be partly replaced with other ingredients, depending on the region and consumer preferences. The most adjuncts are employed in regions of the world where climatic conditions are unsuitable for barley growing and malting. The malt in the grist is substituted mainly with indigenous cereals, such as sorghum in Africa, rice in Asia, and corn in America. In Europe, barley and corn are the most common adjuncts. However, according to the German Purity Law (Reinheitsgebot) sugar and sources of starch other than malt are prohibited for lagers.

In large-scale breweries, adjuncts are mainly used to lower the cost of beer production. Materials such as sugar syrups or solid sugar are widely used to produce high-gravity worts. They increase wort gravity easily and provide wort and beer with a consistent composition.

Novel ingredients provide opportunities for creating unique beer flavors. Altering the sugar and amino acid spectra in the wort changes the flavor profile of beer. The substitution of malt with unmalted materials usually improves the colloidal stability of beers, in particular when syrups or sugar are used, or grains low in proteins such as corn or rice. Due to the lower SN levels in these adjuncts, there is also a favorable decrease in flavor staling precursors in beer. Compared with malt, adjuncts cause fewer Maillard reaction products and Strecker aldehydes, which are generated during malt kilning. Moreover, the replacing malt with adjuncts makes beer lighter, which is usually beneficial. Brewing with unmalted materials can be more sustainable, as reducing the carbon footprint of beer production by avoiding the need for steeping, germination, and above all kilning of malt. Finally, adjuncts associated with a particular region can strengthen its bond to the “*terroir*,” adding to its distinctiveness.

Unmalted raw materials can be divided based on their state, into solids and liquids. Cereals such as barley, corn, rice, wheat, oats, sorghum, rye, triticale, pseudocereals (buckwheat, quinoa, amaranth), teff, cassava, and granulated sugar (sucrose) are solids. With the exception of granulated sugar, they are called mash vessel adjuncts since they require being hydrolyzed in the mashing process with malt or exogenous microbial enzymes. Raw grains lack wholly or predominantly hydrolytic enzymes. Their structures are compact, which hinders enzyme action. Liquid adjuncts include hydrolyzed starch syrups and syrups derived from sugar cane or sugar beet (wort extenders), as well as malt extracts and syrups obtained from hydrolyzed cereals (wort replacements). Malt extracts, both unhopped and hopped, are popular among home brewers and micro-breweries. Similarly to brewing wort, they contain carbohydrates as well as nitrogenous compounds and minerals. Liquid adjuncts and granulated sugar are known as copper or kettle adjuncts. They consist of soluble sugars, which mean they can be added to boiling finished wort. Their use is also a good way to adjust original gravity.

Starch-derived syrups, which are widely used by large-scale breweries, are produced by enzymatic and/or acid hydrolysis of starch from corn, wheat, or rice. Depending on the degree of starch hydrolysis, they can have different compositions of fermentable (glucose, maltose, maltotriose) and nonfermentable (maltotetraose and higher oligosaccharides) sugars. Brewers prefer maltose-rich syrups, wishing to have a sugar composition in their wort similar to that of malt worts. Syrups are classified according to the number of monomers in the sugar chain or their dextrose equivalent value (DE value), which relates to the percentage content of reducing sugars, calculated as glucose. Syrups contain more than 70% solids. They are supplied and stored hot, to prevent crystallization and to make the liquid easier to pump. Syrups should be applied as late as possible in the brewhouse to reduce losses and save energy required for heating and transferring. Like cereals that are low in protein (e.g., corn or rice), syrups can worsen beer foam stability and body and reduce color.

The change in the sugar composition of wort caused by the application of adjuncts can affect yeast growth and fermentation performance. A high concentration of glucose or sucrose in wort, resulting from the use of glucose syrup or granulated sugar, can stunt or even suppress maltose and maltotriose utilization during fermentation. This should be taken into consideration, in particular when stored yeasts are used to ferment wort (Gibson, 2011). The lower yeast growth and fermentation performance observed in worts produced with high quantities of syrups or granulated sugar can also result from raised osmotic pressure and the dilution of essential nutrients, such as amino acids, fatty acids, minerals, or vitamins. Low concentrations of amino acids and small peptides in worts produced with adjuncts can alter the flavor of beer, resulting in a much more delicate taste (Bogdan & Kordialik-Bogacka, 2017). Rice, for example, is said to give beer a dry, light, neutral, and clean flavor (Pliansrithong et al., 2013).

The distinct gelatinization temperatures of starches from different cereals (Table 2.1) cause diverse using of mash vessel adjuncts. Wheat flours and materials precooked outside the brewery, such as flaked maize, wheat, or barley, as well as micronized and torrefied whole grains, can be applied directly to the grist. Maize, rice, and sorghum grits and flours as well as refined starches from these materials require cooking during mashing in the brewery (Meussdoerffer & Zarnkow, 2009). If the starch gelatinizes at temperatures higher than those at which malt enzymes are active, the adjunct must be gelatinized before being mixed with the malt mash. Gelatinization can perform in the mash vessel

Table 2.1 Chemical composition (in % dry matter) and gelatinization temperature for barley, corn, and rice.

Raw material	Starch	Protein	Hemicelluloses	Lipids	Ash	Gelatinization temperature
	[% d.w.]					[°C]
Barley	62.0	10.6–11.8	10.3	2.5	3.1	72–76 (58–62 malt)
Corn	71.8	9.2–10.3	4.2	5.8	1.2	75–82
Rice	75.0	7.4–9.0	2.3	1.5	0.4	65–90

From Annemüller, G., & Manger, H. J. (2013). Processing of various adjuncts in beer production: Raw grain adjuncts-sugars and sugar syrups-malt substitutes. Berlin: VLB.

before mashing the malt grits or in a cereal cooker. It is recommended to apply exogenous α -amylase with cooked grits (in particular rice). It is not recommended to replace more than 30%–40% of malt with corn or rice grits, unless enzyme preparations, in particular α -amylase, are used during mashing (Annemüller & Manger, 2013; Cooper et al., 2016). The use of exogenous enzymes can also be beneficial with other unmalted cereals. The diversity of gelatinization temperatures for rice starch, depending on the variety of rice and its growing conditions, makes using rice problematic for brewers. The rice used in brewing is dehusked and degermed. In the case of corn, germs rich in lipids are removed. When large amounts of rice or corn are used in the grist in a brewhouse with a lauter tun, it can be challenging to obtain a clear wort. Similar problems can occur when processing wheat, which is high in pentosans and tends to increase viscosity (Meussdoerffer & Zarnkow, 2009). To alleviate these problems, coarse grinding of wheat is recommended (Faltermaier et al., 2014). Generally, when large amounts of malt are replaced with cereal adjuncts, mash filters are more suitable than a lauter tun.

Raw barley is cheaper than barley malt and produces beer with similar properties. However, unmalted barley is more difficult to handle. Raw barley is tough, and therefore grinding requires much more energy than in the case of malt. Moreover, to avoid lautering and filtration problems, with larger amounts of barley adjuncts it is often necessary to add microbial β -glucanase. However, with low quantities of raw barley adjuncts, not exceeding 10%–20%, the enzymatic activity of barley malt is sufficient.

It is recommended to apply oats in only small quantities, unless β -glucanase is added (Annemüller & Manger, 2013). This is connected with the high concentration of β -glucans in oats (Kordialik-Bogacka et al., 2014; Schnitzenbaumer & Arendt, 2014). Good quality wort, with up to 20% of oats, can be produced without exogenous enzymes. Acceptable beers can be obtained with even up to 40% oats and without microbial enzymes, but on condition that the oats are hammer-milled (Schnitzenbaumer et al., 2012). However, this reduces the filtration/lautering rate significantly. Oats can be applied with or without the husk. The oat kernel size is much lower than barley, requiring them to be ground at a lower setting in roll mills. Oats are rich in lipids, which can result in poor beer foam stability and rancid off-flavors. Compounding the problem, the lipids are distributed throughout the whole kernel, not only in the germ as it is in corn.

Due to growing numbers of celiacs, craft brewers have become interested in manufacturing of gluten-free beers (Baiano, 2021; Fanari et al., 2017). Gluten-free cereals (sorghum, rice, maize, teff, millet) and pseudocereals (buckwheat, quinoa) can be used to produce beers without any trace of gluten. However, in most countries, the legislative threshold for gluten content in the final beer is below 20 ppm (EC No. 41/2009) (Benucci et al., 2020; Yang & Gao, 2022). Therefore, gluten-free beers can also be obtained from cereals containing gluten. This involves precipitation and enzymatic hydrolysis of gluten proteins and peptides by peptidases isolated from germinated cereals and fungi and/or by applying lactic acid bacteria (Fanari et al., 2018).

8. Water

The impact of water quality on beer is not usually given much consideration. However, the chemical composition and characteristics of the water used are important for achieving the traits of given beer style and brand. Water is, after all, the main constituent of beer, accounting for about 90% of the final product.

Once, the availability and quality of raw water decided the location of breweries and the development of regional beer styles. With the emergence of efficient and reliable water treatment technologies, breweries do not rely solely on the quality of the local raw water. After treatment, water sources previously considered unsuitable for brewing can be used. However, water treatment constitutes an extra cost.

Small breweries tend to use the municipal supply. Larger breweries draw underground water from wells and springs. The water should be drawn from a suitable depth to avoid contamination. Surface water like rivers and lakes can also be used, but its quality may vary and it is susceptible to microbiological contamination. Surface water is also usually low in minerals, which means lower hardness. The origin of the water significantly affects its mineralization, which in turn influences the brewing process and the beer's final taste.

Municipal water usually contains free chlorine, as a consequence of water chlorination by Cl_2 or hypochlorite. Since it is an oxidizing agent, chlorine should not be present in the brew water. Even in the early stages of the technological process, oxidation can have an adverse influence on beer shelf-life. Chlorine from brew water can also react with wort and beer ingredients, generating chlorophenols of a medicinal off-flavor, even at extremely low concentrations. When free chlorine reacts with organic materials in the water, trihalomethanes (THMs) are formed. Not only do THMs impart off-flavors to beer, but they are also carcinogenic. To mitigate the risk of THM formation and ensure the safety and quality of the beer, the use of activated carbon filters is advised. The concentration of THMs should be below 10 ppb (Eumann & Schildbach, 2012).

A brewery's water consumption depends on the brewery equipment and technological process used. It typically ranges from 4 to 10 hL of water per hectoliter (hL/hL) of beer produced (Kubule et al., 2016; Morgan et al., 2021; Olajire, 2012). Small and medium breweries consume more water per hectoliter of beer, while in large-scale breweries water consumption may be as little as 3 hL/hL (carlsberggroup). Rising global temperatures due to climate change lead to heightened water-related challenges, including issues with water availability, dependability, and quality. These challenges necessitate more extensive water treatment measures, resulting in increased water expenses. There has been growing awareness of the need to conserve water in breweries, and many water-saving measures have been implemented. Prioritizing water conservation is vital for craft breweries to protect their brand image and be recognized as an environmentally conscious part of the local community (Baiano, 2021; Morgan et al., 2021; Ness, 2018). Higher costs for water and regulatory requirements will be meaningful drivers to further reduce water consumption, not solely by using less water, but also by reducing quantities of wastewater.

Water is used not only in brewhouses but also for wort cooling, in fermentation and maturation rooms, as well as for beer filtration and packaging. Depending on the purpose, water of different quality is required. According to water standards established by the World Health Organization and European drinking water regulations, the minimum standard for water employed in a brewery is to be potable.

Water for cleaning in place (CIP) processes and the cleaning of returnable bottles and kegs, known as service water, has more rigorous requirements. As these processes are conducted at high temperature, the hardness of the service water needs to be reduced to avoid unwanted fur precipitation and consequently limited cleaning efficiency and heat transmission in heat exchangers, which causes energy losses. To avoid stainless steel corrosion, which occurs particularly at high temperatures, the chloride concentration should be limited to 50 ppm (Eumann & Schildbach, 2012). The

microbiological safety of the service water also very important, for prevention of contamination, quality assurance, equipment longevity, and regulatory compliance.

When beer is produced by high gravity brewing (HGB), dilution water is also needed. As dilution water goes directly into the final beer, it has to be completely devoid of oxygen. The levels of THMs and calcium in the dilute water must be even lower than in the brew water. Any increase in the Ca^{2+} amount in the filtered beer can increase the risk of beer gushing.

9. Brew water quality parameters

Brew liquor needs to meet physical, chemical, microbiological, and sensory standards. The most critical water quality parameters are pH, hardness, and water alkalinity.

The hardness of water is determined by the quantity of calcium and magnesium salts, as well as other cations including iron, manganese, strontium, barium, zinc, and aluminum. Calcium and magnesium are the major cations in natural water. Therefore, total hardness is usually defined as the sum of magnesium and calcium hardness, expressed as calcium carbonate (CaCO_3), in milligrams per liter. The cations in water are associated with anions such as HCO_3^- , CO_3^{2-} , OH^- , SO_4^{2-} , NO_3^- , and Cl^- .

A distinction is made between “temporary” (carbonate hardness or alkalinity) and “permanent” (noncarbonate) hardness. Temporary hardness is caused by a combination of cations (calcium or magnesium) and alkalinity ions of HCO_3^- , CO_3^{2-} , and OH^- . Bicarbonate salts [$\text{Ca}(\text{HCO}_3)_2$ and $\text{Mg}(\text{HCO}_3)_2$] and to a lesser degree carbonates (CaCO_3 and MgCO_3) are the most important. Dissolved calcium or magnesium hydrogen carbonates present in water are converted upon heating into calcium or magnesium carbonates. The carbonates of calcium and magnesium are insoluble and do not completely dissolve back into the water when it is cooled. This type of hardness is called “temporary” since it can be removed easily by boiling, giving softer water.

Permanent (noncarbonate) hardness is usually caused by the presence of calcium and magnesium sulfates and/or chlorides in the water. These become more soluble as the temperature rises, so permanent hardness cannot be simply removed by boiling. However, “permanent” hard water can be softened by other means.

As an easy measure of the suitability of water for beer production, Paul Kolbach elaborated the concept of residual alkalinity. This concept combines in a single term the two critical determinants of pH in water—that is, total alkalinity, resulting mainly from the presence of bicarbonates, and the content of calcium and magnesium. Brew water with a negative residual alkalinity value will cause a beneficial decrease in the pH of the mash, whereas positive values have the adverse effect of increasing the pH, compared with the use of distilled water. Brew water should contain calcium, but not in the form of bicarbonates. Bicarbonates are undesirable since they increase the pH. The ratio of carbonate to noncarbonate hardness in brew water is fundamental. Nonetheless, the amount of other calcium and magnesium salts should also be restricted to some extent.

The minerals in water affect many parameters, such as mash and wort pH, enzyme activity, hop utilization, yeast metabolism and flocculation, and ultimately the sensory properties of beer. However, the majority of minerals present in beer originate from the barley malt. Only a quarter come from the water. Cereals such as wheat, corn, rice, rye, and oats, which are sometimes used to partly replace malt, usually contribute fewer minerals than malt ([Gonzalez-Salitre et al., 2023](#)).

Calcium is an essential component of brew water. Not only does it contribute to lower the mash pH and promote enzymatic activity during mashing, in particular α -amylase; it also promotes protein coagulation and yeast flocculation. Too high calcium in wort can, however, inhibit magnesium uptake by yeast and inhibit fermentation, as magnesium and calcium ions act similarly. Magnesium salts have less of an effect on wort pH, since they are much more soluble. Magnesium is also extremely important for yeast metabolism during fermentation. While calcium is neutral in terms of flavor, magnesium enhances beer's sour, astringent flavor. Sodium, particularly in the presence of chloride ions, rounds out beer flavor and enhances sweetness. However, at higher concentrations sodium, like potassium, gives a "salty" taste. In the presence of sulfates, sodium gives beer an unpleasant harshness and bitterness (Palmer & Kaminski, 2013).

Chlorides contribute to beer fullness, mellowness, and roundedness. However, high amounts give a salty taste. The quantity of chlorides should be lower than 100 ppm to lower the risk of corrosion at elevated temperatures, in particular in the hot brew water reservoir in the brewhouse (Palmer & Kaminski, 2013). Chlorides are not related to chlorine and do not have a disinfecting effect. Nitrates are often present in surface water. This is not a problem in itself, since nitrates do not influence beer flavor. Nonetheless, the concentration of nitrates needs to be controlled, because they may reduce to nitrites, which are carcinogenic. High levels of nitrates may inhibit yeast fermentation activity. Sulfates are a safe option, since they only give beer a dry or harsh astringent flavor at high concentrations.

The concentrations of some ions, including iron, copper, and manganese, in the brew water are also restricted. These transition metals are catalysts for oxidation, which can accelerate staling during beer storage and promote irreversible beer haze. Iron ions gives beer a metallic, astringent, and bloody off-flavor and can exert a toxic effect on yeast.

Water, like malt, has to be selected appropriately for the style of beer. Carbonates are not desirable for pale beers, but good for producing darker, mellower lagers since they raise the mash pH, giving wort with a higher dextrin to maltose ratio. Water from the deep wells in Burton-upon-Trent is an example of good water for brewing full-flavored pale ales. It is high in permanent hardness, due to its high calcium and sulfate, but also has temporary hardness from its high bicarbonates.

10. Water treatment

Hard water must be treated. Carbonate hardness is traditionally removed by the addition of lime (calcium hydroxide, $\text{Ca}(\text{OH})_2$). The bicarbonate is then transformed into carbonate, which is precipitated and sedimented as CaCO_3 . Carbonate hardness can also be reduced by the addition of food grade acids (lactic, sulfuric, hydrochloric, phosphoric, or citric). In Germany, the addition of acids to decrease alkalinity is prohibited. The ionic composition of water can be changed by binding ions to positively or negatively charged sites on a resin packed in a column. This method is also applied to remove bicarbonate ions, in a process called dealkalization. Water treatment has undergone considerable changes in recent decades, with the implementation of membrane technologies, including nanofiltration and reverse osmosis.

11. Hops

Hops are regarded as the soul of beer and have been used for centuries as an additive to confer biological stability, a longer shelf life, and stable foam—but mainly to provide pleasant bitter flavor and aroma to beer. In the past, hops were added to beer primarily because of their antibacterial properties. At the end of the 20th century, as brewing and pasteurization processes were better understood and higher hygiene standards were implemented in breweries, the beer-preserving qualities of hops became less important and hops were added to beer predominantly for their bitter taste. Recently, the craft beer movement has altered the perception of hops again. Craft brewers have become interested in varieties of hops that give beer pleasant aroma (*Amarillo*, *Cascade*, and *Centennial* are some examples) (Baiano, 2021). Bitter hops, previously the most popular, have become less valued. Craft brewers have started to use up to 10 times more hops than most mass-produced brands. Craft brewers experiment with unique hop varieties and quantities to give their beers a special signature. This has encouraged hop growers to breed many new varieties. Finally, craft brewers increased demand for locally grown hops, which have become for craft brewers a successful marketing tool (Reid et al., 2020).

The top three hops producers are the United States (52,450 tons), Germany (47,900 tons), and Ethiopia (46,875 tons) (FAOSTAT, 2019). Hops used in brewing (*Humulus lupulus* L.) are grown in temperate regions between 30 and 50° North and South of the equator, with moderate and humid climate (Mozny et al., 2009). The hop plant is one of the most vulnerable crop plants to climate change (Jupa et al., 2016; Potopová et al., 2021). More frequent droughts have already decreased yields in certain cultivation areas (Potopová et al., 2021). Moreover, climate change causes in hops a reduction in the concentration of α -acids that are ultimate for beer bitterness (Donner et al., 2020).

Hops are one of three species in the Cannabaceae family. *H. lupulus* is dioecious, which means that female and male flowers are in separate plants. Only female plants are cultivated, since only they form cones. Cones consist of yellowish-green over-lapping bracts attached to an axis. Mature hops contain lupulin glands located at the base of the bracts, which are filled with lupulin, a yellowish-brown granular powder (Fig. 2.2). From the brewer's perspective, the lupulin is the most important part of the cone. Lupulin contains various chemicals that can be divided into three groups: resins, oils, and polyphenols. Bitter resins and aroma substances confer the characteristic hop aroma and flavor (Krottenthaler, 2009; Steenackers et al., 2015). The concentrations of resins, essential oils, and other chemicals depend on the variety, cultivation area (altitude, soil composition, climatic conditions), cultivation year, maturation stage, and storage conditions (Alonso-Esteban et al., 2019).

12. Resins

For brewing, the resins and essential oils are the crucial components of hops. Resins constitute 10%–30% of the dry matter of the cone (Sun et al., 2022). They include soft (9%–27.5% d.w.) and hard resins (1.5%–2.5% d.w.) (Sun et al., 2022). The soft resins are composed of α -acids, also known as humulones (humulone, cohumulone, adhumulone, and of minor importance prehumulone and posthumulone), and β -acids, known as lupulones (lupulone, colupulone, adlupulone). The typical level of α -acids in hops is from 1.3% to 12.6% d.w., although the concentration of α -acids can exceed 15% d.w. in very high bittering hop varieties (Klimczak et al., 2023). The α -acids are neither very soluble

**FIGURE 2.2**

Lupulin glands in hop cone.

Author: Żyła M.

nor very bitter, but during wort boiling, they undergo heat-catalyzed isomerization to form iso- α -acids, which are regarded as the primary agents of hop-derived beer bitterness. Depending on the pH of the wort or beer, iso- α -acids are 28–250 times more soluble than α -acids (Lafontaine & Shellhammer, 2019). The thermal conversion of each α -acid results in two epimeric iso- α -acids, differentiated as *trans*- and *cis*-iso- α -acids (Salfer et al., 2020). Iso- α -acids are therefore found in beer as *trans*- and *cis*-isomers of isohumulone, isocohumulone, and isoadhumulone. Not all α -acids isomerize to iso- α -acids during wort boiling. It is assumed that only up to 50%–60% of α -acids will be converted to isomerized form, depending on brewing conditions (Bastgen et al., 2020). Low efficiency of α -acids utilization is connected with the poor solubility of the α -acids in the wort, partial isomerization during boiling, and adsorption of the α -acids and iso- α -acids on the trub (the sediment that forms during brewing) (Jaskula et al., 2008).

Depending on the beer type, the content of iso- α -acids in beers ranges from 10 to 100 mg/L. Iso- α -acids are assumed to cause more than 85% of the bitterness of beer (Krofta et al., 2017). Humulinones, which are formed by oxidation of α -acids during the processing of hops to pellets or hops aging, are other compounds that contribute to the overall bitterness of beer. They are well soluble in water and contribute in particular to the bitterness of cold-hopped beers (Vollmer et al., 2017).

β -acids have much less bittering potential than α -acids. The β -acids are hydrophobic and insoluble in water, in particular at the low pH typical for beer (Korpelainen & Pietiläinen, 2021). Thus, pure β -acids do not contribute to the bitterness of beer (Krofta et al., 2017). However, hulupones, the

oxidation products of β -acids, are soluble in water and show distinct sensory bitterness, which accounts for $84 \pm 10\%$ of iso- α -acids (Krofta et al., 2017). The β -acids are thus transformed into bitter products by oxidation initiated by airborne oxygen during natural aging of hops and wort boiling and not through isomerization, as is the case with α -acids (Sun et al., 2022).

During hop aging, there is a decrease in the content of α -acids and β -acids. The most popular criterion for assessing hops freshness is the hop storage index (HSI). This index reflects changes in hop quality over time, where a high HSI value indicates a greater degree of aging and potential quality degradation. As hops age, the levels of essential acids tend to diminish, while compounds resulting from oxidative degradation increase. These changes can affect the flavor, aroma, and bitterness contribution of the hops to the beer. The HSI value of fresh green hops is in the range of 0.20–0.25 and increases with storage time. However, different hop varieties have different initial HSI. The recommended maximum HSI for dry hopping is 0.5 (Klimczak et al., 2023). Old, badly stored materials have HSI values above 1.00 (Krofta et al., 2017).

The impact of hard resins on beer bitterness is considered minor (Klimczak et al., 2023).

13. Essential oils

Hop essential oils constitute a small part of the dried hop (0.5%–4.0% w/w) (Lafontaine & Shellhammer, 2019). They consist of hundreds of volatile compounds, which can be divided into three chemical groups: hydrocarbons (approx. 70%, mainly monoterpenes and sesquiterpenes), oxygenated compounds (approx. 30%), and sulfur-containing compounds (Mikyška et al., 2023).

Monoterpenes and sesquiterpenes contribute significantly to the raw hop aroma. Regardless of the variety, the monoterpene β -myrcene is the most abundant hop volatile. Sesquiterpenes such as α -humulene, β -caryophyllene, and, in several varieties of hops, β -farnesene are also key representatives of the hydrocarbon group.

The second group of the hop essential oils, oxygenated compounds, is made up of alcohols (linalool, geraniol, β -citronellol, nerol, α -terpineol), esters, ketones, acids, aldehydes, and epoxides (Takoi, 2019). These compounds are more hydrophilic and easier to maintain in wort and beer than hydrocarbons. They confer mostly floral and fruity aromas. The terpene alcohol linalool, a product of β -myrcene hydration, is the most abundant alcohol in most cultivars. Linalool has a very low sensory threshold and floral, lavender-type flavor. Other terpene alcohols in hops are present in smaller quantities than linalool, although some varieties are also high in geraniol (Klimczak et al., 2023). Both geraniol and β -citronellol have an impact on beer flavor, as their thresholds are low. Geraniol imparts a rose-like floral flavor, while β -citronellol contributes a lime-like citrus flavor. In addition, a synergistic effect has been demonstrated among linalool, geraniol, and β -citronellol, enhancing the overall flavor profile (Takoi, 2019). Other compounds present in hops such as aldehydes, ketones, epoxides, and lactones have been relatively little examined.

Among the volatile sulfur compounds, sulfur dioxide (SO_2), hydrogen sulfide (H_2S), thiols (R-SH), and sulfides (R-S-R) are the most abundant (Klimczak et al., 2023).

Generally, floral, fruity flavors are primarily induced by alcohols, esters, and sulfur-containing compounds. Spicy, woody, and herbal flavors are imparted by sesquiterpenes and oxygenated sesquiterpenoids. Green/grassy, herbal, and herbaceous flavors are given by aldehydes (Mikyška et al., 2023). The concentrations, combinations, and threshold levels of hop volatiles in different beer

matrices determine their sensory properties. The synergistic and antagonistic effects of the particular volatiles have an influence on the flavor of beer. Apart from the primary hop odorants in hops, there are hundreds of other minor volatiles, often with a low sensory threshold, which contribute to the complex flavor of beer.

14. Polyphenols

Apart from resins and oils, lupulin includes polyphenols, such as kaempferol, quercetin, catechins, and estrogenically active 8—prenylnaringenin, a derivative of xanthohumol (Almaguer et al., 2014; Astray et al., 2020). In beer, xanthohumol is usually present in the isomerized form of isoxanthohumol, a prenylflavanone (Korpelainen & Pietiläinen, 2021). Hop polyphenols are known for their diverse antioxidant and antimicrobial properties. They also influence beer haze and flavor, adding to the sensation of fullness on the palate. However, these compounds can introduce an astringent taste and result in a harsher and more enduring bitterness. Due to their hydrophilic nature, polyphenols are significantly extracted into both wort and beer (Klimczak et al., 2023).

15. Whole cones and hop products

Hops are a very perishable commodity. After harvesting, the cones have to be dried as quickly as possible, at a mild temperature not higher than 50°C, to decrease the moisture from 80% to about 10%. They should then be stored without light or oxygen access at a low temperature of 1–4°C. Several months after harvesting and even with proper storage, whole hop cones show distinct characteristics. Nevertheless, the hop suppliers also have in their offer fresh hops, which are considered the most unpredictable form of available hops. The addition of fresh hops gives fresh and grassy flavors as well as tropical fruit, stone fruit, earthy, floral, and herbal aromas, which are distinct each year (yaki-machief). Other problems that result from the use of whole cones include separation of spent hops from the hopped wort and considerable wort losses.

Owing to their very short shelf-life, whole and milled fresh cones are rarely used, except by breweries near hop farms. To prolong their shelf-life, whole cones are dried, milled, and compressed into pellets. Pellets are the most often used form of hops, although other hop products are also popular in large-scale breweries. Hop products have many advantages, including better stability, standardized and consistent aroma and bitterness, higher bitterness yield, the possibility of automatization, reduced transport and storage costs, and less dependency on temporal fluctuations (Gomes et al., 2022).

The granulation of hops that occurs during manufacturing Type 90 (T90) pellets preserves the most valuable substances in hops. The name Type 90 refers to the fact that from 100 kg of cones 90 kg of pellets are produced. The chemical compositions of native cones and T90 pellets differ only slightly. When using T90 pellets for hopping, the typical utilization rate of α -acids is about 30%–35%, meaning that 30%–35% of the α -acids added to the wort are converted into iso- α -acids found in the beer. In comparison, with native hop cones, the utilization rate is slightly lower, at 25%–30%. T-90 pellets are both added to the kettle to provide bitterness and hop character and to postfermentation beer to provide aroma and flavor.

Type 45 (T45) pellets are lupulin-enriched hop products. They are produced by freezing the hops at –35°C, milling, and then sieving to separate small particles like lupulin glands. The chemical

composition of T45 pellets differs from that of native hops significantly. The quantity of polyphenols per unit of α -acids is halved and the level of essential oils lower (Zheplinska et al., 2019). The yield of α -acids utilization using T45 pellets is about 10% higher than in the case of whole cones. Type 33 pellets further enriched with lupulin are also available.

Liquid or supercritical carbon dioxide or ethanol can be used to extract the soft resins and oils. Hop extracts, which can be stored for extended periods without altering their composition or sensory characteristics, are free from polyphenols. Polyphenols, by forming complexes with proteins, facilitates the clarification and stabilization of beer. Their absence in wort can also complicate the removal of hot trub. Hop extracts are commonly added to the kettle to provide bitterness to the beer, but they can also be introduced during the whirlpool phase to impart specific flavors and aromas unique to different hop varieties. Using these extracts can lead to improved α -acids utilization.

Pellets and extracts can be further processed to produce iso- α -acids. In this process, catalysts such as magnesium oxide or magnesium hydroxide are employed to α -acids into iso- α -acids in preisomerized form. Using preisomerized products can boost the α -acids utilization yield to between 45% and 80%. These products can be added in the brewhouse, which helps shorten the boiling times, or they can be introduced after fermentation (Bastgen et al., 2020). Preisomerized products exhibit greater stability during storage compared with traditional hop products. However, in Germany adhering to the German Purity Law, the use of isomerized pellets or extracts is prohibited.

Reduced isomerized hop extracts, such as rho-iso- α -acids, tetrahydro-iso- α -acids, or hexahydro-iso- α -acids, can prevent degradation of iso- α -acids by light and improve beer foam. Reduced isomerized α -acids cause a change in perceived bitterness compared with iso- α -acids. Reduced isomerized hop extracts are added to beer before the final filtration. If reduced isomerized extracts are applied to produce light stable beer bottled in clear glass, any traces of iso- α -acids must be removed to prevent the occurrence of a “skunky” off-note. This undesirable flavor is caused by 3-methyl-2-butene-1-thiol (3-MBT), a compound produced when iso- α -acids are exposed to light, leading to this specific flavor defect. 3-MBT has an extremely low sensory threshold, meaning even minute quantities can significantly impact the flavor of the beer (Schönberger & Kosteletzky, 2011).

Modern craft brewers have at their disposal a wide range of hop products, including organic hops, aged hops for brewing traditional Belgian ales, and hop blends (mixtures of different hop varieties to create a specific aroma profile). To increase the hoppy aroma of beer, commercially available pure hop aroma (PHA) extracts are often used. They impart floral, herbal, or woody flavors (Oladokun et al., 2017).

16. Hopping

Hopping can be performed at various stages of the brewing process. Depending on when the hops are added, the beer has different sensory character. This is associated with the different ways particular hop components are extracted and how they change during the brewing process. Hops added during wort boiling (kettle hopping) generally impart bitterness, since the aroma compounds evaporate and are subjected to strong oxidation. Hops with a high α -acid content are therefore added at the beginning of boiling to obtain a high level of isomerization and consequently bitterness. Adding hops toward the end of the boiling process or during the whirlpool stage, a practice known as late hopping, results in a beer with a more pronounced aroma. This method allows for the better retention of volatile hop oils in

the wort, as the shorter exposure to high temperatures prevents the extensive evaporation and oxidation of these aromatic compounds. Late hopping imparts beer spicy, herbal, noble, woody, and fruity notes, which are specific for hop molecules that have undergone oxidation reactions (Rettberg et al., 2018).

Dry hopping, conducted after fermentation, introduces aroma compounds into the beer without significantly increasing its bitterness. This method takes advantage of the fact that the isomerization of α -acids, which contributes to bitterness, requires heat—a condition absent in the postfermentation stage. As a result, the aromatic qualities of the hops can be emphasized, enhancing the beer's sensory profile with minimal impact on its bitterness. Dry hopping has become a standard method in the brewing industry, although it was started by the craft beer revolution (Villacreces et al., 2022). It is mainly associated with top-fermented beers but has also been applied in lagers (Klimczak et al., 2023).

The addition of hops in the cold room gives flavors and aromas similar to those of fresh hops, such as citrus, floral, and pine (Holt et al., 2019). The degree of extraction of particular hop volatiles during dry hopping depends mostly on the solubility of specific volatiles. The compounds from the hydrocarbon fraction (monoterpenes and sesquiterpenes) are normally less extracted into the beer than the oxygenated compounds. Their concentrations are usually below their detection thresholds. The oxygenated derivatives of monoterpenes (e.g., linalool, geraniol, nerol, geranial, and neral) and sesquiterpenes (e.g., humulene epoxides and caryophyllene oxide) can be present in beer at concentrations at or above their detection thresholds (Lafontaine & Shellhammer, 2019; Rettberg et al., 2018). Therefore, the content of monoterpenes and sesquiterpenes in hops is probably more important if hops are added at the beginning of wort boiling, which gives time for their oxygenated derivatives to form. Identifying the contributions of specific hop volatiles to beer aroma is not easy, since these compounds undergo a number of biotransformations during fermentation (Mikyška et al., 2023).

17. Varieties of hops

Brewers have traditionally employed a rather narrow range of hop types. Many European cultivars derive from old, closely related varieties and have very little variation in terms of their genetical, morphological, or ecological profiles (Yan et al., 2019). However, with the boom in craft brewing, there has been increasing interest in experimenting with new varieties of hops.

Particular varieties of hops contain specific aromatic compounds in unique concentrations. Hop varieties are also strongly associated with certain geographical regions. Hops may be classified as American, European, German, Czech, Polish, British, or other countries, depending on their origin. Famous aroma varieties such as *Hallertau*, *Hersbrucker*, or *Saaz* (*Zatec*) are named after the places they are grown. Some hop components are found only in American varieties, and give a different scent compared with the European varieties (Korpelainen & Pietiläinen, 2021). Both American and European hops have similar monoterpene compositions but differ in the sesquiterpene fraction of the oil (Korpelainen & Pietiläinen, 2021). In contrast to American varieties (e.g., *Amarillo*, *Apollo*, *Cascade*, *Citra*, *Mosaic*), the concentration of geraniol in European hop varieties (*Hallertauer Tradition*, *Hallertauer Magnum*, and *Saaz*) is small (Takoi, 2019).

Historically, and especially throughout the 20th century, hops were commonly classified into two main categories based on their primary contributions to beer: aroma hops and bitter hops. However, recent developments have seen the introduction of new hop varieties that serve dual purposes. These versatile hops have been extensively bred, evaluated, and are now widely used in both craft breweries

and larger-scale brewing operations, offering both aromatic qualities and bitterness. Currently, brewers are primarily focused on producing beers with intense hop aromas. In this context, the levels and composition of hop oil volatiles and aroma precursors have become analytical indicators of hop quality.

Craft breweries also prioritize hops grown locally, especially using environmentally friendly methods (Takoi et al., 2017). The local origin of these hops can significantly contribute to the uniqueness and distinctness of the beer produced. The same hop variety grown in different geographical regions can exhibit distinct properties and characteristics (Van Holle et al., 2021). For example, *Cascade* grown in Yakima in the US has a similar chemical composition to *Cascade* farmed in Sardinia in Italy, but contains more terpenes and less esters and polyphenols than *Cascade* cultivated in the Hallertau region (De Simone et al., 2021). The aroma of dual-purpose American variety *Amarillo* grown in different locations varies considerably. *Amarillo* hops from Idaho are less citrusy and floral but more fruity, spicy, and resinous than those from the Washington area. Therefore, the geographic origin and authenticity of hop batches are crucial considerations for brewers when making purchasing decisions (De Simone et al., 2021).

Recently, craft brewers have shown interest in using hop varieties that contain seeds to create innovative products (Yan et al., 2019). Most hop plants grown today are seedless, despite the fact pollination increases yields. There has also been growing interest in the brewing potential of wild hop varieties. These ecotypes contain small quantities of bitter acids and essential oils, making them suitable for use as dual-purpose hops or for dry hopping (De Simone et al., 2021).

18. Unconventional ingredients

Innovations in craft brewing are often associated with using fruits, vegetables, herbs, spices, and other flavoring compounds to alter or enhance the sensory characteristics of the final product, creating a new or personalized beer style or adding localized characteristics (Baiano, 2021). These additions not only produce distinct flavors, but also may introduce high concentrations of bioactive compounds, which are also important for many consumers (Ducruet et al., 2017; Villacreces et al., 2022).

Whole fruits and fruit juices have been among the most widely used and longstanding additions in commercial beer products (Nardini & Garaguso, 2020). In Belgian lambic-style sour beers, whole tart cherries (*Prunus cerasus* L.) are traditionally applied during the maturation of Kriek, and raspberries (*Rubus idaeus* L.) are added for Framboise beers (Leni et al., 2023). Many specialty beers are now manufactured using fruit or fruit juices. An example is Italian Grape Ale (IGA), which contains up to 40% grapes.

Before hops started to be widely used for brewing, fruit beers were commonly manufactured with various herbs and spices, such as bog myrtle (*Myrica gale*), marsh rosemary (*Limonium californicum*), laurel berries (*Laurus nobilis* L.), laserwort (*Laserpitium siler*), anise (*Pimpinella anisum*), caraway (*Carum carvi*), juniper (*Juniperus communis*), St. John's wort (*Hypericum perforatum*), wormwood (*Artemisia absinthium*), or yarrow (*Achillea millefolium*), in various proportions (Klimczak et al., 2024; Verberg, 2018). Modern craft brewers take advantage of this tradition and have renewed interest in the large variety of botanicals that can partially or completely replace hops. Aniseed, bay leaf, boxwood, chamomile, cinnamon, coriander, elderberry ginger, juniper, lavender, lemongrass, nutmeg, peppermint, pepper, rose hips, rosemary, sage, thyme, verbena, wormwood and alder leaves, as well as

beet seed, honey, oak bark, lemon or orange peel, spruce buds and needles, and many other additives are currently used for brewing. However, the wide commercial use of fruits, vegetables, herbs, and spices for beer production has only recently received commensurate attention from researchers. This increased interest has led to numerous studies exploring the use of unusual ingredients in the brewing process, such as chestnuts, cocoa, cornelian cherries, hawthorn fruit, persimmon, mango leaves and mango seed, soursop, orange peel, coriander, carqueja, artichoke, olive leaves, eggplant (cv. Classic), peel extract, purple sweet potato, and herbal extracts such as linden, plantain, sage, chamomile (Borşa et al., 2022; Kordialik-Bogacka, 2022).

19. Conclusion

Each ingredient has a unique role in manufacturing a beer. By understanding its contribution, craft brewers can fully benefit from the potential of the ingredients and create products that are innovative, sustainable, authentic, and high quality, thereby meeting the preferences and expectations of their consumers.

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Brewing of craft beer

3

Antonietta Baiano

University of Foggia, Department of Agricultural Sciences, Food, Natural Resources, and Engineering, Foggia, Italy

1. Introduction

In an increasingly globalized world, populated by standardized and industrialized products, consumers are looking for unique and high-quality products. The craft beer revolution arrived at the end of a dramatic transformation in the beer industry. Industrial brewers chose beer characteristics that could meet the liking of as many consumers as possible, resulting in more homogeneous and milder lager beer. Microbrewers filled market niches left by the standardized industrial beers. “Craft brewery” and “artisanal brewery” are terms used to identify breweries that brewing different types of beer on a small scale, distinguishing them from larger breweries mass-producing beers. Craft beers are characterized by original sensory features, diversity of beer styles, and a higher quality than commercial beers, thanks to the introduction of novel combinations of raw or malted grains, new varieties of hops, new raw materials to modify the beer flavors and taste. The typical properties of craft beers also derive from the production process, which does not include technological operations used in the industry to stabilize the products such as pasteurization and filtration but comprises operations, such as refermentation in bottle, not included in the industrial brewing.

Craft beers are complex combinations of different types of conventional (malted barley, hops, and yeast) and new ingredients (fruits, spices, herbs, etc.) produced using traditional and new brewing techniques, and resulting in a wide range of styles differing also for their geographical origin. As an example, the [Brewers Association \(2023\)](#) classifies beer styles in three main categories, namely ale beer styles, lager beer style, and hybrid/mixed beer styles, which are further subcategorized by geographical origins providing a reference guide also for craft brewers. In-depth information on beer styles is provided in Chapter 4. It is not always easy to fit a beer into a pre-existing beer style since the beer style guidelines try to reflect, as far as possible, historical significance, authenticity, or a high-quality profile and, instead, current commercial examples represent a modern interpretation of the style. This is also the reason why beers belonging to the same beer style can show different physical, chemical, and sensory characteristic to indicate that there are the results of complex interactions among type of yeast/fermentation, raw materials, brewing duration, and adjuncts. Each beer style includes a description of relative intensity of attributes such as color, clarity (haze presence/absence), perceived malt and hop flavors, perceived bitterness, body, fermentation characteristics, and any additional notes together with numerical data that objectively quantify original and final gravity of

wort (as °Plato), alcohol content (% by weight and volume, bitterness (as International Bitterness Unit [IBU]), and color (according to the Standard Reference Method [SRM] and the European Brewing Convention [EBC]). These numerical values do not always coincide with the individual perception of bitterness intensity and with the visual perception of color lightness/darkness or hue.

The following section describes the production process of craft beers, the differences between craft and industrial brewing, the effects of the most important technological parameters on their quality, and the plant solutions used by craft breweries.

2. Brewing of craft beer

The following sections contain information on the brewing steps of craft beer. To facilitate reading and understanding of the text, a generic flowchart of craft beer production is reported in [Fig. 3.1](#).

2.1 Malting

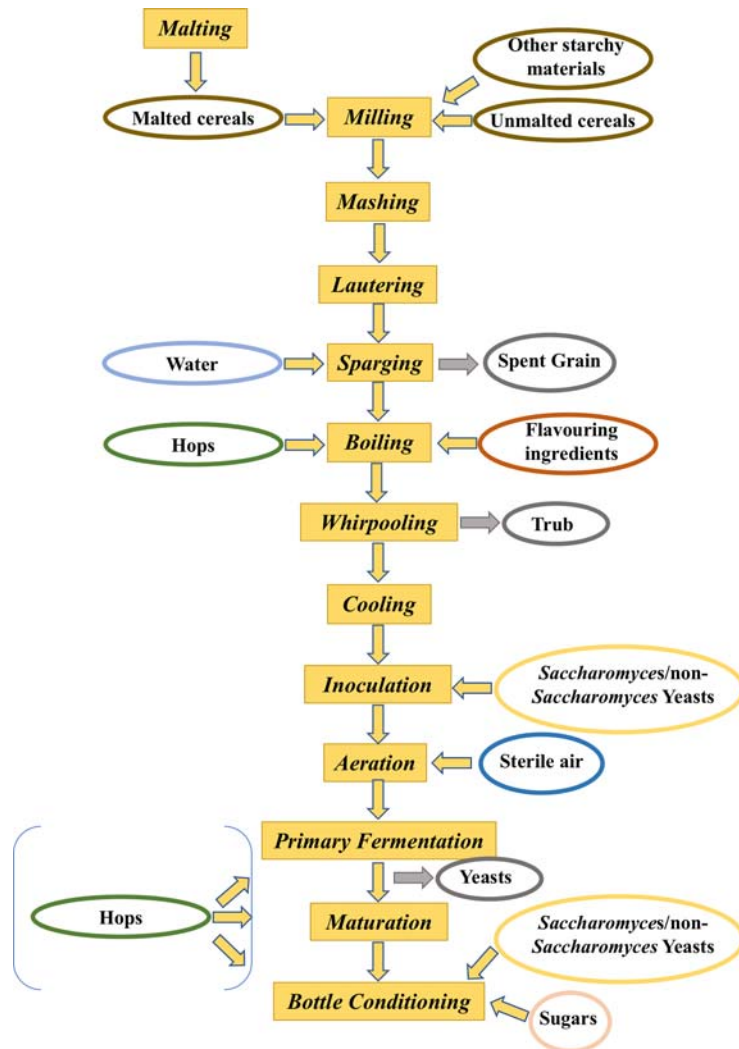
Malting is 5–7 days process consisting of three steps, namely steeping, germination, and kilning, whose aim is to prepare cereals (mainly barley and wheat) for brewing through a partial germination followed by drying. This process softens the grain cell wall and stimulates the synthesis of several enzymes (mainly cytolytic, proteolytic, and amylolytic) that were activated during mashing.

Before steeping, the cereal grains must rapidly precleaned to take away gross impurities and then cleaned, to remove the contaminants that are of similar size as barley/wheat like undersized grains, weed seeds, and broken grains. Barley is graded by grain size into different categories (>2.5 mm; 2.2 – 2.5 mm; <2.2 mm). The grains included in the first two categories are sent to malting while the grains belonging to the third category are used for animal feed ([Rani & Bhardwaj, 2021](#)).

During the steeping stage, the selected dried grains are imbibed in water to increase their moisture from 12% to around 42%–47% ([Schwarz & Li, 2010](#), pp. 478–521). Steeping includes a series of imbibition in water (2–4 days) each followed by an air-rest period, during which the water is drained and carbon dioxide is evacuated to avoid anaerobiosis. An excessive aeration is also to be avoided as it will cause an unwanted growth and a starch loss. The time period between immersions and air rest cycles generally lies between 4 and 12 hours ([Schwarz & Li, 2010](#), pp. 478–521). The steeping temperature is around 12°C for partly dormant grains and 16–18°C for less dormant grains).

During germination, the grain bed is manually turned over to allow adequate airflow and prevent overheating due to grain respiration. Germination occurs under aerobic and humid conditions at 16–20°C for 3–6 days. The embryo starts the biosynthesis of gibberellic acid, which catalyzes the production of the hydrolytic enzymes that depolymerize endosperm cell wall polysaccharides and proteins (grain modification) ([Rani & Bhardwaj, 2021](#)). The main modifications include endosperm softening; translocation of hydrolyzed products to the embryo to provide energy and building blocks for its respiration and growth. The green malt is ready for the following step when the first shoot (acrosipire) has developed to around three of fourth of the grain length ([Schwarz & Li, 2010](#), pp. 478–521).

During kilning, hot air is blown to reduce the moisture content of malt below 5%, thus stopping enzymatic internal modifications, and ensure malt stability. It involves the following three steps ([Rani & Bhardwaj, 2021](#)).

**FIGURE 3.1**

Generic Flowchart of Craft Beer Production.

- free drying/withering phase: It is performed at relatively low temperature (50–60°C) for 12–24 hours to allow water surface evaporation and to reduce moisture from 45% to 12%;
- intermediate/falling rate phase: Temperature is raised slowly up to 75°C for several hours in a stepwise manner to reduce the water content to 4%. The drying rate is lower than during the previous step because it concerns the moisture that moves from the interior of the grains to their surface;

- bound water/curing phase: The temperature is increased above 80°C for 1–2 hours, and then, malt is cooled. This step promotes the formation of melanoidins via the nonenzymatic Maillard reaction and, in turn, the development of ideal color and aroma (Howe, 2020, pp. 1–46; Mallett, 2014). In this step, the type of malt is established (light, dark, caramel, coloring, etc.). The main distinction is between base malt and specialty malt. Base malt, as its name indicates, is the foundation of the beer, representing from 60% to 100% of the starchy materials. Base malt has the highest enzyme activity and provides the bulk of the fermentable sugars to the brew. Specialty malts are kilned at higher temperatures for longer periods of time, thus adding the unique color, flavor, aroma, and body to the beer.

After kilning, the dried rootlets must be removed because they are extremely hygroscopic and malt is aged for 3 weeks to allow a uniform distribution of the residual moisture before packaging.

The brewing industry structure is significantly changing with a marked increase in craft all-malt brewing that is the need for malts with different performance profiles. For this reason, craft brewers are increasingly leaving the global commodity market to turn to locally produced barley malts as a source of novel flavors and a mode of differentiation.

2.2 Milling

Milling of malted and unmalted cereals has an important role in the brewing process since it influences the wort composition and, in turn, the quality of the final product. The function of milling is to expose the starchy endosperm and the embryo with its enzymatic content (Mallet, 2014). If the grains are not properly milled, that is if intact grains or large fragments remain, the results are an incomplete exposure of inner part of the grains, a low conversion of starch to fermentable sugars, and, consequently, a low efficiency of the mashing step (Reinold, 1997). An excessive milling is also undesirable since it could determine the extraction of substances such as phenolic compounds from the malt husk capable to affect physical properties of wort (increase viscosity and trub formation) and sensory properties of the final beer (excessive bitter taste). The preservation of a certain grain husk integrity is fundamental in the proper formation of the filter bed during lautering of the sweet wort since clarification is faster and more efficient the more intact husk fraction is (Reinold, 1997). The most typical method used for milling in craft brewing applications is the roller mill.

Milling can be carried out as dry or wet milling. Dry milling, the most frequently used method in breweries, increases grain comminution and mashing yield but may cause the lauter tun to clog or makes clarification difficult (Mallet, 2014). Wet milling, obtained by adding water up to 45%, makes the husks more resistant and flexible, thus preserving their integrity, but can decrease mashing yield. Wet milling is the most indicated method for unmalted grains, due to their toughness (Bogdan & Kordialik-Bogacka, 2017) but it is also recommended to increase the starch conversion into reducing sugars (Pereira de Moura & dos Santos Mathias, 2018).

2.3 Mashing, lautering/sparging

Mashing is the step in which the malt (cytolytic, proteolytic, amylolytic, etc.) enzymes are activated to breakdown the corresponding macromolecules (b-glucans, proteins, starch, etc.) of malted and unmalted cereals into products such as soluble b-glucans, peptides, amino acids, maltose, glucose,

dextrins, etc., necessary for yeast growth, fermentation, and sensory characteristics of beer. The final wort composition depends on the temperature–time profile adopted. Mashing involves the following phenomena: dissolution of substances that are directly soluble in water; enzymatic hydrolysis followed by dissolution of the products derived from these reactions (Montanari et al., 2005).

Three mashing methods are known: infusion, decoction, and combined. The choice of the suitable mashing mode depends on factors such as the quality of the raw materials, the brewery style to be produced, and the available technological equipment. Infusion methods are characterized by the fact that the whole mash is saccharificated simultaneously, by applying increasing temperatures and variable rest times at the activation temperatures of the various—mainly β -glucanase, protein, and amylase rests—to permit their action. A generic mashing profile is the following: mash in at 45–55°C; protein rest, sometimes absent, in other cases performed at 45°C for 10 minutes or at 50–55°C for 5 minutes; β -amylase rest, performed at 63–65°C for 20–35 minutes; α -amylase rest, carried out at 72°C for 5–20 minutes; mash out at 78°C for 1 minute (Parés Viader et al., 2021). In the decoction method, developed as a result of the use of under-modified or enzymatically weak malts, a portion of the mash (the decoction) is removed for boiling and then returned into the main mash to increase its temperature up to the activation temperatures of the various enzymes. Decoction mashing can involve one-, two-, or three-decoction steps, thus giving origin to single-, double-, or triple-decoction mashes (Montanari et al., 2005). In recent years, the infusion method has become very popular in craft brewing due to the availability of highly modified malts.

The lauter tun process uses grain beds made out of exhausted milled grains to separate the liquid wort and the sparging water from the solid residues and to remove part of proteins and polyphenols, two haze-inducing compounds, from the wort. Lautering occurs in two stages: wort draining, completely subjected to the laws of hydrodynamics, and Brewer's Spent Grain (BSG) washing (sparging), completely subjected to the laws of mass transfer. Lautering is described by the Darcy equation (Kostov et al., 2019):

Where Q is the volume (as m^3) of the filtered liquid for time t through the surface s ; P is the pressure difference (as Pa) on both sides of the filter barrier; r (as m) is the average radius of one capillary; s is the filtration surface (as m^2); z is the number of capillaries in $1 m^2$; h is the dynamic viscosity (as $Pa \cdot s$); l is the average length (as m) of one capillary. According to this law, the factors having the strongest influence on the filtration process are the following (Shopska et al., 2022):

- structure and thickness of the filtration bed: Coarse milling provides a highly permeable layer, but the resulting wort is cloudy; fine milling makes it difficult to filter due to the formation of a dense bed with small capillaries;
- number and size of capillaries affect the filtration rate. Longer capillaries present in a thick filtration bed, slow down the filtration rate. An efficient extraction and high yield of extractives is obtained by using small particles in which the path taken by the substances is minimized and the surface/diameter ratio is maximized to accelerate diffusion;
- temperature: High temperatures reduce viscosity and accelerate the filtration rate but the optimal temperatures are between 75 and 78°C because, at temperatures higher than 80°C, α -amylase is inactivated and the unsaccharified starch acts as an adhesive and increases viscosity.

With sparging, some extra water is added to extract more sugars out of the malt and to get the desired levels of fermentable sugars in the wort. There at least two different sparging methods in craft

brewing: some brewers add water while draining and others only once the wort has been completely drained. An excessive sparging is not desirable since it can determine the extraction of tannins whose bitter taste is different from that of hops (Mosher & Trantham, 2017).

2.4 Boiling

Wort is generally boiled for 90–120 minutes with a 10% evaporation rate (Goldammer, 1999) although, in modern kettles, boiling is shorter (60–90 minutes) and the corresponding evaporation rates lower (5%–9%) (Rourke, 2002).

The major changes occurred during boiling include: nonenzymatic browning; pH reduction; formation of protein-polyphenol complexes (trub); loss of fermentable amino nitrogen (FAN) due to browning reactions and trub precipitation; volatilization of compounds such as dimethyl sulfide; isomerization of α -acids. Fermentation is faster at shorter boiling than at longer boiling or at higher boiling temperatures because the increases in boiling temperature and time favor trub formation and reduces the wort FAN content and pH of wort (Mishra & Speers, 2021).

Beer foam is destabilized by lipids and is stabilized by the interaction between certain beer proteins (Lipid Transfer Protein 1 [LTP1] included) and isomerized hop α -acids. Wort boiling temperatures affect foam stability by affecting LTP1 levels. At high boiling temperatures ($\sim 102^{\circ}\text{C}$), the final beer concentration of LTP1 is low (2–3 $\mu\text{g/mL}$), whereas at low boiling temperatures ($\sim 96^{\circ}\text{C}$), high concentrations of LTP1 are detected (17–35 $\mu\text{g/mL}$) (Van Nierop et al., 2004). Low levels of LTP1 together with high levels of FFA reduce foam stability, while low levels of both LTP1 and Free Fatty Acids (FFA) determine satisfying foam stability.

2.4.1 Hop addition

Hops deserve a separate discussion. Hops can be added during: a) wort boiling; b) at the end of this stage (late hop addition); c) and during or at the end of the fermentation or at the conditioning stages (dry hopping).

When added during boiling stage, at temperature above 80°C , the hop α -acids (humulones, cohumulones, and adhumulones) are isomerized to the corresponding iso- α -acids (iso-humulones, isocohumulones, and iso-adhumulones), which are 28-fold more soluble than the starting α -acids and can better contribute to the foam stability and to the bitter taste. Two epimers are formed during isomerization reactions: cis- and trans-iso- α -acids and their ratio depends on characteristics of the wort matrix: as an example, at pH 5.5 and 7.0, 32% trans-iso- α -acids, and 68% cis-iso- α -acids resulted (Keukeleire, 1991). The cis-isomer exerts higher effects on beer characteristics since it is the most thermodynamically stable configuration. Reduced iso- α -acids such as tetra-iso- α -acids and hexa-iso- α -acids exert stronger foam stabilizing effects than iso- α -acids.

The conversion of α -acids to iso- α -acids during wort boiling is not complete but may range between 40% and 60%. The conversion depends on factors such as extraction of the α -acids from the hops, solubilization of α -acids in the wort, isomerization of α -acids during boiling, adsorption of α -acids and iso- α -acids on the hot trub, and oxidative and nonoxidative breakdown of iso- α -acids during wort boiling. First of all, a compromise should be found between a suitable pH value for the α -acids solubility and the isoelectric point of proteins (pH 5.2) that is also the pH at which mashing is carried out. However, solubility is strongly pH dependent and increases as pH increases. This means that, at higher pH, a greater quantity of α -acids goes from hops to wort. Isomerization is a first-order

reaction, with reaction rate depending on the temperature (Malowicki & Shellhammer, 2005). However, significant degradation reactions of iso- α -acids to humulinic acids (devoid of bitter taste) occur especially at extended boiling times, exceeding two half-lives of the α -acid concentration (Malowicki & Shellhammer, 2005). Degradation of iso- α -acids is controlled by reducing original gravity, temperature, water hardness, and by increasing the pH value. Moreover, hop acids can be adsorbed on coagulated proteins, as a consequence of the formation of ionic bonds with the ϵ -amino group of lysine. The increasing of the mashing-off temperature from 78 to 95°C enables the protein coagulation already during the mashing process, allowing their removal before the boiling step, thus reducing the quantity of trans-iso- α -acids formed.

When hops are added during boiling, most of the hop volatile compounds may evaporate while other nonvolatile compounds (phenolics, as an example) can be degraded.

If a late hop addition is performed, hops provide less bitterness, but greatly contribute to beer flavor.

When dry hopping occurs, nonisomerized α -acids are solubilized into the beer and contribute to its taste in their oxidized forms: oxidized a-acids (humulinones) and b-acids (hulupones). The rise of hoppy beer styles is leading to an ever greater use of dry-hopping.

2.5 Whirlpooling, cooling, inoculation, and aeration

After boiling, the spent hops along with other impurities are separated from the hot wort in a whirlpool vessel. Whirlpooling is a common method used in breweries to separate hop pellets and trub from wort after the wort has been boiled. The wort is pumped into the whirlpool vessel at a velocity of about 4.6 m/second, to cause the wort to spin like a whirlpool. The wort is pumped for 10–20 minutes then it stands for other 20 minutes to allow the hops and trub to form a compact pile in the center of the vessel. The wort is easily separated from the pile by pumping it out of an outlet located on the side of the vessel. The most common design of a whirlpool vessel is a flat-bottomed vessel with a slight pitch toward the drain. This gives a great trub pile but also allows the trub to slide toward the outlet. A most effective whirlpool vessel has a low point in the center and an outlet drain that is a little higher than the low point. This prevents trub from leaving with the wort.

The clarified wort is then recovered and cooled to the fermentation temperatures. In the brewing industry, the cooling process is commonly carried out using plate heat exchangers, due to the fact that they are easy to assemble and clean and easily opened, thus allowing the inspection of the entire heat transfer surface (Mocada et al., 2015). Furthermore, in this way, the possibility of contamination of the wort as a result of contact with the air is avoided. During the heat exchange, incrustations or fouling can result from the caramelized layer of sugars on the surface that reduces the efficiency of the heat exchanger. Yeasts are added to the stream of cooled wort as it enters a flotation tank. In the flotation tank, the cooled wort and yeast mixture is aerated, and additional impurities in the wort adhere to the surface of the air bubbles, thus clarifying the wort. Air, supplied under sterile conditions to avoid wort contamination, is necessary to supply the pitching wort with the dissolved oxygen level necessary for yeast growth, cell viability, and for controlling yeast metabolism during fermentation. However, dissolved oxygen can reduce the final concentration of esters, total SO₂, and higher alcohols such as isobutanol, 2-methyl-1-butanol, 3-methyl-1-butanol, and n-propanol (Verstrepen et al., 2003) although the specific effects are strain-dependent. The different results obtained show that there is a strain-dependent relationship between aeration and concentration of volatile compounds.

From this point on, in absence of real remediation treatments (we will talk about this in more detail below), it is necessary to pay maximum attention to the hygiene of the environment and production plants.

2.6 Primary fermentation and maturation

Although it is possible to obtain different sensory profiles among beers by selecting different types of special malts, adjuncts, hop varieties, water, etc., as highlighted in Chapter 2, fermentation represents the widest opportunity for beer diversification. The choice of one or more yeast strains for wort fermentation and/or beer conditioning affects amount and type of aroma-active compounds found in the final beer (Verstrepen et al., 2003). Pyruvate produced by glycolysis is the source of carbon skeletons for the synthesis of amino acids, in turn are involved in the production of diketones and several aroma compounds such as sulfur-containing compounds, esters, and higher alcohols. Furthermore, yeasts are able to modify the wort phenolic compounds, releasing Volatile Organic Compounds (VOCs).

While fermentation of industrial beers ideally proceeds through two stages, called primary and secondary fermentation, craft beers required a third fermentation that, differently from the first two, occurs in the final containers, mainly bottles and kegs. The fermentation time is another parameter discriminating the two types of beer. Since for industrial beer, time is money, its fermentation time is usually faster (7–10 days) than that of craft beer, which can be prolonged up to 2 months thus giving it a greater sensory complexity and a greater equilibrium among its chemical components.

2.6.1 Primary fermentation

As already known, the principal separation criterion for beer (both industrial and craft) classification relies on the type of brewing process. The primary fermentation is a tumultuous fermentation during which fermentable sugars are rapidly transformed into alcohol and carbon dioxide. This fermentation is also recognized by an abundant foam formation. Ale beers are brewed by top-fermenting *Saccharomyces cerevisiae* strains at temperatures of 15–25°C. These domesticated strains are able to ferment maltotriose, which accounts for 20% of the total fermentable sugars in wort; are unable to produce phenolic compounds such as 4-vinyl guaiacol, that is an unpleasant spicy clove-like compound, and 4-ethylphenol, 4-ethylcatechol and their precursors, 4-vinylphenol and 4-vinylcatechol, the so-called Phenolic Off-Flavor (POF)⁺ phenotype; and tend to float to the top of the vat at the end of fermentation. Lager-style beers are fermented by the cold-tolerant *Saccharomyces pastorianus* at temperatures of 8–12°C. This yeast sediment to the bottom of the fermentation vessel. Detailed information regarding *Saccharomyces* yeasts in brewing is reported in Chapter 5. Spontaneous-style beers are obtained by spontaneous fermentation: originally, wort was exposed to the air, letting it become colonized by wild yeasts and bacteria. After 5–7 days for ale beers and 10–14 days for lager beers, the primary fermentation slows down considerably and the used-up yeast started to drop to the bottom of the fermentation vessel, along with hops, proteins, and other solids.

Besides the traditional yeast strains, the use of aromatic-enhancing microorganisms is rising in craft brewing. This is the case of *S. cerevisiae* yeast strains isolated from other matrices such as wine, spirits (Cachaça spirits, pulche, tequila, and sake), bread or, as an even more extreme choice, of non-*Saccharomyces* yeasts (Chapter 6).

2.6.2 Maturation

It is also known as secondary fermentation or lagering and takes place only after most of the sugars in the wort have been used up. It allows the residual sugars in the wort to ferment, the flocculation phase of the yeast, which settles on the bottom due to the increase in the alcohol content of the beer, the beer saturation with carbon dioxide, and the harmonization of flavors and tastes. The secondary fermentation is carried out at lower temperatures than the primary one and for longer times: around 10°C for 15–20 days for ale beers and around 5°C for 1–2 months for lager beers.

During maturation, several reactions take place. Diacetyl (2,3-butanedione) is a vicinal diketone produced by yeast metabolism whose concentration is used as an indicator of fermentation or maturation quality. Mainly in lager beer, is an undesirable volatile compound if concentration is above 0.10–0.15 mg/L since it is responsible for a stale milk flavor. At the end of the primary fermentation period and during maturation, yeast cells reduce both diacetyl to acetoin and then to 2,3-butanediol and 2,3-pentanedione to 2,3-pentenediol. These compounds have a higher threshold and their presence is acceptable at the concentrations usually found in beer. Acetaldehyde is an “off-flavor” having flavors of green apple/pumpkin peel (Humia et al., 2019). It is produced by yeasts during fermentation as an intermediate compound in the conversion of glucose to ethanol. At the end of the alcoholic fermentation and during maturation the yeast converts most acetaldehyde to ethanol so that residual acetaldehyde falls below its flavor threshold (5–15 mg/L). The presence of active yeasts in the maturation phase is a prerequisite to reduce undesirable volatile compounds.

2.7 Refermentation in bottle or bottle conditioning

As a further distinctive element with respect to industrial beer, craft beers are characterized by a further fermentation that occurs in the final containers. It refers to a method of adding fermenting wort or sugars (dextrose or saccharose) and eventually yeast (if the one remaining from the main fermentation is not vital, taking into account that the concentration of yeast cells should be at least 10^6 CFU/mL) into beer in its final package before capping (Buiatti et al., 2021). It is inspired by the *champenoise* method for the production of sparkling wines, which has been known since the end of the 17th century. Due to their higher fermentation temperatures, top-fermented beers (ales) are typically less carbonated than the bottom-fermented (lager) since at higher temperature the carbon dioxide solubility in beer is lower. Therefore, refermentation in bottle was originally implemented to enhance carbonation and eliminate the flat taste of beer. Beer should contain about 0.5% CO₂, and products containing lower carbon dioxide content are considered flat.

With refermentation, yeasts consume the oxygen incidentally dissolved into beer during filling and ferment the added sugars producing further alcohol and carbon dioxide. Beer results in a greater carbonating, with the consequent positive effects on quantity and persistence of foam and *perlage*, but also in a strongly reducing environment useful to protect it against oxidative damage. The carbon dioxide produced by re-fermentation further acidifies the craft beer, making the bottle environment unsuitable for the development of many microorganisms. Thereby, in absence of stabilizing treatments such as filtration and pasteurization and without the use of additives, refermentation can improve storage stability and prolong the shelf life of craft beer. Furthermore, although beer flavor depends on raw materials used for brewing and yeast metabolism during primary fermentation, during refermentation in the final container, new flavors are generated by the yeast activity (see Chapter 8).

It is important to keep the storage temperature of craft beer constantly low, to avoid the negative effects that can derive from the presence of yeast cells in bottle. In fact, temperature fluctuations could damage yeast cell wall and membrane causing the dispersion of off-flavor compounds and enzymes and the comparison of over-attenuation, and foam weakening phenomena (Buiatti et al., 2021).

2.8 Stabilization treatment of the beer

As anticipated and differently from the industrial beer, the craft one is generally not subjected to stabilization treatments (filtration and pasteurization) of the final product which are instead used in the production of industrial beer. The reason is that filtration treatments could deplete the product of compounds responsible for its unique flavors while thermal treatments, by inactivating the microorganisms and preserving beer from biological degradation, induce oxidation of heat-sensitive volatile compounds. Furthermore, a pasteurization treatment would inactivate the yeast added in the previous step.

In the industrial brewing, the stabilizing effects of filtration depend on cutoff applied, considering that it can be used to remove coarse particles, yeast, and haze-responsible compounds ($0.1/10\ \mu\text{m}$) or to remove all microorganisms thus sterilizing beer ($0.02\ \mu\text{m}$) (Buiatti et al., 2021). The sterilizing microfiltration is often performed in industrial breweries to sterilize beer and fill the beer into sterilized containers (Dilay et al., 2006). Instead, some national legislations allow the application of filtration to craft beers to remove coarse particle and haze-responsible compounds, although the presence of haze is not considered as a negative characteristic in craft beer but, on the contrary, it enhances the naturalness of its production.

Industrial beer production uses additives contained in the positive lists established by national and international legislation. They include enhancers of flavor stability, color regulating agents, taste and flavor regulators, foam stabilizers although the chemical agents used for improvements of flavor stability are the most widely used additives in brewing and they are mainly represented by sulfites. They act as antioxidants since oxygen reacts with many flavor-active compounds in the beer, replacing the typical fresh flavors with cardboard-like, papery, and stale flavors. Instead, no additives are used in craft brewing. Instead, no additives are used in craft brewing although traces of sulfites can derive from ingredients (hops, for example), or can be produced as metabolites by the yeasts.

3. Effects of some technological parameters during processing of craft beer

Parameters such as temperature, time, and steeping degree can influence malting as well as temperature, oxygen concentration, and pH can affect the brewing process and, consequently, safety and quality of the resulting beers.

3.1 Effects of malting parameters

Since malts with high levels of free amino nitrogen and amino acids are essential for the yeast growth and metabolism in wort, the optimization of malting parameters is critical. During malting, the increase in amino acid concentrations is mainly due to the barley respiration while Maillard reaction

occurring in the kilning stage decreases amino acid content. Therefore, the parameters of interests are the following: a) for steeping: duration of alternating wet and dry steps (wet, dry, wet, dry, wet); b) for germination: duration and temperature; c) for kilning: temperature profiles and duration. To increase the total amino acid content the following solutions are appropriate: during steeping, make sure that the dry steps last double that the wet ones; prolong germination for 5 days; during kilning, keep the temperatures below 50–55°C until most of the water has been removed (Nie et al., 2010).

Concerning the production of aflatoxin B1 (AFB1) by *Aspergillus flavus* during malting of wheat grains for craft beer, AFB1 levels increase with increase in the temperature (from 13 to 19°C) or time of germination (48, 72, 96, and 120 hours) but it is not influenced by the steeping degree in the range 41%–47% (Schabo et al., 2020).

3.2 Effects of temperature in brewing

Concerning mashing, the choice of the temperature profile takes into consideration the composition of the starchy materials and the known data on the optimum temperature of the enzyme systems that must be activated and, therefore, it depends on the qualitative and quantitative composition of the must that you intend to produce which, in turn, depends on the type of final product you intend to obtain (Table 3.1 adapted from Shopska et al., 2022). As examples, desired values of FAN in the wort are in the range of 300–350 mg/L, while it is necessary maximize the wort fermentable sugar content. Since FAN concentration is related to the proteolytic activity in the mash, low temperatures of protein rest increase FAN levels. To maximize the sugar content of wort, it is appropriate to maximize the activity of α -amylase since it is the major contributor to the degradation of dextrins to fermentable sugars (Muster-Slawitsch et al., 2014).

Table 3.1 Optimal temperature ranges of the main malted barley enzymes.

Enzyme systems	Optimal temperature ranges (°C)	Enzyme substrates	Reaction products
Maltase	35–40	Maltose	Glucose
Exo-b-glucanase	<40	Cellobiose, Laminaribiose	Glucose
Endo-1,4-b-glucanase	40–45	Soluble high molecular Weight b-glucan	Low molecular weight b-glucan, cellobiose, Laminaribiose
Aminopeptidases	45	Proteins, peptides	Free amino acids
Dipeptidase	45	Dipeptides	Free amino acids
Endopeptidases	45–50	Proteins	Peptides, free amino acids
Lipoxygenase	45–55	Long-chain fatty acids	Fatty acids
Carboxypeptidases	50	Proteins, peptides	Free amino acids
Phosphatase	50–53	Organic phosphate	Inorganic phosphate
Pullulanase	55–65	Limit dextrin	Dextrin
Lipase	55–65	Lipids, lipid hydroperoxides	Glycerine and long- chain Fatty acids, fatty acids

Continued

Table 3.1 Optimal temperature ranges of the main malted barley enzymes.—cont'd

Enzyme systems	Optimal temperature ranges (°C)	Enzyme substrates	Reaction products
Endo-1,3-b-Glucanase	<60	Soluble high molecular Weight b-glucan	Low molecular weight b-glucan, cellobioses, Laminaribiose
b-Amylase	60–65	Starch	Maltose
Polyphenol oxidase	60–65	Polyphenols	Oxidized polyphenols
Peroxidase	>60	Organic and inorganic substrates	Free radicals
b-Glucan solubilase	62–65	Bound b-glucan	Soluble high molecular Weight b-glucan
a-Amylase	65–75	Starch	Melagosaccharides, oligosaccharides

Adapted from Shopska, V., Denkova-Kostova, R., Kostov, G. (2022). Modeling in brewing—A review. Processes, 10(2), 267. <https://doi.org/10.3390/pr10020267>.

An inverse function has been observed between fermentation/maturation temperature and beer bitterness, which is higher at lower (12°C) than at higher (18°C) temperatures. At high fermentation temperature, high concentrations of the majority of the volatile compounds (mainly acetaldehyde, higher alcohols, and esters) have been documented. The effect of the fermentation temperatures on phenolic compounds is not unique, with gallic, chlorogenic, and ferulic acids produced in higher concentrations at higher temperatures while tyrosol, vanillic, and i-ferulic acids are synthesized in higher amounts in beers fermented at lower temperatures (Castro et al., 2022).

3.3 Effects of pH in brewing

pH exerts great effect on the rate of several reactions, solubility, and electrostatic charge of many molecules. Brewing water must be slightly acidic, to avoid the extraction of polyphenols and silica compounds from the malt husk that would be responsible for undesired colloidal instability and astringency. As already said, the reactions occurring during wort preparation are catalyzed by enzymes whose activity is strictly related to their three-dimensional structure that in turn is mainly affected by the occurrence of electrostatic bonding. As a result, enzymes can operate only within a given range around their optimum pH value: b-glucanases (6.0); proteases (5.5); b-amylase (5.5); a-amylase (5.2). Isomerization of a-acids occurring during wort boiling would be favored by high pH values (from 8 to 10) with over 90% conversion. Instead, at the typical wort pH (around 5.2), conversion is around 60% in the kettle that means a final bitterness utilization around 40%. This is the reason for which many brewers chose to use preisomerized hop products (O'Rourke, 2002a, 2002b, 2002c).

During maturation, the nonenzymatic oxidative decarboxylation of a-acetolactate excreted by the yeast into diacetyl occurs in the beer with a higher rate at lower pH values. Since this reaction is the rate-limiting step of the diacetyl removal pathway, the reduction in the pH of the fermenting wort reduces the time of diacetyl removal and leads to a decrease in the diacetyl level (Li et al., 2016).

Since, as described in paragraph 3.4, pH values lower than 4.0 in beer could negatively affect its stability during shelf life, a continuous control and management of pH must be performed through the choice of the ingredients and the setting of the brewing conditions to assure and maintain the buffering capacity of the wort until the obtainment of the final beer. As already said, it is necessary to reduce pH during wort preparation but, although the addition of various compounds is able to realize the same pH of mash, the buffering capacity of the wort can vary greatly. The buffering capacity of the wort increases as the matrix pH increases and the great contribution is given by, in a decreasing order, the following organic acids: acetic, succinic, citric, lactic, fumaric, and pyruvic). Amino acids such as glutamic acid, aspartic acid, and histidine supply a weak contribution to the buffering capacity of the wort at the pH of wort. Phosphate is an ineffective buffer at the wort pH (Li et al., 2016). Based on these finding, yeasts contribute to the buffering capacity of the final beer through the excretion of organic acids during fermentation (Kaneda et al., 1997).

3.4 Effects of oxygen in brewing

The oxygenation of wort causes a reduced formation of esters, as a consequence of the decreased expression of the alcohol acetyl transferase gene ATF1 (Verbelen et al., 2009) but the administration of 25 ppm dissolved oxygen 12 hours after inoculation, is able to reduce the fermentation time of high-gravity wort by 33% compared with the control conditions (Jones et al., 2007). Instead, traditional ale and lager wort require from 6 to 8 ppm dissolved oxygen. Oxygen must be added just prior to fermentation since active yeast has a great capacity to assimilated oxygen before any chemical oxidation can occur. In the absence of active yeast, oxygen can react with wort and beer components to produce oxidized compounds responsible for change in the final product quality such as flavor staling compounds (es. nonenal); oxidized polyphenols responsible for increased astringency, darker color, accelerated chill, and permanent haze formation; oxidized proteins that can polymerize by forming disulfide bridges (O'Rourke, 2002a, 2002b, 2002c).

3.5 Effects of wort osmotic pressure in brewing

Osmolarity of wort can vary in a wide range from that of conventional wort (12 degrees) to that of very high-gravity wort (20°). Osmolarity affects yeast development but induces also chemical and physical changes in the final product. During fermentation of 16° and 20° worts, yeast vitality decreases, yeast proliferation is lowered, and fermentation time increases. Beer brewed at increased osmolarity has high concentrations of diacetyl and pentanedione and lower levels of dimethylsulfide and acetaldehyde, while no differences are highlighted in ester and higher alcohol contents. Increased wort osmolarity does not affect haze of green beer but accelerates beer clarification during maturation (Sigler et al., 2009).

4. Aging and shelf life of craft beer

As a general rule, sealed craft beer is shelf-stable for 6–8 months at room temperature and for more than 12 months under refrigeration. For years, craft beers have been destined to local markets. However, globalization has led to the increase in export of craft beers and to the need to lengthen craft

beer shelf life. The aging of craft beers is a delicate time that can negatively affect beer stability. Beer stability is a complex concept that includes the biological stability (or microbiological stability) and the nonbiological stability, further categorized into flavor and physical.

The biological stability depends on the application of good hygiene practices and on exploitation of the intrinsic beer properties such as the low pH, alcohol concentration, hop acids, reduced oxygen level, and carbonation, which can help to inhibit the microbial growth (Vaughan et al., 2005). During aging in bottle, the yeast cells (survived after fermentation or added to realize a bottle re-fermentation) exert positive effects by synthesizing new flavor compounds and developing a more stable product due to the reduction in the beer oxygen content and the consequent inhibition of beer oxidation. On the contrary, eventual death of yeasts occurring in the bottle can lead to cell autolysis, with release of enzymes and the breakdown of proteins associated with foam formation and stability (Marconi et al., 2016).

Nonbiological stability takes into account all the issues concerning the physical (colloidal, foam, gushing, and light) stability and the flavor evolution. The concept of colloidal stability concerns the haze formation that in turn depends on the presence of complexes between flavonoids and proteins whose formation is favored by high temperature, oxygen, heavy metal ions, and movement (Aron & Shellhammer, 2010; Stewart, 2004). Foam amount and stability not only represent an esthetic characteristic of the beer capable of influencing its liking but they also play a protective role against oxidation and promote aroma exchanges. Gushing is an over-foaming that can be observed when a bottle is opened and that can cause a significant volume loss. This phenomenon is promoted by the presence of fungi, excessively high level of carbonation and iron, prolonged low-temperature storage (Aron & Shellhammer, 2010; Stewart, 2004). Flavor stability is the most critical quality challenge since the flavor profile starts to deteriorate when the production ends with the loss of positive attributes and the development of aging characteristics (Paternoster et al., 2020). These phenomena cause irreversible changes in the aroma and taste, thus limiting beer shelf-life (Rendall et al., 2015). The control and the maintenance of the finished beer pH—at the end of fermentation, it should be in the range from 4.2 to 4.4—is necessary since it affects flavor, and physical and microbiological stability. Although low pH values inhibit the growth of lactic bacteria, pH lower than 4.0 accelerates flavor staling reactions, free radical reactions, degradation of iso-humulones and procyanidins, and colloidal haze formation during storage (Kaneda et al., 1997).

The preservation of craft beer quality during transport and storage until the product reaches the final consumer is a critical issue with temperature, time, oxygen level, and vibrations. Carbonyl compounds are associated with staling off-flavors and, having low threshold, are evaluated to assess the impact of exposure to temperature. The main beer characteristics that can determine increases of their concentrations are the content of oxygen, metal, amino acids, and bound-state aldehydes of bottled beer. During storage, decreases of ester concentrations (due to the presence of yeast-derived extracellular esterases) and alcohols and increases of carbonyl compound levels (2- and 3-methylbutanal deriving from Strecker degradation of amino acids; vicinal diketones synthesized by Maillard reactions; (E)-2-nonenal originated from oxidative degradation of linoleic acid) occurred (Marconi et al., 2016). The storage temperature influenced the decrease in ester components (higher at the highest temperature) but to a lesser extent than storage time. During storage, sensory characteristics such as fruity and malt odor, fullness, acidity, and bitterness decreased during storage while attributes such as cardboard, phenolic, oxidized, and sweetcorn increased. (E)-2-Nonenal, the carbonyl compound responsible for the cardboard flavor of aged beer, is present at a significant lower level in the beers brewed with autochthonous yeasts with respect to that produced with commercial yeast.

Compounds such as furfural and aldehydes from Strecker degradation are known to strongly increased during prolonged storage at temperature around 30°C (Jaskula-Goiris et al., 2011). High storage temperatures are also responsible for the fast degradation of beer trans-iso- α -acid content (Nimubona et al., 2013).

Concerning the impact of vibrations and shocks on beer quality, transport by ship causes: a) browning, whose extent is more pronounced in dark than in pilsner and pale beers; and b) decrease in the initial iso- α -acid content, especially in pale beers (Jaskula-Goiris et al., 2019). The vibrations caused by truck transport are responsible for a strong increase in oxygen uptake and carbonyl compound (aldehydes) concentrations (Paternoster et al., 2019). However, the quality preservation during transport and long-term storage also depends on the type of packaging, with the cardboard crate with plastic foil being the best material to both attenuate vibrations and reduce exposure to temperatures (Paternoster et al., 2017).

Concerning the effects of thermal abuse during storage of craft beer refermented in bottles, it is necessary to reach temperature around 45°C to induce a decrease in beer bitterness and an increase in oxidation with development of brown color and production of high quantity of furans and aldehydes (Valentoni et al., 2022).

Traditional styles of beer that include spontaneous fermentation and/or maturation in oak barrels in their brewing procedures are becoming popular among craft beer producers and consumers. Barrel-matured beers undergo microoxidation; extract oak lactones and aldehydes that impart toasty, coconut, vanilla, and nutty characters; and are “inoculated” by the microbial communities that inhabit the wood (*Brettanomyces* spp., lactic acid bacteria, and acetic acid bacteria).

The aging in barrels that have previously contained wine strongly increases both phenolic content and antioxidant activity of craft beers, especially if wood containers had been used for the aging of red wine (Sanna & Pretti, 2015). Moreover, the aging in oak barrels previously used in Port wine aging is able to increase the craft beer flavor with ethyl esters and oak extractives (Coelho et al., 2019).

5. Plant solutions for craft breweries

Malt is one of the most important ingredients in brewing, and, for a long time, its production has taken place in firms distinct from the breweries and known as malt houses (Fig. 3.2). The latter, by applying economies of scale, are able to achieve profits despite the low added value of malted cereals compared with the unmalted ones, producing large quantities of a limited number of malt types. However, the increased interest toward craft brewing is contributing strongly to development of the craft malt industry. In fact, craft brewers require small quantity of malts but, more in depth, they often require special malts as those deriving from the malting of local grown cereals. This type of request cannot be accommodated by the traditional malt houses that process large quantities of standard barley or wheat varieties. Several plant manufacturing companies have started to design and propose solutions suitable for the production size of both small malt houses and craft breweries that intend to regain control of the production of the starchy raw materials. Often, they offer modular solutions for an easy scalable production. Examples of commercially available mini-malting system are described in Table 3.2.

Some craft brewers started their production using home brew equipment or equipment borrowed from other food sectors, for example for the production of wine and milk, and modified to meet the needs of brewery production (Villacreces et al., 2022). The reason was twofold: limited financial



FIGURE 3.2
A Typical Drum Used in Industrial Malting Plants.

Table 3.2 Micromalting equipment.		
Product name	Company	Description
MINIMALT-5	Zanin (Casale sul Sile, Italy)	It is an automated all-in-one system, capable of processing 500 kg of cereal per cycle. The software allows a tailored management of the process (Zanin, 2023).
RimoMalt	Bühler (Uzwil, Switzerland)	It consists of one cylindroconical steeping unit, one central heating unit, and one unit for germination and kilning built according to a completely modular concept. Each component can be extended, and additional parts can be added to upscale capacity requirements. The starting batch size is 16 tons (Bühler, 2023).
Minimalting	Rauscher Engineering GmbH (Dietersheim, Germany)	In this compact design system, the stages of steeping, germinating and kilning are carried out in one vessel. The batch size is from 100 kg to 5000 kg (Rauscher, 2023).
Small malt making machine	Shandong Senmo Machinery Company (Jinan city, China)	It is an automated machine allowing that steeping, germination, kilning and roasting are performed in the same vessel with a batch handling capacity of 1000 kg barley (Senmo, 2023).
Acro	Sprowt Labs start up (Minneapolis, USA)	It is a small-batch home malting machine and is sized to produce from 4 to 15 kg of finished base and specialty malts. It allows us to automate the sprouting process (Protolabs, 2023).

resources and the lack of plants specifically designed for that type of production. Both problems have been overcome since craft brewing is now a profitable business and more and more leading companies in the sector of industrial systems for beer- and wine-making have expanded their activity to develop lines for craft breweries (Elzinga et al., 2015).

A craft beer production plant can be ideally divided into three parts: equipment for wort preparation; equipment for fermentation; bottling equipment.

The wort preparation requires a milling system and the following series of vessels: one or more mash conversion vessels and a hot wort tank, a lauter tank, a wort boil kettle, and a trub separation vessel (as a whirlpool) (Holbrook, 2020). A wort cooler and an aeration system complete the wort preparation. However, in craft brewers, these vessels are generally equipped to combine more unit operations as in the cases of the mash/lauter tank and the boil kettle/whirlpool (Fig. 3.3). The reason for choosing an option rather than another can depend on several reasons including financial budget, space available, and brewing method. Instead, in an industrial brewery, the choice of combined vessel systems is not common (Villacreces et al., 2022). Another important difference between craft and industrial plants is referred to the preparation of malt substitutes. In the craft breweries, there are processed directly in the mash vessel together with the malted cereals while the industrial brewers are often equipped with separate vessels, the so-called “adjunct cereal cookers,” which are similar to the classic mash vessels.

Equipment for craft beer fermentation includes fermentation and maturation tanks (Fig. 3.4), which are very similar to that used on an industrial scale (Fig. 3.5), with differences in the size and level of control and automation (Clemons et al., 2006).

Unlike most industrial beers, the craft ones are usually not filtered and not pasteurized, and therefore, the plant solutions do not include the corresponding equipment, but only one or more bright beer tanks and bottling (Fig. 3.6), can and/or keg lines. Instead, to achieve a greater microbiological stability of industrial beer and prolong its shelf life, innovative solutions have been proposed. The validity of some of them has been demonstrated at an experimental level. As an example, a cold pasteurization technology based on filtration through silica microparticles (mean size within the 5–50 μm range to preserve the beverage properties) functionalized with natural antimicrobials—that



FIGURE 3.3

Mash/Lauter Tank (on the right) and Boil Kettle/Whirlpool (on the left).



FIGURE 3.4

Fermentation Tanks of a Craft Brewery.



FIGURE 3.5

Fermentation Tanks of An Industrial Brewery.



FIGURE 3.6
Bottling Line.

is, essential oils such as carvacrol, eugenol, and thymol—to entrap and/or inactivate *Escherichia coli*, mesophilic and lactic bacteria, mold and yeast (Peña-Gómez et al., 2020). Another important difference between craft and industrial production plants derives from the increasing production of dry-hopped craft beers that require specific equipment. Examples of commercially available dry hopping equipment are described in Table 3.3.

Table 3.3 Dry/cold hopping equipment.		
Product name	Company	Description
Alhop	Alfa Laval (Lund, Sweden)	It is a stand-alone unit for medium size breweries that covers all dry hopping stages: Introduction of hop pellets; disaggregation and homogenization of hop pellets by means of a cross-flow recirculation loop; flavor extraction; separation of hop solids; recovery of a-acids (Alfa Laval, 2015–23).
Iso-mix External Drive (IMXD) mixer	Alfa Laval (Lund, Sweden)	It enables automation and scale-up of dry hopping techniques for beer volumes from 100 to over 5000 HL. It is located within the cone section of the tank, to operate while the tank is being filled and emptied. It is rotated by an external motor connected to an internal shaft to assure the absence of contact between the circulating fluid and the mixer gears (Alfa Laval, 2015–23).

Continued

Table 3.3 Dry/cold hopping equipment.—cont’d		
Product name	Company	Description
HopMaster	ZIPTECH (Miskolc, Hungary)	It can work with max 5 Kg hop leaves/6 Kg hop pellets. The equipment can be connected to the fermentation vessels or to the bright beer tanks. The equipment architecture results in reduced oxygen uptake and minimized damaging effects. The filter basket allows that only small amounts of fibers go into the beer. The beer flow between the fermentation vessel and the dry-hopping equipment is achieved with the use of overpressure (ZIPTECH, 2023).
HopTower	ZIPTECH (Miskolc, Hungary)	It can work with max 20 Kg hops. The equipment can be connected to the fermentation vessels or to the bright beer tanks. The placement of the inlets allows beer to be pumped into the equipment. The vigorous mixing that results increases the extraction rate of the aroma compounds. The filter cartridge does not allow large particles back into the fermentation vessel. The beer flow between the fermentation vessel and the dry-hopping equipment is achieved with an open impeller centrifugal pump, thus ensuring the colloidal stability of the beer (ZIPTECH, 2023).
HOPSTAR® Dry	GEA Group (Aktiengesellschaft, Düsseldorf, Germany)	It is a fully automated plug-and-play unit that can easily be connected to a cylindroconical tank. It is produced in a number of standard sizes for hop loads up to 750 kg. Hops are manually loaded into the extraction vessel: After that, the automatic dynamic extraction process supported by a low-shear force agitator inside the vessel. A sedimentation step is used to keep the solids into the extractor (Gea, 2023).
HopGun® Pro	BrauKon GmbH (Seeon, Germany)	It is designed for fully automated cold hopping and guarantees the effective dissolving of oils and aromas from hop pellets in the finished beer. It is flexible to use, since hop quantity and treatment time being freely selectable (BrauKon, 2023).
Hop Storm	JSJ Machinery Co. Ltd. (Ningbo, China)	It utilizes a double helix design to create an inner turbulence that dissolves the hop pellets as the beer circulates through the vessel (JSJ Machinery, 2023).

6. Conclusion

The differences highlighted in the production process of craft beer compared to industrial beer are responsible for their unique flavors and smells. The lack of nanofiltration and pasteurization processes requires the management of the entire production process according to extremely rigorous hygiene standards. For these reasons, craft brewers must have acquired solid knowledge of the physical, chemical, biochemical and microbiological aspects to ensure the production of high-quality craft beer.

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Craft beer styles

Ray Daniels^{1,2}

¹*Cicerone Certification Program, Chicago, IL, United States;* ²*Siebel Institute of Technology, Chicago, IL, United States*

Throughout the beer world, participants use a set of words and phrases to quickly describe the attributes of individual beers. Collectively, these words and phrases are known as “beer styles.” They constitute an essential tool for all who are involved with beer: brewers, distributors, retailers, consumers, suppliers and media as well as other observers. Through the use of styles, we can quickly communicate the basic attributes of a beer. Thus, a style-savvy listener, upon hearing a beer’s style name, has a good idea of that beer’s color, alcohol level, and overall flavor profile. In short, styles serve as the language of beer.

1. Two groups of styles

A proper beer style defines a fairly consistent and well-characterized type of beer. Take German Pils, for instance. This style is expected to have a straw-to-yellow color and 4.4%–5.2% alcohol by volume. Furthermore, the aroma and flavor are well characterized, giving a dry, bitter German lager featuring a prominent floral, spicy, or herbal hop aroma (Strong, 2021). Other styles may appear similar in some ways, but Pils can be clearly differentiated from them. Other examples of well-defined styles would include Irish Dry Stout, American Pale Ale, and Belgian Triple.

More than 160 different clearly defined beer styles have been identified, as we will see later in this chapter. Despite this, a huge range of beers fall outside of the well-defined styles. To handle these beyond-style brews, the industry has created additional categories for the classification of beer. These categories are often called styles and treated in much the same way, but they are distinctly different in their character. These “beer categories” typically group beers together based upon a single distinctive trait. For instance, we often find “Coffee Beer” as a beer category. Beers placed in this category have all been made using coffee at some point in the brewing process, so that the finished beer exhibits aromas and/or flavors of coffee. But, while all “Coffee beers” share that trait, they can differ vastly in all other ways. Indeed, if you read the description for this category in the Great American Beer Festival guidelines, you’ll find that all of the quantitative parameters simply say: “Varies with style.” Descriptors are also broadly defined, for example, color is given as “Pale to black” and hop aroma and flavor as “Low to high.”

This sort of description is common for beer categories—and they are a significant contrast to what we see for beer styles. Basically, such a description tells us that as long as the beer meets the one key characteristic for which the category is usually named, a broad range of beer attributes is typically allowed.

Because of the diversity inherent in the commercial beers available around the world today, these sorts of categories are necessary to help organize beers any time we want to analyze or evaluate them. Every set of beer style guidelines in use today includes both well-defined beer styles and the more broadly drawn beer categories. Some guidelines, notably those of the Beer Judge Certification Program, clearly distinguish between styles and categories. Most however simply lump all of the styles and categories together, referring to all as beer styles. Throughout this chapter, we will continue to distinguish between the more narrowly defined styles and the broadly cast categories.

Most compilations of beer styles available today are maintained to facilitate judging of beers in competitions. Competition organizers either create their own style guidelines or they adopt or adapt another set of established guidelines. Brewers submit each beer in a specific style category. During judging, all of the beers of a specific style are assessed against the other beers entered in the same style. Awards are commonly given to the top beers in each style. Examples of such competitions include the African Beer Cup, Australia International Beer Awards, Brussels Beer Challenge, China International Beer Challenge, *Concurso Brasileiro De Cervejas*, *Copa Cervezas de América*, European Beer Star, Great American Beer Festival, Japan Craft Beer Association Competition, and World Beer Cup. We will explore in detail the beer styles and categories used in these competitions in the final section of this chapter.

Before we examine style guidelines and the traits of individual styles, we should first understand the information that goes into designation of a beer style.

2. The basis and organization of beer style descriptions

Any system of beer style descriptions will be based upon the actual universe of beers that are available for sale to the public. The thousands of beer brands and packages available to consumers present an amazing clutter of names and descriptions. While some brands include a style designation in the brand name (e.g., “Sierra Nevada Pale Ale”) or as additional information on the package (e.g., “A Fine Pilsner Beer”), such information may be more suited to selling the beer than to helping consumers understand its actual character.

The ultimate goal of any system of beer styles is to identify groups of similar beers. These groups are then analyzed and characterized in detail so that an exhaustive listing of the style’s traits can be compiled. Often the style takes its name from a designation commonly used by many of the producers (e.g., Porter, India Pale Ale, and Weizen, Blonde), but in other cases, the style name is used more to suggest how a particular style differs from others that are similar (e.g., “American-style Strong Pale Ale”).

While style descriptions are based on analysis of commercial beers, few, if any of the style guidelines in use today, have been compiled from scratch in recent years. Doing so would involve a detailed analysis of tens of thousands of beers. In reality, current beer style descriptions have evolved and developed over several decades and most are refined and edited on a frequent basis to stay abreast of shifts in the marketplace. In this section, we want to review the information that is collected and presented in a beer style description.

Beer classification ultimately depends upon two distinct groups of traits. The first is easiest to assess as it depends upon the measurable physical characteristics of the beer. We call these “quantitative” measurements or traits. The second group of traits depends upon sensory assessment of a beer’s attributes by brewers, beer judges, and others with well-trained palates. These qualitative traits cover a wide range of information about the beer as we will see below.

2.1 Quantitative measurements of beer

The number of quantitative measurements used in characterizing beers is fairly limited, and for our purposes, we'll review only three: alcohol content, bitterness (IBUs), and color.

The most common of the quantitative traits for beer is alcohol content. For most traditional styles, the range of alcohol levels (from minimum to maximum) is 2.5 percentage points or less. For example, the 2021 Beer Judge Certification Program style descriptions list the alcohol range for English IPA as 5.0%–7.5% abv (Strong, 2021). At the narrower end of the range, the same source lists Belgian Pale Ale as ranging between 4.8% and 5.5% abv (Strong, 2021)—a range of just 0.7%.

In modern times, the second most popular quantitative measure has been International Bitterness Units or IBUs. Broadly, IBU levels can vary from 0 to perhaps as high as 100. From a practical perspective, it is rare to see values below 5 or above 75. Style descriptions typically include a maximum and minimum IBU value for each style.

While hops impart a wide range of traits to beer, the IBU value focuses very narrowly on the actual quantity of bitter compounds found in the finished beer. Beyond this numeric value, style descriptions often talk about “perceived bitterness” which is influenced by other attributes of the beer, such as the level of malt flavor and the degree of fermentation.

As previously mentioned, beer color can also be assessed quantitatively. Style descriptions typically list a range of values, from high to low. In some individual styles, the range between the highest and lowest color values may be as low as just 2 SRM (4 EBC) or it may be as wide as 15–20 SRM (30–40 EBC).

Styles tend to display a broader range of quantitative color values when the beer style is at the darker end of the scale than when it is at the paler end. Thus, a “straw” (very pale) style such as American Lager shows a limited range with values between 2.0 SRM and 3.5 SRM. By contrast, a brown beer could have a value anywhere in the range reported for American Brown Ale, from 18 to as high as 35 SRM (36–70 EBC) (Strong, 2021). Thus, there is a strong correlation between darker color and a wider range of reported color values.

The second case where color values for a single style may have a broad range are those styles where the existing commercial examples actually do vary widely in color. An example of this is the style **Weizenbock** where the limited number of well-known commercial examples ranges in color from gold to brown. Because of the relatively small population of examples and since they all fall into a narrow range on all other style parameters, most sources combine them into a single **Weizenbock** style with a large color variation (6–25 SRM) (Strong, 2021).

While quantitative values play a key role in distinguishing between styles, they are sufficient on their own. Take, for example, the BJCP listings for Kölsch and German Pils. The quantitative values for original gravity, final gravity, and alcohol content are 100% identical. Plus, the values for bitterness and color overlap significantly. It would be easy for one particular beer to meet the quantitative parameters of both styles. The question then arises: Why are these considered to be different styles? How do they really differ? To assess these differences, we'll need to rely on sensory analysis of the beers. That's where qualitative criteria come into play.

2.2 Qualitative traits of beer

The qualitative parameters that go into style profiles are drawn from sensory assessments of each beer's attributes. The senses of sight, smell, taste, and touch all play a role in creating the description given in a beer style profile. Typically, the beer style description covers a predetermined set of

characteristics for every beer, as we will see in this section. Let's begin by looking at those qualitative data that correlate to the quantitative values we have already seen before moving on to the overall framework for sensory description of a style.

2.3 Descriptions tied to numbers

Quantitative measures often have a qualitative counterpart—that is a description derived from sensory observation. Qualitative descriptions of quantifiable traits play an important role in understanding the overall character of the beer. This occurs because *perception* of a trait (for instance bitterness) may be somewhat different from what is indicated by a stand-alone quantitative parameter. A good example here is the perception of bitterness in a beer. At first blush, a beer with 60 IBUs might be expected to display a very high level of bitterness. But if it occurs in a beer with high original gravity and alcohol content, the bitterness will likely be substantially balanced by malt flavor and thus perceived as being much lower.

2.4 Putting the senses to work

As a result of the potential differences between perception and what might be interpreted from quantitative values, beer style descriptions tend to provide a thorough, detailed description of all aspects of the style. These descriptions generally assess and describe four areas: appearance, aroma, flavor, and mouthfeel. Within each of these four areas, every aspect of the beer's character is assessed.

Of course, the assessment of appearance is necessarily limited to a short list of items, typically: color, clarity, head/foam formation and retention, and perhaps head/foam color. When relevant to a particular style, other observations may be included such as color from special ingredients.

When it comes to the assessment of aroma and flavor, the following traits are typically assessed and separately noted. Some observations regarding aftertaste may occasionally also be made.

- Description of malt characteristics and perceived levels
- Description of hop aromas/flavors and perceived levels
- Fermentation-related aromas/flavors or effects and perceived level
- Flavors from other ingredients or process steps, their traits and perceived levels
- Balance and overall impression

Finally, the mouthfeel assessment considers weight or thickness as well as textures such as creamy, drying, astringent, etc.

When presented in a set of style guidelines, the information in these four areas of a style can take between 100 and 250 words (in English).

Two typical style descriptions, one brief and one more thorough, are given in exhibit 1 (One BJCP and one BA).

3. The beer styles

Analysis done to prepare this chapter has identified 251 different beer styles and entry categories used by beer competitions around the world. In this section of the chapter, we will discuss these styles and categories with the goal of better understanding how they represent the overall universe of individual beers.

3.1 The analysis

Let us begin by looking at the sources of data and the analysis. This evaluation was performed by examining style guidelines distributed or used by 12 different organizations from around the world. The guidelines examined include the following:

- African Beer Cup
- Australia International Beer Awards
- Beer Judge Certification Program
- Brussels Beer Challenge
- China International Beer Challenge
- *Cerveza Mexico*
- *Concurso Brasileiro De Cervejas*
- *Copa Cervezas de América*
- European Beer Star
- Great American Beer Festival
- Japan Craft Beer Association Competition
- SIBA (Small Independent Brewers Association UK)
- World Beer Cup

Two of these organizations, both based in the United States, each have decades-long histories of maintaining beer style guidelines: Beer Judge Certification Program and the Brewers Association, which runs the Great American Beer Festival competition. Both publish and publicly distribute detailed guidelines that run nearly 100 pages in length. In looking at beer competitions around the world, I found that most lean heavily on one or the other of these sets of guidelines. Some competitions do not list their own guidelines and merely direct brewers to one of these two reference sets. Others use one of the reference sets, but add a handful of additional styles particular to their local or regional brewing scene. Only a couple publish their own unique guideline sets. Yet even in these cases, none is anywhere near as extensive and detailed as the BJCP and GABF sets.

To collect a comprehensive set of international style guidelines and entry categories, the guidelines for each organization were assessed, beginning with the reference sets, BJCP and Great American Beer Festival. Unique styles were found in nine different sets of 13 style guidelines analyzed.

After collecting the full list of unique styles, each was assessed to determine if it identified a clearly defined style or whether it constituted a broader entry category. As discussed earlier in the chapter, entry categories lack the tightly defined traits typical of a style. If a description included certain characteristics, it was classified as a category rather than a style. The following led to designation as a category:

- Entry guidelines that acknowledge that the category includes beers of various styles, including use of the phrases “Varies by style,” “Depending on underlying style,” etc.
- Entry guidelines stating that brewers must do one of the following:
- Identify the base style
- Provide additional information about ingredients, process, brewer’s intentions, etc.

- Very broad quantitative parameters, as defined by¹:
- > 5.0% range for ABV
- > 40 range for IBUs
- > 30 range for SRM

When complete, the analysis yielded 164 unique styles and 87 additional entry categories, thus offering a total of 251 possible classifications for each individual beer.

3.2 Organizing beer styles

As with any large collection, the raw list of styles can benefit from organization according to some broader traits common to many of the individual elements. With beer styles, we commonly start with one of two organizing concepts: fermentation type or geography.

All individual beers can be cleanly classified with regard to fermentation. Each is either lager, ale, hybrid (a combination of ale and lager yeasts and techniques), or mixed (involving non-*Saccharomyces* organisms). As styles are clearly defined, the same is true for them.

Another common approach is to consider the country of origin for each style. This too, is fairly unambiguous, though some confusion can occasionally emerge. But despite the occasional challenge, most styles can be unambiguously assigned to a specific country of origin. Using country of origin as an organizing principle has the advantage of dividing styles into more and smaller groups as compared to using fermentation type. When compared to fermentation type, it also has the advantage of being more familiar to some users than is fermentation type. Many consumers and even retailers and suppliers may not have any idea as to the fermentation type for most styles. But it seems more likely they would know, or have a sense of, the origin of the style. The downside to using geography is that some countries have only one or two styles associated with them. As a result, there must inevitably be an “all others” category.

Of course, other approaches to organizing beer are used as well. One is to divide them by alcoholic strength. Four bands are common: low alcohol, normal strength; elevated alcohol and higher alcohol. A second approach seen in various places is the use of color, for instance: yellow, amber, brown, black.

In the presentation below, we employ both geography and fermentation type to start then add in color and alcohol content. Overall, this divides the styles into groups of four to six styles per grouping.

If we look at our identified style collection by geography, we find the following numbers by country:

Australia 4
 Belgium 20
 Britain 23
 Czech Republic 4
 Germany 32
 Historical 9
 International (various countries): 18

¹If a description gave one broad parameter, say for color, but all other aspects were tightly defined, it was classified as a style. Examples include German-style Rye Ale ([Brewers Association, 2022](#), p. 12B) and South German-style Weizenbock ([Brewers Association, 2022](#)).

Ireland 4
 New Zealand 3
 United States 47
 Total 164

As history has played a key role in the development of beer styles, it is best to start with the countries with the oldest styles, which all come from Europe. We'll then progress to modern New World styles.

3.3 Pilsner and pils

The first well-known pale lager came from the town of Pilsen in the Czech Republic. Now variously known as “Bohemian Pilsner,” “Czech Pilsner” or sometimes just simply “Pilsner,”² the brew surprised the world with its pale color when first produced in 1842. Because of its popularity, it was quickly copied, especially in Germany where the name was shortened to “Pils” to avoid confusion with the original. Today, Czech Pilsner and German Pils are quite different styles (Fig. 4.1).

The **Czech Pilsner** style also known as **Czech Premium Pale Lager** has a rich gold hue compared with the paler straw/yellow of Pils. Alcohol content is similar, though the modern Czech style is more broadly drawn with alcohol from 4.2% to 5.8% abv (Strong, 2021) while the Pils is more constrained at 4.6%–5.3% abv (Strong, 2021). Both exhibit malt flavors similar to white bread or water crackers. While the measured IBU levels are nearly identical, the final gravity of the Czech beers is significantly



FIGURE 4.1 The Pilseners.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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²The Brewers Association prefers to spell it “Pilsener.”

higher, giving it noticeably more body and malt flavor. As a result, the Czech pale lagers have a mellower, richer taste while the **German Pils** is typically perceived as being “crisp” with a light body and a dry, short finish.

3.4 Other Czech lagers

Recent interest in Czech brewing has seen more appreciation for other styles from the country. Remembering that Pilsner fits into the Czech *Premium* Pale Lager style, a second style category recognizes lower-alcohol Czech pale lagers (3.0%–4.1% abv) ([Strong, 2021](#)) and is called simply **Czech Pale Lager**. This style seems to present all the attributes and basic flavors of its higher gravity *Premium* sibling.

Czech Amber Lager varies somewhat in malt flavor from fairly dry breadcrust and American biscuit flavors to sweeter versions with caramel notes. They are full-bodied with moderate bitterness. The alcohol range nearly matches Czech Premium Pale Lager at 4.4%–5.8% abv ([Strong, 2021](#)).

Finally, comes **Czech Dark Lager** in brown to nearly black tones and an alcohol range matching its companions ([Strong, 2021](#)). The recipe commonly includes caramel and debittered roasted malt to give rich complex malt flavors that may include licorice, dark fruits, chocolate, and coffee. As with the Amber, it is full bodied and moderately bittered.

3.5 Normal strength pale German lagers

Pilsner has ultimately inspired nearly every pale lager brewed around the world—though there is a considerable range of differences now. **Munich Helles** shares the color, body, and maltiness of Czech Pilsner, but without the bitterness of either Pilsner or Pils. The result is a malty pale lager popular in Munich and Bavaria.

German brewers also produce a low-calorie lager, or light beer known as **Leichtbier**. Here the alcohol content is considerably lower, running 2.4%–3.6% abv—a full percent or more below either Pils or Helles ([Strong, 2021](#)). But because of the all-malt base, these beers deliver more malt and hop flavor than their adjunct-based American counterparts.

3.6 Elevated alcohol pale German lagers

German brewers produce a significant range of lager beers (See [Fig. 4.2](#)), including three pale styles with alcohol content starting above 5.0%. The first is **Export**, sometimes referred to as a “**Dortmunder**” style. It possesses the color, body, and malt of Munich Helles at a slightly higher alcohol level (5%–6% abv) ([Strong, 2021](#)), a bit more bitterness and added hop aroma and flavor. The net result is essentially a stronger, drier Munich Helles.

The second elevated alcohol lager is the one made famous by the annual Munich Oktoberfest. In Germany, only brewers located within the city limits of Munich can call their beers **Oktoberfest**, so style guidelines sometimes call this style **Festbier**. Once a richly amber brew, the modern Oktoberfests match the color of Helles at a higher alcohol, typically in the range from 5.8% to 6.1% ([Brewers Association, 2022](#); [Strong, 2021](#)). Those brewed elsewhere may be slightly lower in alcohol (down to 5.1% abv), but they maintain the basic malty, full-bodied balance that we first saw in Munich Helles—though with a bit more overall malt and hop flavor to maintain balance.



FIGURE 4.2 Color Spectrum of German Lagers.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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3.7 Amber and dark German and Austrian lagers

The original lagers of Bavaria were amber to dark in color and malt balanced. While the balance is similar to the pale lagers we've seen, the malts bring bread crust, toasty, nutty, and lightly caramelized flavors to the palate. They maintain the clean overall flavor profile typical of lagers with none of the fruitiness characteristic of ales.

Munich Dunkel, once the standard lager of the realm, offers a normal to slightly elevated alcohol content (4.5%–5.6%), attaining a malty balance without any perceived sweetness (17–28 IBUs) (Strong, 2021). Dunkel's color is light brown to brown. A specialty lager, **Franconian-Style Rotbier**, offers similar traits in amber to reddish hues with just a few IBUs more bitterness. Another closely related classic style is **Vienna** lager. Its quantitative parameters match Dunkel and Rotbier closely, but it is made from Vienna malt, rather than darker Munich malt. The result is a paler amber color and lighter flavor to the malt as well as a slightly drier moderate perceived bitterness.

One step up in alcohol from the previous three lagers is **Märzen**—the precursor of the original Oktoberfest brews. With alcohol at 5.6%–6.3% (Strong, 2021) and a bit more restrained bitterness compared with Dunkel, a rich and complex maltiness characterizes most examples.

In general, the dark lagers of Germany steer clear of any of burnt flavors, but the **Schwarzbier** style delivers a roasty flavor profile not found in any other German dark beer. In nearly every other way, the style parallels Munich Dunkel, though it can have higher hop bitterness. This bitterness, combined with the roasty flavors, tends to give Schwarzbier a drier overall flavor profile compared with any other German lager style except for Pils.

3.8 German bock beers: Strong lagers

Under German regulations, beers labeled as “bock” must have an original gravity of at least 16 degrees P (~ 1.065) and those with the label “doppelbock” a minimum of 18 degrees P (1.074). As a result of these higher original gravities, these beers will be higher in alcoholic strength, ranging from 6.3%–7.2% (Strong, 2021) typically for bock and about 6.6% or higher for doppelbock (Brewers Association, 2022). The first style is known as **Dunkles Bock** or “dark bock”—the dark corollary to the Helles Bock seen earlier. As with most German lagers, the flavor profile is malt balanced; this one richly so with complex toasty, crust-like flavors. The style lacks hop aroma or flavor to speak of and has just enough bitterness to keep it from being sweet.

On the pale side is **Maibock**, also known as **Helles Bock**. It starts like Helles and Export, but with a bit more malt owing to the higher gravity. While malt dominates, this is the hoppiest of the bock styles with medium-low hop aroma and hop flavor ranging from low to medium, all from spicy, floral German varieties.

The higher alcohol **Doppelbock** style generally displays a similar toasty malt profile and off-sweet balance. The only difference is that this stronger style allows for pale versions made solely from pilsner or two-row malt. The paler versions will still have a very malt-oriented balance with significant malt complexity, but they will lack the toasty, bread crust notes of the darker versions.

Finally, we have **Eisbock**. This is a fortified beer which cannot be produced under a brewer’s license in some countries. The fortification is achieved by freezing the beer and separating the liquid from the ice—which is solid water. The resulting beer is higher in alcohol with expected range being from 9% to 14% abv (Strong, 2021). Beers in this style will be very full bodied, very rich and sweet usually with noticeable alcohol presence. Hop aromas and flavors are absent and perceived bitterness is very low.

3.9 South German wheat beers

While well-known for its lagers, Bavaria is also famous for wheat beers which are, in fact, ales. Fig. 4.3 These beers are called by the German words for “wheat” (*weizen*) and “white” (*weiss*). Style guides may list them as Weizen, Hefeweizen, Weiss, or Weissbier. Beers by these names are typically unfiltered so that they are hazy with protein and generally still contain some yeast. Only those designated as “Kristal” (crystal) are filtered.

Wheat malt must account for at least 50% of the grain bill under German law and generally comprises 67% or two-thirds of it. These beers typically have a very soft, white bread or even bread dough-like flavor with very little hop character. Indeed, the key flavors of the style come from Weizen/Weiss yeast strains which impart high levels of fruitiness, typically with a banana character. The yeast also produces phenols imparting clove-like flavors to the beer. Thus, banana and clove are the defining flavors of Bavarian-style wheat beers.

The basic **Hefeweizen** or **Weissbier** is yellow to gold in color and runs 4.9%–5.6% abv (Strong, 2021)—though some commercial examples go as low as 4.3% abv. Variations on this include the filtered **Kristal Weizen** which exhibits a lighter body and a bit less bready, yeasty flavor; a low calorie (and lower abv) **German-Style Leichtes Weizen**; and **South German-Style Bernsteinfarbenes Weizen** which uses the same recipe as Hefeweizen/Weissbier but is amber in color. Another related style is **German-style Rye Ale**, sometimes called **Roggenbier**. This style matches the fermentation

German Wheat Beers



FIGURE 4.3 German Wheat Beers.

The glassware shown for each beer is typical of the style, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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traits and most quantitative parameters of Hefeweizen/Weissbier, but it includes at least 30% rye malt for a darker color, appearing dark amber to dark brown ([Brewers Association, 2022](#)).

Brewers also make a **Dunkel weiss** or **Dunkel weizen** style using some dark malts in the same alcohol range, but this style displays some toasty to light caramel flavors. The Dunkel versions appear full amber to mid-brown. Finally, there is a **Weizenbock** style. Like other bock beers, it must be brewed to a specific original gravity ensuring a higher alcohol content running from 6.5% to as high as 9% abv ([Strong, 2021](#)). Here, both pale and dark interpretations are found. The pale ones go from gold to light amber; the dark ones range from amber to mid-brown.

3.10 German rhine ales

Bavaria is not the only place in Germany where ales are popular. Along the Rhine river, two towns, Düsseldorf and Cologne, only 50 km from each other have resisted the movement to lager brewing for hundreds of years. Both have adopted the practice of cold-conditioning or lagering their ale-fermented brews to give them a smoother flavor than most ales. And while the finished beers do not exhibit the same level of fruitiness as other ales, they show clear flavor differences from lagers with discernible fruity fermentation notes. See [Fig. 4.4](#).

Each city brews its own distinct style of ale. In Cologne, the style takes the German name for the city (Köln) and is known as "**Kölsch**." This style matches German Pils on nearly all parameters. Kölsch can be more richly gold-hued while Pils can be the palest straw tone. They also overlap significantly on bitterness, though Pils definitely goes to higher IBUs. The lower IBUs in Kölsch combined with the subtle fruitiness from the fermentation makes the style less crisp and more harmoniously balanced between the elements of malt, hop, and fruit.

German Rhine Ales



FIGURE 4.4 German Rhine Ales.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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The Dusseldorf brewers make a style called **Altbier** (old beer) as ale brewing an older technique than the making of lager. Their beer again matches the original gravity and alcohol content of Pils (and Kölsch) nearly exactly running 4.3%–5.5% abv (Strong, 2021). The Altbier uses some darker malts and though the color varies somewhat, it lies consistently in the amber range. Altbiers do include a firm bitterness with IBU values starting a bit higher than those of Pils and running to an even higher maximum value of 50 IBUs. This top value is attributable imparts an extreme bitterness seen in just one commercial example. Other examples are not nearly so extreme, though all possess a firm bitterness in a well-attenuated beer. Many also exhibit some spicy hop aroma and flavor.

3.11 Mixed fermentation German beers

For most of human history, beers have been fermented by multiple organisms that contributed a range of flavors that we now consider acidic or funky. While pure culture *Saccharomyces* fermentations are now the norm in brewing, styles produced by mixed fermentations—involving other yeast and bacteria—are easily found. Many offer high attenuation, often with high carbonation along with a lactic tartness and consumers find this combination refreshing.

The most resilient of these styles found in Germany is **Berliner Weisse** which was still being produced commercially at the outset of the craft beer movement. This is a tart, lactic style with low alcohol content (2.8%–3.8% abv), no perceived hop character and a very high carbonation level (Strong, 2021). Made from pilsner and wheat malts, it exhibits a bread dough character and some examples show a light fruitiness. A closely related style, adhering to the same tightly defined parameters, is used as the basis for the addition of fruit and other special ingredients and called **Specialty Berliner-Style Weisse**.

A style called **Lepzig-style Gose**, or just Gose, has been resurrected from historical records in recent decades. As with Berliner Weisse, fermentation includes lactic bacteria, but the resulting acidity

is more subdued. The beer is also flavored with coriander and table salt and has a low perceived hop bitterness along with a higher alcohol level: 3.5%–4.7% abv (Strong, 2021). A matching style that accommodates additional flavors from special ingredients is called **Contemporary Gose**.

The **Lichtenhainer** style brings another twist in tart beer with the inclusion of smoked character in a lactic-flavored beer. The bitterness and alcohol content match those of Gose with high attenuation and high carbonation like Berliner Weiss (Strong, 2021).

In contrast to the previous five styles, we have **Adambier** a strong (7.1%–8.7% abv [Brewers Association, 2022]), dark and hoppy historical mixed fermentation brew. Here, *Brettanomyces* yeast strains are expected as a part of the fermentation, but sourness is not required. A bit of smoked malt as well as barrel aging may play a role in production.

3.12 German smoked beers

In a north Bavarian region known as Franconia, several breweries still produce their own malt in the old way resulting in smoke (*rauch*) flavored beers. Some brewers produce more than one style of beer with the smoked malt, so some style guidelines recognize each of them as a style. The beers themselves exhibit the general traits of the underlying styles, but with Beechwood smoke flavor infused into the beer. The most common style is **Bamberg-style Märzen Rauchbier** with the others being **Bamberg-style Weiss Rauchbier**, **Bamberg-style Helles Rauchbier**, and **Bamberg-style Bock Rauchbier**. A number of competitions do not break out these styles individually, preferring to group them together in an entry category for Bamberg-style Rauchbier.

3.13 Kellerbiers

If you visit a brewery taproom in Germany, you may well encounter Kellerbier. This is not a beer style so much as a method of serving beer where immature beer is drawn off of the lagering tank and served without filtration or final carbonation. This technique can be used for any lager and even for weizen/weissbeers, but most breweries only offer one of their regular beers as their Kellerbier. Beer served this way will still have some of the fruity and sulfury flavors that are eliminated during lagering as well as some flavors normally removed during filtration. Finally, it is not as highly carbonated which makes it a bit less dry and softer on the palate overall. Some competition guidelines offer a single entry category called **Kellerbier** and allow brewers to specify the base style. Others offer several different styles of kellerbier to match the beer styles most commonly offered this way: **Kellerbier Hell**, **Kellerbier Dunkel**, and **Keller Pils** (Private Breweries Bavaria e.V., 2022).

3.14 British and Irish ales

Modern bitters trace their origin to the development of India Pale Ale in the 19th century: a highly hopped and well-attenuated pale ale. Fast forward nearly 200 years and we find a set of three very similar styles that start at a low alcohol content and climb steadily upward. All are brewed from pale ale malt with some caramel or crystal malt added and they are typically served on draft with about half the carbonation found in most packaged beers.

At the low-alcohol end of this series is the style commonly known as **Ordinary Bitter** at 3.2%–3.8% abv (Strong, 2021) with firm bitterness and refreshing drinkability. See Fig. 4.5. Next up

Pale English Ales



FIGURE 4.5 Pale English Ales.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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is **Best Bitter** at 3.8%–4.6% abv (Strong, 2021). This level exhibits more malt flavor and body while maintaining good drinkability and medium to high bitterness and perhaps a bit more characteristic English hop flavor and aroma. A related style from Ireland is **Irish Red Ale** has a similar alcohol level (3.8%–5.0% abv) (Strong, 2021) and a maltier balance. Despite this, it can have a dryness from the use of low amounts of roast barley for color.

At the top of the ladder of bitters comes **Strong Bitter**, sometimes known as **Extra Special Bitter**, with alcohol content at 4.6%–6.2% (Strong, 2021). Here malt definitely plays a bigger role with additional hop flavor and aroma common as well. A paler relative to the stronger bitters is found in the style named **English-Style Summer Ale** or **British Golden Ale**. This style drops the crystal or caramel malt and bumps the bitterness up just a bit for a paler, drier, and highly drinkable seasonal quaff. Alcohol runs 3.8%–5.0% abv (Strong, 2021).

UK brewers also produce **English-Style Pale Ale** which is typically a packaged product with fairly standard carbonation levels and similar in most regards to Strong Bitter. The “English-Style” prefix denotes the use of traditional English hop varieties which differentiates it from various New World versions. **English IPA** overlaps the same alcohol content range as Strong Bitter and English-Style Pale Ale, then exceeds them, running from 4.5% to 7.1% (Brewers Association, 2022) but with somewhat higher bitterness levels and again features English hop varieties.

3.15 British strong ales

British brewers offer stronger ales variously classified as **British Strong Ale**, **English Barley Wine**, or simply **Old Ale**. See Fig. 4.6. As a group, they range from a low of 5.5% abv to as high as 12.0%, with the Barley Wine on top (8.0%–12.0% abv) (Strong, 2021) and the other two paralleling each other at

Strong British Ales



FIGURE 4.6 Strong British Ales.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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the lower end. Beyond alcoholic strength, these styles vary a good bit: some are stronger versions of bitters, others stronger versions of mild or brown ale. None, however, tip over into the roasty flavors of porter or stout. The Old Ales are expected to exhibit flavors resulting from aging of the beer, either in bottle or prior to packaging. All are generally bottle-conditioned, though draft versions are seen occasionally.

3.16 Porter

The leading beer of the English industrial revolution, Porter has a 300-year history during which it has changed many times. The biggest change came with the introduction of black malt in the early part of the 19th century. Today, formulations without black malt are known as **Brown Porter**, those with it as **Robust Porter**. The roasty flavors and malt-derived bitterness that come from black malt separate the two. The GABF guidelines offer both types while BJCP specifies just one approach labeled **English Porter** and based upon modern commercial examples. Across the three, alcohol content runs from 4.0% to 6.6% with the lower end seen in modern English Porter (4.0%–5.4%) (Strong, 2021), with historic Brown Porter at 4.6%–6.0% (Brewers Association, 2022) and Robust Porter ranging from 5.1% to 6.6% (Brewers Association, 2022).

3.17 Milds and brown ales

In the twentieth century, a dark ale called **English-Style Dark Mild** was commonly the working man's quaff in Britain. See Fig. 4.7. It was low in alcohol and very lightly hopped with dark malt providing the primary flavor. In recent decades a pale version, **English-Style Pale Mild Ale**, has emerged. Alcohol runs as low as 3.0% (Strong, 2021), but some stronger examples are seen today, up to 4.4% (Brewers Association, 2022). **British Brown Ale** or **English-Style Brown Ale** offers flavors similar to the dark mild in a higher-alcohol formulation, ranging from 4.2%–6.0% abv (Brewers Association, 2022; Strong, 2021). As with most "brown" British beers, roast character from black malt or roast barley are not considered a part of these styles. Finally, one or two commercial examples of **London Brown Ale** may still be found, exhibiting a luscious sweetness that includes caramel and toffee notes with alcohol content ranging from 2.8% to 3.6% (Strong, 2021).

FIGURE 4.7 Dark British Ales.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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3.18 Stouts of all types

Stout first evolved from porter, and now encompasses a range of strengths. Stouts typically exhibit roast grain flavors from roast barley reminiscent of dark roast coffee or dark chocolate. See Fig. 4.8.

The best-known stout is **Irish Stout** also called **Dry Stout**—the style epitomized by an iconic brand from Dublin. Many who have not tried it are surprised at the low alcohol content (usually close to 4.0% abv) and light body. The style also exhibits surprisingly high bitterness levels (up to 45 IBUs), but remains enjoyable in large part due to the low levels of carbonation—a feature that is masked by the foamy nitro pour used in both draft and packaged versions.

The same approximate formulation gets offered in progressively stronger (and meltier) versions: **Irish Extra Stout** (or **Export Stout**) at 5.0%–6.5% abv (Strong, 2021), and **Foreign Extra Stout** at 6.3%–8.0% abv (Strong, 2021). At the top of this ladder comes **Imperial Stout** (8.0%–12.0% abv (Strong, 2021)), though this style has its roots primarily in Britain. At the top end, body is rather full and the malt flavor rich and complex with bitterness more likely perceived as moderate rather than high.

With alcohol content at 6.5%–9.5% (Brewers Association, 2022; Strong, 2021) **Baltic Porter** is an interpretation of Imperial Stout that has a long history in the Baltic countries. Unlike the original British and Irish stouts, this one is fermented by lager yeast and uses debittered black malts. The result is malty and sweet with flavors of “caramelized sugars, dark sugars, and licorice” (Brewers Association, 2022) Hop flavor and aroma are very low with low to medium-low bitterness.

Sweeter formulations of stout come in **Sweet/Cream Stout** (4.0%–6.0% abv (Strong, 2021) and **Tropical Stout** (5.5%–8.0% [Strong, 2021]). The sweet flavor of Milk Stout is commonly assured by the addition of lactose sugar. **Oatmeal Stout** (4.2%–5.9% abv [Strong, 2021]) may be fuller bodied and tend toward sweetness or drier and lighter, often with a grainy note, depending upon the brewer.

3.19 Scottish and scotch ales

The brewers of Scotland produce four specific styles that are uniquely found in their country. They consist of three low-to-moderate strength ales analogous to the English bitters and one strong ale. Recipes may include a small portion of roast barley to impart a faint reddish to amber color along with a faintly toasty or toffee note that is characteristic of the styles. Caramel malt may be used in addition

Global Stout Collection



FIGURE 4.8

Global Stout Summary.

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or as an alternative. The lower-alcohol styles each follow a similar formulation as they get progressively stronger, starting with **Scottish Light** (2.5%–3.3% abv) (Strong, 2021) then **Scottish Heavy** (3.3%–3.9% abv) (Strong, 2021) and finally **Scottish Export** (3.9%–6.0% abv) (Strong, 2021).

At the high end is **Strong Scotch Ale** sometimes known simply as **Scotch Ale** and commonly called and labeled as **Wee Heavy**. Beers in this style run from 6.5% up to 10% abv (Strong, 2021), exhibiting sweetness and considerable richness featuring caramel and toffee flavors.

3.20 Traditional Belgian and French ales

Among the most popular Belgian ales is a refreshing and moderate alcohol (4.5%–5.5% abv) (Strong, 2021) wheat beer called **Witbier** (white beer). Like German Weizens, it has very low hop bitterness,

flavor, and aroma as well as a hazy appearance. It also offers unique fruit and spice flavors but they come from the use of spices including coriander and Curacao orange peel rather than phenol-producing yeast. See Fig. 4.9.

Saison is perhaps the driest and hoppiest of Belgian brews, though the use of European noble hops creates a complex and well-integrated overall flavor profile. Here, phenol-producing yeasts contribute peppery flavors rather than the clove notes found in weizens. Formal style descriptions start with **Classic Saison** (Brewers Association, 2022, style 84) or **Modern Saison** (Brussels Beer Challenge, 2023) which runs 5.0%–7.0% abv and gives the color as gold to mid-amber with the abovementioned flavor traits. But the style has a history of variability and has served as a palette for innovation among craft brewers in recent decades so there are now a number of variant styles. The Brussels Beer Challenge lists **Traditional Saison** as having lower-alcohol levels (3.5%–5.5% abv) (Brussels Beer Challenge, 2023) plus flavors from mixed fermentation. Brussels Beer Challenge also list a **Double Saison** with alcohol at 7.0%–9.0% abv (Brussels Beer Challenge, 2023) which parallels the Modern Saison flavor description. GABF includes a **Specialty Saison** (Brewers Association, 2022) category and the European Beer Star a similar one named **New Style Saison**. These two style guidelines expand the color range up to light brown and alcohol up to around 9.0% while allowing the additional ingredients like spices and herbs as well as mixed fermentation traits.

The French **Bière de Garde** style offers a malty counterpoint to Saison along with unique flavor attributes. The name translates as “beer to keep” and traditional examples are stored (aged) for a period before consumption. Beers in this style are made a bit stronger in alcohol, ranging from 6.0% to 8.5% abv (Strong, 2021). Producers have long made beers in various colors, including blonde, amber, and brown and the malt flavors will vary with the color.

3.21 Modern Belgian ales

These straw-to-amber ales result from modern brewing techniques and cover various alcohol levels. See Fig. 4.10. At 4.8%–5.5% abv (Strong, 2021), **Belgian Pale Ale** is an all-malt ale in the gold to amber range lacking in phenolic yeast flavors, mixed fermentation traits, or assertive hop character. A half step up in alcohol content (5.1%–6.3% abv) (Brewers Association, 2022) is the gold to amber **Speciale Belge** where both yeast phenols and hop flavor/aroma occur at low to moderate levels over a

FIGURE 4.9 Traditional Belgian Ales.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer’s recipe, glass dimensions, and lighting conditions.

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**FIGURE 4.10 Modern Belgian Ales.**

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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subdued malt character. Next is **Belgian Blond Ale** at 6.0%–7.5% abv (Strong, 2021), a straw-to-gold ale that offers a palate rich with clean bready malt flavors plus fruity, spicy yeast notes. At the top of this group is **Strong Golden Ale**, a very pale and highly attenuated strong beer with alcohol at 7.5%–10.5% abv (Strong, 2021). It offers a similar flavor profile to Belgian Blond Ale, but with a lighter mouthfeel and fermentation flavors long in fruitiness and largely lacking in peppery phenols.

3.22 Monastic ales

Belgium has long been home to a number of Trappist breweries and other monastic-related brews described as “Abbey ales.” See Fig. 4.11 These producers focus on a handful of styles in a range with increasing alcohol content and each style is broadly named for its relative alcoholic strength. **Belgian Single** (Strong, 2021) starts it off at 4.8%–6.0% abv, a moderately bitter blond ale with high levels of fruity and spicy yeast character along with notable hop character.

Belgian Dubbel comes next at 6.0%–7.6% (Strong, 2021) abv red-amber to brown in color with matching toasty malt flavors as well as notes reminiscent of dark fruits. Despite low bitterness, it finishes quite dry due to the use of sugar for as much as 20% of the extract. In the middle is arguably the most popular style of the group, **Belgian Triple** at 7.5%–9.5% abv (Strong, 2021). Here again, the color turns a golden pale and the peppery phenolics and fruity esters take center stage. Cracker-like malt supports these and alcohol makes its presence known. As with the Dubbel, the body is light and finishes fairly dry thanks to the use of sugar for a portion of the extract.

Pushing the alcohol content just a bit higher is the style known as **Belgian Dark Strong Ale** often known as **Belgian Quadrupel**.³ This full amber to brown brew clocks in at 8.0%–12.0% abv (Strong, 2021) with now-familiar flavors: rich and dark malt and fruity notes countered by the spicy phenols from the yeast. As with the others, sugar reduces the body and sweetness and the resulting beer is often seductively smooth.

³The GABF guidelines divide these into two separate styles.

Monastic Ales



FIGURE 4.11 Monastic Ales of Belgium.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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3.23 Belgian mixed fermentation styles

Belgium harbors age-old methods of beer production in a number of working breweries around the country. They produce a range of styles using methods that include spontaneous fermentation, fruit additions, wood- and barrel-aging, blending, and Champenoise bottle-conditioning.

Three of these styles will all generally be produced by any lambic brewery: Lambic, Gueuze, and Fruit Lambic. See [Fig. 4.12](#) All three come from the same initial brew of raw wheat and malted barley, with the outcome dependent upon aging, blending, and fruiting over several years. **Lambic** is the young, unaged product of the process, typically served within 6 months to a year old. It is tart and somewhat funky, generally served from a pitcher in the brewery taproom with little to no carbonation. Alcohol content runs 5.0%–6.5% abv ([Strong, 2021](#), p. 47).

After up to 3 years aging and blending of various years and batches, lambic with no fruit added is called **Gueuze**. This beer will be bottle conditioned and runs 5.0%–8.0% abv. The beer is well-carbonated and refreshingly tart with a pleasant blend of flavor notes resulting from *Brettanomyces* yeast as well as lactic and other bacteria (barnyard, leather, earthy, goaty, hay) ([Strong, 2021](#)).

The final type, **Fruit Lambic** results from the addition of fruit, commonly raspberries or cherries, to the aging beer which then undergoes additional fermentation. Blending is used to create final batches for bottling. Brewers' approaches to this style vary from quite dry and tart to nearly sweet.

Two Flanders styles share similarities, but ultimately diverge. **Flanders Red Ale** (sometimes Oud Rood) delivers tart, fruity flavors from a Munich or Vienna malt base at 4.6%–6.5 abv ([Strong, 2021](#)). Much of the character results from aging in oak vessels which harbor acetic bacteria and *Brettanomyces* species. **Oud Bruin** (4.0%–8.0% abv [[Strong, 2021](#)]) delivers more caramelly malt flavor and

Mixed Fermentation Belgian Styles



FIGURE 4.12 Mixed Fermentation Belgian Styles.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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body along with fruitiness from a pilsner malt base plus dark caramel and even roast malts. The brown ales commonly age in stainless tanks, though non-*Saccharomyces* organisms participate to add a tangy sour flavor to the finished beer.

3.24 New world pale lagers

European brewers who emigrated to North America in the 1800s, were compelled to replace a portion of the malt in their recipes with corn or rice to achieve beer clarity. Since that time, adjuncts have been a standard part of new world pale lagers, but myriad variations now exist as shown by the many style categories offered.

The **American Lager** style sets parameters common among the related styles: straw to light yellow in color, very low to low bitterness, very low to low malt flavor, with a generally clean lager fermentation profile. This central style weighs in at 4.2%–5.3% abv ([Strong, 2021](#)). The closest relation is **American Light Lager**, a reduced calorie variation, which generally shares the same traits but with a lower-alcohol content 2.8%–4.2% abv ([Strong, 2021](#)). Another close variant is the stronger **Malt Liquor** at 6.3%–7.6% abv ([Brewers Association, 2022](#)). Original gravity and bitterness both increase to maintain the same basic balance with some slight malt sweetness expected.

A version of American lager popular prior to 1920, once called Classic American Pilsner, now goes by **Preprohibition Lager** ([Strong, 2021](#)) or **American Pilsner** ([Brewers Association, 2022](#)). This formulation contains less adjunct than the styles above, thus exhibiting a moderate malt flavor. Hops provide substantial balance and bitterness (25–40 IBUs) ([Strong, 2021](#)) giving it a flavor profile more akin to modern European pilsners, though sometimes with greater alcohol (4.5%–6.0% abv). ([Strong, 2021](#)).

The Great American Beer Festival style guidelines list “Contemporary” variations of the main light lager styles: **Contemporary Light Lager**, **Contemporary American-Style Lager**, and **Contemporary American Pilsener**. These styles use less adjunct than their noncontemporary counterparts and may, in fact be all malt in their formulation. Unlike similar international styles, they do not allow for the use of sugar as an adjunct and the values for color and bitterness remain very low, in line with the other American lager styles.

While American competitions no longer offer a specific style for it, the Brussels Beer Challenge still includes a category for **Ice beer**. This highly attenuated pale lager generally matches the flavor profile of American Lager but with less body and perhaps slightly more alcohol and bitterness ([Brussels Beer Challenge, 2023](#)).

Brewers around the world make beers similar to American-style Lager but with less adjunct or, often, none at all. Though still mild compared with German Pils and Czech Pale Lager, these beers generally offer more malt and hop flavor than the American-style versions. They go by the style name of **International Pale Lager** ([Strong, 2021](#)) or perhaps **International Pilsner** ([Brewers Association, 2022](#)) with alcohol ranging up to 6.0% (4.5%–6.0%) ([Strong, 2021](#)). Compared with American Lager, they bump the color up a notch toward gold and the bitterness up another 5–7 IBUs as well. Guidelines also now recognize a **New Zealand-style Pilsner** style that employs hops from that region which are said to convey flavors of “lime, white grapefruit, gooseberry, honeydew melon, with a light green bell pepper or grassy aspect” ([Strong, 2021](#)).

The GABF offers an “**Australasian, Latin American or Tropical-Style Light Lager**” style which differs little from their American-style Lager save for the allowance of sugar as an adjunct and tiny increases in color and bitterness. The Australian International Beer Awards offers a clearly different **Australian Lager** style which not only allows the use of sugar, but places the color in the gold to light amber range and specifies that hop aroma must not be derived from traditional European noble hops. It also specifies higher bitterness (up to 30 IBUs) and offers greater leeway for fruity esters compared with the US version, though the alcohol content basically matches American Style Lager at 4.0%–5.0% abv ([Australian International Beer Awards, 2023](#), pp. 21–33).

The final entry in the pale lagers category comes from the Brussels Beer Challenge, called simply **Hoppy Lager**. It offers much higher hop character and bitterness than any of the abovementioned pale lager styles yet retains the straw to light gold color. Malt character climbs to “medium” to balance the additional hop usage and with that, alcohol rises to as high as 7% abv ([Brussels Beer Challenge, 2023](#)).

3.25 Amber and dark new world lagers

American style guidelines recognize three separate amber lagers. **American Amber Lager** exhibits low to medium-low toast or caramel malt flavors and low to medium-high hop traits including bitterness (up to 30 IBUs). Alcohol runs 4.8%–5.4% abv ([Brewers Association, 2022](#)). A uniquely American beer style, dubbed **California Common**, results from warm fermentation using lager yeast in wide flat vessels similar to a coolship. It offers medium-level toasted/caramel malt flavors and medium to medium-high bitterness (up to 45 IBUs) while keeping hop flavor and aroma low to medium-low and abv similar to the American Amber Lager ([Brewers Association, 2022](#)). Finally, American brewers make a hoppier version of the German Festbier or Märzen, dubbed simply **American-style Märzen/Okttoberfest** with 5.0%–6.1% abv ([Brewers Association, 2022](#)). **International Amber Lager** matches the American version pretty closely, with perhaps a bit more malt flavor,

the clear floral/spicy presence of European noble hops and a slightly broader alcohol range: 4.5%–6.0% abv (Strong, 2021). All of these styles reach medium amber color (14–15 SRM) and most will be no paler than light amber (6–8 SRM), though the Märzen/Oktoberfest range goes to light gold (4 SRM) (Strong, 2021).

Dark lager styles exhibit some chocolate or roast notes and medium-dark amber to brown color. **American-Style Dark Lager** exhibits low to medium-low malt flavors with very low to low hop traits, including bitterness. Alcohol runs 4.1%–5.6% abv (Brewers Association, 2022). **International or European Dark Lagers** show more malt flavor ranging from low to medium and more bitterness running medium-low to medium-high, but hop flavor and aroma remain low. Alcohol remains in a tighter range: 4.8%–5.3% abv (Brewers Association, 2022; Strong, 2021).

3.26 New world wheat and blond ales

American Wheat Ales have long been popular with American craft brewers, though the style's flavor profile lacks the clove and banana notes characteristic of German Weizens. Wheat malt typically accounts for about one-third of the grist and hop traits vary from low to medium-high depending upon the brewer. Typically these are easy-drinking beers with bready, doughy malt flavors. Alcohol runs from as low as 3.5% up to 5.6% abv (Brewers Association, 2022; Strong, 2021). Some guidelines distinguish between clear versions versus those with a yeast haze (Brewers Association, 2022). **Golden Ale** or **Blonde Ale** is similar in balance and impression, but with little or no wheat malt, no yeast haze, and no more than medium bitterness at the high end. Both styles are straw to gold in color (Brewers Association, 2022; Strong, 2021). **Dark American Wheat Beer** matches the alcohol of these styles but increases the malt flavor with caramel, toast, chocolate, and even some roast notes. Bitterness is typically more restrained. See Fig. 4.13.

Cream Ale may employ both corn and sugar adjuncts to create an ale that is similar to American Lager, though with just a bit more malt and hops. Sometimes characterized by the presence of a corn-like flavor. Alcohol and color match the range seen across American Lager and American Wheat Ale.

3.27 American pale, red, and amber ales

Beers falling from about 5 to 18 on the SRM color scale (10–36 EBC) can vary in both tint and hue to meet a range of descriptors including pale, red, and amber. The variations give rise to a range of styles which vary primarily by appearance and alcoholic strength. The “pale ale” styles run gold to light amber (~7–10 SRM, ~14–20 EBC) and hop flavors take the lead, with bitterness running from medium to very high across them. **American Pale Ale** is the classic with medium to medium-high perceived bitterness and high hop flavor and aroma in a beer that runs 4.4%–5.4% abv (Brewers Association, 2022). **Juicy or Hazy Pale Ale** drops the bitterness to low to medium adding grains like oats to create a hazy appearance but maintaining very high levels of hop flavor and aroma. **American-Style Strong Pale Ale** increases alcoholic strength to 5.6%–7.0% (Brewers Association, 2022), upping perceived bitterness to high and the **Juicy or Hazy Strong Pale Ale** matches this strength, but drops the bitterness and adds haze-causing ingredients.

The Amber/Red Ales provide more malt emphasis across a range of alcoholic strengths. **American Amber/Red Ale** runs 4.4%–6.1% abv (Brewers Association, 2022) and while bitterness runs medium to medium-high, hop flavor remains relatively low and malt flavor rises to medium-high or high

Popular American Styles



FIGURE 4.13 Popular American Styles.

The glassware shown for each beer is typical of the region where the style originates, but other types of glasses may frequently be used. The beer color shown in each picture is within the range for the style, but actual beers may look different as beer color is influenced by several factors including the brewer's recipe, glass dimensions, and lighting conditions.

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perception. **Strong Red Ale** steps the alcohol up to 6.1%–7.9% abv ([Brewers Association, 2022](#)) and brings hop flavor and bitterness up considerably while maintaining medium to medium-high maltiness. This trend continues with **Imperial Red Ale** running 8.0%–10.6% abv ([Brewers Association, 2022](#)) with high hop flavor and aroma, very high bitterness but still showing medium to medium-high malt flavor. A final style here is **American-Style Extra Special Bitter** which broadly follows the Strong Bitter category described earlier but allows hop flavors and aromas of any type rather than requiring them to be from typical British varieties.

3.28 Other new World pale ales

Various pale ale styles have been defined outside of Europe and America in recent decades, but the first, **Australian Sparkling Ale**, has been around for more than 150 years. It has a mild, malty sweetness countered by medium to medium-high bitterness and alcohol in the range from 4.5% to 6.0% abv. It is highly attenuated and well carbonated to make a dry and refreshing ale (Strong, 2021). The GABF lists an **Australian-Style Pale Ale** and a **Classic Australian-Style Pale Ale**, both ranging from 4.0% to 6.0% abv (Brewers Association, 2022, p. 58B). The Classic style has low malt flavor, low to medium hop flavor and bitterness and perceptible fruity esters in an overall mild flavor profile. The nonclassic pale ale offers more malt and hop character and specifies that a wide range of New World hop flavors is allowed. Both run straw to light amber in color.

New Zealand-Style Pale Ale matches the general traits of Australian-Style Pale Ale with hop flavors typical of New Zealand-grown hops including tropical fruit, passion fruit, stone fruit, cut grass, or diesel (Brewers Association, 2022). GABF guidelines also list an **International Pale Ale** which is broadly drawn with regard to hop flavor (low to high) and the type of hops flavors expected. Bitterness should be medium to low and alcohol 4.4%–6.6% abv (Brewers Association, 2022). Both Australian International Beer Awards and European Beer Star list a **New World Pale Ale** or just **New Style Pale Ale** where brewers should enter pale ales made with emerging hop varieties having traits not described in the other pale ale categories.

3.29 Americanized India pale ale

We previously described the modern descendant of the original India Pale Ale, English IPA. American craft brewers seized on this style in the 1990s and made it their own with American hop varieties—often applied to extreme levels. Today, numerous variants exist and the first step to understanding them is to assess alcoholic strength. **American-Style India Pale Ale (IPA)** runs 5.5%–7.5% abv (Strong, 2021) with low to medium malt flavor and prominent to intense American hop flavor and aroma with matching bitterness levels. This sets the default alcohol level for IPAs generally. A lower-alcohol version, dubbed **Session India Pale Ale (Session IPA)** provides similar balance and flavor emphases, but in a lower-alcohol range, from 3.0% to 5.0% or even lower (Strong, 2021). At the higher alcohol end is **Double IPA** or **Imperial IPA** where alcohol content ranges from 7.6% to 10.6% (Brewers Association, 2022, style 66). Sugar is sometimes used as an adjunct in this style to reduce malt flavor and maintain the dominant character of the hops.

Both the **Hazy or Juicy India Pale Ale** and the **Hazy or Juicy Imperial IPA** styles offer substantially reduced bitterness and include haze-making grains such as oats while maximizing hop flavor and aroma, typically from American varieties.

New Zealand-Style India Pale Ale (Brewers Association, 2022) and **Pacific IPA** (Brussels Beer Challenge, 2023) exhibit the aromas and flavors of hops from these regions. An **Argentine IPA** (*IPA Argentina*, in Spanish) follows suit with hops from that country, but leans more toward modern English IPA with a bit more malt character (Strong, 2021; Copa Cervezas De América, 2023).

Many subtypes of IPA have color names. **White IPA** is a hybrid with either Belgian Witbier or German Weizen featuring spicy notes from fermentation or spice additions. **Red IPA** has the maltiness of Amber Ale with the hopping and alcohol of IPA. **Brown IPA** includes the dark caramel, chocolate, toffee, or dark fruit malt flavors of brown ale, but with the hop-forward balance and lighter mouthfeel

of IPA. **Black IPA** offers just a hint of the roasty, chocolate flavors found in porters and stouts, maintaining hop dominance in the flavor profile in a black-colored beer.

Some other variants of IPA include **Belgian IPA** (Brussels Beer Challenge, 2023; Strong, 2021) which should display some phenolic/spicy fermentation notes; **Rye IPA** which includes rye malt to create a “bready, peppery” character (Strong, 2021); and **Brut IPA** which is highly attenuated and carbonated to mimic the character of champagne.

Finally, we have lager variants. **India Pale Lager** (Brewers Association, 2022) or perhaps **Cold IPA** (Brussels Beer Challenge, 2023) removes the fruity fermentation esters from the flavor profile but otherwise maintains the balance of IPA. Some versions use adjuncts to achieve higher attenuation and dryer finish.

3.30 American dark ales

Most Americanized ales were adapted from British or Irish styles, starting with **American Brown Ale**. This style incorporates medium-level caramel and chocolate-like flavors but balances them with significant hop notes and medium to high bitterness. Alcohol ranges 4.3%–6.2% abv (Strong, 2021). Compared with this, **American Porter** exhibits higher bitterness and a bit more color accompanied by a bit more burnt character. Alcohol can also be a touch higher (4.8%–6.5% abv) (Strong, 2021, p. 36), and hop aroma and flavor can vary from high to low. An **American-style Imperial Porter** is also found with alcohol running from 7.0% to 12.0% abv (Brewers Association, 2022). The emphasis here is on caramel and chocolate malt flavors with both hop flavor/aroma and bitterness generally kept no higher than a medium level.

Moving from brown to black beers, we start with **American-style Black Ale**, which offers the color of stout but the malt flavor of porter, where acrid, burnt notes are not allowed. Perceived bitterness is medium-high to high and alcohol falls within that of American-Style Stout at 6.3%–7.6% abv (Brewers Association, 2022). **American-Style Stout** exhibits the roasty, acrid, and burnt notes of roast barley along with medium to high hop flavor, aroma, and bitterness—typically with American hop varieties. Alcohol runs 5.0% up to at least 7.0% and to 8.0% abv in some guidelines (Strong, 2021; GABF). At the high alcohol end of this collection, we find **American-Style Imperial Stout** running from 7.0% or 8.0% up to 12% abv (Brewers Association, 2022; Strong, 2021). This style displays high levels of malt flavor with roast, chocolate, and dried dark fruit notes as well as medium-high to high hop aroma, flavor, and bitterness.

3.31 Strong American ales

On the heels of Imperial Stout, we find three additional related strong beers, starting with **American Strong Ale** (Strong, 2021). This amber beer exhibits both maltiness in the flavor and bitterness in the finish but lacks the hop aroma and flavor that would otherwise result in classifying it as an IPA of some sort. Alcohol ranges from 6.3%–10.0% abv. **American Barley wine** tops the alcohol range for American amber ales at 8.0%–12.0% abv (Strong, 2021). It exhibits high hop flavor and aroma from American hop varieties and high bitterness as well as high maltiness with no dark malt or roast flavors. As this style is typically all-malt, it generally exhibits a good deal of malt sweetness beneath all the hops. Finally, we have **Wheat Wine** (sometimes “**Wheatwine**”) a strong American style where wheat

malt is typically used for 50% or more of the grist. It matches the alcohol level of American Barley wine with a bit less bitterness and paler color. Bready, doughy malt flavor is common and banana/clove fermentation flavors are not appropriate ([Strong, 2021](#)).

3.32 Smoke-flavored American styles

Inspired by the early history of Porter, American craft brewers started making **Smoked Porter**. Some guidelines list this as a distinct style while others direct these beers into a broader category for classic-style smoked beers. In any case, the smoke flavor can vary in character and intensity, but the base beer broadly follows Porter guidelines with a malty balance and alcohol ranging from 5.1% to 8.9% abv. Another Americanization of a modern British style is **Peated Scotch Ale**. While the Scots swear peated malt has never been used in making beer, American brewers are quick to seize on any chance to add new flavors. Examples are described as being “aggressively malty” ([Brewers Association, 2022](#)) with all hop traits being low to none. Alcohol ranges from 6.60% to 8.50% abv ([Brewers Association, 2022](#)).

3.33 American historical styles

Certain historical styles have been revived, both in the Americas and elsewhere and a few are sometimes listed in style guidelines. We earlier covered Preprohibition Lager in the section titled New World Pale Lagers. There is also a **Preprohibition Porter** often made with adjuncts, ranging in color from light brown to nearly black and in alcohol from 4.5% to 6.0%. They exhibit low hop character and low to moderate bitterness ([Strong, 2021](#)). **Kentucky Common** is a novel American style once particularly popular in the state it is named for. It is brewed with a third or more corn adjunct in the grist but retains enough malt flavor for that to be the dominant trait. It can be any shade of amber with malt flavors that match: bread crust, caramel, toast, etc. It is highly carbonated with alcohol ranging from 4.0% to 5.5% abv ([Brewers Association, 2022](#); [Strong, 2021](#)).

3.34 Emerging international styles

Italian Grape Ale, sometimes just **Grape Ale**, is a style that originated in Italy and now brewers in other grape-growing regions have begun to make it as well. It adds grape juice to a fairly neutral beer to create pleasant blends of the flavors found in both beverages. Often the color remains pale to gold but some examples can be fairly dark (Copa Cervezas De América 2023). The style displays low bitterness and usually low hop flavor and aroma as well. Alcohol content runs from 6.0%–8.5% abv ([Strong, 2021](#)) and perhaps even up to 12.0% (Copa Cervezas De América 2023).

Catharina Sour is a fruited, kettle-soured beer from Brazil. It is most similar to Berliner Weiss but stronger at 4.0%–5.5% abv ([Strong, 2021](#)) and includes fresh, often tropical fruit added late in fermentation. As with Berliner Weiss, bitterness is very low and color is pale, running from straw to gold (Copa Cervezas De América 2023; [Concurso Brasileiro De Cervejas 2023](#), p. 17).

Dorada Pampeana, or Golden Pampas Ale in English, is a blond/golden ale style with origins in Argentine homebrewing which is now offered by some commercial brewers. The classic recipe calls for pilsner malt to take the lead in the flavor of the finished beer, with Cascade hops providing a muted counterpoint. Alcohol runs 4.3%–5.5% abv (Copa Cervezas De América 2023, X1; [Strong, 2021](#)).

3.35 Chinese specialty styles

The craft beer movement in China has developed two specialty beer styles.⁴ The first is **Chinese Style Wheat Beer**, an easy-drinking wheat ale of modest to normal alcohol content (3.5%–6.0%) ([China International Beer Challenge, 2023](#)) which exhibits the flavors of Chinese specialty spices such as osmanthus, tea, and jasmine. Hops generally do not play a significant role beyond a mild bitterness. The second is **Chinese Style Barley wine** which matches British Barley wine in many respects: caramel and toffee malt flavors take the lead with low to medium-low bitterness and alcohol runs 8.0%–12.0% abv ([China International Beer Challenge, 2023](#)). What distinguishes this style is the use of a yeast strain traditionally used to ferment Chinese yellow rice wine or liquor. The resulting beers exhibit flavors characteristic of those traditional Chinese beverages.

3.36 European historical brews

In recent decades, a number of historical styles have been revived. They are mentioned here for completeness. The first is **Breslau-Style Schoeps** ([Brewers Association, 2022](#)) which uses up to 80% wheat and has been produced in both pale and dark versions. Malt flavors dominate with bread and American biscuit flavors in pale versions and toasted, nutty flavors in the dark ones. Caramel notes are not seen. Alcohol runs 6.0%–7.0% abv ([Brewers Association, 2022](#)). Most of the grist for **Dutch-Style Kuyt Beer** ([Brewers Association, 2022](#)) comes from oats (up to 45%) and wheat (20%) to give a gold-to-copper hazy beer. Bitterness is medium-low to medium; hop flavor and aroma are low and come from European noble hops. Alcohol is 4.7%–7.9% abv. **Finnish-Style Sahti** often employs rye as well as barley in the grist and is flavored during brewing with juniper. Hop traits are low to very low across the board. Low alcohol versions have been seen in the past but current commercial examples run significantly higher, at 7.0%–8.5% abv. It is generally fermented with baker's yeast which leads to banana and clove flavor notes ([Brewers Association, 2022](#)).

Piwo Grodziskie, just **Grodziskie** or, in German, **Grätzer** is a low-alcohol beer brewed entirely from oak-smoked wheat malt ([Brewers Association, 2022](#); [Strong, 2021](#)). Hops provide medium-low to medium bitterness, but low to no hop aroma and flavor. The smoked wheat flavors dominate the flavor profile and alcohol runs 2.7%–3.7% abv. **Swedish-Style Gotlandsdricke** combines traits of Sahti and Grodziskie, using Birchwood smoked malt and baker's yeast. Hop bitterness is medium-low to very low and hop flavor and aroma are low to none. Alcohol ranges 5.5%–6.5% abv ([Brewers Association, 2022](#)).

4. From styles to categories

The list of styles we have reviewed clearly demonstrates the tremendous variety of flavors achieved with the basic ingredients of beer. We have seen that distinctions between styles can be drawn on differences both broad and fine. These differences are quite real and discernible by both experts and consumers, so when a number of producers make beers that follow a given formula, it makes sense to highlight the distinction through recognition of a new style. While it may be hard to believe, the 164 styles we have reviewed still only represent a portion of the overall beer flavor universe. To cover the rest requires a look at the broad range of entry categories.

⁴There are also several Chinese specialty beer categories in addition to these styles.

4.1 Entry categories

When classifying beers, many will not conform to the established styles. When this happens we look for similarities that can be used to define groups of beers, often a special ingredient (e.g., coffee) or process (e.g., barrel-aging). The resulting groups are often referred to as “styles,” but in fact they are just categories of beers that share a key trait.

If you read the details of category descriptions, you’ll often find they give very broad criteria. For instance, rather than actual values, the quantitative traits may just say “Any,” or “Varies by Style.” Alternately, they may give a range of values that basically match the full range of values possible in beer such as “0–100 IBUs” or “2–40+ SRM” for color. Broad parameters like this are a hall mark of entry categories.

With categories, the directions for entry often dictate that brewers must specify either a base style or “additional information” about ingredients and process to inform the judges about what to expect. This is another indication of an entry category rather than a style. Let’s examine the range of beer categories in use.

4.2 Free-style approach

European Beer Star has a “Free-Style Beer” category that basically allows the brewer to submit any beer that does not fit into their established 72 styles and submit a statement to describe what it is along with it. The Japan Craft Beer Association has refined this further designating a set of 10 specialty categories divided by ale/lager, color, and ABV to cover all possibilities as listed below. These approaches may overlap with some of the other specialty categories listed later, but they represent a unique approach to handling beers that fall outside the well-defined styles ([Japan Craft Beer Association, 2024](#)).

- Free-Style Light Lager
- Free-Style Dark Lager
- Free-Style Light Ale
- Free-Style Dark Ale
- Free-Style High-Alcohol Light Lager
- Free-Style High-Alcohol Dark Lager
- Free-Style High-Alcohol Light Ale
- Free-Style High-Alcohol Dark Ale
- Tax-Saving “Happoshu”
- Other Beer-Flavored Alcohol Beverage

4.3 Entry categories with Chinese influences

Chinese Style Grain Beer (Uses Chinese grain.)

Chinese-Style Spice Beer (Uses Traditional Chinese spices and/or Chinese medicinal herbs.)

Chinese-Style Flower Beer (Uses osmanthus, jasmine, chrysanthemum, orchid)

Chinese-Style Fruit Beer (Uses Chinese fruits jujube, lychee, wolfberry, mulberry, plum)

Chinese-Style Special Processed Beer (Production methods or tools from Chinese traditions.)

Other Chinese-Style Beer

Chinese Style Traditional Tea Beer (Includes flavors from Chinese tea. Overlap w Chinese Wheat beer, a catch all for those that do not fit there.)

4.4 Omission beers

Gluten-Free Beer

Nonalcohol Beer

Reduced Alcohol Ale or Lager

Low Carb

4.5 Cellared beers

Aged Beer

Old Ale

4.6 Nontraditional ingredients

Alternative Grain Beer

Alternative Sugar Beer

American Fruit Beer

Belgian-Style Fruit Beer

Brazilian Fruit Beer (Using fruit from Brazil.)

Brazilian Herb and Spice Beer (Using herbs and spices from Brazil.)

Chili Beer

Chocolate Beer

Coffee Beer

Coffee Stout or Porter

Field Beer

Fruit and Spice Beer

Fruit Wheat Beer

Ginjo Beer or Sake-Yeast Beer

Japanese Green Tea Beer

Herb and Spice (and Vegetable) Beer

Honey Beer

New World Lager (New world hop varieties in traditional lager style.)

Other Tea Beer

Pastry Stout (Contain coffee, choc, sweeteners)

Pumpkin Beer/Autumn Seasonal Beer

Rye Beer

Specialty Beer

Specialty Fruit Beer

Specialty IPA

Specialty Spice Beer

Strong Red Fruit Beer (Made with red fruit such as cherry, raspberry, strawberry, etc. >6% abv.)

Winter Seasonal Beer

Yuzu Beer

4.7 Nontraditional process

Experimental IPA

Experimental Beer

4.8 Non-*Saccharomyces* fermentation agents used

Wild Beer

Wild Specialty Beer

American Sour Ale

Fruited American Sour Ale

Brett Beer

Mixed-Culture Brett Beer

Contemporary Belgian Style Spontaneous Fermented Ale

Other Belgian-Style Sour Ale

4.9 Wood- and barrel-aged

Brazilian Wood-Aged Beer

Experimental Wood-Aged Beer

Wood- and Barrel-Aged Pale to Amber Beer

Wood- and Barrel-Aged Dark Beer

Wood- and Barrel-Aged Strong Beer

Wood- and Barrel-Aged Strong Stout

Wood- and Barrel-Aged Sour Beer

Fruited Wood- and Barrel-Aged Sour Beer

4.10 Belgian-style or Belgian-related beers

American-Belgo-Style Ale

Belgian-Style Table Beer

Belgian-Style Session Ale

Other Belgian- or French-style Ale

Other Belgian-Style Strong Specialty Ale

Other Belgian-Style Abbey Ale

4.11 Special smoked beers

Classic-Style Smoked Beer

Specialty Smoked Beer

Other Smoke Beer

4.12 Miscellaneous specialty beers

Brut Beer (Very dry method-champagne beer without fruit)

Other Hoppy Lager
 Commercial Specialty Beer
 Mixed-Style Beer
 Session Beer
 Other Strong Beer
 Fresh Hop Beer
 Historical Beer
 Happoshu (Must contain LESS THAN 25% malt by weight of dry raw ingredients.)

5. Conclusion

This chapter demonstrates how the wide range of beers produced around the world are organized and categorized in order to facilitate both communication and analysis. A catalog of approximately 160 well established traditional beer styles sits at the core of this system. These styles are tightly defined with regard to both physical attributes and flavor components. Beyond this, beers are organized by more broadly defined categories which often focus on a specific distinctive ingredient or process element (coffee beers and smoked beers, for instance). Beers in these categories may also reference one of the classic beer styles which it closely matches.

This survey of international beer style guidelines shows that the vast majority of beer styles and categories are recognized all over the world. To supplement that, in nearly every region of the world there are novel categories and styles which center on ingredients unique to the local region or culture. Through these styles and categories, everyone involved with beer from maltsters and hop growers, through brewers and commercial representatives, retail staff and finally consumers are provided with a useful catalog of beer types. These styles and categories provide the lexicon of beer. More importantly, they provide access to the world of beer for any and all to enjoy.

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Conventional brewer's yeast: *Saccharomyces*

5

Angela Capece¹, Diego Libkind², Julieta Burini² and Patrizia Romano³

¹Department of Agricultural, Forest, Food, and Environmental Sciences (DAFE), University of Basilicata, Potenza, Italy;

²Centro de Referencia en Levaduras y Tecnología Cervecería (CRELTEC), Instituto Andino Patagónico de Tecnologías Biológicas y Geoambientales (IPATEC, CONICET - UNComahue), Bariloche, Argentina; ³Universitas Mercatorum, Department of Economy, Rome, Italy

1. Introduction

The key point of beer production, in addition to the raw materials, malt, hops, and water, is based also on the activity of yeasts, which play the most important role by fermenting the wort, metabolizing the sugars, and producing the compounds that define the peculiar characteristics of the final product. Yeasts of the genus *Saccharomyces* are predominantly involved in alcoholic fermentation processes (Stewart, 2016) and play an important role in the characteristics of the beer, not only by the good fermentation yield efficiency, but also by affecting aroma and flavor (Capece, Romaniello, Siesto, & Romano, 2018).

In the early 1880s, Emile Christian Hansen from Carlsberg Laboratory in Copenhagen was the first to succeed in isolating pure yeast strains and in November 1883, the Old Carlsberg Brewery used for the first time one of Hansen's isolates, Unterhefe Nr. 1 (*Saccharomyces carlsbergensis*, now reclassified as *Saccharomyces pastorianus*) as starter culture for the production of their lager beers. Given the excellent results, this technique was quickly adopted by other breweries, resulting in the widespread use of pure culture *Saccharomyces* as starters. Auguste Joseph François de Bavay, a Belgian brewer employed in Melbourne, isolated and used Australian No. 2 strain, which probably represents the first pure culture used in ale brewing (Steensels & Verstrepen, 2014).

The broad biodiversity reported among *Saccharomyces cerevisiae* strains and the willingness of craft beer producers to experiment with a variety of different yeast strains represent a wealth of possibilities to create beers with unique attributes and flavor profiles.

2. General characteristics of *Saccharomyces*

The *Saccharomyces* genus is composed of ascosporogenic yeasts that produce ovoid, spherical, or elongated cells, with a large diameter of 5–10 µm and a smaller diameter of around 5 µm. The cellular structure is similar to higher eukaryotic cells, and it is composed by a cell wall, nucleus, mitochondria, endoplasmic reticulum, Golgi apparatus, vacuoles, microbodies, secretory vesicles, and a complex

extracellular and intracellular membrane network. *Saccharomyces* cells require high water activity (a_w), with a minimum a_w of about 0.65; in general, *Saccharomyces* strains grow well at temperatures ranging between 20 and 30°C and pH between 4.5 and 6.5. Vegetative reproduction occurs by multilateral budding, while sexual reproduction involves the formation of an ascus containing from one to four spores.

The life cycle of budding yeast is similar to that of any sexual eukaryote, alternating between haploid and diploid states, containing one and two sets of chromosome complements, respectively. The classical description of the budding life cycle is haplo-diplobiontic. Under nutrient-rich conditions, diploid cells proliferate mitotically by asexual process called budding, in which a daughter cell is initiated as an outgrowth from the mother cell, followed by nuclear division, cell-wall formation, and finally cell separation. As in other eukaryotes, yeast cells age, and therefore, each mother cell usually forms no more than 20–30 daughter cells, and its age can be determined by the number of bud scars left on the cell wall. In industrial fermentation processes, yeast asexual reproduction is advisable to guarantee the conservation of the genotype and maintain stable the fermentation behaviors. Diploid cells, exposed to a poor carbon source and nutrient conditions, such as lack of amino acids, undergo meiosis and sporulation by producing four haploid spores enclosed within an ascus. Sporulation is a mechanism used by yeasts to survive in adverse situations and spores have great resistance to extreme environmental conditions. When the spores are placed in nutrient-rich conditions, they germinate and begin to grow as haploids. When two haploid cells mate and fuse, they yield a diploid cell, which contains two sets of each chromosome. Spores can be individually isolated and propagated as haploid spore clones or mated to one another to form a diploid.

Saccharomyces yeasts are facultative anaerobe, which means that they can grow with or without oxygen. In the presence of oxygen, they undergo mitochondrial electron transport chain and oxidative phosphorylation where glucose is converted to carbon dioxide (CO_2), H_2O , and energy. In anaerobic conditions, as in alcoholic fermentation, these yeasts do not grow efficiently, because the energy is derived from glycolysis only and sugars are instead converted to the intermediate by-products such as ethanol, glycerol, and CO_2 .

Yeasts live in a wide variety of ecological niches, in particular on plant and fruit surfaces, where they can carry out their metabolism and fermentation activity in consequence of the presence of necessary nutrients and substrates. Regarding the *Saccharomyces* genus, it includes eight primary biologically distinct yeast species, also defined as “natural,” namely *S. cerevisiae*, *S. paradoxus*, *S. uvarum*, *S. mikatae*, *S. kudriavzevii*, *S. arboricola*, *S. eubayanus*, and *S. jurei*, and two species resulting from intra-specific hybridization, namely *S. pastorianus* and *S. bayanus* (Borneman & Pretorius, 2015; Naseeb et al., 2017). All these species have been isolated from many different substrates, including those of alcoholic fermentation and those related to natural niches such as trees. Among the species of the *Saccharomyces* genus, *S. cerevisiae* is by far the most studied and most important for the production of beer. The species *S. eubayanus*, *S. kudriavzevii*, and *S. uvarum* are also interesting for low temperature fermentations, but there are also applications and interest for the species *S. pastorianus* (typical of low-fermentation lager beer) and *S. paradoxus*.

In the brewing sector, the genus *Saccharomyces* can traditionally be divided into two groups: ale and lager yeasts, also known as the top fermenting (*S. cerevisiae*) and bottom fermenting yeasts (*S. pastorianus*, previously known as *S. carlsbergensis*), respectively (Stewart, 2016).

2.1 Nutritional request

On a nutritional level, *Saccharomyces* yeasts are not particularly demanding, their growth being supported by the existence of basic compounds in the medium. In addition to hydrogen and oxygen, the main elements that yeasts must find in the substrate are carbon, nitrogen, phosphorus, sulfur, potassium, and magnesium. They also require microelements, which act as cofactors for the numerous enzymes involved in various metabolic activities, namely calcium, iron, manganese, cobalt, copper, molybdenum, and zinc. Finally, growth factors are also essential for the development of yeasts and include vitamins (biotin, pantothenic acid, nicotinic acid, and thiamine), nitrogenous bases (purines and pyrimidines), amino acids, fatty acids, and sterols. Depending on the chemical-physical characteristics of the substrate, the availability of the various nutrients and the environmental conditions, the yeasts modulate their metabolism showing different development kinetics and consequently releasing compounds at various concentrations which directly influence the organoleptic characteristics of the product.

The cells absorb dissolved sugars, simple nitrogenous matter (amino acids, ammonium ions, and small peptides), vitamins, and ions through their plasma membrane. Subsequently, they employ a series of reactions known as metabolic pathways (glycolysis, biosynthesis of cellular constituents, etc.) and use these nutrient materials for growth and fermentation.

2.2 Carbon metabolism

Carbon metabolism involves a network of interrelated pathways that provide energy in the form of adenosine triphosphate (ATP) molecules, precursors for biosynthesis of many cellular compounds and cofactors used in redox reactions. Yeasts of the genus *Saccharomyces* are able to utilize a wide variety of sugars as sources of carbon and energy. In particular, brewing yeasts can utilize various carbohydrates (glucose, sucrose, fructose, maltose, galactose, raffinose, and maltotriose), and the major distinguishing difference between ale and lager strains is the capability of lager yeasts to ferment melibiose (Lodolo et al., 2008). These yeasts preferentially utilize monosaccharides (glucose and fructose), but they metabolize also disaccharides (sucrose, maltose) and trisaccharides (maltotriose) synthesizing specific hydrolytic enzymes. The most abundant fermentable sugars in brewer's wort are maltose (50%–60%), maltotriose (15%–20%), glucose (10%–15%), followed by sucrose, and fructose. Sugary raw materials, which come from cereal starches, require enzymatic prehydrolysis. The uptake pattern of carbohydrates first follows the path of the simplest sugars (the monosaccharides glucose and fructose), followed in increasing order of complexity by the disaccharides (maltose) and trisaccharides (maltotriose), excluding dextrins (Stewart, 2006). Before complex sugars can be used by the cell, they must be hydrolyzed to simple sugars and then transported across the cytoplasmic membrane. Glucose and fructose are transported across the cell membrane by specific transporters and then enter the cell thanks to transport proteins (permeases) through a facilitated diffusion process. In general, only when half of the wort glucose has been taken up by the yeast, the uptake of maltose and maltotriose will start, with a slower uptake rate for maltotriose than for maltose. Utilization of maltose and maltotriose requires the active transport of the sugar across the plasma membrane by different transport systems and, subsequently, its hydrolysis by cytoplasmic glucosidases (maltases). One of the most important characteristics of *Saccharomyces* yeasts is that the sugars, once transported inside the cell, can be used both in aerobic and anaerobic conditions. Aerobic metabolism (respiration) occurs in

the presence of oxygen and is much more efficient (more energy is produced from the same substances), while anaerobic metabolism (fermentation) occurs in the absence of oxygen and is less efficient. Respiration essentially produces water and CO₂ as final products, whereas the final products of fermentation are ethanol and CO₂ (as well as many other minor by-products). Really, the yeast ability to conduct respiration or fermentation depends not only on the oxygen availability but also on the sugar concentration. In any case, both metabolic processes have in common the glycolysis, which occurs in the cell cytoplasm without the involvement of molecular oxygen. Glycolysis consists in the transformation of 1 mole of glucose (or fructose) into 2 moles of pyruvic acid, with a net gain of energy in the chemical form of 2 moles of ATP (adenosine triphosphate), and of reducing power in the form of 2 moles of NADH (nicotinamide adenine dinucleotide). The reactions that make up glycolysis take place in the cytoplasm of cells and each reaction is catalyzed by a specific enzyme. The efficient use of sugars by *Saccharomyces* depends both on the ability of membrane transporters to bring them into the cell and on the main glycolytic enzymes. The process as a whole is catabolic, in which more complex and energetic molecules are transformed into simpler and less energetic molecules, with accumulation of energy. Therefore, glycolysis transforms sugars into pyruvates and represents the starting point of yeast metabolism in both aerobic and anaerobic conditions.

Pyruvate is the heart of carbon metabolism and occupies a central position in many metabolic pathways. Its fate is closely linked to the possibility for *Saccharomyces* yeasts to undertake respiratory or fermentative metabolism.

In respiration (aerobic conditions), pyruvate is transported inside the mitochondria where it is decarboxylated by the enzyme pyruvate dehydrogenase with the formation of acetyl-coenzyme A (Acetyl-CoA), CO₂, and reducing power (NADH). Then, Acetyl-CoA is completely oxidated in the Krebs cycle to CO₂, energy, and intermediates to promote yeast growth. As a result of respiration, cells produce 6 molecules of CO₂, 6 molecules of H₂O, and 38 molecules of ATP from one glucose molecule.

Fermentation is the anaerobic decomposition of organic compounds (primarily carbohydrates) that involves the pyruvic acid as the final electron acceptor. The absence of oxygen (anaerobic conditions) prevents the normal Krebs cycle and aerobic cellular respiration which indirectly regenerate NAD⁺, essential for the glycolysis of other glucose molecules. As a consequence, the fermentation is essential for NAD⁺ generation. As a result of alcoholic fermentation, cells produce 2 molecules of ethanol, 2 molecules of carbon dioxide, and 2 molecules of ATP from one glucose molecule. In fermentation process, glucose is slowly utilized to produce the energy required just to maintain the yeast cell live.

2.2.1 The “Pasteur effect” and the “Crabtree effect”

In general, *Saccharomyces* yeasts in aerobic conditions are able to perform respiration of all carbon sources, while in the absence of oxygen (anaerobiosis) they conduct alcoholic fermentation. Their respiration and fermentation processes are regulated not only by oxygen concentration, but also by environmental factors, mainly glucose concentrations. Therefore, depending on the variations in environmental conditions, different mechanisms are activated to regulate the two metabolic processes, which are referred to as the “Pasteur effect” and the “Crabtree effect,” from the name of the scientists who first described them.

Saccharomyces in the presence of limited concentrations of sugars can carry out both respiration and fermentation in relation to the availability of oxygen. Aeration induces a significant increase in the cell biomass production and a drastic reduction in the ethanol concentration. This phenomenon,

known as the Pasteur effect, is primarily attributed to the competition for pyruvic acid of the enzymes pyruvate decarboxylase, involved in fermentative metabolism, and pyruvate dehydrogenase, involved in respiratory metabolism. The lower affinity for pyruvic acid of pyruvate decarboxylase compared with pyruvate dehydrogenase could contribute to the inhibition of fermentation caused by respiration. Since sugar respiration is energetically more favorable than fermentation (38 ATP vs. 2 ATP), most yeasts use fermentation only when respiration is compromised, for example when oxygen availability decreases. In particular, the inhibition of glycolytic enzymes largely explains the “Pasteur effect”. In fact, it causes an accumulation of phosphorylated hexoses at the cytoplasmic level. This accumulation inhibits the activity of membrane transporters and glycolysis. This phenomenon occurs when the glucose concentration in the medium exceeds a critical threshold value, above 0.8 mM (Verduyn et al., 1984). In this case, in response to sugar level, the yeast switches to a mixed respiration-fermentative metabolism, resulting in the production of ethanol. This phenomenon is independent of the presence of oxygen and is known as the “Crabtree effect” or glucose catabolic repression (De Deken, 1966).

The yeasts expressing this trait are called “Crabtree positive” yeasts in contrast to “Crabtree negative” yeasts, that lack fermentative products and under aerobic conditions biomass and carbon dioxide are the sole products (Fig. 5.1).

This phenomenon of glucose repression in *Saccharomyces* is due to the limited respiratory capacity of yeast cells in the presence of excess intracellular glucose and is the result of multiple factors, including the types of sugar transport and the regulation of enzymatic activity involved in respiration.

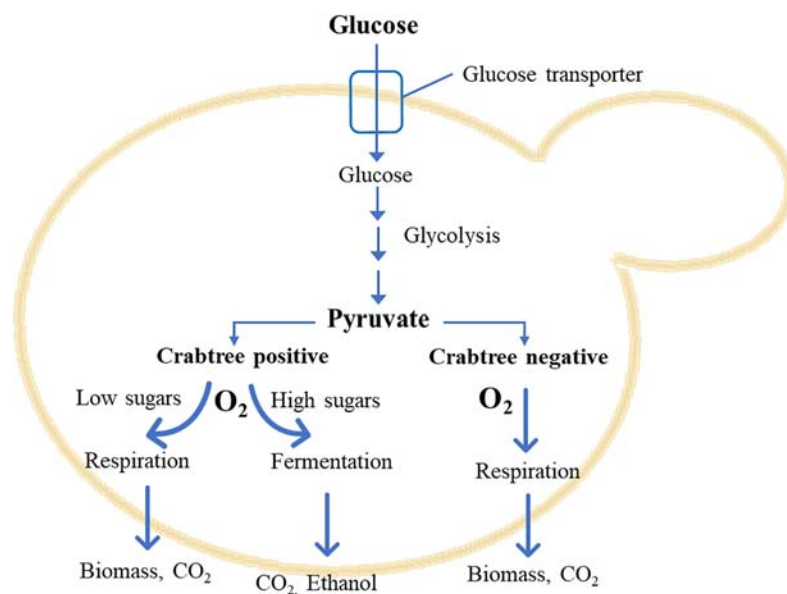


FIGURE 5.1

Crabtree effect in yeasts.

In “Crabtree positive” yeasts a glucose catabolite or glucose itself causes the repression of the genes that encode the respiration enzymes, causing profound changes in the yeast cell. *Saccharomyces* yeasts used in brewing are “Crabtree positive,” simultaneously generating adenosine triphosphate (ATP) from respiration and fermentation pathways depending on the environmental conditions. When glucose concentrations are high enough, enzymes used in respiration are repressed and fermentation takes over respiration. Once ethanol is formed, it is secreted into the growth media as well as various other by-products of fermentation such as glycerol, acetic acid, hydrogen sulfide, higher alcohols, and esters.

2.3 Nitrogen metabolism

In addition to carbon, nitrogen is of fundamental importance for the growth of yeasts during alcoholic fermentation. In wort, the main nitrogen source for the synthesis of proteins, nucleic acids and other nitrogenous cell components are amino acids, ammonium ions and, to a lesser extent, di- and tri-peptides. These substances are formed partly during malting and partly during mashing, due to the action of barley proteases.

All yeasts are capable of using ammonium and amino acids. Ammonium, which is the form of nitrogen preferentially assimilated by yeasts, is transported into the cell through a facilitated diffusion mechanism thanks to the action of specific permeases.

Most of the wort free amino nitrogen (FAN) is used by the yeast for the formation of proteins (structural and enzymatic) necessary for yeast growth and for other functions such as osmoregulation. The level and composition of wort FAN also has a huge impact on the formation of higher alcohols, esters, vicinal diketones and hydrogen sulfide, due to the role of amino acid metabolism in the formation of these flavor compounds.

For the construction of cellular proteins, especially during the logarithmic phase of growth, yeast needs amino acids. Therefore, yeast removes the amino groups from the wort amino acids through transamination and utilizes them to create its own amino acids.

Amino acids from wort are utilized by yeast in a characteristic sequence that appears to be independent of the fermentation conditions, but dependent of the yeast strain and by environmental changes. Thus, the amino acids are classified into groups according to the order in which they are taken up from the wort. Jones & Pierce (1964) established a unique classification of amino acids according to their rates of consumption during typical ale brewing wort fermentations. The amino acids in group A are absorbed quickly (such as arginine, asparagine, aspartate, glutamate, glutamine, lysine, serine and threonine), followed by the slower disappearance from wort of group B amino acids (such as histidine, isoleucine, leucine, methionine and valine), while the amino acids in group C (such as alanine, glycine, phenylalanine, tyrosine, tryptophan as well as ammonia) are only absorbed after the complete removal of group A amino acids. Group D consists exclusively of proline which is little used having no free amino groups and therefore it cannot be transaminated.

Yeast can synthesize most of the amino acids needed to build cellular proteins using ammonium ion and wort amino acids, with glutamate playing a vital role; glutamate can serve as an amino group donor to produce different amino acids. Yeast can also degrade amino acids to obtain ammonium ions. Ammonium ions and glutamate are used directly in biosynthesis and are therefore the preferred nitrogen sources for yeast growth.

The absorption and utilization of exogenous nitrogenous wort compounds and their synthesis intracellularly are controlled by main factors, which are the total wort concentration of assimilable nitrogen; the concentration of individual nitrogenous compounds and their ratio; the competitive inhibition of the uptake mainly of amino acids.

2.4 Flocculation

Following the definition reported by [Stewart et al. \(2013\)](#), the requirements of an acceptable brewer's yeast strain are the following ones: "in order to achieve beer of high quality, it is axiomatic that not only must the yeast be effective in removing the required nutrients from the growth/fermentation medium (the malt extract—the wort), able to tolerate the prevailing environmental conditions (e.g., ethanol toxicity) and impart the desired flavor and stability to the beer, but the microorganisms themselves at the end of fermentation must be effectively removed from the wort by flocculation, sedimentation, centrifugation, and/or filtration after they have fulfilled their metabolic role."

Indeed, one of the desirable characteristics of any brewer's yeast is the ability to have a strong and complete flocculation. Yeast flocculation is defined as the asexual, homotypic (involving in the interactions only one type of cell), reversible (flocs can be dispersed by the action of EDTA or specific sugars, like mannose), and calcium-dependent aggregation of yeast cells to form flocs containing thousands or even millions of cells ([Fig. 5.2](#)).

Upon formation, these aggregates can rapidly separate from the fermenting wort by sedimentation (lager yeasts), or by rising to the surface (ale yeasts).

The ability of some yeast strains to aggregate facilitates enormously cell separation; thus, flocculation process is of considerable importance for the brewing industry, as it provides an effective, environment-friendly, simple and off-cost process of cell separation from green beer at the end of fermentation, which does not require energy input. The use of flocculent yeasts for fermentation, reducing the energy consumption associated with cell separation, increases the efficiency of the process.

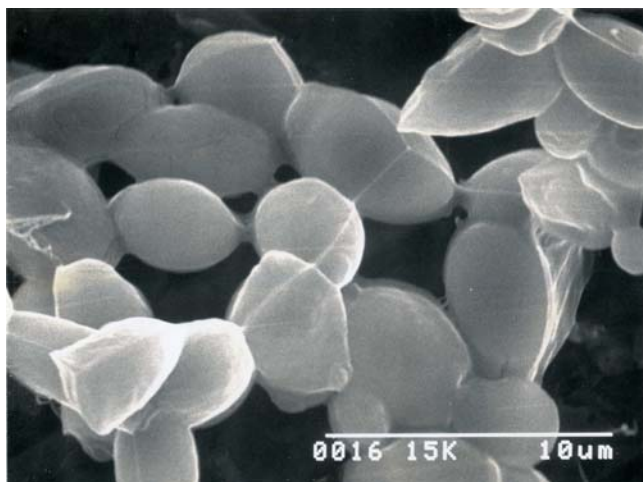


FIGURE 5.2

SEM image (6000×) of *Saccharomyces cerevisiae* flocculent cells.

The flocculation properties of brewer's yeast strains are very important because the number of yeast cells suspended in wort during both primary and/or secondary fermentations affects many process parameters, such as wort fermentation velocity, beer flavor, maturation, cell crop formation, and filtration. The ideal brewer's yeast should exhibit strong flocculation toward the end of fermentation as a premature flocculation, that is cell flocculation before the complete wort attenuation, causes sluggish fermentations (so called "hanging"), with potential production of off-flavors (Stratford, 1992). Furthermore, this flocculation behavior should also be maintained during consecutive cycles of fermenting, cropping, storing, and repitching.

Following the great interest generated by the cost-saving nature of ale yeasts that generate self-clearing beers at the end of fermentations, the genetic, biochemical, and physiological aspects of yeast flocculation were widely investigated.

Different hypotheses have been proposed to explain flocculation mechanisms in *S. cerevisiae* (Soares, 2010). The prevalent hypothesis of flocculation is the lectin-like theory (Miki et al., 1982). According to this model, a specific protein (lectin), present only in flocculent yeast cells, is secreted into the outer extremities of the cell wall and the N-terminal part of this protein recognizes and interacts with carbohydrate residues of alfa mannans (receptors) present in the cell walls of neighboring yeast cells, both flocculent and nonflocculent. Calcium ions are necessary for the activation of the lectins as they enable the lectins to achieve their active conformation.

Among the several factors affecting yeast flocculation, the first one is the strain genetic background. Like any other protein, flocculins are encoded by specific genes, the so-called *FLO* genes. *FLO1* is the most studied (Stewart, 2018) but there are at least nine *FLO* genes—*FLO1*, *FLO5*, *FLO8*, *FLO9*, *FLO10*, *FLO11*, *FLONL*, *FLONS*, and *Lg-FLO*, encoding flocculin proteins.

In *S. cerevisiae* cells, two main flocculation phenotypes were described: Flo1 and NewFlo. In yeast cells belonging to Flo1 phenotype, the flocculation is specifically inhibited by mannose and derivatives, while in NewFlo phenotype (containing the majority of brewery ale strains) the flocculation is reversibly inhibited by mannose, maltose, glucose, and sucrose but not by galactose. Strains belonging to Flo1 and NewFlo phenotypes also display different sensitiveness to culture conditions, such as temperature, pH, and nutrient availability (Soares & Seynaeve, 2000).

By considering that yeast flocculation is a cell surface property, different studies investigated the impact of cell surface charge and cell surface hydrophobicity (CSH), on yeast flocculation. A positive correlation between CSH and flocculation has been reported (Jin et al., 2001), indicating that CSH is partially responsible for triggering flocculation in brewing yeast strains (Suzzi et al., 1994). Indeed, other researchers report that the correlation between the increase in CSH and flocculation of brewing yeast cells is most likely due to the presence of active flocculins on yeast cell surface (Van Holle et al., 2011). Other than the genetic background, the flocculation is affected by the physiological environment, which includes the pH, availability of appropriate metal ions, and nutrients during the growth phase. For example, the pH can influence the cell surface charge, with a direct effect on the flocculation phenotype. Furthermore, changes in pH may also modify the ionization of functional groups in flocculin proteins, with modification of their conformation.

The physical environment is another factor affecting flocculation. The hydrodynamic conditions (relevant to liquids in motion) might be favorable and promote collision rates between cells. Factors able to facilitate cell-cell contact, such as factors increasing cell walls' hydrophobic character (cell surface hydrophobicity) and decreasing the repulsive negative electrostatic charges on cell walls (cell surface change), result in stronger flocculation.

3. *Saccharomyces* for ale beer

Ale yeasts are strains of the species *S. cerevisiae* that traditionally conduct “top fermentations” where yeasts congregate on the surface of the fermenting wort. Such yeasts exhibit flotation and have the ability to trap CO₂ bubbles to form a yeast “head” at the top of fermentation vessels. In traditional open fermenters, they rise to the surface of the vessel, facilitating their collection by skimming, and their use by repitching into the next fermentation. The hydrostatic pressure in modern cylindroconical fermenters, many of which may contain up to 10,000 hL of fermenting beer, tends to overcome this tendency of ale yeast, which collects in the cone of the tank.

Ale yeasts are more diverse than lager yeasts and generally, the type of beers produced are closer to ancient ones (Harrison & Albanese, 2019). Ale yeast is the oldest microorganism actively employed by mankind to transform raw materials to produce food and drinks, with archeological evidence of brewing and wine-making dating back to the Pre-Pottery Neolithic A (9.500–8.800 BCE). This yeast species has been isolated in innumerable locations worldwide and it is the most commonly used worldwide for numerous purposes other than beer production, such as bread, cider, or wine production (Carrau et al., 2015), together with scientific study or research. Indeed, *S. cerevisiae* may be considered perfect as a model organism; it is the first eukaryotic organism to have its genome sequenced (Goffeau et al., 1996), first genetically modified organism (GMO) approved for a food application (Aldhous, 1990), the first eukaryote to be synthetically engineered at the whole-chromosome scale with the first synthetic eukaryotic chromosome (Annaluru et al., 2014).

Ale and lager yeast strains exhibit differences in genetic, biochemical, fermentative, and physiological characteristics.

The phylogenetic relationships of the ale yeast strains used to ferment different types of beers, as well as their relationships with non brewing strains of *S. cerevisiae*, emerged quite recently. By using microsatellite markers, Legras et al. (2007) gave the first indication that wine and ale beer strains were genetically distinct. The investigation of a comprehensive collection of ale-type beer yeasts by using complete genome sequences showed that these yeasts were fundamentally distinct from other industrially relevant strains of *S. cerevisiae* (Gallone et al., 2016; Gonçalves et al., 2016). Most of the *S. cerevisiae* ale strains clustered into two independent lineages, called Beer 1 (which consists of three separate Belgium/Germany, Britain, and the United States strains), and Beer 2 (which includes yeasts originating from Belgium, the United Kingdom, the United States, Germany, and Eastern Europe).

Furthermore, it has been demonstrated that the majority of beer yeasts actually used are genetically different from wild strains (Gallone et al., 2016) and these yeasts are the result of the evolution during centuries in a highly selective niche (Gallone et al., 2018).

In general, ale strains are characterized by large scale variations in genome structure, including changes in ploidy and large segmental duplications or copy number variations. While wild *S. cerevisiae* strains are generally diploids, most ale strains are tetraploid or more than diploid, with aneuploidies that prevent them to perfectly match the diploid or tetraploid status (Peter et al., 2018). Although transient, aneuploidy and polyploidy can provide an adaptive advantage under selection, but this characteristic reduces the ability of ale yeasts to sporulate. The yeast ability to sporulate is a relevant trait in harsh environments, but this is not the case of wort, the nutrient-rich medium where most ale beer strains were isolated. However, continuous growth of ale yeasts in wort selects against this characteristic.

Another genetic characteristic typical of brewing yeasts is the high level of domestication. Indeed, domestication is most pronounced in *Saccharomyces* brewing strains as the industrial niche in which these yeasts live allow only limited genetic admixture with wild strains and minimal contact with natural environments. The domestication resulted in the available commercial strains specifically adapted for the production of beer, being specific selected for many characteristics important at industrial level, including sugar utilization, stress tolerance, flavor production, among others (Preiss et al., 2018).

Ale and lager strains can be differentiated not only by their different genomic constitutions, but also by their metabolic behavior. Ale yeast achieves best yields and high quality beer during fermentation at relatively high temperatures (18–27°C). Other physiological traits of these yeasts are the relatively high fermentation rate and the production of higher total amount of volatile compounds than the bottom fermenting yeasts (lager yeasts). Top fermenting *S. cerevisiae* strains produce intense flavors, and they are the focus of many craft- and microbreweries. Generally, top fermenting yeasts produce beer with a characteristic fruity aroma, in consequence of higher production levels of esters and higher alcohols compared with lager yeast strains. Furthermore, in lager yeasts, the higher proportion of esters produced remains inside cells, while these compounds remain in the fermenting medium for ale yeasts, and this further explains the higher complex flavor combination of ale beers. Different studies (Lasanta et al., 2021; Paszkot et al., 2023) report that conducting fermentation at higher temperatures leads to beers with higher concentrations of volatile compounds, including higher alcohols that can be converted to esters.

Regarding the physiological characterization, ale yeasts have the ability to take up a wide range of sugars, for example, glucose, fructose, mannose, galactose, sucrose, maltose, maltotriose, and raffinose (although only in part). Assimilation of melibiose and trehalose are not common characteristics of ale strains and could even be criteria for differentiation with lager strains. The ability of ale strains to ferment maltotriose is another evidence for domestication of these yeasts; this sugar accounts for 20% of the total fermentable sugars in brewer's wort, but is not normally present in high concentrations in natural yeast environments.

Another characteristic of ale strains, supporting the distinct genetic make-up of these yeasts, is the inability to produce volatile phenols, compounds with spicy, leathery, medicinal and clove-like aromas, also known as POF (Phenolic Off Flavors). This trait is correlated to the presence of *PAD1* and *FDC1* genes, which represent a detoxification system used by the cells to protect themselves against phenolic acids normally present in natural environments (such as bark) and in a less extent in barley. *FDC1* decarboxylates the ferulic and p-coumaric acids of the cereal cell walls into 4-vinylguaiacol and 4-vinylphenol, while *PAD1* provides the cofactor of *FDC1*, necessary for its function. These genes are functional in biofuel or nonindustrial strains, while have mutations or premature stop codons in brewing yeasts (Gallone et al., 2016), representing another effect of domestication phenomena in the specific industrial environment. It was reported that *PAD1* and *FDC1* genes were inactivated in *S. cerevisiae* strains used for German and British style beers, but not in lambic and most wheat beer strains (Gonçalves et al., 2016), such as in traditional German “weizen” beers, also called Weissbier, a wheat beer produced worldwide with mixed barley malt and wheat malt. Although 4-VG has been historically considered an off-flavor, it is characteristic of this type of beers and POFs are desired flavor components; as a consequence, in strains fermenting these beers *PAD1* and *FDC1* genes are functional, similarly to the ancestral state found in wild strains and other industrial lineages, such as wine yeasts (Gonçalves et al., 2016).

3.1 *Saccharomyces cerevisiae* variants

Certain strains of *S. cerevisiae*, such as *S. cerevisiae* var. *diastaticus* and *S. cerevisiae* var. *boulardii* are considered varieties within the species, and have unique characteristics relevant to the production of differential beers.

Unlike most *S. cerevisiae* ale yeasts domesticated for beer, *S. cerevisiae* var. *diastaticus* has the ability to hydrolyze the dextrins in the wort into fermentable sugars, presenting a high degree of attenuation (>90%, when for brewing yeasts the normal is between 65% and 85%; Walker & Walker, 2018). They have STA genes (STA + phenotype) that allow the yeast to produce and secrete the glucoamylase enzyme with the mentioned hydrolytic capacity. Also, *S. cerevisiae* var. *diastaticus* has functional PAD1 and FDC1 genes, related to the production of volatile phenols (POF + phenotype), imparting dominant spicy and clove-like flavors in beer (Latorre et al., 2020; Meier-Dörnberg et al., 2018). This yeast is generally considered a dangerous contaminant agent in breweries due to the production of phenols and its super-attenuating capacity (which generates a secondary fermentation with excessive formation of carbon dioxide in bottles, cans or kegs, as well as changes in flavor, sedimentation and increased turbidity); however, its application for the production of beers with distinctive sensory characteristics, as well as low-carbohydrate beers (in line with the consumer's lifestyle, increasingly focused on well-being and health) are reversing its bad reputation (Meier-Dörnberg et al., 2018). Indeed, one strain in which the gene responsible for these undesirable traits was attenuated gave interesting results, as it was reduced the production of undesirable flavors and CO₂, by obtaining beers with different physical and chemical properties (Krogerus & Gibson, 2020; Meier-Dörnberg et al., 2018). This yeast is used to produce the Belgian Saison-style beers, which are highly carbonated, fruity, and spicy; actually, these beers are highly appreciated by the consumers.

Another variety within the *S. cerevisiae* species whose potential in the brewing industry has been studied is *S. cerevisiae* var. *boulardii*, a yeast known and studied for its probiotic capacity which, supplied in adequate quantities in unpasteurized and unfiltered beer, promotes health benefits for the consumer (Capece, Romaniello, Pietrafesa, et al., 2018; Senkarcinova et al., 2018). So far this is the only yeast widely used as a probiotic, although the probiotic potential of other yeasts is being studied (Iorizzo et al., 2021). It has properties that make it a potential probiotic agent, such as survival at body temperature (37°C), resistance to stomach and bile acids, and survival in the competitive environment of the intestinal tract. Also, *S. cerevisiae* var. *boulardii* was shown to be able to ferment sugars in brewing wort, has suitable resistance to alcohol and growth in the presence of iso-alpha-acids, making its application in brewing production possible (Capece, Romaniello, Pietrafesa, et al., 2018; Senkarcinova et al., 2018). In studies using *S. cerevisiae* var. *boulardii* as the only starter culture, as well as in co-cultures, beers with higher antioxidant activity, lower alcohol content and sensory characteristics similar to those obtained with brewer's yeasts were obtained, resulting in a very promising tool to increase the healthy quality of beers (Capece, Romaniello, Pietrafesa, et al., 2018; Senkarcinova et al., 2018).

3.2 *Saccharomyces cerevisiae* from other sources

Many traditional fermented foods and beverages (such as sourdough, wine, cachaça, chicha, tequila, mezcal, and sake) can contain a mixture of native yeasts, including wild *S. cerevisiae* strains or

Table 5.1 *Saccharomyces cerevisiae* strains from nonbrewing environments used in beer production.

Isolation source	Strain code	Property	References
Wine environments	41CM	Good fermentative aptitudes at 12°C and 20°C	Testa et al. (2022)
	G520	High production of esters, higher alcohols and acids	Postigo et al. (2021)
	DBVPG 2170, DBVPG 2187 and L951	High levels of volatile compounds related to fruity and flowery aromas, bottle refermentation	Canónico et al. (2014)
Baker/Sourdough/Flour	DBVPG 1058	Fermentative ability and volatile profile	Rossi et al. (2018)
	S38	Good fermentative performances, interesting aromatic properties	Marongiu et al. (2015)
	Y17	High stress tolerance	Catallo et al. (2020)
	S-42	High content of volatile substances	Mascia et al. (2015)
Bioethanol distilleries	P4	High production of esters, alcohols, and terpenes, IGA beer	Pietrafesa et al. (2023)
	LBTL419, LBTL207	High tolerance to stressful conditions, distinctive organoleptic profile, high-gravity brewing	Christofoleti-Furlan et al. (2020)
	LBGA-287	Flavor complexity	Lorca Mandujano et al. (2022)
Cachaça distilleries	LBCM1045	Good fermentation performance at low temperatures, high flocculation capacity, acetate esters production	Araújo et al. (2018)

domesticated for that specific beverage. Among such strains isolated from various substrates, there may be considerable phenotypic diversity caused by fluctuations in gene expression levels and selection due to environmental stress; and this variability can be assessed for its application in brewing innovation (Rossi et al., 2018). For example, strains of *S. cerevisiae* isolated from sourdough, as well as cachaça, and other fruits and fermentative environments, showed promising results for the production of beer (Araújo et al., 2018; Catallo et al., 2020; Marongiu et al., 2015), expanding and contributing to the diversity of strains that can be used (Table 5.1).

4. *Saccharomyces* for lager beer

Lager beer styles dominate more than 90% of the market and are fermented with a yeast that evolved as a highly successful interspecies hybrid in brewing environments—an unintended result of artificial selection. This domesticated yeast, now known as *S. pastorianus* (synonym of *S. carlsbergensis*), has a complex taxonomy that has been extensively reviewed elsewhere (Sampaio et al., 2017). This hybrid yeast likely emerged in Central Europe around the same time lager beer originated in Bavaria during

the 15th century. Recent research suggests its origin may be traced to a Munich brewery between 1600 and 1615 (Hutzler et al., 2023).

The hybrid nature of *S. pastorianus* has been recognized since the early 1980s, but it was only decades later, with the discovery of the wild yeast *S. eubayanus* (Libkind et al., 2011), that the origins of lager yeast became clearer. *S. pastorianus* resulted from the hybridization of two progenitors: an *S. cerevisiae* ale yeast, the classic brewing yeast, and *S. eubayanus*. This hybridization combined beneficial traits for lager brewing from each progenitor, such as the cold tolerance of *S. eubayanus* and the superior fermentation characteristics of *S. cerevisiae*. Over the past 400 years, intensive domestication through selection and repitching during beer fermentation further refined these traits (Gallone et al., 2019; Hutzler et al., 2023; Okuno et al., 2015).

This unique hybridization and domestication history, coupled with the industrial standardization of lager production in 19th century in Germany (Gallone et al., 2019; Hutzler et al., 2023), has led to today's commercial lager yeast strains having significantly restricted genetic diversity compared with ale strains. Consequently, this limits the range of flavors and aromas found in lager beer.

The *S. pastorianus* strains can be divided in only two genetically distinct subgroups, the Group 1 or “Saaz” strains and Group 2 or “Frohberg” strains (Gibson et al., 2013). Group 1 includes well-known strains from the Carlsberg brewery in Denmark, and also strains used to produce the Saaz type beer, typically in regions now in the Czech Republic. Group 2 contains strains used to produce the Frohberg-type beer, strains from the Netherlands (Heineken and other breweries), and the most used lager strain in the world, Weihenstephan TUM 34/70 from Bavaria (Sampaio et al., 2017). The genomic analysis reveals notable variations between the two groups. However, recent evidence indicates that both Group 1 and Group 2 lager strains originated from a single hybridization event (Salazar et al., 2019). The observed differences among modern strains are mainly due to distinct evolutionary paths following a population bottleneck caused by the isolation and propagation of pure cultures during the industrialization of brewing yeast. For instance, a representative strain of Group 1 (*S. carlsbergensis* CBS 1513) has a triploid genome comprising diploid *S. eubayanus* and haploid *S. cerevisiae* content. Meanwhile, the Weihenstephan strain, a representative of Group 2, is tetraploid, featuring two diploid genomes of *S. eubayanus* and *S. cerevisiae*, along with unique chromosome translocations. These genetic differences translate into distinct brewing characteristics: Saaz strains, which grow better at low temperatures (10 °C), exhibit weaker fermentation performance compared with Frohberg strains at higher temperatures (22 °C). Saaz strains also do not ferment maltotriose as effectively and produce lower levels of esters (Gibson et al., 2013). Variations in cryotolerance and fermentation performance are associated with the relative amount of *S. eubayanus* DNA retained in their genomes. All *S. pastorianus* strains have inherited their mitochondrial DNA from *S. eubayanus* (Baker et al., 2019; Okuno et al., 2015). Recent studies have demonstrated the critical role of mtDNA in temperature tolerance in both synthetic and industrial hybrids, highlighting its importance for the hybrid's adaptation to cold beer brewing conditions (Baker et al., 2019).

4.1 *Saccharomyces eubayanus*

Saccharomyces eubayanus was formally described in 2011 and was found to share 99.6% genetic similarity with the non *S. cerevisiae* parental genome of the lager hybrid (Libkind et al., 2011). This species, in its pure form, is exclusively found in natural environments, particularly associated with trees such as those of the genus *Nothofagus* (samples include bark, leaves, soil, and fungal stromata of

Cyttaria spp.) in the forests of Patagonia, Argentina, and Chile (Eizaguirre et al., 2018; Nespolo et al., 2020). It has also been isolated from China, Tibet, New Zealand, and North America (Langdon et al., 2020), but not from Europe until recently, when two strains were discovered on soil samples of a university campus in Dublin, Ireland (Bergin et al., 2022).

The phylogeography of *S. eubayanus* is quite intricate, with phylogenetic analyses revealing that isolates are grouped into two primary clades, PA and PB (Peris et al., 2016). Clade PA is further subdivided into two lineages, PA-1 and PA-2, while clade PB is divided into at least four lineages: PB-1, PB-2, PB-3, and the Holarctic clade (Langdon et al., 2020; Nespolo et al., 2020; Peris et al., 2016). In addition, some admixed isolates have been identified (Langdon et al., 2020; Peris et al., 2016). Genetically, strains from Tibet appear to be the closest relatives to the *S. eubayanus* portion of the *S. pastorianus* genome, but the significantly higher rate of *S. eubayanus* isolation and the greater genetic diversity found in Argentina and Chile support the hypothesis that Patagonia is the primary center of radiation for this species (Langdon et al., 2020; Nespolo et al., 2020; Sampaio et al., 2017).

The case of *S. eubayanus* is an example for the successful transfer of a native yeast to the industry. Regarding its fermentative capabilities, *S. eubayanus* ferments glucose and maltose in brewing wort, but no maltotriose, with average attenuation values of 65%–68% (Burini et al., 2022; Eizaguirre et al., 2018; Gibson et al., 2013; Mardones et al., 2020); to date, wild strains of *S. eubayanus* have not been able to assimilate maltotriose, but only through genetic modification strategies (Baker & Hittinger, 2019; Brouwers et al., 2019). This species presents low flocculation evidenced by the suspended biomass and, as mentioned in the previous section, it is capable of fermenting at low temperatures (Gibson et al., 2013; Krogerus et al., 2015; Mertens et al., 2015). Perhaps the most characteristic aroma and flavor associated with *S. eubayanus* beers is the clove and/or smoky flavor derived mainly from 4-vinylguaiacol and 4-vinylphenol (4-VF) (Burini et al., 2022).

4.2 Development of novel lager yeasts

The intense selection of lager yeast has led to a reliance on a few closely related strains from Group 1 and Group 2 lineages for global lager beer production. This lack of diversity, the discovery of *S. eubayanus* (Libkind et al., 2011), and the availability of hundreds of well-characterized distinct *S. cerevisiae* strains (Gallone et al., 2016; Goncalvez et al., 2016) sparked interest in developing new lager yeasts. Initial efforts focused on creating de novo “lager-like” hybrids by mating *S. eubayanus* with *S. cerevisiae* ale strains, either using non-GM approaches (Krogerus et al., 2015; Mertens et al., 2015) or plasmid-assisted methods (Alexander et al., 2016). Given *S. eubayanus*'s role in cryotolerance, other cold adapted species like *S. mikatae*, *S. kudriavzevii*, *S. arboricola*, and *S. uvarum* have also been hybridized with *S. cerevisiae* to create new cryotolerant strains for beer and wine (Nikulin et al., 2018).

While de novo hybrids from *S. eubayanus* and *S. cerevisiae* have shown the ability to ferment at lager temperatures (10–13 °C), they often inherit undesirable traits from wild *S. eubayanus*, such as phenolic off-flavors (POF) and limited or none maltotriose consumption. Adaptive laboratory evolution has been used to address these limitations and remove undesirable traits (Diderich et al., 2018; Baker et al., 2019; Baker & Hittinger, 2019), as well as genetic engineering, such as CRISPR-Cas9 to eliminate the POF phenotype (Mertens et al., 2019). However, in most beer markets, genetic modification faces consumer and industry resistance limiting commercial use of these new strains. To circumvent this, Turgeon et al. (2021) achieved to isolate mating-competent meiotic segregants from

domesticated *S. pastorianus* and use them to create non-GM lager hybrids with desirable traits. The hybrids exhibited suitable fermentation performance and phenotypic attributes for industrial lager production and have been released for commercial use very recently. Since diverse parental strains were used, the genetic and phenotypic diversity was expanded while maintaining lager-like qualities. These yeasts are considered within a third group (Group 3) of lager yeasts broadening the lager strain clade and diversifying industrial lager production.

5. Other *Saccharomyces* not included in the previous sections

Among other *Saccharomyces* yeasts related to beer production, we can distinguish those belonging to the species *S. cerevisiae*, but different from the traditional ale brewing yeasts. Here, we can mention strains of *S. cerevisiae* originating from historical beers with very particular characteristics and/or from specific geographical regions (which is why they have gone through different domestication trajectories—e.g., farmhouse yeasts), as well as the already addressed variants of the species with different attributes (e.g., *S. cerevisiae* var. *diastaticus* or *S. cerevisiae* var. *boulardii*). In addition, we can mention strains of *S. cerevisiae* isolated from natural environments (wild) or isolated from other traditional drinks (e.g., cachaça, chicha, and sake), as well as saccharomycetic yeasts that include species other than *S. cerevisiae*, such as the already named *S. jurei*.

5.1 Norwegian farmhouse yeasts

Norwegian farmhouse yeasts or kveik (a term for “yeast” in the dialect of western and southern Norway) are those that have been used for generations in traditional brewing on Norwegian farms. These are mixed cultures which are usually composed of several strains of *S. cerevisiae* (and sometimes bacteria) that underwent a different domestication process than traditional brewing varieties, and have been recognized as a genetically and phenotypically distinct group (Preiss et al., 2018). A first study of comparative analysis of the genome of six kveiks indicated that they are tetraploid *S. cerevisiae* yeasts genetically separate from other ale yeast, with one haplotype forming a distinct group within the “Beer 1” clade and the other forming a distinct group between the “Asia” and “Mixed” groups (Preiss et al., 2018). Traditional Norwegian brewing practices, dating back to the early 17th century, may have been a determining factor in the long-term adaptation and evolution of kveik yeasts that resulted in their distinctive phenotypic qualities relevant to brewing (Preiss et al., 2018). In general, these yeasts are highly flocculent, highly tolerant to ethanol, thermotolerant, have enhanced accumulation of trehalose (associated to stress resistance), do not produce phenols (POF-), are capable of utilizing maltose and maltotriose, and exhibit a high fermentation rate (Foster et al., 2022; Preiss et al., 2018). Farmhouse brewers brewed typically only 1–4 times per year and maintain the kveik stored dry in perforated wood elements (“yeast logs” or “yeast rings”) between uses. They also have the traditional practice of inoculating yeast into wort at 28–40°C (well above the temperatures typically used for commercial ale yeast), being associated with these yeasts having high thermotolerance; in this regard, a differential characteristic is that kveiks can be used in fermentations at high temperatures without the production of off flavors (particularly higher alcohols) (Garshol, 2020; Preiss et al., 2018). In addition, traditional Norwegian brewers use a much lower inoculation rate than typical for beer production (approximately 10% of the usual inoculation rate), even in high gravity worts,

without production of off flavors or obvious sensory changes; and prefer a short fermentation time, often only 1–2 days, before transferring to barrel with the majority of brewers using farmhouse yeast in Norway reporting fermentation times shorter than 50 hours (Preiss et al., 2018). Organoleptically, kveik are capable of producing esters in concentrations above the perception threshold: ethyl caproate (pineapple, tropical), ethyl octanoate (pineapple, tropical, apple, cognac), and ethyl decanoate (apple) (Garshol, 2020; Preiss et al., 2018), which contribute to the fruity flavor and aromatic complexity associated with these yeasts in brewing.

Farmhouse brewing yeasts are not restricted only to Norway. Related strains were also isolated from farmhouse breweries in other geographical regions including Lithuania, Latvia, and Russia, as well as African rural brewery strains from Ghana. Preiss et al. (2024) propose to refer to this group of yeast strains as “European Farmhouse” brewing yeast, which shows a higher expected admixture of Asian fermentation strains and is closely related to ale yeast strains. Within this group, geographical origin of the strains could also be distinguished based on genome sequence, with strains from western Norway, eastern Norway, and the Baltic region clustering separately based on genome-wide SNPs.

5.2 *Saccharomyces jurei*

Saccharomyces jurei is the latest addition to the genus and is currently being studied for beer production. The strains studied presented resistance to high sugar concentrations and relatively high fitness at low temperature (Naseeb et al., 2018). *S. jurei* was also able to ferment wort, with the production of ethyl hexanoate (red apple, anise) and 4-vinylguaiacol (spice, clove). It is noteworthy that maltotriose fermentation by this wild species was evidenced (Giannakou et al., 2021; Hutzler et al., 2021).

6. Management of *Saccharomyces* starters in brewing

6.1 Yeast repitching

The “repitching of yeast” is a practice in which brewers collect yeast at the end of a fermentation cycle, maintained it in refrigeration (when not used that day) and use it to inoculate (or “pitch”) a new batch. It is worthy of note that brewing is the only major alcoholic beverage process that recycles its yeast (Stewart, 2017, pp. 241–258). Indeed, at the end of brewery fermentation, a common practice is to harvest the yeast from the fermentation vessel and store it for a short time before re-inoculating (repitching) the culture into a fresh batch of wort. The main purpose of this practice is to make easier the brewing process, preserving at the same time the sensory or taste profile of the beer. It is, therefore, important to jealously protect the quality of the cropped yeast because its use for subsequent wort fermentation has a significant effect on the quality of the beer resulting from subsequent wort fermentations. Indeed, fermentation performance of brewing yeast is dependent on its ability to adapt to all the environmental stresses encountered during brewing (temporal changes of oxygen concentration, osmotic potential, low pH, ethanol concentration, carbon and nutrient limitation, and temperature shift), but especially during serial repitching over a number of fermentations (Gibson et al., 2007). After pitching, the yeast grows exponentially after a brief lag phase, rapidly depleting the available oxygen, with formation of an anaerobic environment. In this step, also the wort sugars and assimilable nutrients are quickly consumed resulting in carbon and nutrient deficit; in these conditions,

the cells remain in a dormant state, a common response of yeast cells to nutrient depletion. Pitching yeasts recovered from a previous fermentation is in stationary phase. During re-pitching, yeast biomass is inoculated into aerated wort; in this phase, the yeast cells are undergone to a rapid change from anaerobic into aerobic conditions, to osmotic pressure exposition and change from starvation to nutrient-rich conditions. As a consequence, the yeast has to adapt quickly to carry out a desirable fermentation performance. Furthermore, oxidative stress is reported to be one of the key stressors throughout the brewing process.

Theoretically, the culture might be used for an unlimited number of times, but in practice yeast cells are usually not used for an excessive number of cycles and when the death rate reaches an unacceptably high level, the yeast is discarded. The permanence of dead brewer's yeast cells in the fermented beverage can impart unpleasant flavors, such as a "yeasty" taste, to the final beers (Xu et al., 2014). The intracellular compounds released can influence the flavor, colloidal, and foam stability of the beer.

Usually, the yeast is employed for 8–15 cycles, but the precise number of times a yeast culture can be reused is dependent from different factors, such as the individual strain characteristics, the quality of cropped slurry, the type of fermentation, the demand, and in some instances, company policy. Generally, the number of repitchings is reduced with increasing wort gravity. Although the high variability observed among the different strains, it was reported that ale strains are more susceptible than lager strains to recurrent repitching in high gravity wort ($>16^{\circ}\text{P}$) (Stewart, 2009).

However, an excessive number of repitching can negatively affect yeast fermentative performance and, consequently, the aromatic profile of the final beer (Kalayu, 2019). Evidence shows that accumulation of dead cells and the decreasing accumulation of neutral lipids in cells could explain the decline in the fermentation performance of yeasts during serial repitching (Buhligen et al., 2014). On the contrary, one of the key stressors throughout the brewing process is the oxidative stress. The imbalance between reactive oxygen species (ROS) generation and elimination results in the ROS accumulation in the cells, which can lead to protein inactivation, lipid peroxidation, damage to nucleic acids and to the mitochondria of yeast cells. Damage occurs to the cells gradually and affects the yeast vitality, with a decrease in fermentation performance (Wang et al., 2019).

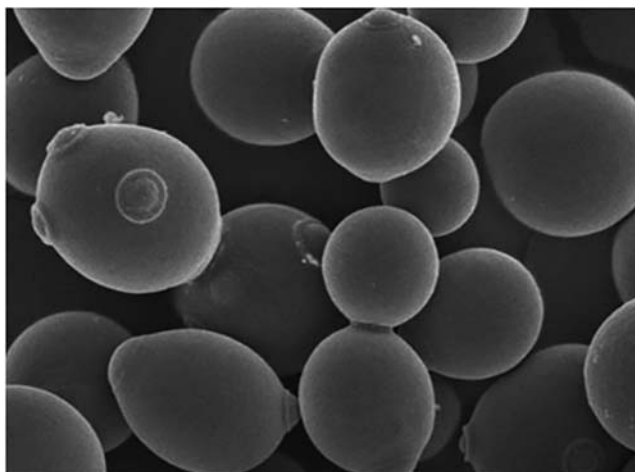
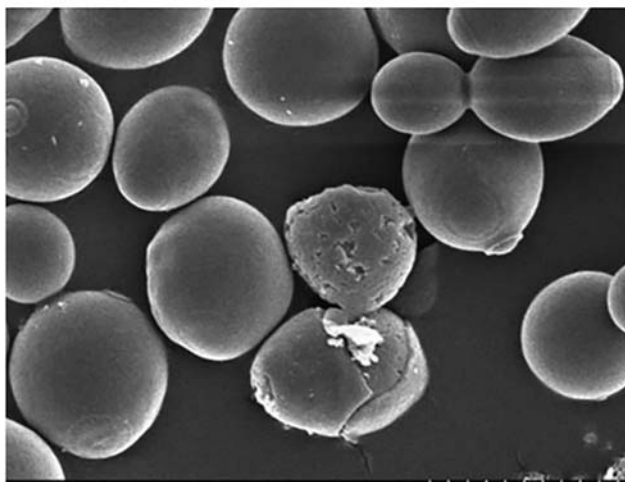
The repeated exposure of yeast to physical, biological, and chemical stresses occurring during serial repitching can lead to both reversible and irreversible damage to the cells (Fig. 5.3A and B), with progressive loss in cell viability occurring with increasing pitch number (Kalayu, 2019). Indeed, long storage periods and physiological stress during the brewing process can seriously impair the vitality of yeast slurries, resulting in loss in viability.

It has been reported that genetic mutations can occur during yeast recycling. Although such changes to the population genome may not always be immediately obvious, the extended serial repitching increases the possibility of accumulating viable variants which can compete with the original culture. The accumulation of variants, in addition with the possibility to select cells during cropping, may cause the transmission of certain characteristics to the subsequent generations. These events would lead to genetic drift within the culture and are supported by evidence of instability within brewing yeast genome. While it is well known that many brewing yeasts undergo genetic drift, in the absence of selective pressures, some yeast strains may be particularly resistant to such changes, suggesting that the capacity to serially repitch a yeast culture is intrinsically related to the genetic stability of the yeast strain employed.

FIGURE 5.3

SEM images (7000 $\times$) of *Saccharomyces cerevisiae* cells during serial beer fermentation. (A) Initial batch; (B) after five cycle repitching (see Wang et al., 2019).

Data from Wang, J., Ding, H., Zheng, F., Li, Y., Liu, C., Niu, C., & Li Q. (2019). Physiological changes of beer brewer's yeast during serial beer fermentation. *Journal of the American Society of Brewing Chemists*, 77(1), 1–11. <https://doi.org/10.1080/03610470.2018.1546030>.

**a****b**

Genetic modifications observed during repitching can be responsible of strain viability reduction and modification in the aromatic profiles of the beer. It is well known that optimal conditions for beer production depend not only on the quality of the wort and the technological conditions of the process, but also on the physiological state of the yeast. Strong and physiologically stable yeasts are essential for producing high-quality beer, and serial repitching of brewer's yeasts causes a decline in the quality of yeast, thus compromising the fermentation processes. Different studies report mutations and traits

that accumulate in ale yeasts over repitching (Krogerus et al., 2021). Modifications in other yeast traits, such as flocculation, were also observed, though not linked definitively to these mutations.

In consequence of reduction in yeast cell viability increased potentiality of microbial contamination occurrence (cross-contamination with other brewing cultures, wild yeast or bacteria) and potential changes in flavor and aromatic profiles, brewers stop repitching after about 8–10 cycles and restart brewing with a fresh yeast culture.

6.2 Formulation

In the production of beer, the choice of yeast can have a profound impact on the final product. The increase in craft beer production has led to the need to use of suitable yeast strains within small, regional microbreweries that are often limited in terms of equipment and hygienic safeguarding required for yeast propagation. Therefore, they typically purchase “ready-to-use” yeast from manufacturers. These yeasts are available in dry or liquid form, in variable volume and strain depending on the needs of the brewery. Commercial yeast strains commonly used in craft breweries can be found in dry formulations (active dry yeast), where the water content of the yeast biomass is reduced to less than 8% (Jenkins et al., 2011) or in liquid suspension, where the yeast biomass is kept in rich media or wort. Active dry yeast (ADY) is a dehydrated form of yeast commonly used by both homebrewers and commercial breweries. Yeast drying is a controlled process that subjects harvested cells to low temperatures and low humidity, thereby slowly removing water from the cells without damaging them. A very fine powder is obtained which preserves cell vitality and the ability to ferment. Dry yeast is usually packaged in small sachets or packets, making it easy to measure and add to the wort. Dry yeast can be inoculated into the wort or rehydrated according to the manufacturer’s instructions. Rehydrating yeast generally provides a higher number of viable cells than sprinkling it into the wort. An advantage of dry yeast is that it typically has a high cell count, which means a higher concentration of viable yeast cells in each packet, ensuring a robust fermentation process. Another important advantage of dry yeast is its long shelf life, it can be stored for months or even years without refrigeration, making it a convenient option for homebrewers, who can stock up on dry yeast and always have it available whenever they want to brew beer. It is also easier to transport and does not require cold shipping, thus reducing delivery costs. Hence, one of the significant advantages of using dry yeast is also its cost-effectiveness. In addition, dry yeast is highly stable and resistant to temperature changes, making it a popular choice for brewers who may not have precise temperature control during fermentation. Its ease of use, the absence of the need to oxygenate the first fermentation of the beer and the absence of the obligation to mix the beer produced with first-generation yeast, are the main advantages of ADY. A disadvantage of dry yeast is that there are not as many strains available. When a more specific flavor profile is required, one of the myriad yeast strains available in liquid form can be used.

Liquid yeast is a live culture of yeast cells suspended in a liquid medium and is typically sold in vials or smack packs. Liquid yeast requires more careful handling and storage, as it is more sensitive to temperature and it requires refrigeration to maintain its viability. The liquid culture can be added directly to the wort to ferment it and obtain beer. Otherwise, a yeast starter can also be prepared by adding the liquid yeast culture in small batch of wort to increase the cell count before pitching the yeast into the main batch of wort. Liquid yeast offers a wider range of specialty strains for specific beer styles, allowing brewers to create beers characterized by unique flavor profiles. Many brewers believe that the extra effort is worth it for the unique characteristics that liquid yeast can bring to their beers.

Liquid yeast is recognized as imparting complexity and diversity of flavors compared with active dry yeast. One of the great benefits of using liquid yeast is the ability to cultivate and harvest yeast from previous brews, creating a self-sustaining brewing system. By reusing yeast, brewers can develop unique homemade strains, adding a touch of character and texture to their beers. This allows precise control over the flavor and aroma profiles of the final product. Although liquid yeast offers numerous benefits, it requires proper storage and handling as it has a shorter shelf life and is more sensitive to temperature changes than ADY. This means that brewers need to ensure they have the necessary equipment and knowledge to maintain the viability of the yeast culture. From the cost perspective, liquid yeast tends to be more expensive due to the quality control measures involved in the production of liquid yeast. Furthermore, using liquid yeast may require additional preparation steps, such as creating a yeast starter, to ensure a healthy fermentation process, and the need to oxygenate the wort for proper fermentation (which also involves additional equipment). Although using liquid yeast can add some complexity to the brewing process, many brewers find the end result worth the effort. As a result, the use of liquid yeast continues to gain popularity among homebrewers and professional brewers alike. Its ability to impart different flavors and aromas, combined with the ability to create unique house strains, makes liquid yeast an invaluable tool for pushing the boundaries of brewing.

Regarding fermentation speed and efficiency, dry yeast typically starts fermenting quickly and completes fermentation in a shorter period than liquid yeast. On the contrary, liquid yeast may take longer to activate and may lead to a slightly longer fermentation process.

Both dry and liquid yeast have their advantages and disadvantages and the choice depends on the brewer's preferences and the specific style of beer being produced. Some brewers prefer the convenience and reliability of dry yeast, while others appreciate the flexibility and complexity offered by liquid yeast.

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Nonconventional yeasts in craft beer brewing

Maurizio Ciani, Laura Canonico and Francesca Comitini

Department of Life and Environmental Sciences (DiSVA), Polytechnic University of Marche, Ancona, Italy

1. Introduction

Saccharomyces cerevisiae and *Saccharomyces pastorianus* are defined as conventional yeasts in both craft and industrial beer production (see chapter 5). In recent years, several studies have introduced the use of other yeast species for brewing. Indeed, over time, the awareness has grown that the choice of yeast to use in the beer production is fundamental to obtaining a product with distinctive characteristics that consumers appreciate. The growing demand for beers with distinctive aromatic characteristics and unique sensory properties have driven research toward biotechnological studies with fermenting yeast at the center of the experiments. Yeast in craft beers became the maximum expression of bioflavor, able to give a characterizing impact to each product. Therefore, in the last 15 years, beer has transformed from a convivial drink with low quality standards, to a drink with character.

Following the request of the market, the most relevant topics concern the improvement and diversification of beer flavor, the development of no-alcohol and low-alcohol beers as well as functional and fortified beers. In this context, the selection of new starter strains carried out within non-*Saccharomyces* species and generically defined nonconventional yeasts may be of interest. The production of quality beer is linked to the activity of the yeast and it is due not only to the good fermentation efficiency but above all, to the ability to produce volatile and nonvolatile compounds (esters, higher alcohols, acids) that may enhance the quality of the final product. The research, based on applied microbiology studies, has unequivocally demonstrated that the selection of indigenous nonconventional yeasts is a suitable strategy to produce beer with desirable characteristics, such as low ethanol and increased organoleptic levels. Certainly, the distinctive role of yeast in the quality of the finished product is closely linked to other factors, first of all, the substrate. The selection protocol of a new starter strain and the beer styles in which the starter will be used need to be considered.

Nonconventional yeasts represent a great source of biodiversity able to diversify and produce new beer styles. The use of nonconventional yeasts may improve the analytical and aromatic profiles of the beer and reduce the alcohol content, opening quality products to the market that promote the well-being and health of the consumer. As previously mentioned, beers are classified into two large categories: Ale and Lager. This classification is the most used and simple as both categories include styles and substyles, each with its own fermentative, organoleptic, and aromatic characteristics. In addition to the classic categories and styles, the so-called specialty beers are appearing in the market: products that

do not fall into conventional styles such as for example, low-calorie, low-alcohol, and/or nonalcoholic (NABLAB) and gluten-free beers. In this regard, the brewers introduced several innovations in the production process such as the use of different hop varieties, special malts, spontaneous and preparation of blends, among others. In these specific and innovative types of beers the use of appropriate nonconventional strains finds its maximum expression, giving satisfactory results, even more appreciated by the audience of consumers, expert or not. An example that can help to understand this concept is the selection of *Brettanomyces* spp. strains that may be used for primary, secondary, bottle fermentation, and barrel maturation facilitate the beer production process by adapting to extreme conditions. *Brettanomyces* strains allow us to obtain beer with a peculiar aromatic taste, and its contribution is characteristic for spontaneous fermentations, the description and the use of this yeast species is covered in the chapter 7.

It is well known that nonconventional yeasts have low primary metabolic efficiency, with a consequent limited fermentation performance, but, because of their secondary metabolism they can affect the sensory characteristics of the final products. Volatile compounds from nonconventional strongly contribute to the flavor and aroma profile. Below, it will illustrate the characteristics and potential uses of the main nonconventional yeast strains in the production of craft beers.

2. Nonconventional yeasts: Use and role in craft beer production

Although there are several commercial starter strains in the brewing sector, the requirement of new starter strains is crucial to characterize the different craft beers and the various microbreweries. Following the winemaking development, in the brewing sector is growing interest in selected nonconventional yeasts for their ability to give unique bioflavor in different craft beers.

The most prominent and dominating beer flavor due to nonconventional fermentation is related to specific profiles and amounts of alcohols, esters, aldehydes, and organic acids. However, the production of most of them by nonconventional yeasts is a large part of the potential to be developed to enrich the fermentation final products. Besides that, nonconventional yeasts, unlike *Saccharomyces*, produce and excrete a wide variety of extracellular enzymes that specifically contribute to exalting the aroma and sensorial profile of final products, making them different from their industrial counterparts. These enzymes interact with aromatic precursors making them aromatically active, thus influencing the aromas of fermented beverages. For this reason, the use of nonconventional yeasts can impact the flavor of craft beer. For example, microbial enzymes belonging to dehydrogenase or dehydratase are involved in terpenoid biosynthesis or biotransformation. Moreover, the contribution of yeast in craft beers is certainly relevant since the product is unpasteurized, unfiltered, and sometimes re-fermented in a bottle.

For all these reasons, brewers interested in meeting consumers' expectations of more healthy products and in expanding their assortment of beers must increase their production of nonalcoholic beers (NAB) or low-alcoholic beers that are taking over the market. Regarding the healthy products, functional beers are obtained by adding beneficial health value, intended either as functional ingredients or functional fermenting yeasts. An absolute novelty is represented by probiotic beer among the functional beers, obtained by incorporating probiotic microorganisms. Moreover, functional beers represent a new class of nontraditional beers, with healthy profiles due to functional ingredients added during or after the traditional brewing process, which contain bioactive compounds that can bring

health benefits to the consumer. To assess this potential benefit, total phenolic content, phenolic profile, antioxidant activity, and volatile profile are usually increased. To reach this goal, botanical addition of plant derivatives from quinoa, goji berries, eggplant, mango, or sweet potato or extracts are made, as an additional source of bioactive compounds and increased antioxidant activity in the special beer segment.

2.1 Nonconventional yeasts and bioflavor

The valorization of aromatic compounds with biological methods occurs through the metabolic pathways and the diversification of yeasts for the promotion of the so-called bioflavor. This approach may include microbial bioconversion of flavor precursors, using different nonconventional yeasts, that produce the required compounds (Postigo et al., 2022; Canonico et al., 2019.) as well as using the genetic modification by interspecific *Saccharomyces* hybrids (Mertens et al., 2015). For this reason, there is a growing interest in the use of different nonconventional yeasts that can improve or modify the analytical and aromatic profiles of craft beers. Some researchers have evaluated nonconventional yeasts coming from different environments to diversify the aroma and flavor characteristics of beer (Satora and Pater, 2023). Co-fermentation or sequential fermentation with *Saccharomyces* starter strains (*S. cerevisiae* or *S. pasteurianus*) could be a strategy to achieve an improvement or diversification of the bioflavor.

The choice of nonconventional yeast strain based on the strain-species performance combined with the modalities of the fermentation process as well as raw materials used determines the overall features of the final craft beer. Table 6.1 summarizes the most relevant bio-aromatic characteristics in beers produced with nonconventional yeasts.

2.2 Nonconventional yeasts and NABLAB beers

Among consumer demand, particularly in the craft beer market, the request for nonalcoholic/low-alcohol beers (NABLAB) is constantly increasing. New flavors and styles, but also nonalcoholic beers are the main drivers of product diversification in craft beer production. For these reasons, brewers interested to meet the consumers' expectations toward more healthy products must increase the production of nonalcoholic beer (NAB) or low-alcoholic beer.

The definition of NAB, depending on the regulations of the countries, generally can contain 0%–0.5% ethanol (Okaru & Lachenmeier, 2022). In Germany, the USA, and in China, this limit is no more than 0.5% v/v; in Spain, the maximum value is 1% v/v alcohol; while in France, is of 1.2% v/v alcohol. In Islamic countries, where alcohol consumption is prohibited by law, nonalcoholic beers (NABs) contain less than 0.5% v/v alcohol and represent an alternative to other nonalcoholic beverages because these provide some of the main bioactive components of traditional beer. Physical techniques or arrested fermentation could be used to brew NABLAB but in both cases there may be the loss of aromatic compounds. Several nonconventional yeasts have been studied for producing nonalcoholic beer with promising results: *Lachancea fermentati* (Bellut et al., 2020); *Wickerhamomyces anomalus* (De Roos et al., 2018; Spitaels et al., 2015), *Saccharomycodes ludwigii* (used commercially for this purpose for many years); *Zygosaccharomyces rouxii*, *Zygosaccharomyces bailii*, *Zygosaccharomyces kombuchaensis* (De Francesco et al., 2015; Sohrabvandi et al., 2009); *Cyberlindnera fabianii* (*Candida fabianii*), *Cyberlindnera mrakii* spp. (Bellut & Arendt, 2019; Liu & Quek, 2016; Methner et al., 2022);

Table 6.1 The most relevant and characterizing bioflavors in beers brewed with nonconventional yeasts.

Nonconventional yeasts	Bioflavors compounds	References
<i>Brettanomyces bruxellensis</i> / <i>Dekkera bruxellensis</i>	Significant esters production: Ethyl acetate (solvent-like aroma), ethyl caprate, ethyl caprylate and ethyl lactate (sour flavor)	Crauwels et al., 2015; Steensels et al., 2015; Serra et al., 2019; Michel, Meier-Dörnberg, et al., 2016
<i>Debaryomyces hansenii</i>	Increase in isoamyl alcohol (alcoholic, banana), hexanol, isoamyl acetate (banana aroma), ethyl octanoate (sour apple aroma), ethyl hexanoate (sweet apple aroma)	Spitaels et al., 2015; Holt et al., 2018; Tokpohozin et al., 2016; Canonico et al., 2020
<i>Lachancea thermotolerans</i>	Ethyl butyrate and ethyl acetate (solvent-like aroma)	Domizio et al., 2016; Canonico et al., 2020; Zdaniewicz et al., 2020; Bokulich et al., 2012
<i>Pichia kluyveri</i>	Increase in isoamyl acetate (banana aroma), isoamyl alcohol (alcoholic, banana), ethyl butyrate, ethyl hexanoate (sweet apple aroma), and ethyl octanoate (sour apple aroma), ethyl acetate (solvent-like aroma)	Bokulich et al., 2012; Holt et al., 2018; N'guessan et al., 2011; Saerens & Swiegers, 2014a, 2014b
<i>Saccharomyces ludwigii</i>	Production of ethyl acetate (solvent-like aroma), isoamyl acetate (banana aroma) and 4-vinylguaiacol (strong medicinal, clove-like aroma), but high quantities of amyl alcohols (alcoholic, solvent) and higher alcohols floral (fruity or herbal aromas)	De Francesco et al. (2015)
<i>Torulaspora delbrueckii</i>	Increase in β -phenylethanol (rose aroma), n-propanol (alcohol, sweet), iso-butanol (solvent), amyl alcohol (alcoholic, solvent), and ethyl acetate (solvent-like aroma), ability to convert hop monoterpene alcohols into linalool	Canonico et al., 2016; Bellut et al., 2018; Einfalt, 2021
<i>Wickerhamomyces anomalus</i>	Increase in ethyl propanoate (rum and pineapple), phenyl ethanol (rose aroma), 2-phenylethyl acetate (roses and honey aroma), and ethyl acetate (solvent-like aroma)	Basso et al., 2016; Holt et al., 2018
<i>Zygosaccharomyces rouxii</i>	Increase in ethyl acetate, amyl alcohols, isoamyl alcohols, and other esters and higher alcohol	De Francesco et al., 2015; Sohrabvandi et al., 2009
<i>Hanseniaspora guilliermondii</i> , <i>Hanseniaspora opuntiae</i>	Ethyl acetate and phenyl ethyl acetate, producing a beer with a pleasant “honey” aroma	Liu et al. (2022)
<i>Williopsis saturnus</i> var. <i>mrakii</i>	Higher levels of acetate esters, terpenes and terpenoids	Liu & Quek, 2016
<i>Cyberlindnera fabianii</i> and <i>Pichia kudriavzevii</i>	Decrease in high alcohols and volatile esters (ethyl acetate, 3-methylbutyl acetate, methyl propyl acetate, phenylethyl acetate, ethyl hexanoate and ethyl octanoate)	Van Rijswijck et al. (2017)

Table 6.1 The most relevant and characterizing bioflavors in beers brewed with nonconventional yeasts.—cont'd

Nonconventional yeasts	Bioflavors compounds	References
<i>Pichia anomala</i> and <i>Zygoascus meyeræ</i>	Production of 4-vinylguaiacol, β -phenylethyl alcohol, and isoamyl alcohol	Larroque et al. (2021)
<i>Hanseniaspora vineae</i>	Increase in esters production (in 2-phenylethyl acetate (roses and honey aroma)	Larroque et al. (2021)
<i>Candida tropicalis</i>	Increase in higher alcohols (floral, fruity, or herbal aromas) and acetaldehyde (green apple and cut grass), and lactate, and reduced aroma-active compounds	Alloue-Boraud et al. (2015)
<i>Mrakia gelida</i> and <i>Mrakia blollopis</i>	Increase in higher alcohols (floral, fruity or herbal aromas) and acetaldehyde (green apple and cut grass) and lactate, and reduced aroma-active compounds	De Francesco et al. (2018)

Hanseniaspora vineae, *Hanseniaspora valbyensis* (Bellut et al., 2018; Larroque et al., 2021); *Torulaspora delbrueckii* (Canonico et al., 2017; Methner et al., 2022); and *Pichia kluyveri*, *Kluyveromyces lactis* (Holt et al., 2018; Saerens & Swiegers, 2014a, 2014b). In Table 6.2 are summarized the species of nonconventional yeasts evaluated, and the main results obtained. Physical techniques or arrested fermentation could be used to brew NABLAB but in both cases there may be the loss of aromatic compounds. For this reason, the use of biological methods, based on the use of nonconventional yeasts as starter cultures, could be a valid tool for producing NABLAB. Indeed, these yeasts generally exhibit low capacity to ferment maltose (and maltotriose), the most abundant reducing sugars, in wort, but combining, at the same time, aroma enhancement aromatic complexity improvement (Bellut & Arendt, 2019; Canonico et al., 2021; Yabaci Karaoglan et al., 2022).

Table 6.2 Nonconventional yeast species used for the production of NABLAB beers and related ethanol produced (% v/v).

Nonconventional yeasts	Ethanol content (% v/v) NABLAB beer
<i>Lachancea fermentati</i>	<1.3
<i>Saccharomycodes ludwigii</i>	<0.4
<i>Torulaspora delbrueckii</i>	0.5–2.7
<i>Pichia kluyveri</i>	<0.2
<i>Wickerhamomyces anomalus</i>	<0.2
<i>Zygosaccharomyces rouxii</i> , <i>Zygosaccharomyces bailii</i> , <i>Zygosaccharomyces kombuchaensis</i>	<0.5
<i>Hanseniaspora vineae</i> , <i>Hanseniaspora valbyensis</i>	<0.5
<i>Mrakia gelida</i>	<1.5
<i>Candida shehatae</i>	<0.5
<i>Candida zemplinina</i> (<i>Starmerella bacillaris</i>)	<1.5
<i>Cyberlindnera fabianii</i> (<i>Candida fabianii</i>) <i>Cyberlindnera mrakii</i> (<i>Williopsis saturnus</i> var. <i>mrakii</i>)	0.6–1.7
<i>Pichia kudriavzevii</i>	<1

2.3 Nonconventional yeasts and low-calorie beers

There is increasing concern about the negative impact of the modern food system on the health of both us and our planet. In particular, foods and beverages that contain high levels of salt, sugar and fat, or ethanol have been linked to unhealthy food. Also in brewing, consumers want to look for a healthy aspect. For this reason, low-calorie beers are an example of a new segment recently introduced on the market.

Beer with a low-calorie content has achieved great interest due to the problem of obesity, which accounts for a growing market segment. This type of beer can be made by special mashing and collection of the wort with large amounts of fermentable sugars, or by inoculating microorganisms that can hydrolyze the more complex sugars, to reduce the concentrations of residual sugars in the final product. This is the case for *Brettanomyces* yeasts, which use complex sugars and lead to low-calorie, but slightly more alcoholic, beers. Regarding the production of beverages with lower alcohol contents but providing superior sensory quality, new methods have been studied (Sohrabvandi et al., 2010). In this regard, nonconventional yeasts represent a very attractive alternative, both for the null processing of extra costs and for the advantage of avoiding involuntary extraction of flavor compounds. *Wickerhamomyces subpelliculosus* (formerly *Pichia subpelliculosa*) and *Cyberlindnera saturnus* (formerly *Williopsis saturnus*), for example, have shown interesting results in the production of low-alcohol beers with acceptable flavor profiles. In Table 6.3 are reported the main nonconventional species and their use in the production of low-calorie beers.

2.4 Nonconventional yeasts and functional beers

Functional beers are products obtained by adding beneficial health value, intended either as functional ingredients or functional fermenting yeasts. Nonconventional yeast could be widely used to produce or transform some beneficial compounds. For example, melatonin, which is a sleep-regulating hormone in mammals that has antioxidant properties, can be produced in beer during alcoholic fermentation by the appropriate yeast (Maldonado et al., 2009). Within functional beers, there are gluten-free beers

Table 6.3 Nonconventional yeast and healthy characteristics.

Species	Healthy characteristics
<i>Brettanomyces bruxellensis/anomalus</i>	Low-calories beer
<i>Wickerhamomyces subpelliculosus</i>	Low-calories beer
<i>Cyberlindnera saturnus</i>	Low-calories beer
<i>Hanseniaspora uvarum, Hanseniaspora vineae</i>	Gluten free
<i>Torulaspora delbrueckii</i>	Gluten free
<i>Wickerhamomyces anomalus</i>	Gluten free
<i>Saccharomyces cerevisiae</i> var. <i>boulardii</i>	Probiotic
<i>Lachancea thermotolerans</i>	Potential probiotic
<i>Kazachstania unispora</i>	Potential probiotic
<i>Pichia kudriavzevii</i>	Potential probiotic
<i>Saccharomyces cerevisiae</i>	Potential probiotic

with functional properties for consumers with the condition known as celiac disease, which is gluten-sensitive and immune-mediated enteropathy. In this regard, there are promising nonconventional yeasts such as yeast from the genera *Hanseniaspora*, *Pichia*, *Torulaspora*, *Wickerhamomyces*, and others, that can offer diversified enzymatic and bioconversion.

Beer is a world-famous beverage and the choice of fermentable cereals along with the addition of plant extracts or specific functional ingredients along with the removal of certain compounds (e.g., alcohol, gluten, or carbohydrates) can create a tailor-made beverage suitable for every type of consumer. Although some of the functional additions to beers have already entered production, others are still the subject of different research.

Starting from the concept that among the functional foods/drinks in the dairy sector, kefir (fermented by a consortium of bacteria such as *Lactobacillus*, *Leuconostoc*, *Lactococcus*, and *Acetobacter* and yeasts like *Kluyveromyces*, *Candida*, *Saccharomyces*, *Torulaspora*, *Kazachstania*, *Lachancea*, and *Yarrowia*), undoubtedly represents a halter product (Rodrigues et al., 2016) reported the production of beer with kefir. This innovative product is made with added kefir (fermented with *S. cerevisiae*) resulted in increased values of phenolic content in the drink. The increased health-beneficiary synergistic effect between kefir and beer should be further investigated concerning the potential health-promoting activities of probiotic beers. Other similar symbiotic communities of bacteria and yeasts (kombucha, boza, koumiss, and borde drinks) could also be investigated and developed as health-beneficiary, functional beverages, like craft beer. Again, worts produced from different cereals (winter wheat, oats, barley, and combinations of stated cereals) were subjected to fermentation using some *S. cerevisiae* yeast strains and/or the lactic acid bacteria (*Lactobacillus casei* subsp. *paracasei*) isolated from kefir grains. All the cereals and their mixtures showed satisfactory results in the sensory analysis of the produced beers/beverages and were described as pleasant and/or fruity.

Citrus winter wheat malt appeared as very promising for several beverages with acceptable sensory properties, and nonhopped low-alcohol beverages with a fresh citrus aroma and acidic flavor are a result of fermentation using a mixed yeast and bacterial culture (Schneedorf, 2014; Mikyška et al., 2015). Indeed, a novelty functional beer is represented by probiotic beer obtained by incorporating probiotic microorganisms exploiting the evidence that craft beer is an unpasteurized and unfiltered matrix, thus representing a potential vehicle for delivering probiotics. Chemical or biological acidification via *Lactobacilli* and bifidobacterial fermentation secure the microbial stability of the final beer. However, excessive levels of acetic acid could be produced if *Bifidobacteria* are incorporated.

Moreover, since viability is crucial for the efficacy of probiotics, attention must be paid to the sensitivity of probiotics to hop bitter acids, which can inhibit the survival of Gram-positive lactic acid bacteria (Suzuki, 2011; Vriesekoop et al., 2012). Probiotics are not only bacteria; indeed, *S. cerevisiae* var. *boulardii* is a probiotic yeast strain that was explored for specialty unfiltered and unpasteurized probiotic beer (Capece et al., 2018; Mulero-Cerezo et al., 2019; Paula et al., 2021). Studies have shown that foods and drinks with live probiotics are more effective in providing health effects than products containing inactive probiotics. Craft beer with live yeasts can be considered a new tool for beneficial health effects (Comitini et al., 2023).

The use of several non-*Saccharomyces* genera (*Lachancea*, *Kluyveromyces*, *Torulaspora*, *Metschnikowia*, *Kazachstania*, *Brettanomyces*, *Pichia*, *Candida*, *Hanseniaspora*, *Rhodotorula*, and *Rhodospiridiobolus*) was found to have great potential for use in the production of craft beers with probiotic characteristics (Canonico et al., 2021). Craft beer is generally unfiltered or unpasteurized; thus, the yeast cells remaining after fermentation can provide a probiotic character to the beer and can

be considered beneficial to health. For this reason, craft rather than industrial brewing would be more appropriate since viability is crucial for the efficacy of probiotics (Capece et al., 2018). However, it should be noted that the presence of yeasts in beer can give beer a unique taste but reduce its shelf life as during storage, cell lysis can negatively affect its quality. For these reasons, strain selection is fundamental in the production of craft beers (Iorizzo et al., 2021).

Another strategy to obtain beer with functional characteristics could be the combination of functional or probiotic yeasts with functional ingredients (Zannini et al., 2022). Among them, promising ingredients could be legumes such as chickpeas, lentils, and soy as an important source of protein for human nutrition (Canonico et al., 2022). The goal of functional craft beer could be reached using wort enriched with hydrolyzed protein and fiber sources and by using probiotic yeasts (Canonico et al., 2022). Table 6.3 reports the main nonconventional species and their use in the production of functional craft beers.

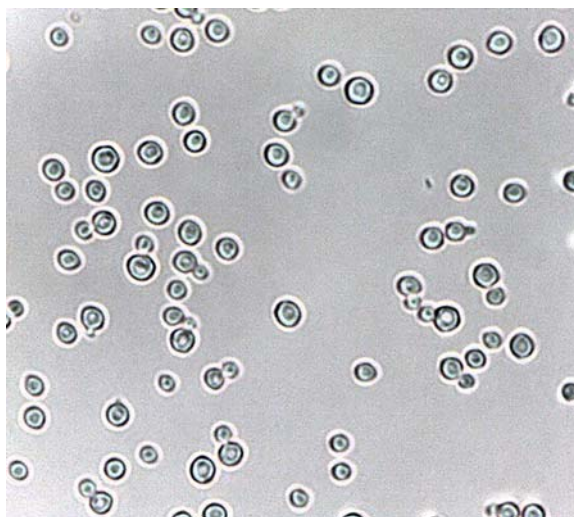
3. Most relevant species of nonconventional yeasts in craft beer production

3.1 *Torulaspora delbrueckii*

Among the nonconventional yeasts *Torulaspora delbrueckii* (Fig. 6.1) is a species of winemaking origin and is used in controlled wine fermentation processes because it reduces volatile acidity and shows a good production of fruity flavors (Ciani et al., 2010). In craft brewing, has received attention due to its ability to ferment maltose (strain dependent), produce ester compounds, and bio-transform the monoterpenoid flavor compounds of hops (e.g., reduction in geraniol to citronellol). Some authors assume that *T. delbrueckii* is also one of the main contributors for the fermentation of Bavarian wheat beers (Michel, Meier-Dörnberg, et al., 2016; Tataridis et al., 2013). This species is an osmotolerant and

FIGURE 6.1 Micromorphological of the nonconventional yeast *Torulaspora delbrueckii*.

Optic microscope 1000 magnifications produced ourselves.



grows well at low temperatures (cryotolerant). The ability to consume maltose—the main sugar in the wort—is conditioned by the activity of the enzyme maltase, as well as by maltose transporters also present in this species, highlighting that such biochemical apparatus can be inhibited by high concentrations of glucose in the medium. Interestingly, the ability to metabolize sugar maltotriose was also proven in this microorganism.

Ethanol tolerance and fermentation efficiency are also strain-dependent features. Although it has been reported that *T. delbrueckii* can tolerate up to 11% (v/v) ethanol in winemaking conditions, recent evidence demonstrates that many strains are only able to grow in concentrate beer wort with at most 5% ethanol in the medium (Michel, Kopecká, et al., 2016). Tolerance to iso- α -acids from hops is another important trait that yeast should display in brewing conditions, once that compound has inhibitory effect against some microorganisms, including gram-positive bacteria and yeasts.

The main feature in the bioflavor enhancement of this yeast is the volatile profile characterized by β -phenyl ethanol (“rose” flavors), n-propanol, iso-butanol, amyl alcohol (“solvent brandy” aroma), and ethyl acetate (Basso et al., 2016; Pires et al., 2014).

Beer produced by pure cultures of *T. delbrueckii* and by mixed fermentations of *S. cerevisiae*/*T. delbrueckii* were characterized by “fruit/citric” and “fruity/ester” notes and had “full-bodied” attributes (Canonico et al., 2016; Michel, Meier-Dörnberg, et al., 2016).

Another relevant application of *T. delbrueckii* in craft beer production is related to the production of NABLAB beer since some yeast strains could not ferment maltose/maltotriose, producing at the same time good flavors so that they may be suitable for low-alcoholic beer production (Michel, Meier-Dörnberg, et al., 2016; Yabaci Karaoglan et al., 2022). In pure fermentation using specific substrates as recycled brewer’s spent grains BSG *T. delbrueckii* may be an interesting species to produce NABLAB beers due to its aromatic imprinting.

3.2 *Lachancea thermotolerans*

The main goal of the use of *Lachancea thermotolerans* in craft beer is the production of sour beer. Indeed, among several nonconventional yeast species that showed lactic acid production, *L. thermotolerans* yielded other than relevant lactic acid production an excellent attenuation and sensory characteristics, positioning it as viable alternative to lactic acid bacteria for sour beer production avoiding the refermentation process (Osburn et al., 2018). *L. thermotolerans* exhibited good competition in co-cultures with *S. cerevisiae*, dominating the fermentations with 1:20 *S. cerevisiae*/*L. thermotolerans* ratio. Both pure and co-cultures affected significantly the beer aroma determining a general reduction in acetaldehyde content in all mixed fermentations and a significant reduction in pH value due to the production of lactic acid. Regarding the aromatic composition of the resulting beers, an enhancement of ethyl butyrate and ethyl acetate can be highlighted (Canonico et al., 2020; Domizio et al., 2016). Recently, *L. thermotolerans* was proposed as a strategy to enhance functionality of craft beer (Canonico et al., 2021, 2022) and also to recycle brewers spent grains (BSG) for nonalcoholic and low alcohol beer (NABLAB) (Canonico et al., 2023, 2024). This strain of *L. thermotolerans* showed interesting probiotic features and potential development for the production of functional beers since this species is presumption of safety (QPS).

3.3 *Pichia* spp.

Wickerhamomyces anomalus (formerly *Pichia anomala*) (Fig. 6.2) is considered the most important nonconventional yeast in brewing process. This yeast can produce low concentrations of undesirable aroma compounds, and it is characterized for its resistance to unfavorable environmental conditions. *W. anomalus* produces ethyl propanoate, phenyl ethanol, 2-phenyl ethyl acetate and especially ethyl acetate (Passoth et al., 2006) influencing the taste and aroma profile of beer, imparting an interesting fruity character or an unpleasant solvent-like character, depending on the concentration values. Regarding to its ability to ferment maltose, the results are variable. For these reasons, *W. anomalus* could be used in brewing process in sequential or co-inoculation with *S. cerevisiae* starter strains (Walker, 2011; Lee et al., 2011).

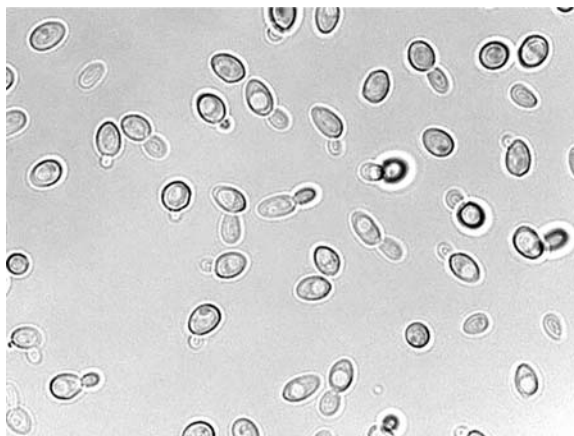
Pichia kluyveri is another nonconventional yeasts which has shown to contribute with fruity banana flavors. However, this species also produces compounds that gives undesirable flavors, problem that can be solved with a sequential fermentation with a *Saccharomyces* yeast before the concentration of these substances becomes too high (Holt et al., 2018). Moreover, *P. kluyveri* has been discovered for producing low-alcohol or alcohol-free beers owing to its limited ability to ferment glucose while significantly changing hop compounds into positive flavor compounds (Saerens & Swiegers, 2014a, 2014b). This species produces average levels of esters and high quantities of higher alcohols. The formation of flavor-active compounds, such as average levels of esters and high quantities of higher alcohols, and fast fermentation ability of this species might indicate a brewing potential for novel alcohol-free beer, although to date the commercial use of this strain has not been implemented.

The yeast *Pichia kudriavzevii* has been used to produce light craft beer, with low bitterness, and neutral characteristics (Santos et al., 2022). This yeast produced higher alcohols 1-octanol and 1-pentanol, responsible of orange, rose and bread aromas among others.

A selected strain of *P. kudriavzevii* yeast was used as the starter culture to elaborate a craft Belgian-style pale ale based on a strictly reproducible brewing process using malted barley as the only substrate. This yeast was extremely efficient in the consumption of sugars and the generation of ethanol. The barley employed in the brewing process complied with the established standards. Higher alcohol was identified in the resulting beer. The main one, isoamyl alcohol, has been reported to impart fruity

FIGURE 6.2 Micromorphological of the nonconventional yeast *Wickerhamomyces anomalus* (formerly *Pichia anomala*).

Optic microscope 1000 magnifications.



flavors to some beverages. A medium level of ethanol was found in the beer (5.2% v/v) (Nieto-sarabia et al., 2023). Different species belonging to *Pichia* genera were tested with the aim to reuse a brewers spent grains (BSG) to obtain another beer with low or no alcohol with promising results (Canonico et al., 2023, 2024).

3.4 *Hanseniaspora* spp.

The information on the use of yeasts belong to *Hanseniaspora* genus strains in beer production is still limited. A few *Hanseniaspora* spp. strains have been isolated from spontaneously fermented beer, but its role in this context is still unclear (Spitaels et al., 2014) (Fig. 6.3).

The species *Hanseniaspora valbyensis* and *Hanseniaspora vineae* have been evaluated for their application in the production of nonalcoholic beer (Bellut et al., 2018), and the *H. vineae* strain has been tested for its ability to acidify the wort in the production of sour beer (Osburn et al., 2018).

H. uvarum selected strains might be applied as co-starters for producing beer up to 10% (v/v) ethanol (Matraxia et al., 2021). Some strains showed better growth phases than the commercial *S. cerevisiae* strain, indicating that the tested strains possess useful characteristics for use as co-starters in the production a wide range of beer worts, including highly hopped worts. Actually, it has been not reported maltose fermentation for the *Hanseniaspora* genus and genes for maltose assimilation were not identified in public available genomes of *Hanseniaspora* spp (Cadez & Smith, 2011; Čadež et al., 2019). Some strains of *H. uvarum* were able to produce considerable amounts of acetic acid characteristic well noted in winemaking process. Regarding glycerol production, the trial showing the highest amounts was that involving the sequential inoculation, while no substantial differences were detected for ethanol production. The most abundant VOC category was alcohols that are known for enhancing flowery, solvent like or alcoholic flavors; however, their presence above certain threshold could be undesired (Eßlinger, 2009). The second most abundant VOC class has been represented by esters, whose level depends on the yeast strain-specific activity of synthesis and breakdown enzymes (Pires et al., 2014). The main aroma-active ester was ethyl acetate, a secondary metabolite of alcoholic fermentation, responsible of fruity aroma of beers and whose perception threshold in beer is equal to 25 mg/L. Ethyl hexanoate and isoamyl acetate, which are liable for banana, apple, and anise aroma, showed values below the perception threshold (Meilgaard, 1975).

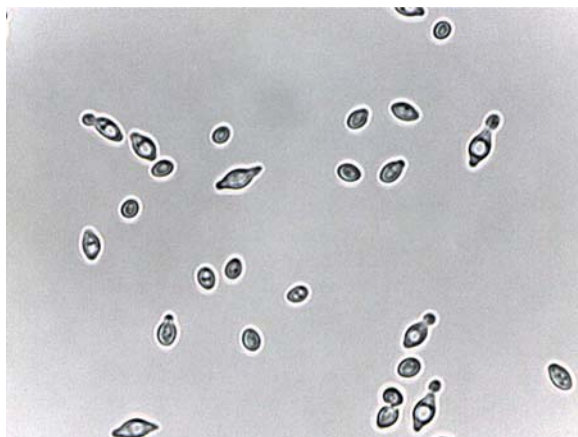


FIGURE 6.3 Micromorphological of the nonconventional yeast *Hanseniaspora uvarum*.

Optic microscope 1000 magnifications.

3.5 *Schizosaccharomyces* spp.

The genus *Schizosaccharomyces* is currently composed of five species, such as *Schizosaccharomyces pombe*, *Schizosaccharomyces octosporus*, *Schizosaccharomyces japonicus*, *Schizosaccharomyces* and *Schizosaccharomyces osmophilus*. The application of yeasts belonging to the genus *Schizosaccharomyces* has been widely studied in the field of winemaking mainly for their ability to decompose malic acid in alcohol via malo-alcoholic fermentation (Benito, 2019). These yeasts could be proposed for beer fermentation. Indeed, malo-alcoholic fermentation could be considered a valid approach to preserve fresh aromatic properties, imparting freshness notes to beverages (Cecchini et al., 2006).

3.6 Other nonconventional yeasts

Other nonconventional yeast species were considered and tested in craft beer production. *Zygorhizula florentina* species evaluated in pure and mixed fermentation with *S. cerevisiae* commercial starter US-05 showed an increase in higher alcohols, the isoamyl acetate and α -terpineol content. *Saccharomyces ludwigii*, which was described as unable to catabolize maltose showed low ethanol production ranging from 0.51% to 1.36% (v/v) while *Z. rouxii*, exhibited an ethanol production of 0.93% (v/v) with positive aromatic characteristics (De Francesco et al., 2015).

Another genus, *Cyberlindnera* was screened for the production of fruity nonalcoholic beer (Bellut & Arendt, 2019). Others nonconventional yeasts investigated in the production of NABLAB beers are *Kazachstania servazzii*, *Kluyveromyces marxianus*, *Pichia fermentans* together with *T. delbrueckii* isolated from sourdough cultures, for cold contact fermentations to evaluate their ability to reduce wort aldehydes at a fermentation temperature of 1.0 °C but only *T. delbrueckii* strain showed promising results (Methner et al., 2022).

P. kudriavzevii and *Meyerozyma guilliermondii* MUS122 were promising nonconventional yeasts isolated from a Belgian wheat beer sludge. They were suitable to produce volatile compounds such as ethyl acetate, 2-phenyl ethanol, and isoamyl alcohol, with characteristic fruity notes and floral nuances to the aroma profile of mixed fermentations with brewer's yeast (Postigo et al., 2022).

In another investigation together with *P. anomala* (now *W. anomalus*), *Zygoascus meyeriae* was selected for the production of 4-vinylguaiacol, β -phenyl ethyl alcohol, and isoamyl alcohol at levels significantly above the sensory threshold making it interesting for wheat and blond craft beer styles (Sampaioles et al., 2023).

Together with *T. delbrueckii*, *Candida zemplinina* (reclassified as *Starmerella bacillaris*) and *Z. bailii* species were tested in mixed fermentation with selected *S. cerevisiae* strain as starters for production of different beer styles (Pilsner, Weizen, and Amber) evaluating different fermentative medium on starter behavior. A significant wort–starter interaction was found. *C. zemplinina*/*S. cerevisiae* mixed fermentation showed a promising starter for the production of low-alcohol beers with pleasant organoleptic characteristics in all fermentation media tested (Capece et al., 2021). Further other species of nonconventional yeasts were investigated for their potential in nonalcoholic beer production. In addition to more investigated *T. delbrueckii*, *P. kluyveri*, and the species above described, other species such as *Starmerella bombicola*, *Lindnera saturnus*, *Lindnera jadinii*, *Z. rouxii*, and *Candida pulcherrima*. *S. bombicola*, an important industrial producer of biosurfactants, was able to produce nonalcoholic beer even at 20 °C after 10 days of fermentation with neutral and no negative impact on organoleptic properties of the beer. More interesting organoleptic properties were found in

beers fermented with *L. jadinii* and *L. saturnus*, which produced banana-flavored beers with low alcohol content. These strains could be a promising nonconventional yeast to produce nonalcoholic beers (Gibson et al., 2017). Others nonconventional yeasts related to production of nonalcoholic beer are *C. pulcherrima* and *Z. rouxii*. This last species exhibited a resistance to ethanol and hop (De francesco et al., 2018) with relevant features in brewing process.

4. Safety aspects on the use of nonconventional yeasts

The use of nonconventional yeasts in craft beer as a natural tool for innovation has attracted great interest from both researchers and commercial brewers and in craft beer production, a wide diversity of nonconventional yeasts coming from different environments was investigated. In this regard, the safe use of nonconventional yeasts is an important aspect that needs to be evaluated and investigated. Indeed, there is only a short list of yeasts that are surely and generally recognized as safe (GRAS/QPS) for use in food production (Ricci et al., 2017), and further testing may be necessary to ensure consumer safety. The fundamentals of a safety assessment include an accurate taxonomic species identification using up-to-date methods, along with a literature study regarding the yeast species in question. The strain-specific safety concerns that should be assessed involve pathogenic potential, antifungal resistance, production of biogenic amines and possible allergic reactions (Miguel et al., 2022).

Certain nonconventional strains, some of them already commercialized as starters for enology or brewing, could release some amino acids providing yeast assimilable nitrogen to *S. cerevisiae* during mixed fermentations. In the yeast–yeast interactions, high levels of histidine, phenylalanine, or tyrosine could increase the synthesis of biogenic amines with a risk to the health of the consumer. One strategy to reduce these amines is the identification of yeast strains with low capacity to produce amines. For example, the strain of *L. thermotolerans* proposed by Benito et al., (2016) in mixed fermentations with *S. cerevisiae* have produced amounts of biogenic amines similar to or lower than those produced by *S. cerevisiae* (<0.85 and < 0.50 mg/L for putrescine and histamine, respectively). On the contrary, the ancient role of yeasts in making beverages safer support and testify for their safety role. Considering these aspects and the innovative fermentation strategies in the brewing sector, it is advisable to implement the guidelines for the use of nonconventional yeasts in brewing.

The expansion of functional beer will likely continue to grow and new technologies based on added ingredients and/or microorganisms to enhance flavors and diversify products will continue to attract different types of people with different preferences, medical conditions, or religious views. The propagation of functional beers will likely create a new type of health-conscious customer who enjoys enjoying a traditional beer with a twist. Since the possibilities are endless, the results of microbiological research on the role of nonconventional yeasts may represent the basis of knowledge to bring quality products to the market, at affordable production prices.

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Spontaneous fermentation

7

Renan E. Araujo Piraine¹ and Matthew L. Bochman²

¹University of São Paulo, Ribeirão Preto, Brazil; ²Indiana University, Molecular & Cellular Biochemistry Department, Bloomington, IN, United States

1. Introduction

Fermentation is widely used in industrial processes and at the amateur level in individual households all over the world to produce everything from pharmaceuticals to pickled vegetables to alcoholic beverage alcohol, the latter of which is our focus here. When most people think of ethanolic fermentation to make something like beer, they envision a brewer stirring a vat of steaming grain and adding hops and yeast to make their favorite lager, pilsner, stout, or pale ale. However, fermentation far predates man, and even this bucolic vision of controlled fermentation by a brewer is a recent invention (Rogers, 2014).

The Earth is approximately 4.5 billion years old, and fossil evidence points toward the emergence of life around 3.7 billion years ago (Pearce et al., 2018). This environment contained little atmospheric oxygen, so anaerobic metabolic pathways, such as fermentation, were utilized by early life for energy production. Modern *Homo sapiens* evolved only 200,000 years ago (Galway-Witham and Stringer, 2018), and archeological evidence points toward humans harnessing fermentation around 10,000 years ago (Dietrich et al., 2012; Rogers, 2014). No one knows the exact circumstances of this marriage between man and microbe, but the Paleolithic Hypothesis states that early humans gathered sugar-rich fruits or berries such as grapes. The ripe grapes at the bottom of the collection vessel would burst under the weight of the grapes on top, releasing their juices and allowing the naturally resident yeasts on the grape skins to spontaneously ferment the sugar. If curious humans tasted this aromatic liquid, they would find it to be pleasantly flavored, slightly carbonated, and mildly intoxicating, and they would likely want more. Putting two and two together, these clever humans would ultimately start to purposefully press grapes into juice, allow the “magic” of fermentation to happen, and *voilà*, wine is born (McGovern, 2019; Rogers, 2014). It was not until thousands of years later that scientists identified yeasts (and bacteria) as the biological agents of beverage fermentation and introduced the concept of pure culture brewing, that is, the inoculation of a sugar-rich substrate like beer wort with a single defined species of yeast, to truly modernize already industrial processes like beer production (reviewed in [Lengeler et al., 2020]).

Thus, despite being the original type of fermentation that humans harnessed, spontaneous fermentation in beverage alcohol production largely disappeared by the beginning of the twentieth

century with the advent of controlled pure culture brewing. However, spontaneous fermentation lived on in enclaves all over the world where traditional beverages continued to be produced, even in the face of the global dominance of beverages made by modern methods. More recently, the craft beer revolution, which took root in the US but spread globally, has rekindled interest in these traditional styles of beer produced by spontaneous fermentation (Tonsmeire, 2014). As such, they have made a comeback, and new styles based on traditional techniques are being developed.

In this chapter, we focus on traditional and modern versions of these beers made by spontaneous fermentation. We first define spontaneous fermentation, and then briefly cover various styles of beer made by this process, touching on some methodological differences in their production. The microbiology and chemical compositions of these beverages are discussed, and their risks are also considered. It should be noted that other types of beverage alcohol, such as “natural” or “low-intervention” wines, are also made by spontaneous fermentation, but they are not covered herein for the sake of simplicity. Regardless, much of the information discussed below is applicable to these other spontaneously fermented beverages.

2. Defining spontaneous fermentation

In general, the beer industry has a problem with definitions because so many variations on the same theme of a fermented malt-based beverage are possible, blurring the lines between styles. For instance, does a product made by fermenting any malted grain count as beer, or does it have to be malted barley? Similarly, can only ale and lager yeasts—both *Saccharomyces* species (Hittinger et al., 2015)—be used to make beer, or can any yeast species capable of ethanolic fermentation be used? On the finer-grained side of things, among popular categories of beer like the India pale ale (IPA), how does one accurately define what constitutes an IPA with the modern and ongoing proliferation of types (e.g., West Coast IPA, East Coast IPA, and New England IPA)? Even the Beer Judge Certification Program style guidelines have difficulty in this instance, grouping traditional English IPAs, American IPAs, and Double IPAs into completely separate categories despite their shared roots (Strong & England, 2021). Spontaneous fermentation shares in this dilemma. How does one define spontaneous fermentation?

Most literally, a spontaneous fermentation would be one that occurs with no human interference, such as a ripe apple dropping from a branch and fermenting via the action of wild yeasts at the base of a tree. That definition fails for spontaneously fermented beer though because humans are preparing the fermentable substrate in the context of beer wort. In that sense then, a spontaneously fermented beer should be one in which a brewer makes wort but has no impact on the introduction of fermenting microbes (Tonsmeire, 2014). Here too is where things get muddled because, by and large, brewers sanitize equipment but do not sterilize it (Smith, 2019), allowing resident microbes to proliferate in wort. Indeed, because breweries themselves are environments inherently enriched for brewing yeasts, an open container of uninoculated wort can easily be colonized by a house strain of yeast for fermentation.

Quibbling aside, in interviews with American craft brewers, such as Logan Rice of Kane Brewing Company and Brandon Boldt of Primitive Beer, who produce beer via spontaneous fermentation, the most agreed upon definition for what constitutes spontaneous fermentation is to define it by omission. In other words, these beers undergo no intentional inoculation directly by the brewer. For instance, for wort that is cooled in a *koelschip* (or coolship; see below), the brewer purposefully pumps wort into a

sanitized, open-air vessel, but then allows microbial colonization to occur via the ambient environment rather than intentional inoculation (Tonsmeire, 2014). After the initial fermentation from these ambient yeasts and bacteria, a brewer may expose the beer to a defined culture of one or more microbes for secondary or ongoing fermentation, but the primary fermentation was spontaneous.

While on the topic of definitions, it should also be noted that the terms “spontaneous” and “sour” are often and incorrectly used interchangeably with regard to beer. Although many spontaneously fermented beers contain lactic acid, not all of them are perceived as sour or tart because the concentration is below the sensory threshold of the human palate. They may go through an early phase of fermentation in which the culture is dominated by lactic acid bacteria (LAB) but without perceivable amounts of lactic acid produced because another suite of microbes then takes over the fermentation. Alternatively, the LAB may create enough lactic acid to make the beer taste sour 24 h into the fermentation but which is later bio-converted into other compounds by subsequent microbes dominating the culture, yielding nonsour beer 6 months into fermentation. Therefore, much like all sour beers are not made by spontaneous fermentation, not all spontaneously fermented beers are sour. A range of spontaneously fermented beers, both sour and not, are described below.

3. Traditional beer styles produced by spontaneous fermentation

3.1 Lambics and lambic-inspired beers

Most of the styles described today for classifying beers refer to beers using the inoculation of specific strains of yeast (e.g., ales vs. lagers) and, in some cases, bacteria. However, some styles or types of traditional beers, usually strongly linked to the region of their production, are known for the use of spontaneous fermentation in their production, such as Belgian lambic beers. Lambic sour beers were originally produced only in regions around Brussels, in the Senne Valley, and in periods of the year with lower temperatures (between October and March) (Bongaerts et al., 2021; Bossaert et al., 2021). One of the first mentions of the term “Lambic” for beers was observed in 1559, being interpreted as a reference to the village of *Lambeek* (near Brussels) and also to “*alambics*,” facilities in which both spirits and beers were produced (Patterson, 2014; Verachtert & Derdelinckx, 2014). The recipes for making the wort for these beers usually include malted barley (60%–70%), unmalted wheat (30%–40%), and aged hops (to avoid undesired bitterness but maintain the bacteriostatic properties of the hops) (Verachtert & Derdelinckx, 2014).

For the production of wort in traditional lambics, many brewers use the technique of *turbid mashing* or *slime cooking*, in which the temperature of beer wort is slowly increased, allowing each enzyme present in the malt to act in its best temperature range (Bongaerts et al., 2021). In addition, a portion of the wort (only the liquid) is removed from the total amount of wort and heated separately, resulting in the inhibition of the enzymes present in that portion. This is a way of gelatinizing and converting some of the starch present in unmalted wheat into simpler fermentable sugars, while at the same time maintaining a significant portion of maltooligosaccharides (complex carbohydrates) that will only be metabolized in the final stages of fermentation (Bongaerts et al., 2021; Tonsmeire, 2014).

In the lambic production method, after mashing, the beer wort is transferred to open and shallow metal vessels (known as *cooling tuns* or *coolships*) and kept overnight for cooling, at which point microorganisms present in the air microbiota will inoculate the wort and initiate fermentation (Spitaels et al., 2014). These coolships are usually kept in wooden-roofed rooms, where the steam from the hot

wort condenses and falls back into the cooling liquid. Then, after being transferred to wooden casks, fermentation will take place at room or cellar temperature, between 15 and 25°C, where the microbiota present in these barrels will also participate in the fermentation (see Fig. 7.1) (Spitaels et al., 2015; Spitaels et al., 2014). Maturation occurs in the same wooden casks, lasting between 1 and 3 years before bottling. The final result is a sour, cloudy beer with an alcohol content of around 5%, low carbonation, and reduced foam (Patterson, 2014). Some characteristics evolve with aging. Bitterness and lactic acidity soften over time, while complexity and dryness increase. The flavors present in lambics are diverse and are categorized as barnyard, earthy, goaty, horsy, fruity, oak, honey, white wine, and others (Straka & Hleba, 2022; Witrick et al., 2020). Although different sensory characteristics can be found in lambic beers, the balance between them is extremely important, always seeking pleasant aspects for the consumer (Tonsmeire, 2014).

If lambic beers are blended, gueuze or fruit lambic beers are the product. Gueuzes are created from a blend of Lambics with different maturation periods, in which “young” (1-year-old) and “old” (2- to 3-year-old) beers can be used as a base, with bottles being stored as long as 10 years (Bongaerts et al., 2021; Straka & Hleba, 2022; Verachtert & Derdelinckx, 2014). The months with milder temperatures, from autumn to spring, are the ideal period to start lambic production because higher summer temperatures for fermentation can favor bacteria and wild yeasts to ferment and acidify the wort in a very aggressive and undesirable way (Spitaels et al., 2014; Straka & Hleba, 2022; Tonsmeire, 2014). Lambic-producing brewers reuse barrels that give rise to the best beers for decades but discard, reform, or sterilize barrels that originally yielded lower-quality beers. This is one of the methods applied by brewers that results in selective pressure in the factory environment, benefiting the presence of specific microorganisms and, consequently, consistency in the production of sour beers despite their spontaneous fermentation origins (Tonsmeire, 2014).

Although the most famous spontaneously fermented beers are the Belgian lambics, brewers around the world apply their techniques (or those adapted from lambics) in the production of beers with unique characteristics, which, like lambics, also have great geographical appeal and are related to the place of origin. In the United States, brewers have used the knowledge gained in the production of lambics to prepare American Cooship Ales, seeking to create a similar style of sour beers based on the recipes and production methods already used in European spontaneously fermented beers (Bokulich et al., 2012). While American Coolship Ales are not strictly linked to any centenary or traditional roots in their production methods, freedom of creation, implementation of contemporary methods, and new ingredients are key features that American brewers utilize to produce beers that use spontaneous fermentation (Carriglio et al., 2022; Tonsmeire, 2014). The original American beer of this type was produced and marketed in the country only in 2007, demonstrating that investment in production lines for spontaneous fermentation beers is something recent, which may be due to cost, time, losses, and the complexity of establishing a product for a demanding consumer market (Carriglio et al., 2022). Exploring all of the characteristics of the process, seeking predictability, and using techniques that prevent significant losses in production are the keys to success in implementing spontaneously fermented beer production lines.

Another difference from Belgian beers is that American breweries that produce spontaneously fermented beers (normally) also produce other conventional styles with *S. cerevisiae* inoculations, which may favor the presence of *Saccharomyces* in the fermentation environment due to cross-contamination and selective pressure to maintain its presence in the first moments of wort fermentation (Bokulich et al., 2012; Bossaert et al., 2021; Carriglio et al., 2022). There is still no

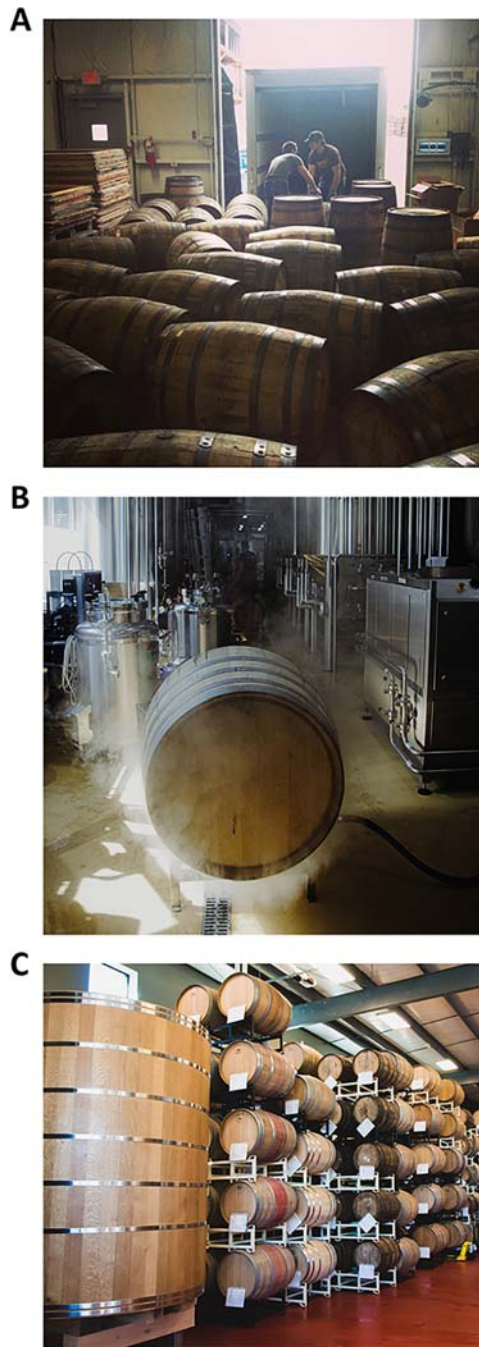


FIGURE 7.1 Barrel aging beer.

(A) Empty barrels being unloaded at a brewery. (B) A barrel at Varvar brewery, Kyiv, Ukraine, waiting to be filled. (C) Racked barrels containing aging beer.

a) Image courtesy of Matt Wisley, Upland Brewing Company. b) Image courtesy of Yuri Gryaznov. c) Image courtesy of Matt Wisley, Upland Brewing Company.

well-established definition for the expected sensory profile of a beer classified as an American Coolship Ale, but the aromas and flavors are generally those found in Belgian lambics, such as those referring to lactic and acetic acids, for example (Carriglio et al., 2022). One of the “arts” that has also been acquired by American brewers from the Belgian lambic tradition is the technique of blending beers from multiple fermenters or batches, aiming for uniformity of the product through the elimination of beer from barrels that have unpleasant aromas or characteristics that differ from the expected product. Although the application of the spontaneous fermentation method for beer production is recent in the United States, excellent beers have been commercially available for more than a decade throughout the country (Tonsmeire, 2014), a result of the positive exploitation of this brewing technique.

3.2 South American chicha

Chicha, one of the oldest drinks in South America, is a beer produced from the spontaneous fermentation of malted cereals such as corn and rice. In the Andes, since pre-Hispanic times, *chicha* has been consumed at celebrations such as social, political, and religious events (Grijalva-Vallejos et al., 2020; Hayashida, 2008; Piló et al., 2018). This is a beverage prepared mainly using corn, but its name is also generically used for other fermented or nonfermented beverages made from different raw materials, which can confuse its classification as beer (Piló et al., 2018).

For the production of *chicha de jora*, yellow maize grain is kept submerged in water for one or more days, then removed from the water, and allowed to germinate for a period of approximately 13 days. After 2 days under the sun for drying (the end of the malting stage), the grains are ground, and the flour is used to produce *chicha*. The flour is initially mixed with cold water, then transferred to hot water to boil for 20 min. The resulting wort from this process is strained and kept in open vessels, where it is fermented by indigenous microorganisms for 2 days until obtaining a yellow, clear, or cloudy beer with low foam formation (Piló et al., 2018). The fermentation process is interrupted, and the final product is obtained when the sweetness is reduced and a marked flavor resulting from spontaneous fermentation by bacteria and yeasts becomes predominant in the beverage (Grijalva-Vallejos et al., 2020). This is just one of the examples of *chicha* production, and the brewing methods may vary according to the region where it is produced and the ingredients used (Hayashida, 2008). The ancestral processes of *chicha* production also involved chewing corn as a source of starch-degrading amylases from human saliva. However, changes in the current processes of preparation and use of ingredients made it possible to eliminate this step (Grijalva-Vallejos et al., 2020; Torres-Guardado et al., 2022).

3.3 African sorghum beers

Sorghum-based beers are among the most traditional and consumed alcoholic beverages in Africa, with a high socioeconomic impact because they are produced and sold mainly by women (Sawadogo-Lingani et al., 2008). Colored varieties of sorghum (e.g., red or brown grains) are preferred for making these beers because consumers have shown a preference for beers with a different color than traditional imported lagers (Lyumugabe et al., 2012). Differences in sorghum varieties also impact the biochemical aspects of beverage production, as they affect the amounts of substrate available and, consequently, the microbiota involved in the fermentation process (Jespersen, 2003). Depending on the geolocation in Africa, different names are used for spontaneously fermented beers, such as *pito* (Ghana

and Nigeria), *dolo* (Burkina Faso and Togo), and *amgba* (Cameroon), among others (Lyumugabe et al., 2012). According to Apiwe Nxusani-Mawela, senior consultant brew master and brewer at Tolokazi Beer in South Africa, spontaneously fermented beers have been brewed for centuries and are part of the country's beer culture. She also points out the importance of traditional beers during apartheid, when "black people were not allowed to have white man's liquor," and these were the brews that sustained many people. In addition, many women started homebrewing during this period to sell to the mineworkers, a fact that influenced the rise of *Shebeen queens*, that is, women that sold homebrewed and home-distilled alcohol in illicit bars known as *shebeens* (Rogerson & Hart, 1986). It should be noted that some of the beers made from sorghum in African countries use the inoculation of starter cultures in their process, either commercially available or obtained from previous fermentations, which is not the object of this chapter on spontaneous fermentation.

Although there are several variations in the types of beers produced by spontaneous fermentation on the African continent, the production process of these beers is mainly based on malting, souring, boiling, mashing, straining, and alcoholic fermentation (Jespersen, 2003), with some decantation steps that can intersect this process (Sawadogo-Lingani et al., 2021). The souring step, before the mashing, occurs through primary spontaneous fermentation, in which the objective is mainly the action of native LAB to acidify the wort, and then, for the alcoholic fermentation, a new spontaneous fermentation takes place (Jespersen, 2003). During the mashing step, local plant extracts can be used as flocculants, clarifying the wort and consequently decreasing the turbidity of the beer (Sawadogo-Lingani et al., 2021). However, the final product is usually an opaque beverage with suspended solids (starch and dextrin residues), low alcohol content (<4.5% ABV), a pH between 3.3 and 4.0, usually fruity and yeasty aromas derived from fermentation, and is considered a refreshing drink (Djameh et al., 2019; Lyumugabe et al., 2012). These sorghum beers are consumed during their active fermentation period, with no concern for postfermentation processes, resulting in a maximum shelf life of 72 h (Lyumugabe et al., 2012).

Traditional beers are still less attractive than imported conventional lagers and ales (which use malted barley in the process), as they are generally produced in an artisanal way with rudimentary equipment and poor hygiene control, resulting in a beverage with organoleptic variations, low levels of ethanol, inconsistencies in product quality, and storage problems (Djameh et al., 2019; Lyumugabe et al., 2012). Traditional homemade beers produced with corn flour and malted millet have already been associated with cases of poisoning in Africa and Asia and can even cause death due to the presence of contamination by microorganisms, their spores, and toxins in the ingredients used for production (Gudo et al., 2018; Nikander et al., 1991). In some cases, even with boiling or the addition of yeast, pathogens can survive, and toxins can withstand the conditions found in beer (Kim et al., 2014). Therefore, minimal controls related to hygiene, storage care, the origin of raw materials, and a well-established production process are of great importance in the production of spontaneously fermented beers that are safe for consumption. A discussion of the risks of spontaneous fermentation is included in a subsequent section of this chapter.

3.4 Burgeoning spontaneous styles and lost styles

In countries like Australia and Brazil, spontaneous fermentation has been explored for the production of beers that seek a niche market that is still underexplored in these regions. Like American Coolship Ales, these budding spontaneously fermented styles begin with inspirations from the lambic tradition

but seek to incorporate local uniqueness. Although Brazil is currently one of the great powers in the production of wild beers with terroir, the manufacture of spontaneously fermented beers is visualized in the domestic market as a possible future in the short term. Indeed, efforts by innovative breweries in this field have sought to implement terroir in beers, creating a unique identity for the drink in these places. In Brazil, for example, there is a long history of traditionally fermented foods and beverages (Lima et al., 2022) that informs the production of modern craft beers to generate this terroir.

In contrast to the blending of old and new traditions above, some historical beer styles that used spontaneous fermentation disappeared altogether with the lager revolution. This is the case of *Jopenbier*, a style of European origin in which the beer was produced from a wort with an initial density above 1.100 g/cm^3 that was dark in color and viscous from a long boiling time, and the subsequent spontaneous fermentation lasted months (Kizik, 2021). This is just one example of a long list of “lost” beer styles, many of which utilized spontaneous fermentation in their production (Kenney, 2021). One benefit of the craft beer revolution is a renewed interest in these styles and, in some cases, a revival of these brewing traditions, with the German style of *Gose* beer being a recent and successful example (Allen, 2018).

4. The spontaneous beer production process

The processes used to make spontaneously fermented beers are as varied as the beer styles themselves. As stated above, many modern types of spontaneous beer are based on the traditional Belgian lambic production process, often employing turbid mashing, coolships, and wooden vessels for aging (Bongaerts et al., 2021; Tonsmeire, 2014). However, as also highlighted in the previous section, local styles such as chicha are made quite distinctly from lambics. Thus, we do not dwell on the generalities of spontaneously fermented beer production here. Instead, we highlight below one aspect of the production process that is optional but highly studied and can impact the final product.

4.1 Hopping: An optional step during production of traditional spontaneously fermented beers

An elective step during the production of traditional spontaneous beers is hopping. The acidity present in many beers produced by spontaneous fermentation can often increase the perception of bitterness resulting from the use of hops, and for this reason, the use of hops is questioned in spontaneously fermented beers. Adding hops in the final stages of boiling (i.e., *dry hopping* to provide flavor and/or aroma to the beer [Klimczak et al., 2023]) can also be avoided, as the volatile aromatic compounds will dissipate during the long period of fermentation and maturation (Tonsmeire, 2014).

In the process of producing lambic beers, the hops are added during boiling or while the kettle begins to be filled, and thus prior to the start of boiling. In this type of beer, brewers use aged hops, seeking to maintain the acidic character of these beers because the bitterness originating from fresh hops can negatively interfere with this aspect related to taste. The decrease in bitterness in aged hops is due to the oxidation of the alpha-acids (compounds responsible for bitterness), increasing the concentration of humulinones rather than humulones, which does not affect the inhibitory and bacteriostatic power of the hops against harmful bacteria (Bongaerts et al., 2021). The oxidation of hop resins can result in the development of unpleasant aromas and off-flavors characterized by cheese-like

descriptors. However, the long boiling period (4–5 h) to which the wort is subjected eliminates this odor (De Keersmaecker, 1996). Although there is a desire to apply the traditional method of lambic production using aged hops by brewers worldwide, in practice, it is observed that the use of fresh hops in recipes is not harmful to the manufacture of lambic-style beers (Verachtert & Derdelinckx, 2014).

Other traditional styles of spontaneously fermented beer production do not include hops in their workflow, as observed in the case of traditional African beers (Lyumugabe et al., 2012). In addition, such use of hops may even be prohibited by law to maintain the name “traditional beer,” as is the case in South Africa, which describes the prohibition under the Liquor Products Amendment Act (2021). It is likewise not common to use hops in the production of traditional chicha, demonstrating that the flavors and their appearance are the exclusive result of the methods of production, the preparation of the wort, and the microorganisms involved in the spontaneous fermentation (Hayashida, 2008).

5. Microbiology of spontaneously fermented beer

Using lambic beers as a classic example, the ingrained vision for how spontaneous fermentation begins is that microbes in the air settle into the beer wort being cooled overnight in coolships to naturally inoculate the substrate (Straka & Hleba, 2022). It was thought that this Old World style could only be made in and around Brussels due to the unique and desirable makeup of these airborne microbes in the fertile Senne Valley (Tonsmeire, 2014). However, recent research has debunked this viewpoint. First, the modern environment in Belgium is quite different from that during and before the 19th century, and human impacts are known to change native microflora (Cavicchioli et al., 2019), yet lambic beers continue to successfully be produced. Second, spontaneously fermented beers made in the style of lambics have successfully been produced in other geographies, including across the North American continent (Tonsmeire, 2014). Indeed, it is now understood that the majority of the microbes that ferment lambics come from the wood structure and wooden casks of the breweries themselves (Straka & Hleba, 2022). However, beer can spontaneously ferment when exposed to the ambient environment away from manmade structures, and insect vectors like fruit flies may play a role in transporting microbes from, for instance, orchards to cooling wort (Christiaens et al., 2014).

Regardless of the overall contributions of the various sources of spontaneous inoculation, it is clear that the spontaneous fermentation microbiome is diverse, dynamic, and complex (Bokulich & Bamforth, 2013, 2017; Straka & Hleba, 2022). Hundreds of individual bacterial and yeast species have been identified during spontaneous fermentation and in the resultant beer, differences which have been attributed to variations in geography, wort recipe, and individual brewers (Bokulich & Bamforth, 2013, 2017; Piraine et al., 2021; Spitaels et al., 2014; Straka & Hleba, 2022). However, there tends to be less variation when one considers the microbes from the genus level, and even different styles of spontaneously fermented beer undergo similar patterns of microbial community development. Thus, spontaneous fermentation can generally be separated into four phases, each of which is dominated by a different consortium of microbes. Below, we focus on the predominant yeasts and bacteria found in each phase. However, modern analytical methods indicate that bacteriophages/viruses, archaea, and other fungi are also present (Piraine et al., 2021). Their contributions to the fermentation process and ultimate product are largely uncharacterized, so they are not considered here.

5.1 Phase 1—Enterobacteria

Over the first 1–4 weeks of spontaneous fermentation, facultative anaerobic bacteria in the Enterobacteriaceae family dominate the microbial community in the young beer (De Roos et al., 2018; De Roos et al., 2018b; Martens et al., 1991). The genera *Citrobacter*, *Enterobacter*, *Escherichia*, *Hafnia*, *Klebsiella*, *Proteus*, and *Serratia* are routinely found, among others. This is also the phase at which oxidative yeasts, such as in the *Candida*, *Cryptococcus*, *Debaryomyces*, *Hansenula*, *Pichia*, and *Rhodotorula* genera (Bokulich et al., 2012), can be found. Acetic acid bacteria (AAB), belonging to *Acetobacter* and *Gluconobacter* genera, such as in kombucha fermentation (Yang et al., 2022), can also briefly thrive in this early step.

All of these organisms collectively metabolize the monosaccharides in the wort using mixed-acid fermentation pathways to create organic acids (e.g., acetic, formic, lactic, and succinic acids) (Van Vuuren & Priest, 2003). The Enterobacteria phase generally ends as the beer acidifies to $\text{pH} \leq 4.5$ and the ethanol concentration increases to $\sim 2\%$ ABV, environmental stresses that decrease the viability of the microbes that predominated this phase of spontaneous fermentation and allow a new wave of microbes to flourish (reviewed in [Straka & Hleba, 2022]). However, next-generation sequencing of DNA sampled from spontaneous fermentations shows evidence of Enterobacteria and oxidative yeasts much later into the fermentation and/or in the final beer (Piraine et al., 2021). Regardless, it should be noted that the sequencing technology is sensitive enough to detect the genomes of dead organisms or spores and does not speak to the potential viability of any of these organisms in the samples.

5.2 Phase 2—Main fermentation

Following the initial phase of spontaneous fermentation, the main fermentation or ethanol production phase begins. Unlike during the first few weeks, yeasts—specifically in the *Saccharomyces* genus—dominate this stage of the process. These include the common ale yeast *Saccharomyces cerevisiae*, as well as other *Saccharomyces sensu stricto* species such as *S. bayanus*, *S. kudriavzevii*, and *S. pastorianus* (Bongaerts et al., 2021; Straka & Hleba, 2022). Other yeasts found in the main phase include strains in the *Hanseniaspora* and *Kazachstania* genera (Bongaerts et al., 2021; De Roos et al., 2018). However, in a survey of mixed cultures used to make sour beers and recovered from the beers themselves, which included multiple samples from spontaneous fermentations, *Saccharomyces* yeasts were completely absent in some cases (Piraine et al., 2021). The reasons for this are unknown, especially given the sensitivity of the deep sequencing techniques used as described above, but it has been found that nontraditional (i.e., non*Saccharomyces*) yeasts can both fully attenuate wort to produce high-quality beer and act as a primary souring agent if the case of heterofermentative species like *Lachancea* spp. and *Schizosaccharomyces japonicus* (Barry et al., 2018; Osburn et al., 2018; Peepall et al., 2019; Piraine et al., 2022).

Regardless of the identity of the yeast species present during the main fermentation phase, this is when the bulk of the ethanol is produced and nutrients become depleted, accompanied by a considerable drop in the specific gravity of the beer over several weeks (Bokulich et al., 2012; Bongaerts et al., 2021; Straka & Hleba, 2022). At the same time, the yeasts are creating secondary metabolites such as esters, higher alcohols, and organic acids that impact the organoleptic profile of the beer (De Roos et al., 2018b; Hirst & Richter, 2016; Pires et al., 2014). There is also evidence of metabolism or

bioconversion of organic acids produced during the Enterobacteria Phase by yeasts during the Main Phase of fermentation (De Roos et al., 2018).

5.3 Phase 3—Acidification

After the two comparably rapid preceding phases, the Acidification Phase of a spontaneous fermentation can last up to 10 months (De Roos & De Vuyst, 2019; Straka & Hleba, 2022). During this time, *Lactobacillus* and *Pediococcus* (especially *P. damnosus*) genera dominate the culture in the beer, where they produce organic acids, chiefly lactic and acetic acid, dropping the pH below 4 and often below 3.5 (Bokulich et al., 2012; Bongaerts et al., 2021; Spitaels et al., 2015; Vanoevelen et al., 1977). The relative ratios of the Lactobacilli and Pediococci are affected by the type of beer being produced, with the use of aged hops—as is traditional in lambics—more strongly inhibiting *Lactobacillus* spp. than *Pediococcus* (Bokulich et al., 2012; Vriesekoop et al., 2012). The Acidification Phase also sees the activity of AAB that were last active during the Enterobacteria Phase, again creating acetic acid. This occurs at the liquid-air interface in casks due to the slow ingress of oxygen through the wood and/or when oxygenation occurs during the mixing of one or more beers (Bongaerts et al., 2021; De Roos & De Vuyst, 2019; De Roos et al., 2018; De Roos et al., 2018b). Acetic acid is found in all spontaneously fermented beer and can be a desirable part of the flavor profile above its sensory threshold (Macías et al., 2012; Spitaels et al., 2015; Zhang et al., 2012). However, it is also harsher to the palate than comparable concentrations of lactic acid, so care is taken to avoid high concentrations of acetic acid.

The Acidification Phase is also when *Saccharomyces* yeasts are replaced by *Brettanomyces* spp. as the predominant eukaryotes in the fermentation (De Roos & De Vuyst, 2019), though *Brettanomyces* yeasts also play a major role in the Maturation Phase (see below). In fact, some researchers consider the Acidification and Maturation Phases to be a single step in the process (Bokulich et al., 2012; De Roos & De Vuyst, 2019; Spitaels et al., 2015; Spitaels et al., 2014; Straka & Hleba, 2022). Regardless, other yeasts have also been found during this phase, though their effects on the beer at this step are uncharacterized (Vanoevelen et al., 1977). We cover the impacts of *Brettanomyces* in greater detail below.

5.4 Phase 4—Maturation

The last phase in spontaneous fermentation is often the longest, being an extended period of beer maturation lasting many months to years (De Roos et al., 2018; Straka & Hleba, 2022; Tonsmeire, 2014). At this point, the beer is nutrient-depleted, acidic, and high in alcohol, so organisms capable of surviving these stressors and metabolizing the remaining long-chain carbon sources dominate the spontaneous fermentation microbiome. However, the Maturation Phase may also include blending of mature and younger beers, such as in the creation of Gueze (Verachtert & Derdelinckx, 2014), or bottle conditioning to carbonate the beer (Rogers et al., 2016). Both processes can introduce carbon sources and/or oxygen to stimulate microbial growth. As above, oxygen ingress can also occur through the porous surface of wooden casks used for extended maturation. This oxygen at the liquid-air interface can stimulate the formation of a pellicle or biofilm of organisms such as oxidative yeasts (Fig. 7.2) (Bongaerts et al., 2021; De Roos & De Vuyst, 2019; De Roos et al., 2018; Vanoevelen et al., 1977).



FIGURE 7.2 Pellicles on beer.

(A) A wild yeast pellicle growing on beer in a miniature fermenter. (B) A pellicle created by *Pichia membranifaciens* in a contaminated tank. (C) A pellicle that formed on a Solera barrel-aged sour beer refermenting on raspberries.

a) Image courtesy of Wild Pitch Yeast, LLC. b) Image courtesy of Steve Llewellyn, Function Brewing. c) Image courtesy of Bill Riley.

Anecdotal reports suggest that the pellicle protects the beer from off-flavor production in the presence of oxygen at this late stage of fermentation, but evidence to corroborate this hypothesis is lacking.

In any event, as stated above, *Brettanomyces* yeasts are important during maturation. Strains of *B. bruxellensis* are common, though *B. anomalus*, *B. custersianus*, and *B. naardenensis* are also found

(Bongaerts et al., 2021; De Roos & De Vuyst, 2019; De Roos et al., 2018; Piraine et al., 2021; Vanoevelen et al., 1977). Even in fermentations dominated by *B. bruxellensis*, recent data indicate that multiple individual strains or subspecies may be involved in a single beer (Piraine et al., 2021), and various strains of *B. bruxellensis* that yield distinct organoleptic profiles are commercially available. Thus, just like the many categories and uses of individual strains of *S. cerevisiae* ale yeast, *B. bruxellensis* should not be considered a single monolithic ingredient in terms of beer production.

That said, most *Brettanomyces* strains are known for producing “Brett character” in beer, which is described as a mousy taint with greater or lesser degrees of barnyard, horsy, leathery, medicinal, smoky, and/or clove-like notes (Oelofse et al., 2009; Pires et al., 2014; Snowdon et al., 2006). The organoleptic compounds are produced as secondary metabolites from the consumption of long-chain malt sugars (e.g., maltotetraose and maltopentose), alcohols, and organic acid precursors (Crauwels et al., 2015; Gamero et al., 2014; Kheir et al., 2013; Kumara & Verachtert, 1991; Oelofse et al., 2009; Steensels & Verstrepen, 2014). In contrast, *Brettanomyces* yeasts also create fruity and floral organoleptic compounds via bioconversion of compounds from hops, wort carbohydrates, and even the exopolysaccharides produced by *Pediococcus* (Daenen, 2008; De Roos & De Vuyst, 2019; De Roos et al., 2018; Straka & Hleba, 2022). These *Pediococcus*-derived exopolysaccharides increase the viscosity of the beer in the form of a “ropey” slime, and beers containing these compounds are often described as “sick” (Tonsmeire, 2014). Not all spontaneously fermented beers experience this “sickness,” but some producers consider it a desirable phase of the process due to the pleasant organoleptic compounds that *Brettanomyces* and other microbes produce from the exopolysaccharides (Dimopoulou & Dols-Lafargue, 2021).

5.5 Other microbiological considerations

The above description of the microbiology of spontaneous fermentation is skewed toward beers created in the traditional Belgian lambic style or at least influenced by those methods. Due to their highly attenuated nature, high alcohol content, and low pH, such beers tend to be very shelf stable when packaged and stored correctly, with the stability of gueuze being up to 10 years (Spitaels et al., 2015). However, not all spontaneously fermented beers are an homage to lambic, nor do they all undergo extended aging, such as *chicha* (Grijalva-Vallejos et al., 2020; Hayashida, 2008; Piló et al., 2018). The phases of such fermentations and the predominant microbes found therein consequently vary, but much less research has focused on the microbiology of these beverages to date. However, with the decreasing price of deep sequencing needed to perform microbiomics analyses and the growing global interest in regionally spontaneously fermented beers, we believe that this gap will soon be narrowed.

Indeed, next-generation sequencing was previously used to characterize the microbiomes of malted barley samples and wort inoculated with these grain samples during sour mashing (Bochman & Young, 2016). Sour mashing is a process by which beer wort is inoculated with grain, incubated at an optimal temperature for grain-resident LAB to produce lactic acid, and then boiled to sterilize it for subsequent ethanolic fermentation by yeast (Springer, 2015; Tonsmeire, 2014). As such, beers produced by sour mashing undergo a spontaneous fermentation that consists only of an Acidification Phase, followed by a controlled Main Fermentation Phase initiated upon controlled yeast inoculation. Under the conditions used by the Blue Owl Brewing for sour mashing (Bochman & Young, 2016), the LAB *Weissella cibaria* dominates the mash microbiome, regardless of grain age (2015 vs. 2016 harvest) and place of

origin (malt from North America, South America, and Europe tested). This contrasts with lambic acidification dominated by *Lactobacillus* and *Pediococcus* spp. (Tyakht et al., 2021), though the physiochemical output is the same: lactic acid. *W. cibaria* was first described in 2002, originating from Thai fermented foods (originally named *W. kimchi*) and European sourdough cultures (Fusco et al., 2015) but until recently had never been associated with beer.

6. Chemical features of spontaneously fermented beers

The chemical profiles of spontaneously fermented beers are as varied as the beers themselves, and they change over the four-phase fermentation process described above. Therefore, producing a wholistic picture of this space is impossible. Instead, we here focus on the finished beers and largely cover commonalities found among the various examples of this style. This includes aroma and flavor compounds, many of which were mentioned above in the context of the organism that produce them.

Chief among the chemical constituents of spontaneously fermented beers are lactic and acetic acid. These are largely derived from the actions of LAB and AAB, respectively. However, yeasts capable of heterolactic fermentation can also produce high levels of lactic acid (Osburn et al., 2018), and *Brettanomyces* can produce acetic acid during the Maturation Phase of spontaneous fermentation in the presence of oxygen. These organic acids yield the tart flavor of sour beers (Li & Liu, 2015), but they can also impact the overall production of spontaneous beers. Indeed, organic acid stress is detrimental to yeast fermentation (Narendranath et al., 2001a, 2001b; Thomas et al., 2002), and high levels of lactic acid can inhibit refermentation during the bottle conditioning traditionally used to carbonate packaged beer (Rogers et al., 2016). Likewise, ethanol is a common and desired component of spontaneously fermented beers, but it is also a microbial stressor. This is true even for *S. cerevisiae*, one of the most ethanol-tolerant organisms known (Casey & Ingledew, 1986). Ethanol nevertheless impacts its growth rate (Canetta et al., 2006), viability (Pascual et al., 1988), and fermentative capacity (Fernandes et al., 1997), which again can affect the rate and success of bottle conditioning (Rogers et al., 2016).

The other two large common categories that the chemical components of spontaneous beers can be divided into are aromatic and flavor compounds. Beer aroma is primarily dependent on volatile compounds from the grains and hops used in the beer, as well as from microbial metabolism (Humia et al., 2019). These volatiles are mainly alcohols, aldehydes, carbonyl compounds, esters, and terpenes. Of these, aromatic esters in beer have been extensively studied for decades. These compounds are synthesized by yeasts as secondary metabolites, perhaps as a means of attracting insects for yeast dispersal (Christiaens et al., 2014). Different yeast species and individual strains of a single species are known to produce different spectrums of esters (Holt et al., 2018; van Rijswijk et al., 2017; Verstrepen et al., 2003). The aromatic esters commonly found in spontaneously fermented beers include ethyl acetate (buttery aroma), ethyl caproate (waxy and fruity), ethyl caprylate (sour apple), isoamyl acetate (banana), isobutyl acetate (fruity), phenylethyl acetate (rose and honey), and ethyl octanoate (fruity, honey, and flowery aroma) (Dack et al., 2017; Kobayashi et al., 2008; Renger et al., 1992; Verstrepen et al., 2003). Much like the organic acids, each ester has a sensory threshold concentration below which it cannot be detected. However, research shows that subthreshold concentrations of these compounds can also impact the organoleptic perception of the beer due to synergistic effects between various esters (reviewed in [Humia et al., 2019]). The malts used in the fermentation can also affect

aromatic esters due to Maillard reaction products found in darker malts (Dack et al., 2017). In this case, the use of dark malts tends to decrease ester levels.

The other aromatic compounds can also be produced by the action of yeasts during fermentation, and their biosynthetic pathways are often linked. In other words, increases or decreases in ester production can impact higher alcohol levels and vice versa. For instance, mixed cultures of *S. cerevisiae* and *Torulaspora delbrueckii* have been shown to generate higher isoamyl acetate levels at the expense of lower levels of isoamyl alcohol during fermentation compared with either yeast fermenting wort alone (Toh et al., 2018). This is particularly germane to spontaneous fermentations, which always involve a mixed population of microbes, and indicates caution when considering published work that focuses on monocultures.

As smell and taste are intimately linked, many aromatic compounds also impact beer flavor. For instance, the fruity esters described above can also lend a fruity note to the flavor profile (Dack et al., 2017; Kobayashi et al., 2008; Renger et al., 1992; Verstrepen et al., 2003). Among flavor compounds in beer though, the carbonyl diacetyl is among the most heavily researched. Its levels are generally low— ≤ 0.1 ppm, with a perception threshold of 0.05–0.1 ppm—in mature beer because while it has a toffee-like flavor at low concentration, higher concentrations yield a buttery stale milk aroma and flavor (Humia et al., 2019; Rosca et al., 2016). That said, diacetyl has been found to be absent in lambic beers (De Roos et al., 2018), and its contribution to spontaneously fermented beers overall is less well characterized than for clean beers.

One class of flavorant that is common to both clean and spontaneous beers, though, is phenolics. These compounds can add a bitter or astringent flavor that can help to balance the sweet malt base of a beer at low concentrations (in addition to hop bitterness) or be off putting at high concentrations (Bettenhausen et al., 2018; Humia et al., 2019; Moura-Nunes et al., 2016). Phenolics also have antioxidant capacity, which can be related to the stability of the finished beer and may even have beneficial health effects (de Gaetano et al., 2016). The most commonly reported phenolic compounds in beers are amino phenolics, flavonoids, phenolic acids, proanthocyanidins, and tannins, which can be derived from malts, hops, and even wooden casks and barrels (Montanari et al., 1999; Zhao et al., 2010).

Finally, when discussing the organoleptic properties of spontaneously fermented beers, it would be remiss of us to not focus on the effects of *Brettanomyces* spp. As mentioned above, these yeasts are often, though not always, found in spontaneous fermentations and contribute a common “Brett character” that can add to the complexity of well-made spontaneous beers or overwhelm the aroma and flavor profiles with barnyard-like and other unpleasant notes (Oelofse et al., 2009; Pires et al., 2014; Snowdon et al., 2006). *Brettanomyces* spp. can generate esters like ethyl acetate (pineapple-like aroma at low concentration, solvent-like at high concentration) and phenols (generally related to “Brett character”), as well as be a source of tetrahydropyridines (De Roos & De Vuyst, 2019; De Roos et al., 2018; Snowdon et al., 2006). The latter, which can also be produced by LAB, are a class of flavor compounds that lend an unpleasant and lingering “breakfast cereal” note to spontaneously fermented beers that lack sufficient maturation or that experienced unintended oxygen exposure during packaging (Moulis et al., 2023; Steensels, 2015; Tonsmeire, 2014). Thus, while *Brettanomyces* yeasts are often an important player in the production of spontaneously fermented beers, they can also be a spoilage organism if a particularly aggressive strain is involved.

7. Risks and best practices to conduct spontaneous fermentation for beer production

Having covered the microbiology and chemical constituents of spontaneously fermented beer that one may encounter commercially, it is worthwhile also noting the potential risks associated with spontaneous fermentation from these same points of view. This is critical because due to the high alcohol and antimicrobial nature of most beer styles, it is often stated that no pathogens have ever been found in beer. This bit of folk knowledge is quite wrong, and one only needs to go as far as the common ale yeast *S. cerevisiae* to find a culprit. Three decades of research shows that *S. cerevisiae* can cause local and systemic infections (Fadhel et al., 2018; Munoz et al., 2005; Perez-Torrado & Querol, 2015; Ramos et al., 2023; Ventoulis et al., 2020). Such infections are especially prevalent in immunocompromised patients, but even this situation is globally more common due to severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) causing coronavirus disease 2019 (COVID-19) and associated cases of *S. cerevisiae* coinfection (Ramos et al., 2023; Ventoulis et al., 2020). Regardless, such infections have remained rare, hopefully due to sick individuals wisely avoiding beer consumption.

Other microbes more commonly associated with pathogenesis can be a danger to the brewers rather than the consumer. This is because it is not uncommon for brewers to sample an ongoing fermentation to gauge flavor development. If this occurs during the early Enterobacteria Phase of the fermentation, then exposure to enterics such as *E. coli* can cause intestinal illness (Martens et al., 1991; Spitaels et al., 2014). Some producers of spontaneously fermented beers preacidify their wort to pH 4.5 to shorten or eliminate the Enterobacteria Phase because this low pH inhibits the growth of enterics (De Roos et al., 2018; Spitaels et al., 2015; Tonsmeire, 2014).

Such controlled wort acidification may also inhibit the growth of *Clostridium botulinum*, the causative agent of botulism (Rawson et al., 2023). *C. botulinum* produces toxins that are extremely potent at low concentrations, causing paralysis and respiratory failure. Furthermore, it is a spore-forming organism, and brewing steps such as boiling that destroy microbial cells are insufficient to kill *C. botulinum* spores. Although there are few if any known cases of botulism poisoning from commercially made beer, this is a valid concern for beers produced by amateurs. For instance, there are reports of botulism outbreaks in US correctional facilities where incarcerated individuals brewed and consumed illicit “hooch” in their prison cells (see [Marlow et al., 2021] and references therein).

7.1 Biogenic amines (BAs)

BAs are low-molecular-weight nitrogen compounds, usually formed via microbial decarboxylation of free amino acids (ten Brink et al., 1990), and the Gram-positive LAB found in spontaneously fermented beers are a known source (Nalazek-Rudnicka et al., 2021). Common BAs found in beer include agmatine, β -phenylethylamine, cadaverine, histamine, putrescine, spermine, spermidine, and tyramine (Izquierdo-Pulido et al., 1996a; Izquierdo-Pulido et al., 1996b; Nalazek-Rudnicka et al., 2021). Human monoamine oxidases and diamino oxidases help to metabolize these compounds, but high levels can be toxic. Similarly, people taking monoamine oxidase inhibitors (MAOIs) as antidepressants become increasingly susceptible to the effects of BAs at lower concentrations (Tailor et al., 1994). Thus, over 3 decades of research exists concerning the safety of BAs found in food and drink. Much of this work focuses on spontaneously fermented beers due to the use of BA-producing microbes in their

production, while the remainder covers the effects of beer contamination by spoilage microbes that likewise produce BAs.

Early reports indicate that spontaneously fermented beers contain the highest levels of the BAs histamine and tyramine compared with more widely consumed lager-style beers ([Izquierdo-Pulido et al., 1996c](#)). These levels were generally ≥ 4 -fold higher than in “clean” beers, though it is difficult to place an exact toxicity value on each BA because many factors affect their impacts on humans ([Izquierdo-Pulido et al., 1996c](#); [ten Brink et al., 1990](#)). Another study of 297 Belgian beer similarly found that samples from spontaneously fermented beers tended to have the highest levels of histamine and tyramine, as well as cadaverine ([Loret et al., 2005](#)).

At this point, it is clear that BAs represent a real health risk, but their levels in spontaneously fermented beers—at least those types that have been studied to date—are only a concern to consumers who use MAOIs. We believe that as laws for nutrition labeling on beers go into effect and more craft brewers establish quality control/quality assurance programs, analyses of BA levels in beers may become as standard as monitoring for shelf stability and microbial contamination. Beers produced by spontaneous fermentation should especially undergo such measurements of BAs.

7.2 Repitching and cross-contamination

Repitching in a brewery is simply the harvesting of a yeast slurry or mixed culture from one beer and using it to inoculate another ([Kalayu, 2019](#)). Based on our definition of spontaneous fermentation above, repitching has no place in the production of spontaneously fermented beers. However, brewers who make a variety of spontaneously and nonspontaneously fermented beers may decide to repitch a mixed culture from a particularly excellent spontaneous beer in an effort to emulate its qualities in a more controlled manner. An obvious risk to this process is that the mixed culture can become contaminated by spoilage organisms or cross-contaminated with other brewery-resident yeasts (e.g., the house ale strain), but this holds true for any manipulation of cultures in the brewery. If a brewer can make a variety of clean beers without cross contaminating their Belgian ale strains and English ale strains, for instance, then they can likewise use rigorous sanitation and the skills of their craft to properly handle mixed cultures.

The perhaps less obvious risk to repitching a spontaneous culture is that it is a dynamic community of microbes that changes during the course of the many phases of a spontaneous fermentation. The organisms responsible for the beneficial attributes of the early stages of spontaneous fermentation may no longer be viable in the mixed culture at the end of fermentation and thus not available to impact the organoleptic qualities of the beer into which the culture was repitched. For those microbes that are still alive, their relative proportions in the mixed culture also only represent the final stage of fermentation, and whatever balance of microbial metabolism yielded the original beer is now no longer in play. Beer is also a highly selective growth medium for microbes ([Bokulich et al., 2015](#)). Therefore, even a fully active, balanced mixed culture will change as it is exposed to wort in subsequent fermentations. This is not to say that mixed cultures cannot be repitched to make excellent beer. We simply highlight that the process comes with risks against which the benefits of the process need to be weighed.

8. Conclusions

From the original beer style to near extinction to their 21st century resurgence, spontaneously fermented beers have a long history and evolving place in the beer pantheon (Rogers, 2014; Tonsmeire, 2014). Diverse human cultures and brewing traditions have led to a multitude of specific types of spontaneous beers fermented by diverse microbial cultures of yeasts and bacteria. Although some of these beers, such as the Belgian lambics, have been intensively studied (Bongaerts et al., 2021; Carriglio et al., 2022; De Roos & De Vuyst, 2019; De Roos et al., 2018; De Roos et al., 2018b; Kumara & Verachtert, 1991; Martens et al., 1991; Vanoevelen et al., 1977; Witrick et al., 2020), most have only recently begun to be investigated or still await their introduction to research laboratories. Our hope is that with a renewed interest in spontaneous fermentation, the economic viability of this sector of the beer market (Philemon, 2023), and decreasing costs of things like deep sequencing used to profile the mixed cultures that underpin spontaneous fermentation, we will see a proliferation of research in this area in the near future. Questions still remain concerning the variability between the mixed cultures that ferment different types of spontaneous beers, how these cultures change over the course of the fermentation, how they respond to different aspects of the production processes used, and how these changes ultimately impact the sensory profile of the finished beers, among many others. Such studies will be informed by investigations of defined mixed cultures and likely even by the big data, machine learning, and artificial intelligence revolutions that are occurring in science. We envision two outcomes of such efforts. The first will enable better process control, even in the face of the uncertainty of spontaneous fermentation, allowing brewers to repeatably and reliably create and market spontaneous beers. Indeed, such efforts are what drove the industry to the pure culture brewing dominance that we see globally today. However, a second scenario will likely involve an amalgamation of both spontaneous and pure culture brewing, wherein the field has learned enough about the microbes, their interactions, and their organoleptic outputs in spontaneous beers to enable breweries to create bespoke mixed cultures for defined outcomes. In this case, a brewer could choose from a bank of bacterial and yeast strains with a known goal in mind (e.g., a highly attenuated, mildly tart beer with floral aromatics and tropical fruit flavors) and design a defined mixed culture to reach that end goal.

Acknowledgments

We thank Brandon Boldt, Shannon and Kris Brown, Jay Goodwin, Jared Miller, Apiwe Nxusani-Mawela, Logan Rice, Diego Rzatki, and Jeff Young for help researching this topic and for providing stimulating conversations about spontaneously fermented beer. Similar thanks go out to the moderators and contributors of the online Milk the Funk community for being an accessible depot for real-world data and experience concerning spontaneous fermentation. This work was supported by funds from Indiana University and crowdfunding campaigns on [Experiment.com](#) (DOI: 10.18258/7099 and 10.18258/13495). MLB is the cofounder of Wild Pitch Yeast, LLC.

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Aroma of craft beer

8

Nils Retthberg

Versuchs- und Lehranstalt für Brauerei in Berlin (VLB) e.V., Research Institute for Raw Materials and Beverage Analysis, Berlin, Germany

1. Introduction: Aroma of craft beer

There exist countless definitions and interpretations on the nature of craft brewing and craft beer. According to the Brewers Association, a key feature of craft beer is innovation, which means that craft brewers interpret historic beer styles with unique twists and develop new styles beyond mainstream beer. Craft beer is typically made from traditional ingredients such as malted barley and hops, while local and sometimes nontraditional ingredients are added for distinctiveness (Baiano, 2021; Jaeger et al., 2020). The styles of beer brewed by craft brewers appear countless: The 2022 Brewers Association Beer Style Guideline lists more than 150 beer styles that broadly divide Ales, Lagers, and so-called all origin hybrid/mixed Lagers. The Ale category accounts for close to 100 individual styles ranging from British origin Ale styles, over North American origin Ale styles, to German and Belgian origin Ale styles. The Ale category includes the popular American Pale Ale (PA) and the American Indian Pale Ale (IPA), as well as associated styles such as the American Double IPA, the Hazy (New England) IPA, and the Session IPA. The modern IPA is a hop forward beer style within the broader category of PA (Oliver & Colicchio, 2011) and has clearly become the signature style of craft brewing worldwide (Cantwell, 2018). Regarding the BA definition/style guidelines and the commercially available beers in the above-named style categories, there exists a considerable overlap between their sensorial and chemical characteristics (Table 8.1), most importantly all the above-listed styles are strongly influenced by hops (Table 8.1). Within this chapter, the term hoppy Ales is used when general characteristics of these styles are discussed.

The success of craft brewing around the world and the success of hoppy Ales are closely connected. Compared with pale Lager beer, which is the signature style of industrial beer productions, there exist significant sensorial and chemical differences. Pale Lager beers typically feature a thirst quenching, crisp character with moderate to (very) low bitterness. On the one hand, brewers acknowledge that brewing clean Lager beers is challenging, because any small defect will affect aroma. On the contrary, within this style, differentiation between products is very difficult to achieve and to some consumers brands are interchangeable. In contrast, the flavor of hoppy Ales can vary greatly from brand to brand and from brewery to brewery. Hoppy Ales are simply much more complex and the variation in recipes offers countless opportunities to modulate aroma and flavor. In hoppy Ales, for example, varying

Table 8.1 Comparison of chemical and sensorial characteristics of PA, IPA, and session IPA based on the 2022 Brewers Association Beer Style Guideline.

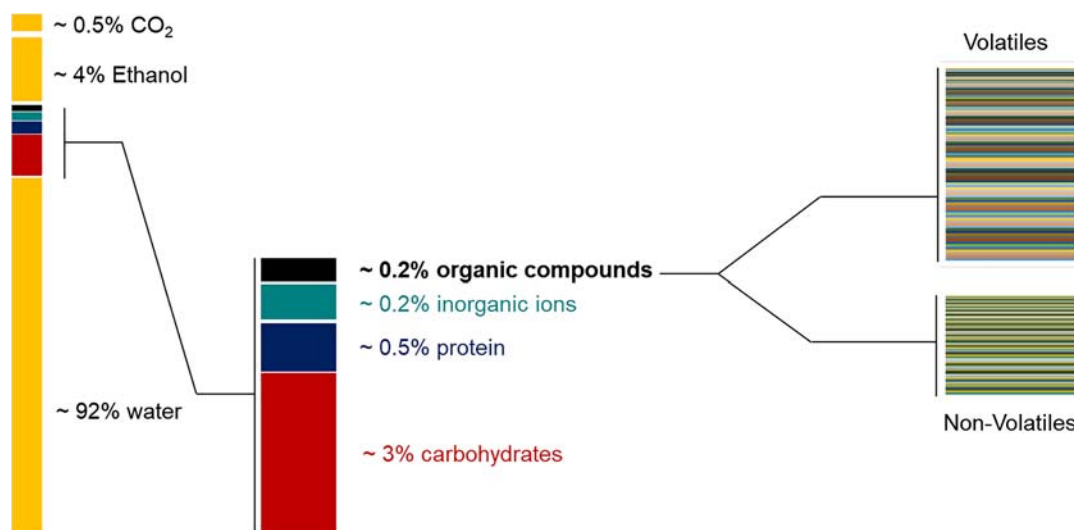
	PA	IPA	Session IPA
Original gravity	11–12.4°Plato	14.7–17.1°Plato	2.1–12.9°Plato
Alcohol by volume	4.40%–5.40%	6.30%–7.50%	0.50%–5.00%
Color	Straw to light amber, 8–14 EBC	Pale to copper, 8–24 EBC	Straw to copper 6–24 EBC
Malt character	Low caramel malt aroma/character. DMS should not be present	Medium-low to medium intensity malt attributes are present in aroma and flavor. DMS should not be present	Low to medium maltiness should be present in aroma and flavor. DMS should not be present.
Body	Medium, final gravity 2.1–3.6°Plato	Medium-low to medium, final gravity 2.5–4.1°Plato	Low to medium, final gravity 1.3–4.6°Plato
Fermentation characteristics	Fruity esters may be low to high. Diacetyl should not be present.	Fruity esters are low to high. Diacetyl should not be present.	Fruity esters are low to medium. Diacetyl should not be present.
Hop character	High, exhibiting a wide range of attributes including floral, citrus, fruity, sulfur, diesel-like, onion-garlic, catty, piney, resinous, etc.	High, exhibiting a wide range of attributes including floral, citrus, fruity, sulfur, diesel-like, onion-garlic, catty, piney, resinous, etc.	Hop aroma and flavor are medium to high, exhibiting a wide range of attributes. Overall hop character is assertive.
Analytical bitterness	30–50 IBU	50–70 IBU	20–55 IBU
Perceived bitterness	Medium to medium-high	Medium-high to very high	Medium to medium-high

degrees of residual sugars/extract are balanced with acidity and with hop bitter compounds to create a suitable baseline for a range of hop and malt-derived aromas. Depending on wort quality/gravity, fermentation temperature (typically 18–22°C), and the yeast strain employed, significant concentrations of ethanol, higher alcohols, and esters might complement the complex flavor profile of hoppy ales.

Even though this chapter is called “Aroma of Craft beer,” the chemical complexity of beer, the multiplicity of beer styles, and their interpretation by (craft) brewers do not allow a comprehensive summary of “craft beer” and maybe not even for hoppy Ales. Still, an attempt is made to establish a link between the perceived flavor and aroma of hoppy Ales with the most relevant and aroma-active VOCs (volatile organic compounds) typically found in those beers.

2. Beer flavor and aroma

Per definition (ISO 5492, 2008) the term flavor stands for the combination of the olfactory, gustatory, and trigeminal sensations perceived during food consumption. Beer flavor results of a complex

**FIGURE 8.1**

Overview on the main substances and substance classes of beer. The organic compounds of beer collectively account for less than 0.2% by weight; they can be divided into volatile and nonvolatile compounds.

interplay between multiple of volatile and nonvolatile compounds present in a carbonated, hydro-ethanolic solution. Fig. 8.1 illustrates a simplified, schematic overview on the main substances and substance classes of beer.

Beer contains more than 90% (by weight) of water, around 4% (by weight) of ethanol, and around 0.5% (by weight) of carbon dioxide. This said, ethanol is by far the most abundant VOC in beer and has a large influence on beer aroma. The inorganic ions in beer collectively account for only around 0.2% (by weight); they are derived from minerals introduced by water, barely, adjuncts, malt, and hops. The organic compounds of beer that were not metabolized by yeast upon fermentation are commonly referred to as residual extract. This fraction includes (higher molecular weight) carbohydrates, peptides, and proteins. Together, they account for approx. 2%–5% (by weight) of beer, contribute to sweetness, mouthfeel, and are important for beer foam formation. The remaining organic compounds (for the sake of simplicity called “organic compounds” in Fig. 8.1) that are neither protein nor carbohydrates collectively account for less than 0.2% (by weight). This quantitatively minor fraction of compounds contains >1000 different molecular species and largely determines beer flavor and aroma. Regarding their chemical properties and contribution to beer flavor, these organic compounds are typically divided into volatile and nonvolatile compounds. The nonvolatile compounds such as polyphenols, hop bitter acids, and organic acids affect beer taste. Their presence in beer is typically linked with sensory descriptors such as astringency, bitterness, and sourness. Descriptors like mouthfeel and sweetness are, as mentioned above, tied to the composition and concentration of carbohydrates (Langstaff & Lewis, 1993). While residual simple sugars such as glucose, fructose, and maltose contribute to sweetness, polysaccharides such as beta-glucans and arabinoxylans affect viscosity, palate fullness, and mouthfeel. As schematically shown in Fig. 8.1, the group of volatile organic

compounds (VOCs) appears more complex when compared to the nonvolatile fraction. The array of VOCs in beer includes compounds that are finally responsible for aromas in beer, that is, beer aroma. Aroma stand(s) for the sensory attribute(s) perceptible by the human olfactory organ via the back of the nose (Jaeger et al., 2020). To be detected by the receptors in the olfactory system, VOCs need to be released from the beer during drinking. This release will depend on the properties of the VOCs themselves, but also on beer temperature, carbonation, nonvolatile composition of beer, etc. The array of VOCs that contribute to beer aroma results from multiple chemical and biochemical reactions occurring during brewing (i.e., boiling, fermentation, and maturation) and raw material production (i.e., malting). The number of VOCs in beer is hard to estimate. Beer likely contains 800 to >1000 VOCs that can broadly be grouped into alcohols, esters, ketones, (fatty) acids, aldehydes, terpenes, terpenoids, phenols, and volatile sulfur compounds. Per definition, all VOCs have some volatility, but this can range from low molecular weight compounds (i.e., acetaldehyde) that exist as permanent gases at room temperature to compounds (i.e., caryophyllene oxide) with very little vapor pressure and molecular weights up to 300 Da (Mottram & Elmore, 2003).

The key challenge in estimating the contribution of a single compound to the characteristic aroma of beer (or any other food), or even of a particular beer style or brand, is that aroma perception is multidimensional: The perceived aroma depends on multiple factors including the concentration of a VOC in a food item/beverage, the sensitivity of the human olfactometric system toward a specific VOC (odor threshold), its odor character and vapor pressure, the interaction with other (nonvolatile) components, and/or synergism with other volatiles (Dietz et al., 2020). The complexity of beer aroma perception relates to the construction of the human olfactory system in which volatiles released during food consumption compete to bind to an array of odorant receptors (The Nobel Prize in Physiology or Medicine, 2004). The odor perception for an isolated VOC can vary with concentration, and the odor quality can be strongly affected even when only traces of a second VOCs are present. Even though there exist several thousand food aroma compounds, it is evident that not all these VOCs contribute (equally) to food aroma. Already 60 years ago, Rothe & Thomas (1963) made an initial attempt to scale the relative importance of aroma compounds by setting the concentration of an aroma compound in a food item in relation to its odor threshold. The quotient of concentration and threshold was called odor activity value (OAV), an abbreviation that is still used in flavor chemistry today. Considering the above said, VOCs with high flavor thresholds, even when present in good quantities, might have comparably little contribution to food/beer aroma. Vice versa, aroma compounds with low flavor thresholds, even when only present in trace amounts, might be responsible for the characteristic aroma of beer styles and brands. The higher the OAV of a substance in a food item, the more important its contribution to aroma. VOCs with high OAV are called key aroma compounds or key odorants. The weak point in the OAV concept is the determination of the flavor/aroma thresholds. It is typically carried out sensory analysis using ethanolic solutions spiked with the VOC of interest, whereby the abovementioned matrix effects are not considered. Another approach to identify key aroma compounds (in beer) is the so-called aroma extract dilution analysis (AEDA) (Acree et al., 1984; Grosch, 1994). It involves the isolation and concentration of aroma compounds into an aroma extract, followed by the analysis of different dilutions of this isolate by gas chromatography-mass spectrometry (GC-MS) and gas chromatography olfactometry (GC-O). GC-MS enables the identification and quantification of aroma compounds, while running stepwise diluted isolates is used to identify the most

odor-active regions/VOCs by GC-O. Put simple, VOCs that can be still detected by human accessors in GC-O after multiple dilutions of the beer isolate are very aroma active and are therefore referred to as key aroma compounds or primary odorants. Compared with the overall research in the field of brewing science, the identification of key aroma compounds by AEDA accounts for very few, but important publications. One of the most significant works in this field was published by [Fritsch & Schieberle \(2005\)](#), who identified the primary odorants of Bavarian Pilsner type beer. What can be seen from this work is that volatiles derived from malt (i.e., beta-damascenone), from hops (i.e., linalool), from wort boiling (i.e., 4-hydroxy-2,5-dimethyl-3(2H)-furanone), and from fermentation (i.e., ethanol and acetaldehyde) are significant contributors to beer flavor. [Kishimoto et al. \(2006\)](#) compared the odor-active compounds of un-hopped beer and beers brewed using with different hop varieties (Saazer, Hersbrucker, and Cascade). They found that linalool, geraniol, ethyl 2-methyl butanoate, ethyl 3-methyl butanoate, and ethyl 2-methylpropanoate were mainly responsible for the hop character of beer. [Steinhaus & Schieberle \(2000\)](#) compared the most odor-active compounds in fresh and dried hop cones (variety Spalter Select). They found that trans-4,5-epoxy-(E)-2-decenal, Linalool, and myrcene were the most potent odorants. [Su & Yin \(2021\)](#) reported myrcene, methyl octanoate, Geraniol, and linalool as being the primary aroma compounds of Cascade and Chinook hops. [Neiens & Steinhaus \(2018\)](#) confirmed that Myrcene, Linalool, and 2- and 3-methylbutanoic acid are important variety-independent hop aroma compounds. In addition, they reported that the aroma of Huell Melon hops (fruity, cantaloupe-like odor note) is influenced by Ethyl 2-methylbutanoate, Ethyl 2-methylpropanoate, and Propyl 2-methylbutanoate, while the fruity and minty odor notes in Polaris hops are associated with high amounts of 3-methylbutyl acetate and 1,8-Cineole. In terms of malt aroma, [Beal & Mottram \(1994\)](#) found that 2- and 3-methylbutanal, maltol, and some (unidentified) pyrazines were most potent in exhibiting malty, cereal-like aroma. Their findings were later confirmed by [Fickert & Schieberle \(1998\)](#) who identified 2- and 3-methylbutanal being the key odorants of a caramel malt and also by [Fechir et al. \(2021\)](#) who focused on the contribution of different specialty malts on the volatile profile of lager beer. The determination of key aroma compounds requires specific equipment (GC-MS, GC-O), reference standards, and (analytical) experience. This said, it can neither be performed by brewers themselves nor in high throughput commercial laboratories. Works that are dedicated to identifying OAV in brewing raw materials, beer, and other beverages have an important contribution to our understanding of beer aroma. They assist to shorten the list of 800–1000 VOCs in beer to a list of approx. 100–200 compounds with significant contribution to aroma. This finally enables brewers, researchers, or commercial laboratories to perform quantitative analysis of selected VOCs, to collect data for specific styles, brands, products, etc., as well as to correlate these data with results from sensory evaluation. In this view, the determination of OAV or key aroma compounds can be practically seen as the first step to understand beer flavor. Given the fact that this type of analysis cannot be done in high throughput, the contribution of raw materials, the brewing process, or beer storage conditions on beer flavor, is typically evaluated by combination of sensorial and chemical analysis. Analysis data are then treated with different multivariate statistical methods, aiming to find correlations between sensory scores (for certain attributes) and the concentration of aroma compounds. Recent examples for this approach were reported by [Schubert et al. \(2022\)](#) (hoppy Ales) and [Lafontaine et al. \(2020\)](#) (nonalcoholic beer).

3. Chemical and sensorial characteristics of pale ales and India pales ales

Taste and aroma are the most important attributes of beer and determine its success in the marketplace (Sega et al., 1967). Fresh beer aroma varies as a function of the raw materials, the brewing recipes, but also the equipment used for brewing and the downstream processing including packaging. Table 8.2 summarizes the fundamental chemical and sensorial characteristics of relevant hoppy Ale styles, namely PA, IPA, and Session IPA. As mentioned above, hoppy Ale production typically uses malted barley as carbohydrate source. Malted barley represents a high-quality ingredient which can be processed without necessity of exogenous enzyme addition and or complex mashing schemes. Next to pale-colored base malts, the grist of hoppy Ales features small percentages of specialty malts covering a wider color range (i.e., Munich malt, Caramel malt, Crystal malt). Sometimes wheat (malt) and oats are used to balance flavor (mouthfeel) and haze. As can be seen from Table 8.2 malt character of hoppy Ales might rank from low to medium. The VOCs that are formed during malting and their contribution to fresh and aged beer aroma are discussed below. Next to the composition of the grist, the grist-liquor ratio varies within the range hoppy Ales. Original gravity of IPA typically ranges between 14°P–17°P. The residual extract after fermentation can range from only 2°P in dry IPAs to up to 4°P. This variation

Table 8.2 Summary of basic physiochemical analysis data for a selection of hoppy ales including fermentation by-products and DMS.

	Unit	Median	Min	Max
Bitterness	IBU	60	35	72
Color [EBC]	EBC	26.5	17	170
pH		4.6	4.5	4.9
Original gravity	% mas	16.9	12.8	21.7
Alcohol by volume	%	7.1	5.4	9.9
Apparent degree of attenuation	%	78.6	68.5	86.5
Apparent extract	% mas	3.61	2.34	5.62
Acetaldehyde	mg/L	1.3	0.7	2.7
Ethyl acetate	mg/L	15.8	0.3	28.8
1-Propanol	mg/L	19.3	2.2	125.3
Isobutanol	mg/L	14.8	1.3	28.3
Isoamyl acetate	mg/L	0.5	<0.1	1.2
2-Methyl-1-butanol	mg/L	7.4	1.0	11.9
3-Methyl-1-butanol	mg/L	39.6	5.6	83.2
Phenethyl alcohol	mg/L	14.8	1.7	36.0
Phenylethyl acetate	mg/L	0.07	0.02	0.13
DMS	µg/L	81	22	180

Data as published by Schubert, C., Lafontaine, S., Dennenlöh, J., Thörner, S., & Rettberg, N. (2022). The influence of storage conditions on the chemistry and flavor of hoppy ales. Food Chemistry, 395, 133616. The fermentation by-products are reported in mg/L but were corrected to 12°P original gravity.

affects body and mouthfeel and is therefore very important for IPA flavor. What can also be seen from Table 8.2 that the fermentation characteristics might vary within a style. This is not surprising given the range of commercially available yeast strains and the variability of their characteristics such as attenuation, flocculation, optimal fermentation temperatures, and flavor profile. From a brewer's perspective, the formation of yeast-derived flavor compounds and/or the ability of yeast to refine beer flavor by interaction with hop/botanical/fruit-derived compounds offers a great opportunity to diversify craft beer aroma. Still in IPA, fermentation-derived aroma compounds should be balanced to not interfere with hop character (see below).

Bitterness a key quality attribute of hoppy Ales. Bitterness intensity ranges from medium-high to very high (Table 8.2) which is reflected by an analytical bitterness ranging from 50–70 IBU (Tables 8.2 and 8.3). Beer bitterness mainly relates nonvolatile hop resins. Historically (Wöllmer, 1916), the hop resins were defined as the fraction of hops that could be extracted by organic solvents, more specially a methanol/ether mixture. The hop resins were then further divided into the hexane soluble soft resins (i.e., α - and β -acids) and the hexane insoluble hard resins (i.e., xanthohumol). In Lager beer production, hop extracts or pellets are typically added during the kettle boil. The hop α -acids (AA) isomerize into iso- α -acids iso-alpha acids (IAAs). IAAs are the main contributor to bitterness in Lager beer, whereas α -acids are only present in trace amounts and do not largely influence beer bitterness (Fritsch & Shellhammer, 2007). In contrast, dry-hopped Ales typically contain a much wider range of hop soft and hard resin constituents that contribute to bitter quality, intensity, time course, and astringency. The enrichment of hop resins in dry-hopped beer is because large amounts of hops (0.5–6 kg/hL) added to these beers at different stages of the brewing process. By adding large

Table 8.3 Concentration [g/kg] of 16 selected hop aroma compounds in Hallertauer Perle, Mosaic, Cascade, and Citra hops.

Analyte	Hallertauer perle [g/kg]	Mosaic [g/kg]	Cascade [g/kg]	Citra [g/kg]
α -pinene	0.82	1.27	0.98	1.09
β -pinene	6.26	9.22	9.58	9.40
Myrcene	432.24	663.58	541.24	727.82
2-MBIB	3.20	9.65	6.76	3.76
Limonene	1.60	2.88	2.53	2.35
cis-Linalool oxide	0.04	0.04	0.15	0.02
trans-Linalool oxide	<0.01	<0.01	0.01	0.01
Linalool	3.53	6.13	5.58	7.66
α -Terpineol	0.07	0.30	0.12	0.20
Citronellol	<0.01	<0.01	<0.01	<0.01
Nerol	0.59	1.09	0.79	0.70
Geraniol	1.18	8.50	4.51	5.02
Geranyl acetate	0.05	0.35	13.87	1.13
β -Caryophyllene	119.60	35.51	48.71	45.00
α -Humulene	374.25	86.46	124.77	73.94
Caryophyllene oxide	1.26	0.93	4.89	0.79

quantities of hops to cold beer, hard and soft resins are extracted along with hop-derived VOCs, which is why hoppy Ales next to IAAs contain significant levels of AA and oxidized AA (humulinones), but also detectable levels of β -acids and oxidized β -acids (Oliver & Colicchio, 2011). A key consideration in using hops in the production of hoppy Ales is to balance bitterness and hop aroma. By adding hops to the kettle boil, it contributes to bitterness but imparts only very little aroma. By adding hops to cold beer, hops contribute to bitterness, astringency, and aroma.

4. Sources of aroma compounds in hoppy ales

Beer aroma is strongly dependent on the raw materials and relative ratios the raw materials stand to each other. In addition, brewing equipment and technology, downstream processing, and packaging will affect the organoleptic properties and sensorial stability of beer. Water is by far the main ingredient of beer. Water itself should be neutral in aroma and flavor, but should contain minerals (i.e., Calcium, Magnesium, and Potassium) that positively affect the brewing process, the mineral balance in beer, and finally flavor. Calcium, for example, stabilizes malt enzymes during mashing, can lower the mash pH, might precipitate oxalates, and enhances hop character. While water should influence beer aroma only indirectly, the volatile compounds of malt and their fate during wort boiling and fermentation, the aroma compounds that are formed by yeast metabolism, and the aroma compounds that are derived from hops, have a direct contribution to beer aroma.

5. Volatile compounds derived from malt

Overall beer quality is strongly influenced by the starch source available for brewing. Across the world, several (local) raw materials such as barley, wheat, maize, rice, and sorghum are used for brewing. Sugar syrup is used to achieve high attenuated dry beers or simply to compensate for raw material-induced fluctuations in gravity. On the one hand, the usage of local raw materials can be used to distinguish brands from products in the mass market and on the contrary might have a large influence on production costs. While product diversification and exclusivity are key to craft brewing, production cost are especially important in industrial beer production where margins are low. Each of the abovementioned raw materials varies in their mechanical, and physio-chemical properties. Therefore, the usage of raw materials other than (barley) malt requires dedicated brewing techniques and dedicated brewhouse equipment (e.g., cereal cookers) to convert starch into fermentable sugars and to separate spent grain from wort. IPA production historically used pale barley malt as the main ingredient. Using malted barley enables brewers to produce well-fermentable wort that contains all required nutrients for a healthy fermentation. Also, rather simple brewhouse equipment (two-vessel brewhouses, for example) and infusion mashing might be used. Thus, next to flavor, processability is likely the reason that (pale) malted barley is still the main starch source in IPA brewing today. Malt quality parameters most relevant to brewing include extract, friability, homogeneity, beta-glucan content, viscosity, total and soluble nitrogen, as well as free amino nitrogen (FAN). These parameters influence wort/beer filterability, wort quality (i.e., nutrient content), and extract yield. As discussed below, wort composition influences yeast metabolism and physiology and is therefore important for beer aroma. Next to pale base malts that supply enzymes to the mash and nutrients to the wort, the production of craft beer including IPA, also takes advantage of dark and roasted specialty malts. Malts

such as Vienna malt, Caramel malt, Munich malt, or Cristal malts are used as so-called strategic ingredients. They are incorporated at low percentages, typically < 5–10% of the total grist to add color and aroma (Gu et al., 2023). During malt production, the intensity of barley germination in combination with thermal processing of green malt (kilning, roasting) is most significant in creating distinct malt aroma through the Maillard reaction. Concerning food aroma, the Maillard reaction is likely the most important reaction in food processing and preparation and can be broadly characterized by three key stages: It starts with the initial reaction between reducing sugars, such as for example glucose, fructose, or maltose, and an amino acid. This reaction yields a so-called N-glycosylamine that subsequently undergoes a rearrangement into a so-called Amadori product. The Amadori product (chemically speaking a glycosylamine) can then either undergo dehydration or sugar fragmentation and after elimination of the amino group an array of Maillard reactions products (MRPs) is formed by either dehydration, fragmentation, polymerization, or Strecker degradation. Depending on the concentration of reducing sugars and amino acids in green malt and the intensity of killing and/or roasting different volatile and nonvolatile MRPs are formed. The volatile MRPs include, but are not limited to, aldehydes (e.g., 2- and 3-Methylbutanal), furans (e.g., 2-acetylfuran), and Pyrazines (e.g., 2-methylpyrazine, 2,5-dimethylpyrazine, and 2,6 dimethylpyrazine). While 2- and 3-Methylbutanal exhibit a sweet, bready, and malty aroma, Pyrazines remind of roast-like smells that range from pleasantly nutty to burnt/earthy. The concentration of Aldehydes, Furans, and Pyrazines changes with malt color. Pale malts such as pilsner malt (5 EBC) contain very low levels ($\mu\text{g/kg}$) of Furans and no detectable levels of Pyrazines, while Aldehydes are present in the mg/kg range. The concentrations of Aldehydes in Munich malt (20 EBC) are approx. by factor 10 higher when compared a Pilsner malt. In addition, low levels ($\mu\text{g/kg}$) of both Furans and Pyrazines can be found. In Chocolate malt (900 EBC) the concentration of Pyrazines and Furans can be very high (several 100 $\mu\text{g/kg}$), while the levels of most aldehydes are comparable to those of Munich malt. Clearly, consumers and professional tasters record that malt aromatic qualities influence beer flavor. Still, it is not yet completely understood which volatile compounds contribute to pleasant malty aromas in beer and how these volatiles make their way into the final product. Unlike to dry-hopping, where hops are extracted by fermenting/fermented beer, malt is added at the very beginning of the brewing process and its soluble constituents undergo significant processing. Recent findings (Fechir et al., 2021; Prado et al., 2023; Reglitz et al., 2022) show that the direct transfer of malt aroma compounds into beer plays a minor role. Depending on the chemical structure, reactivity, and volatility some aroma compounds are evaporated in the brewhouse or, in the case of aldehydes, are biochemically reduced into less flavor active alcohols during fermentation. Others, such as the Pyrazines, remain largely stable during boil, fermentation, and downstream processing. It seems plausible that nonvolatile Maillard reaction intermediates, such as 3-deoxyglucosone (3-DG), play an important role in malty beer aroma development and instability (Nobis et al., 2021), but more research in this field is needed to gain a proper chemical understanding. To date, there exist very limited published analysis data on MRPs in IPAs, which likely relates to the fact hop aroma compounds (see below) were simply considered as being more important/interesting. Regarding the abovementioned MRPs analyses conducted on commercial hoppy Ale samples by VLB show that aldehydes are the main MRPs in PAs, IPAs, and Double IPAs. Dennenlöhner et al. (2020a) reported that Ales and nonalcoholic beers (NABs) contain high levels of aldehydes when compared to pale Lager beer. They also found that the aldehyde profile of Ales differed significantly from the aldehyde profile of pale Lagers and NABs, which relates to the usage of specialty malts, and in terms of

NAB, to production technology. Schubert et al. (2022) compared the aldehyde profiles across a range of hoppy Ales with beer colors from 17–170 EBC and OGs from 8–22°P. They confirmed that 2-methylpropanal (sweet, herbal, and malty aroma) is by far the most abundant aldehyde in hoppy Ales, followed by 2-Furfural (caramel, bready, and coffee aroma). Regarding the influence of malt aroma compounds on the flavor of hoppy Ales, Schubert et al. (2022) did not confirm a significant correlation between the malt character determined by quantitative descriptive analysis (QDA) and aldehydes levels determined by gas chromatography. This observation relates to the fact, that the aroma of hoppy Ales is complex and hop-derived VOCs might cover malt character in some fresh IPAs.

In addition to the abovementioned MRPs, malt, wort, and hoppy ales contain a range of linear alkanals, alkenals, and alkadienals that are formed by oxidation of (poly) unsaturated fatty acids during malting and mashing. Lipid-derived aldehydes are key aroma compounds of many foods and are also relevant for beer aroma. Among them, (E)-2-nonenal has received most attention in brewing science as it was reported to exhibit papery/cardboard flavor to aged Lager beer already decades ago (Drost et al., 1990). Today, cardboard notes are not typically found in sensory evaluations of aged (craft) beer. This is likely due to the fact that low mashing in temperatures (45–60°C), which favor enzymatic oxidation of unsaturated fatty acids by lipoxygenases, are avoided today. Also, manufacturing practices that minimize oxygen ingress during processing and packaging are implemented in many breweries. In fresh hoppy ales, Schubert et al. (2022) found that the concentrations of (E)-2-nonenal (and several other lipid-derived aldehydes) showed very little variation across a range of beer made from variable grist bills. This stands in contrast to their observation made for Maillard-derived aldehydes, which were, for example, positively correlated with wort gravity and beer color. As shown in Table 8.2 hoppy Ales typically aim at a medium-low to medium intensity of malt-derived aroma and in addition, DMS should not be present. DMS is one of the most potent aroma compounds in green malt and contributes to the characteristic aroma of mash and unboiled wort. The aroma of DMS is described as corn-like and beer styles with little flavor complexity can be negatively affected by DMS concentrations of >70 µg/L. In hoppy ales, DMS levels can easily exceed 150 µg/L without being detected by trained panelists. It is important to note, that dry hopping can increase DMS levels in beer.

6. Volatile compounds derived from fermentation

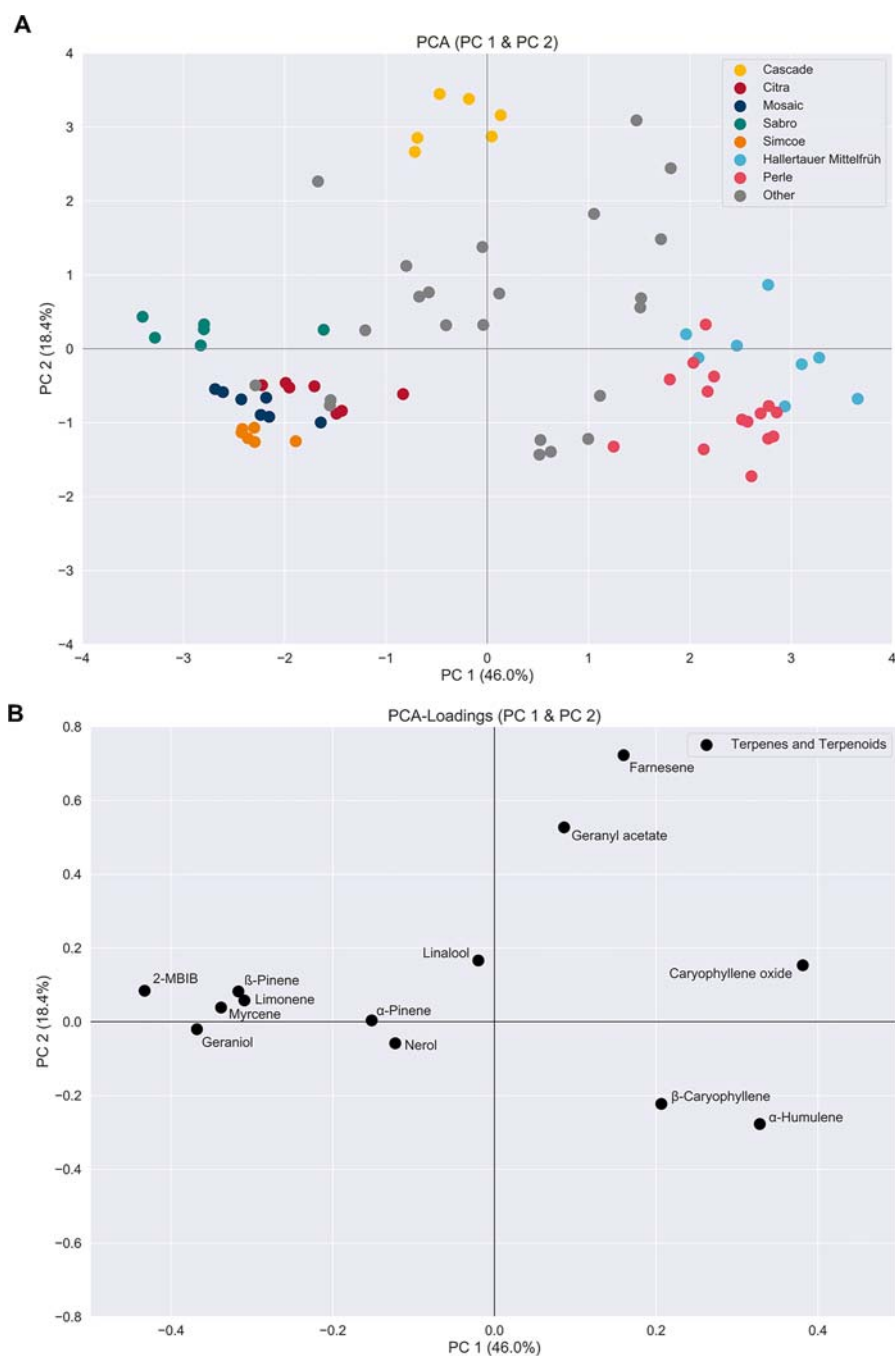
Ethanol and carbon dioxide are the main products of beer fermentation and contribute to beer flavor and aroma by adding warming-alcoholic notes, mouthfeel, and sparkling character to beer. Yeasts' growth on wort utilizes nutrients such as sugars, free amino nitrogen (FAN), and fatty acids, removes undesirable wort flavors, and shall finally result in a desirable mixture of fermentation by-products. Brewing yeasts feature a versatile metabolism. They can adapt to a variety of conditions, and environmental factors such as nutrient supply, aeration, temperature, and pressure will influence their physiology. Hence, final beer characteristics and aroma (within the category of hoppy Ales) vary because of the yeast strain employed by brewers, the characteristics of the cold wort to which the yeast is added, and finally the fermentation and maturation profile (pitching rate, aeration, time, temperature, etc.). In contrast to other beer types/styles (e.g., Belgian Ales, German Wheat beer) the aroma profile of hoppy Ales is typically not dominated by yeast-derived VOCs. Instead, yeast-derived VOCs shall

support/showcase the fresh hop character. This means, as for many other beer styles, that elevated concentrations of acetaldehyde (green apple aroma), diacetyl (buttery/butterscotch aroma), and volatile sulfur compounds (sulfury, drain, mercaptan aroma) should not be present. The levels of esters (sweet, fruity, floral, sometimes perfumy aroma) and higher alcohols (sweet, warming aroma) should be balanced to complement, but not mask, malt, and hop-derived VOCs. To achieve low acetaldehyde levels, measures that ensure healthy, viable yeast are effective. Operations that avoid excessive aeration during pitching, oxygen ingress while transferring green beer from fermentation to maturation or when dry hopping should be quite effective. Diacetyl production is strongly depended on yeast strain and levels of around 100–120 $\mu\text{g/L}$ can cause objectionable beer flavor defects. The reduction in diacetyl to acetoin and 2,3-butanediol is ensured by keeping viable yeast in suspension during late fermentation. Green beer should not be cooled down prior to the diacetyl target has been achieved. While acetaldehyde and diacetyl accumulate early in fermentation and are then reduced, higher alcohols accumulate throughout fermentation. Their maximum concentration is reached once wort amino acids fall to their minimum (day 5–6 of a 7-day fermentation). As can be seen from [Table 8.3](#) the concentrations of higher alcohols can vary greatly in hoppy ales. 1-propanol (alcoholic, sweet, weak fusel and tequila-like aroma), 3-methyl-1-butanol (alcoholic, fusel, ethereal, cognac, fruity, and banana-like aroma), and phenethyl alcohol (sweet, floral, fresh and breadly aroma) are typically the main higher alcohols and collectivity can reach concentrations from 50–150 mg/L. To some extent, higher alcohol concentrations correlate with ethanol content and higher ABV beers can therefore have a boozy character. Next to yeast strain selection, factors influencing yeast growth (high FAN, aeration, fermentation temperature) can be used to balance higher alcohol synthesis. The application of tank top pressure will also affect higher alcohol formation and is therefore very commonly used in industrial brewing. Next to the higher alcohols, acetate esters, and ethyl esters represent important beer flavor compounds. Even though beer can contain more than 100 different esters resulting from different biochemical reactions or dry-hopping, most brewers are familiar with ethyl acetate (fruity, solvent-like aroma), isoamyl acetate (banana-like aroma), and ethyl hexanoate (fruity, apple-like aroma). With up to 20 mg/L ethyl acetate is the most abundant ester in beer, while most of the other esters do typically not exceed 1 mg/L. Ester synthesis is linked to lipid biosynthesis/metabolism, which is why limiting wort aeration is likely the most practical measure to lower ester levels in beer.

Fermentation by-products serve important roles in yeast metabolism, yeast growth, and yeasts' response to nutrients and oxygen, etc. The nutrients taken up by the yeast are mainly derived from the starch source, most typically malt. In hoppy Ale production, some brewers add hops toward the end of the primary fermentation, typically when diacetyl concentrations fall below 100 ppb. By adding hops to beer containing active yeast cells and yeast enzymes some hop-derived compounds can be altered and contribute to beer flavor and aroma. These alterations are commonly referred to as “biotransformation,” which chemically includes reactions such as the reduction in carbonyl compounds or ethers into alcohols and diols, the isomerization of monoterpene alcohols nerol, geraniol, and linalool into alpha-terpineol, the conversion of geraniol into beta-citronellol, the release of glycosidically bound aroma precursors (e.g., linalool), the release of peptide/amino acid bound thiols, transesterification, or hydrolysis of esters, as well as the conversion of acids into (ethyl) esters ([Rettberg et al., 2018](#)).

7. Volatile compounds derived from hops

Hoppy beer flavor and aroma is key to the success of hoppy ales such as IPA. Hop aroma in beer can vary greatly in terms of intensity and quality and is commonly described with terms such as floral, fruity, citrus-like, woody, beery-like, piney, and resinous. Hop aroma in beer is basically determined by hop oil constituents that are extracted from hops into wort or beer. In addition to the extraction, as mentioned above, some hop-derived compounds undergo biotransformation throughout the brewing process, and thereby the intensity and character of hop aroma can change (Dennenlöhner et al., 2020b; Rettberg et al., 2018). The most prominent example for a biotransformation is surely the formation of Citronellol from Geraniol during fermentation by which the aroma shifts from floral to citrus-like notes (Takoi, Koie, et al., 2010, Takoi, Itoga, et al., 2010). Brewers can adjust hop aroma in beer choosing specific hop varieties, by the quantity of hops they add, by the type of hop product they use, by the timing of the hop dosage, and the dry hopping technique they use. Currently, more than 250 different hop varieties are marketed and used in the brewing industry. Their chemical composition, that is, their content of alpha-acids and oil, as well as the composition of the oil, vary and primarily relate to plant genetics. Factors such as growth location, harvest year, harvest maturity, and hop processing can contribute to some variation (Lafontaine et al., 2021), which is why hop selection shortly after harvest is a must have event for many brewers. The content of hop oil is correlated with the content of AA, and ranges between 0.5% and 3.0% hop dry matter. As a side note, this explains why next to weight, AA content is typically used to adjust dry hop dosage. Qualitatively, there exist significant differences in the volatile compounds found in different hop cultivars (Fig. 8.2), which represents the results of a principal components analysis (PCA) of the major volatile compounds determined in a range ($n = 80$) of hop varieties from the US, Germany, and Australia, illustrates those differences. Next to the so-called “new world” hop varieties that are used in the production of hoppy ales, for example Cascade (US), Citra (US), Mosaic (US), Sabro (US), and Simcoe (US), two classical “old world” German aroma hop varieties (Hallertauer Mittelfrüh and Hallertauer Perle) are shown. The hops pictured as “others” are for example, Polaris (GER), Mandarina Bavaria (GER), Columbus (US), Galaxy (AUS), Vic Secret (US) Bravo (USA), and Chinook (USA). They are included to reflect the chemical diversity of VOCs in hops. The PCA scores plot in Fig. 8.2A illustrates how chemically similar or different the VOC profiles of hop varieties (i.e., their oils) are and how variable the VOC profile within a certain variety is. The samples of Hallertauer Mittelfrüh and Perle scatter in the right quadrants of the plot. Their VOC profiles are similar when compared to each other, but different when compared Cascade, Sabro, Citra, Mosaic, and Simcoe. The latter four varieties are chemically similar, at least with respect to the VOCs that were targeted in this analysis. Fig. 8.2B represents a so-called PCA scores plot. This plot illustrates which variables, in this case hop VOCs are responsible for the separation or clustering of hop varieties in Fig. 8.2A. Put simple aroma compounds that appear close to each other in PCA are positively correlated (e.g., α -Humulene and β -Caryophyllene), and aroma compounds that appear very far away from each other on the PCA are negatively correlated (e.g., farnesene and geraniol). VOCs that appear on the outer areas of the PCA (e.g., 2-MBIB and Caryophyllene Oxide) vary in relative abundance in the dataset and therefore separate/distinguish samples. VOCs that appear around the origin of the PCA (e.g., Nerol and Linalool) have comparably little contribution to the separation of the hop varieties. The separation of hop varieties in PC1 is strongly affected by their Farnesene content. Hop varieties that contain Farnesene appear on the upper half of

**FIGURE 8.2**

Principal component analysis of VOC in hops. (A) PCA scores plot illustrating the two-dimensional separation/clustering of hop varieties based on their VOC. (B) PCA loadings plot illustrating the VOC responsible for the clustering/separation shown in (A)

the PCA, hop varieties that do not contain Farnesene appear on the lower half. Farnesene is not typically detected in beer in levels $>1 \mu\text{g/L}$ and therefore has low/no contribution to aroma. In hop oil analysis it has been historically used as a marker compounds in variety identification. Linalool, one of the most important VOCs in hops and hoppy beer, sits very close to the origin of the PCA.

This indicates that Linalool has little influence on the separation of the varieties in this dimension, or in other words, the concentration of Linalool is comparable across the varieties pictured in Fig. 8.2. In contrast, aroma compounds that appear on the far left or right-hand side of the PCA differ strongly across the varieties tested. These are for example 2-MBIB, Geraniol, Myrcene, but also of Humulene, Caryophyllene, and Caryophyllene oxide. VOCs that are present only in a limited number of hop varieties are called “varietal” aroma compounds. Examples for these are thiols and esters, their contribution to beer aroma is discussed below. The PCA in Fig. 8.2 gives an overview on relative similarities and differences between hop varieties. Table 8.3 lists the concentration of 16 hop aroma compounds across four selected hop cultivars. While Hallertauer Perle is not frequently used for hoppy Ale production, Mosaic, Cascade, and Citra are key to many IPA recipes. Table 8.3 (again) illustrates that the difference in Linalool between the four hop varieties is relatively small (3.53–7.66 g/kg) while large differences for geraniol (1.18–8.50 g/kg) and geranyl acetate (0.05–13.87 g/kg) can be seen.

In summary, the illustrations and data shown above shall underline that the VOC profiles of hops are variety depended, and therefore, hop variety selection has a major influence on beer aroma. In the production of hoppy Ales, large quantities (several kg/hL) of hops are added late in the boil, during wort transfer into whirlpool, and to green—or end-fermented beer (dry hopping) (Rettberg et al., 2018). Dry hopping emerged as a key technique of craft brewers since the cold extraction of hops leads to distinct beer flavor profiles in which the characteristics of a hop variety are (partly) transferred into beer. One of the key challenges in understanding hop aroma in beer relates to the fact that the aroma of hops and the hoppy aroma of beer differ. On the one hand, this offset can be explained by the fact that hop volatiles in beer are accompanied by malt and fermentation-derived aroma compounds.

On the contrary, the complexity of hop oil itself, the (bio-) chemical modification of some hop constituents during beer production, and the limited solubility of some VOCs (e.g., Myrcene) are of importance. The concentration of hop aroma compounds in beer depends on which compounds might or might not be transferred into wort and beer, to which extend some of these compounds are removed during processing, and which alterations occur by the action of yeast/yeast enzymes (Takoi, Koie, et al., 2010). It is beyond the scope of this chapter to discuss all these factors in detail, but next the hop quality itself, the dry hopping technique (static vs. dynamic), the alcohol content and beer temperature at the point of hop addition, as well as the timing of dry hopping (with or without yeast) have a main influence on aroma of dry-hopped beer. The addition of hops during maturation can result in resinous and piney beers, while dry hopping in presence of yeast can result in more subtle, fruity aroma. Next to the biotransformation of some hop-derived VOCs or VOC-precursors, yeast will adsorb terpenes such as Myrcene and therefore (positively) alter beer aroma. Next to biotransformation a further advantage of adding hops while yeast is present is, that yeast can metabolize oxygen that is introduced by dropping and this can potentially result in more stable beers. However, if hops are added early during primary fermentation, considerable losses of hop-derived VOCs might occur. VOCs will be stripped of with CO_2 , and some terpenoids (e.g., Geraniol) will be absorbed by the yeast and will then be converted into Sterols (Lafontaine et al., 2021).

To provide a brief overview on typical concentration of hop-derived VOCs in Ales, Table 8.4 summarizes the minimum, maximum, and median concentrations ($\mu\text{g/L}$) of 16 hop-derived aroma

Table 8.4 Typical concentrations of hop aroma compounds in Ales.

Analyte	Ale minimum [μg/L]	Ale median [μg/L]	Ales maximum [μg/L]
α-pinene	<1.0	<1.0	5.5
β-pinene	<1.0	<1.0	28.9
Myrcene	2.6	23.4	2126.1*
2-MBIB	<1.0	50.5	584.4
Limonene	<1.0	4.1	669.5
<i>cis</i> -Linalool oxide	<1.0	7.8	46.6
<i>trans</i> -Linalool oxide	<1.0	6.6	36.2
Linalool	17.2	166.3	2035.0
α-Terpineol	9.2	20.3	213.0
Citronellol	1.6	13.4	142.1
Nerol	<1.0	4.9	99.7
Geraniol	4.9	20.9	756.0
Geranyl acetate	<1.0	<1.0	59.2
β-Caryophyllene	<1.0	<1.0	443.0
α-Humulene	<1.0	2.6	764.1
Caryophyllene oxide	<1.0	<1.0	34.7

compounds in Ale (Neiens & Steinhaus, 2018). What can be seen from the row “Ale minimum” in Table 8.3 that some compounds, such as β-Caryophyllene, which are present in any commercial hop variety (cf. Table 8.3), are not necessarily present in beer. Esters, such as 2-MBIB and Geranyl acetate, can be found in large quantities in some hop varieties, but are not present in others. If hop varieties that contain these esters are used for dry hopping, these esters can consequently be found in beer. Compounds that can be detected in any Ale are Myrcene, Linalool, α-Terpineol, and Geraniol. The concentration of Myrcene in hops is very high (up to 70% of the total oil) and even though Myrcene is not well beer soluble, traces of Myrcene will always be detected in dry-hopped beer, but will not contribute to aroma. Linalool, α-Terpineol, and Geraniol are well soluble in wort and beer, and their transfer rate from hops to beer is therefore, depending on the dry hopping regime, in the range of 80%–100%. Depending on the dry hopping regime very high levels of some hop-derived VOCs can be found in beer. As can be seen from Table 8.4 the row “Ale maximum” in Myrcene and Linalool might exceed concentrations of 2000 μg/L. The row “Ale median” in Table 8.4 provides the median concentration of each aroma compound. The median is the value separating the higher half from the lower half of the data population and is therefore not (unlike the arithmetic mean) distorted by a small proportion of extremely large or small values.

What can be taken from the median concentrations is that Linalool (floral, fresh lavender aroma), 2 MBIB (fruity aroma), Myrcene (spicy, resinous, earthy aroma), Geraniol (rose and slightly citrus-like aroma), and α-Terpineol (lilac aroma) are typically the most prominent hop-derived aroma compounds in Ale. It might be difficult and somewhat abstract, to link the above-listed concentrations and concentrations to beer aroma. It might therefore be helpful, to reference the concentrations of some hop-

derived VOCs in two well know IPAs: For Lagunitas IPA, 2-MBIB is in the range of 20 µg/L, Linalool and Geraniol are both at around 270 µg/L, and the concentration of Citronellol is in the range of 50 µg/L. For Goose Island IPA, 2-MBIB is comparable to the concentration in Lagunitas IPA, Linalool is in the range of 350–400 µg/L, Geraniol around 260 µg/L, and Citronellol at around 70 µg/L. The level of Myrcene does not exceed 50 µg/L in both beers, Caryophyllene and Humulene range between 2–5 µg/L.

Terpenes and terpenoids are main constituents of hop essential oil and beer terpenoids are also well extracted into beer during dry hopping. It is therefore not surprising that early research in the field of hop aroma focused on these substance classes. With the increasing importance of hoppy Ales and the introduction of hop varieties with more complex aromas, the focus of hop and beer analysis was enlarged toward aroma compounds that are responsible for the characteristic aromas of certain hop varieties. Next to hop varieties that contain high levels of geraniol and geranyl esters (Cascade, Bravo, Mosaic, Huell Melon, Sabro, VIC Secret, Amarillo, etc.), it was found that esters, ketones, and Sulfur compounds discriminate hop varieties (Kishimoto et al., 2006; Takoi et al., 2016). While thiols, which are discussed below, are minor constituents of hops (Reglitz & Steinhaus, 2017) and beer (Dennenlöhner et al., 2020c), esters represent a major share of hop essential oil (15%) (Biendl et al., 2015) and some esters, for example 2-methylbutyl isobutyrate (2-MBIB), are found in concentrations of >500 µg/L beer (Fickert & Schieberle, 1998; Rettberg et al., 2020). In chemistry, compounds formally derived from an (in) organic acid in which at least one hydroxyl group is replaced by an (alcohol derived) alkoxy group are named ester. Given the multiplicity of acids and alcohols present in hops and hoppy beer, it is not surprising that both matrices also contain a wide range of methyl-, acetate-, ethyl-, isobutyl, and geranyl-esters. Among the multiplicity of esters found in beer two groups of esters appear of prior ranking importance: The first group includes esters which are mostly formed from hop-derived short-chain acids or esters during fermentation (Kishimoto et al., 2006; Nielsen, 2009; Rettberg et al., 2014) or upon beer storage (Vanderhaegen et al., 2007; Williams & Wagner, 1978). These are ethyl isobutyrate, ethyl 3-methylbutyrate, ethyl 2-methylbutyrate, ethyl-4-methylpentanoate, and phenethyl 3-methylbutyrate (Dennenlöhner et al., 2020a; Prado et al., 2023). Second, hoppy beer contains esters originating from hop plant metabolism that are transferred into beer as hops are added. This group includes 2-MBIB, isobutyl isobutyrate, and isoamyl isobutyrate (Gu et al., 2023; Schubert et al., 2022) as well as a series of geranyl esters such as methyl geranate, geranyl propionate, geranyl isobutyrate or geranyl acetate. Concentrations of all abovementioned esters in beer have been reported by Schubert et al. (2023). The concentration of hop-derived esters in beer is variable and as for the other hop-derived VOCs depends on hopping rate, hop variety, and time of hop addition. In addition, the age of a beer sample plays an important role. As described below, some esters (e.g., 2-MBIB) are hydrolyzed during beer storage, while others (e.g., ethyl 3-methylbutyrate) are formed from esterification of 3-methylbutanoic acid and ethanol. The quantitative changes of esters in beer go along with aroma changes, which might be undesired. Sulfur containing aroma compounds contribute to a minor fraction of hops and beer (Takoi, Itoga, et al., 2010). Earlier literature was rather concerned with the unpleasant aromas (Cheesy, brunt rubber, Onion, Garlic, Cooked vegetable, etc.) that are associated to thioesters, thiophenes, polysulfides, methyl sulfides, or episulfides in hops and beer (Lermusieau & Collin, 2003). Doubtless, there are some hop varieties (e.g., Apollo, Columbus, Summit, etc.) that, when added in large quantities, can contribute to undesired sulfury notes. Still, in the last 10–15 years the focus has shifted toward polyfunctional thiols that are associated with pleasant fruity flavors in certain hop varieties and the beers produced thereof (Steinhaus & Schieberle, 2007). Thiols are not detectable with the industry standard methods for hop essential oil analysis and their analysis in both hops beer

requires sophisticated analytical assays (e.g., [Lafontaine et al., 2021](#)). Due to their remarkably low odor thresholds in beer (approx. 1.5 ng/L for 4-mercapto-4-methyl-2-pentanone (4 MMP), approx. 5 ng/L for 3-mercaptohexylacetate (3 MHA), and approx. 55 ng/L for 3-mercapto-1-hexanol (3 MH)) ([Kishimoto et al., 2008](#); [Lafontaine et al., 2020](#)) thiols can impact beer aroma. Their presence is associated with blackcurrant (4 MMP), grapefruit (3 MH), and passion fruit (3 MHA) aromas perceivable in many popular hoppy ales. It is important to note that 4-MMP and 3-MH can be found in hops, while 3-MHA is formed from 3-MH in presence of yeast. [Table 8.5](#) reports the concentrations of 4-MMP and 3-MH in selected hop samples from the US, Australia, and Germany of the 2022 harvest. As for any other hop aroma compound/oil constituent, thiol levels can vary from harvest year to harvest year, from growing region to growing region, but also from lot to lot. Typically, by far the highest 4-MMP levels can be found in Citra hops. Depending on harvest year, Citra might contain as much as 50 µg 4-MMP per kg hops. Then, there exist several varieties that contain less 4-MMP when compared to Citra, but still very high levels (approx. 10–25 µg 4-MMP per kg hops) when compared to other varieties. This group comprises of, for example, US-grown Mosaic and Simcoe, or Australia-grown Ella, and Galaxy.

What can also be seen in [Table 8.5](#) is that some varieties contain detectable levels of both 4-MMP and 3-MH. In most cases the 4-MMP concentration is higher than the 3-MH concentration, but other than that the concentration of these two volatiles do not correlate. An exception from the above said is the German variety Hallertauer Blanc, which contains very high levels of 3-MH and moderate levels of 4-MMP. Interested readers should refer to the work of [Reglitz & Steinhaus \(2017\)](#), which gives an overview on the 4-MMP levels across a wider range of hop cultivars and harvest years.

Next to free 4-MMP and 3-MH hops also contain cysteine and glutathione-bound thiols. These molecules are typically referred to as thiol-precursors, they are nonvolatile and odorless. While 4-MMP in hops is predominantly present in its free form, 3-MH is predominantly present in its

Table 8.5 Concentration of 4-MMP and 3-MH in selected hop cultivars. Please note that the thiols levels might vary from harvest year to harvest year, from growing region to growing region, but also from lot to lot. The information in this table might be used to differentiate hops that are typically high/low in 4-MMP or 3-MH.

Beer	4 MMP [µg/kg]	3 MH [µg/kg]
Citra (US)	33.0	13.6
Mosaic (US)	15.0	<0.5
Simcoe (US)	11.6	<0.5
Amarillo (US)	1.7	<0.5
Cascade (US)	0.6	<0.5
Centennial (US)	0.2	1.0
Apollo (US)	3.4	3.1
Eureka (US)	6.5	<0.5
Topaz (AU)	0.4	<0.5
Galaxy (AU)	12.2	6.0
Ella (AU)	25.6	7.0
Hallertauer Blanc (DE)	3.4	31.3

Table 8.6 Concentration of 4 MMP, 3 MH, and 3 MHA in 12 beer samples produced by means of different hop varieties.

Beer	4 MMP [ng/L]	3 MH [ng/L]	3 MHA [ng/L]	Hop variety
DE-SHIPA	<1.0	18.0	<1.0	Huell melon
DE-pale ale	<1.0	21.0	<1.0	Cascade, centennial, willamette
US-IPA	9.9	31.4	3.2	Galaxy, mosaic, topaz
US-IPA	8.0	49.5	<1.0	Amarillo, citra, mosaic, comet
US-IPA	33.3	36.5	1.0	Nugget, cascade, chinook, mosaic, simcoe
US-IPA	86.1	40.0	<1.0	Citra, mosaic
US-IPA	6.0	32.0	<1.0	Magnum, citra, mosaic, simcoe
US-IPA	1.1	44.1	<1.0	Azacca, millennium, galaxy,
US-IPA	6.7	69.8	3.8	Citra, centennial, apollo, ekuanot
US-double IPA 1	39.1	18.6	<1.0	Azacca, mandarina bavaria, mosaic
US-double IPA 2	75.9	130.4	9.5	Citra, centennial, ekuanot, nugget
US-NEIPA	189.1	68.1	14.1	Sabro, simcoe, citra
US-NEIPA	106.6*	37.3	<1.0	Cascade, citra, galaxy, mosaic, simcoe
US-NEIPA	238.9*	119.6	<1.0	Cascade, citra, mosaic, simcoe

bound form. Bound 3-MH is also present in barley and barley malt, which is why the widely used term “hop varietal thiols” for 4-MMP, 3-MH, and 3-MHA is inaccurate. There exist several publications that report thiol precursors in different hop varieties (e.g., [Gros et al., 2012](#); [Roland et al., 2016](#)). To enhance thiol levels in beer by releasing volatile thiols from odorless precursors, suitable yeast strains might be selected ([Gros et al., 2013](#); [Krogerus et al., 2021, 2023](#); [Michel et al., 2019](#)) Similar to [Table 8.4](#), which gives an overview on the concentrations of the main hop oil constituents in hoppy Ale, [Table 8.6](#) illustrates thiol concentrations across a range of hoppy Ales. The 4-MMP levels in these beers range from not detectable (<0.1 ng/L) up to 239 ng/L beer in a New England Style IPA. The concentration of 3-MH in beer is typically less variable and ranges from 18 ng/L in a single-hop IPA to 130 ng/L in a double IPA.

As can be seen from [Table 8.6](#) beers that do not contain 4-MMP might still contain 3-MH. This can be explained by the occurrence of 3-MH and 3-MH precursors in malt. Internal analyses results from VLB validated that a Lager beer bittered by addition of hop extract during the boil contains a baseline of approx. 15–25 ng/L 3-MH.

8. Flavor (in) stability and aroma changes in IPA

To meet consumer expectations, the desire of any brewer is to achieve reproducible and sufficiently stable beer flavor. Unfortunately, fresh beer is not in an equilibrium state, and except for brewpubs that sell their beers shortly after production, weeks or rather months can elapse between beer packaging

and consumption. During this period, the beer might be subjected to heat, light, and mechanical stress that all can trigger certain chemical reactions. These reactions can result in a) loss of fresh beer flavor and b) the appearance of an undesired stale character. Whether a) or b) are finally more important is hard to determine. It depends on the sensorial characteristics and chemical composition of a given beer and the conditions the beer stored under (Drost et al., 1990). In light Lager beer, one can safely conclude that the emerging stale character (more malty, sweet, papery, oxidized, etc.) is largely related to the formation of some trace level VOCs, such as aldehydes. The flavor and aroma of (hoppy) Ales are typically much more diverse and complex when compared to those of Lager beer. It is therefore not surprising that also the flavor and aroma changes in Ales appear more complex (Drost et al., 1990). With the commercial success of hoppy Ales, distribution of craft beer became less local and iconic craft beers are today sold (inter-) nationally. Hence, flavor stability has also become a concern of craft brewers who observed that hoppy Ale flavor can sometimes change very rapidly (weeks rather than months). Until recently, brewing science offered only very limited information concerning hoppy Ale flavor stability. One of the first published studies that characterized flavor changes in hoppy Ales was conducted by Barnette & Shellhammer (2019). They investigated the effect of storage temperature and beer oxygen levels on the flavor of dry-hopped ales. Their data showed that the aroma changes in dry-hopped Ales were linked to storage temperature: under cold storage conditions, which can be considered ideal to maintain beer freshness, Ale aroma changed from fresh pineapple, dank-cannabis, hoppy, and citrus to more tea, herbal, and spicy. An increase in storage temperature and elevated dissolved oxygen resulted in a loss of hop character and the appearance of malty aroma. Interestingly, their instrumental analysis of terpenes and terpenoids showed that the concentration of these primary hop-derived VOCs was not correlated with the loss of hop aroma found in sensory analysis. Therefore, the authors hypothesized that an increase in aldehydes and/or decreases in minor hop volatiles (i.e., esters) were responsible for aroma changes. Their conclusions were mostly confirmed by Kemp et al. (2021) who showed that IPAs stored at 3°C for 10-month retained higher tropical and citrus qualities when compared to their ambient stored counterparts. Again, storage at elevated temperatures resulted in beers with detectable malty, alcoholic, and undesired sweeter qualities. Instrumental analysis of hop-derived volatiles confirmed that terpenes and terpenoids were largely stable throughout storage, while hop-derived esters decreased upon storage. The instability of hop-derived esters was also reported by Rettberg et al. (2020) and Schubert et al. (2022, 2023). Rettberg et al. (2020) showed that the decrease in 2-MBIB (and other esters) is driven by both chemical and enzymatic reactions and that ester hydrolysis largely contributes to hoppy Ale instability. Schubert et al. (2022) presented the most comprehensive published study on hoppy Ale stability, in which a wide range of instrumental analyses were applied to fresh, naturally aged, and forced aged Ales. The results of instrumental analysis were correlated with quantitative descriptive analysis and confirmed that upon Ale storage, significant increases of staling aldehydes were observed, while the concentrations hop-derived VOCs (i.e., linalool and geraniol) were largely stable throughout storage. Ales with higher hop aroma intensity appeared more stable since they suppressed “oxidized” aromas. Dry Ales with higher attenuation were perceived less sweet and bold when they were aged. Unfortunately, there exists little published information on how thiols, more specifically their stability, influence aroma changes in beer. It is known from wine research that thiol instability triggers undesired aroma changes. Biendl et al. (2021) studied the stability of hop-derived thiols in New England IPAs. They found that the concentration of 4-MMP and 3-MH decreases significantly (–50% after 3 months) even when beer is stored cold. These findings

were confirmed by [Rettberg \(2024\)](#) who found that under ambient storage conditions the 4-MMP half-life time is only in the range of 6–12 weeks. Unfortunately, since no sensorial analysis was performed in both studies, no conclusion regarding the thiol reduction, the changes of other hop-derived VOCs, and aroma changes could be drawn.

Therefore, to comprehensibly illustrate some key aspects of Ale instability an unpublished dataset taken from a current research project is presented below. It includes the results from instrumental analysis and quantitative descriptive analysis by a trained panel. To first visualize sensorial changes in IPA, [Fig. 8.3](#) shows two spiderweb diagrams. Each of the spiderwebs includes the tasting results from an IPA which has been tasted fresh, after 24 weeks ambient storage and after 24 weeks of cold (4°C) storage. [Fig. 8.3A](#) shows the sensorial changes determined by quantitative descriptive analysis (QDA) using a generic beer tasting scheme targeting main beer quality attributes. What can be taken from [Fig. 8.3A](#) is that upon storage the scores for some attributes change and that there, not surprisingly, exists difference between the storage conditions. In this example, the ambient stored IPA had a less intense hop character, lower quality, taste, and balance, and received high scores for oxidation. [Fig. 8.3B](#) displays the sensory analysis results using a tasting scheme targeting differentiation of specific hop attributes.

What can be seen here is that some attributes are (significantly) affected by beer storage and by the storage conditions. Clearly, scores for Citrus and green-grassy decrease, while a slight increase in woody/aromatic scores is noted. Bitterness quality and aroma quality remain constant when beer is stored cold, while warm storage favors lower scores for the attributes. Even generalization is difficult in beer flavor chemistry, it appears that beers with intense hop aroma are somewhat less prone to changes in aroma. This can be explained by the fact, hop-derived VOCs have the potential to cover emerging stale aromas to some degree. This seems to explain, why brewers have historically targeted to balance hop and malt character. Upon aging, the quality of hop aroma changes. How rapidly these

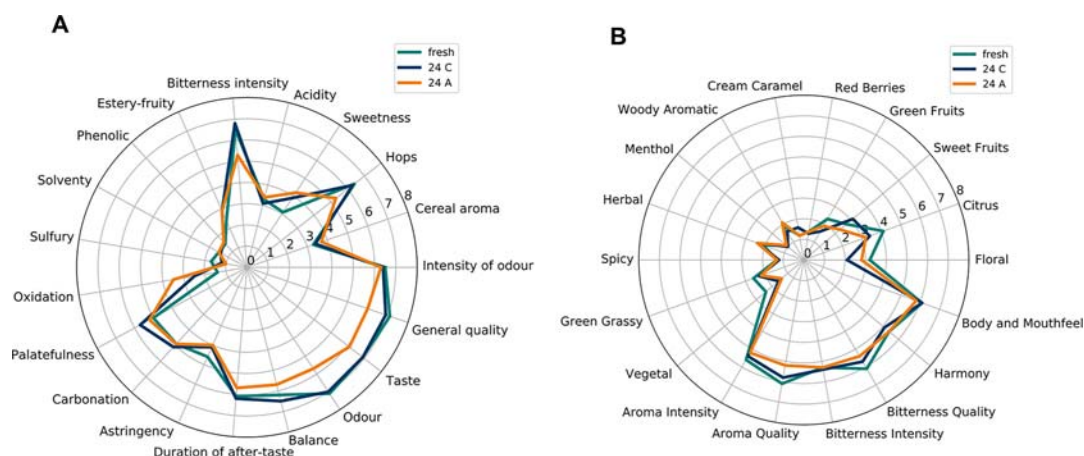


FIGURE 8.3

Spiderweb diagrams comparing the result of quantitative descriptive analysis of three IPA samples (fresh, stored for 24 weeks (ambient or cold)). (A) Sensory scores using a generic tasting scheme to scale basic beer quality. (B) Sensory score using a tasting scheme with attributes to describe hop character in beer.

shifts can be perceived sensorially and to which extent these reduce beer quality depends on fresh beer aroma and on how (long) the beer is stored. To gain a better understanding of aroma changes and to (potentially) implement measures to improve flavor stability, it is important to identify VOCs or groups of VOCs that are correlated with aroma changes. [Fig. 8.4A and B](#) illustrate the chemical and sensorial changes of an IPA upon aging. The changes are quite typically observed when hoppy Ales age, which is this example was chosen. [Fig. 8.4A](#) represents PCA scores plot in which the chemical and sensorial differences are displayed in a two-dimensional space. This plot explains >80% of the variation in the underlying dataset. Fresh samples and aged samples stored under different conditions (cold vs. ambient) and for different durations (fresh, 4, 16, and 24 weeks) are well separated. Fresh samples and samples that were stored for only 4 weeks are chemically and sensorially similar, while when samples are stored for 16 or 24 weeks the storage temperature has a significant impact on aroma and VOC concentration. This means, the temperature at which the beer is stored is highly influential and (by far) more important than the actual storage duration. [Fig. 8.4B](#) represents a PCA loadings plot in which the results of sensorial and chemical analyses are combined. The position of certain analytes or attributes in the plot indicates the relative abundance (VOC concentration or sensory score for a specific attribute) in fresh and aged samples. Looking at [Fig. 8.4B](#) shows that Aldehydes (e.g., 2-Methylpropanal and 2-Furfural) are more abundant in aged beer and are correlated with “Oxidation,” “Sweetness,” and “Cereal” aromas. Fresh or moderately aged (4 weeks) beer is characterized by higher levels of thiols (4-MMP, 3-MH, and 3-MHA), “Sulfury,” “Balance,” and “Hops” as well as by higher scores for “General quality,” and “Odor.”

[Fig. 8.4A and B](#) illustrate the chemical and sensorial changes for one selected IPA. The changes in VOC levels and sensory are somewhat specific for the IPA tested, but still represent changes commonly observed when comparing fresh/aged IPAs. The stability of beer toward aroma changes generally depends on many factors. Intrinsic beer flavor stability depends on the balance of pro- and antioxidants, which is influenced by raw materials, recipe, and oxygen uptake during production and filling. In the context of beer flavor stability also the thermal load during production is commonly listed to influence aroma changes/stability. Depending on the setup of a brewery, the product range, and desired shelf-life beer might be pasteurized after filtration or packaging. Pasteurization aims to prevent microbial spoilage of beer; in terms of aroma, this additional heating step is not generally regarded as beneficial for beer flavor and aroma. In terms of the aroma of hoppy Ales, a recent study ([Schubert et al., 2024](#)) found that pasteurization can be useful to stabilize the aroma of hoppy Ales. The stabilization can be attributed to the inactivation of yeast cells or enzyme that lead to the decrease in hop-derived esters. Lastly, beer packaging also plays a role in beer stability, oxygen uptake during the filling and light protection during distribution and storage are most important. Glass bottles and aluminum cans are the preferred packaging materials for beer. Brewers that use both types of packaging can sometimes observe the influence of pack type on beer aroma. The influence of pack type might come from oxygen uptake during filling, the influence of light during distribution, and the scalping of aroma compounds by either crown or can liners. Compared with the effect of temperature and time, the influence of pack type might be small, yet detectable shows analysis data for selected, hop-derived VOCs and aldehydes from a single-hop Pale Ale that has been filled in both cans and bottles and has then been stored for 24 months at room temperature. What can be taken from [Table 8.7](#) is that the behavior of most aroma compounds (increase over time, decrease in time) appears independent of pack type and mostly compounds whose formation (aldehydes) or oxidation (thiols) linked to oxygen uptake tend to reflect small differences.

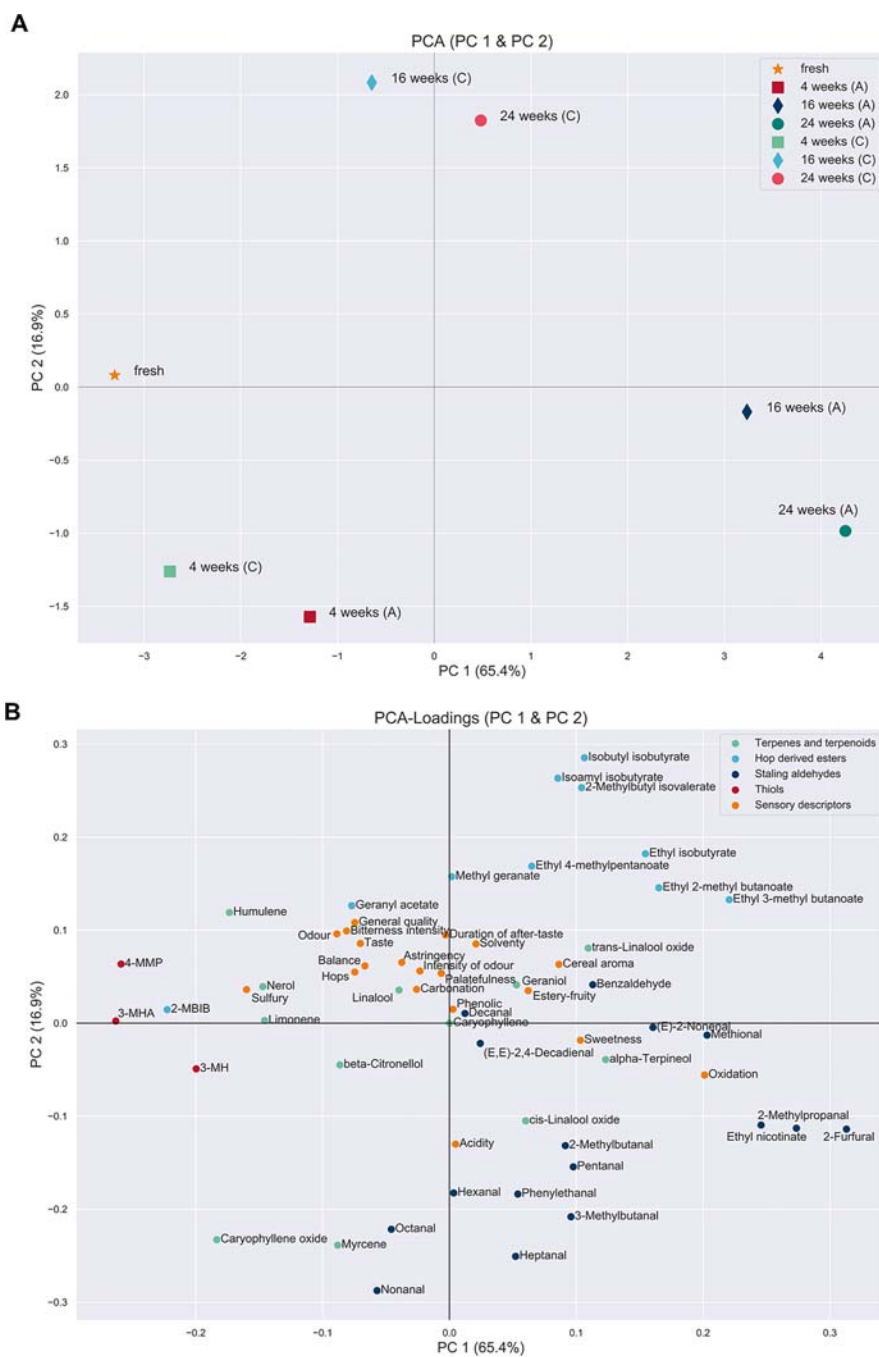


FIGURE 8.4

Principal component analysis of illustrating the chemical and sensorial changes in IPA stored for 24 weeks ambient (A) and at 4°C (C). (A) PCA scores plot illustrating the two-dimensional separation of beer over time. (B) PCA loadings plot illustrating the VOCs and sensorial attributes responsible for the clustering/separation shown in (A).

Table 8.7 Influence of pack type and storage time on the concentration of selected aroma compounds in hoppy ale.

	Bottle	Bottle	Bottle	Can	Can	Can
	Fresh	3 months	6 months	Fresh	3 months	6 months
4-MMP (ng/L)	9.2	2.7	<1.0	9.6	1.7	<1.0
Myrcene (μg/L)	19.7	4.9	4.2	25.0	12.6	11.8
Linalool (μg/L)	48.6	41.7	41.1	49.1	47.9	45.9
α-Terpineol (μg/L)	11.1	14.5	23.8	11.7	14.1	23.9
Citronellol (μg/L)	8.3	6.7	7.3	8.5	7.6	7.7
Geraniol (μg/L)	9.9	6.3	6.2	10.3	7.3	6.2
2-MBIB (μg/L)	18.8	18.1	12.5	17.3	16.5	13.0
Ethyl isobutyrate	10.2	15.1	11.6	11.1	14.8	13.5
2-Methylpropanal (μg/L)	6.6	17.1	27.3	4.65	12.67	22.70
2-Furfural (μg/L)	40.6	140.1	433.3	30.47	139.40	408.65
Methional (μg/L)	3.0	7.3	6.3	2.33	5.04	5.76
E-2-nonenal (μg/L)	0.05	0.10	0.10	0.05	0.08	0.06

9. Summary and conclusion

Aroma is a main quality characteristic of beer and is therefore very important for brand recognition and identification. Beer aroma results from a complex interplay between beer VOCs, the human olfactory system, but is also influenced by the nonvolatiles compound in beer that might favor retention or release of aroma compounds during drinking. The nonvolatiles compounds in beer, such as (poly) saccharides, AA, IAAs, humulinones, and polyphenols, modulate beer sweetness, mouthfeel, bitterness, and astringency. The VOCs in beer quantitatively represent a minor fraction, but qualitatively they comprise a complex group of compounds of >1000 substances ranging from ng/L to mg/L levels. Their overall concentration and their relative ratios in beer relate to beer style/brand and are finally responsible for its aroma. Raw materials and brewing regime allow to manipulate the concentrations of VOCs in beer and thereby to achieve beers with great variation in aroma. While, for example, industrial Lager beers feature light and crisp flavors, craft beer production aims for diversification, new, and sometimes extreme aroma and flavor sensations. The most relevant style category in craft brewing is the Ales category, which includes the popular American Pale Ale (PA), the American Indian Pale Ale (IPA), as well as associated styles such as the Hazy IPAs and the Session IPAs. These styles, sometimes with regional twists and local raw materials, are today brewed by craft brewers all over the world. The modern IPA is a hop forward beer style and in the last decade has become the signature style of craft brewing worldwide.

This book chapter has briefly summarized the key characteristics of IPA aroma and has highlighted the origin of aroma compounds. Measures to influence the concentrations of relevant aroma compounds by raw material choice as well as brewing technology were discussed. Well-balanced IPA aroma is characterized by the ratios of malt-, fermentation-, and hop-derived VOCs. Hop aroma can vary greatly in terms of intensity and quality; it is commonly described with terms such as floral, fruity,

citrus-like, woody, beery-like, piney, and resinous. The aroma compounds that are responsible for these aromas primarily are terpenoids, esters, and thiols. In hoppy ales, the concentration of terpenoids such as Linalool and Geraniol commonly ranges from 20 µg/L to >500 µg/L and esters such as 2-MBIB can be found in comparable concentrations. In contrast, thiols such as 4-MMP and 3-MH are only present in ng/L ranges but can still strongly and pleasantly affect beer aroma. Hop varieties such as Citra and Simcoe that contain elevated levels of these thiols are desired by many craft brewers. To brew hop aromatic beers, brewers have the choice between approx. 250 hop varieties that differ in their oil content, oil composition, and resin content. In single-hop beers, only one hop variety is used to add aroma and bitterness, while in most cases brewers chose combinations of different hops to boost aroma complexity. Next to hop variety choice, the quantity of hops added to wort/beer, the timing of addition, and how the solid hop matter is brought in contact with beer to finally extract hop resins and aroma are of importance. Adding hops to beer that contains viable yeast cells or yeast enzymes hop aroma changes, which occur due to biotransformation and the removal of nonpolar VOCs by yeast or through stripping with CO₂. Dry hopping in presence of yeast might be less effective regarding the quantitative transfer of hop components into beer, from a qualitative perspective it can achieve fruitier and more citrus-like beers in which the resinous and piney character might be reduced.

Even though IPA aroma is driven by hops, malt aroma and fermentation-derived aroma compounds are still important. They add aromas to beer, which cannot be brought in by hops. To support and showcase hop aroma, a balance between hop-derived VOCs and malt/fermentation-derived VOCs is needed. Specialty malts introduce for example Aldehydes, Furans, and Pyrazines that exhibit malty, caramel-like, or roast aroma to beer, even if they are only added at very low percentages of the overall grist. IPAs are fermented relatively warm (15–20°C). These elevated fermentation temperatures can result in high levels of fermentation by-products. To avoid this, brewers tend to use yeast strains that genetically do not produce excessive levels of esters and higher alcohols. Next to choosing a suitable yeast strain, yeast management and operations in the cellar aim to produce clean base beers, with low levels of acetaldehyde and diacetyl. Even though hop aroma compounds can compensate one or the other of flavor, producing clean base beers, that is, base beers without any aroma defects, is important to meet consumer expectations on the long run. As with any other beer styles, hoppy Ales in general and IPA in specific are subject to the chemical and sensorial changes when they age. Typically, IPAs become slightly sweeter, more estery and malty upon storage. Hop aroma becomes less intense and its character changes, bitterness might become harsh, astringency increases. Among the hop-derived aroma compounds, terpenoids are comparably stable, while ester- and thiol levels decrease strongly and rapidly when beer is stored ambient.

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Craft beer safety: Control of critical points of the production process

9

Dolores González de Llano, Magaly Rodríguez-Saavedra and M. Victoria Moreno-Arribas

Institute of Food Science and Research (CIAL), CSIC—UAM, C/ Nicolás Cabrera 9, Madrid, Spain

1. Introduction

Beers are one of the most consumed beverages worldwide, ranking third in popularity after tea and coffee. It is also the most consumed alcoholic beverage, with an annual production of almost two billion hectoliters by 2022 ([statista.com](https://www.statista.com)). The global beer market is primarily dominated by industrial beers, which are predominantly produced by multinational companies in large-scale breweries following standard procedures and actions focused on hygiene and product safety. This was done to limit spoilage risks and variations in the final product ([Rodhouse & Carbonero, 2019](#); [Vrellas & Tsiotras, 2015](#)). In recent years, craft beer has emerged as a popular choice among consumers because of its distinctive organoleptic properties, which are based on numerous aromas and flavors, and are perceived as high-quality.

The brewing process involves the preparation and boiling of wort, cooling, fermentation, maturation, filtration and/or stabilization, and packing. Craft beer is typically produced at local micro-breweries, where cereals, primarily malted barley, are brewed and fermented and hops are added for flavor. This process does not involve filtration or pasteurization. Nevertheless, the production of craft beer adheres to a nonstandardized process, resulting in the development of novel styles incorporating unconventional starch-rich ingredients, such as corn or rice, and the incorporation of fruits for distinctive industrial purposes. Beers are classified into numerous styles based on their properties, including alcohol content, color, bitterness, clarity, flavor, and ingredients. Craft breweries are primarily engaged in the production of traditional ales, lagers, and other beer styles that do not align with these two main categories. They compete in the market, based on the quality and diversity of their products.

Craft brewers have endeavored to produce beers with superior sensory quality and safety. However, as most craft beer is still manufactured globally under traditional open, nonsterile, and simple equipment conditions, the risk of spoilage and hazards associated with metabolic by-product formation is heightened. The beer production and storage processes may be affected by microbial, chemical, and physical contamination, which may have detrimental effects on consumer health ([Awuchi et al., 2022](#); [Ciont et al., 2022](#); [Rodhouse & Carbonero, 2019](#)). From a hygienic standpoint, microbial contamination can differentiate between industrial and craft beers, which are rarely pasteurized or filtered. Consequently, microbial spoilage of beer represents a

significant concern for brewers, as it can lead to potential safety hazards, the production of undesirable aromas and flavors in the final product, and substantial financial losses (Rodríguez-Saavedra et al., 2020). Many spoiled microorganisms are highly adaptive and can grow in a beer-hostile environment, reducing the shelf life of beer and producing haze, sedimentation, acidification, turbidity, ropiness, and off-flavors (Suzuki et al., 2008). The metabolism of spoiled microbiota during beer fermentation can also result in the formation of chemical spoilage compounds, including biogenic amines (BA) and other nitrogen-derived metabolic hazards, such as ethyl carbamate (EC) and nitrosamines. Furthermore, fungal contamination of raw beer materials can also result in the production of mycotoxins, which have adverse effects on human health (Awuchi et al., 2022; Schabo et al., 2021). Given their potential toxic effects, BA (e.g., histamine) and mycotoxins have attracted special attention from the FDA and EFSA agencies (EFSA, 2011; EFSA, 2019). New brewing styles with intrinsic factors that are less stressful for potential microbial contaminants (lower alcohol and higher hop content), as well as the introduction of new raw materials (fruits and cereals) with their respective microbiota, can also favor beer spoilage. Agricultural practices, including the extensive use of pesticides (insecticides, fungicides, and herbicides) and the dynamic use of phytosanitary products under changing climatic conditions, could cause the presence of toxic residues derived from barley and hops, which could affect beer safety and quality (Pérez-Lucas et al., 2023). Other forms of contamination that can adversely affect beer quality and safety include the presence of heavy metals (Tian et al., 2022) and other emerging contaminants, such as plastic particles (Jiménez-Arroyo et al., 2023). Although the maximum tolerable level of contaminants in beer varies by country legislation (Avîrvarei et al., 2023; Ciont et al., 2022), beer safety is a global concern for the brewing industry, which must comply with hygiene legislation, consumer preferences, and sustainability, as well as the expectations of all authorities, who are focused on food and beverage safety and human health.

To ensure food safety, adherence to good manufacturing practices (GMP) is required, which often requires the implementation of a safety management system and good hygiene practices. In addition, the principles of hazard analysis of critical control points (HACCP) must be observed. Regarding the brewing process, HACCP places greater emphasis on preventive measures than on end-product evaluation for control. Besides identifying and preventing hazards, precautions are recommended for beer control to protect beer and consumers from hazards and contaminants (Awuchi, 2023; Kourtis & Arvanitoyannis, 2001). Furthermore, specific regulations and standardized analytical methods are required for all the identified contaminants. Consequently, clinical studies are required to elucidate the health and environmental implications of physical, chemical, and biological contaminants in beer, as well as the synergistic effects of some chemical contaminants in beer, and to analyze their economic impacts (beer production, health costs, etc.).

This chapter presents a comprehensive review of the potential contaminants in beer, their sources, and their implications for human health, including control practices and current risk reduction or elimination strategies. It specifically addresses the unique characteristics of craft brewing and the potential associated contamination risks. First, after a review of common beer-spoilage microorganisms and the potential for more diverse and variable microbiota associated with craft beer, the steps most susceptible to contamination that are considered CCP in craft production are described (Section 2). Next, the most commonly identified chemical contaminants, both of biological/natural origin and

added chemicals, are detailed at several points throughout the beer chain, as well as undesirable physical and emergent hazards (Section 3). Subsequently, the principal benefits and barriers associated with HACCP in the production of safe craft beer are identified and discussed, along with a summary of the operating conditions and CCP (Section 4). A summary of the identified gaps and future directions is provided in Section 5.

2. Microbial hazards and control points

One of the most significant factors contributing to the large-scale production of beer is its high microbiological stability, which is attributed to the hostile environment created by most microorganisms. The combination of intrinsic factors that aid in antimicrobial effects is ethanol content, hop-bittering compounds, low pH, elevated carbon dioxide, low oxygen concentration, and lack of nutritive substances. These factors contribute to the microbiological stability of beer, resulting in a few cases of microbiological incidents in breweries with high standards of hygiene (Rodríguez-Saavedra et al., 2020; 2021a). Ethanol and hops interfere with essential cell membrane functions of microorganisms. Low pH hinders enzyme activity. The lack of nutrients and oxygen starves many potential pathogens. Elevated dissolved carbon dioxide lowers pH, inhibits enzymes, affects cell membranes, and creates an anaerobic environment (Menz et al., 2011). Conversely, extrinsic factors, such as acidification of malt, mashing, wort boiling, pasteurization, filtration, and cold storage, represent stages of the brewing process that reduce potential contamination and proliferation of microorganisms (Menz et al., 2011). However, not all of these factors apply to beer-style processing (Turvey et al., 2017). In contrast to large-scale beer, craft beer is more susceptible to microbial spoilage because it is rarely pasteurized or sterile filtered. Moreover, craft breweries frequently lack the resources of well-resourced microbiological laboratories, and cold storage is not guaranteed throughout the distribution and retail chains (Bose et al., 2021). In contrast, large-scale breweries rely on stringent practices to limit spoilage risk and variation in packaged beer. There is considerable variability among the different beer styles regarding their susceptibility to microbial spoilage. Beers with low acidity, bitterness, alcohol content, and added sugars are particularly prone to contamination (Vaughan et al., 2005). The microorganisms present depend on the beer-style and production process. Currently, craft breweries produce various beer styles within the same factory; however, they often lack knowledge regarding which styles are the most susceptible to microbial contamination. Furthermore, low-alcohol and alcohol-free beer, as well as novel beer styles incorporating novel raw materials and their respective microbiota to achieve a distinctive beer aroma and taste, or beers with modified selective microbiological parameters, are potentially more susceptible to microbial spoilage (Avîrvarei et al., 2023).

2.1 Lactic acid bacteria and other fermentative bacteria associated with beer spoilage

The high incidence of beer spoilage in the craft beer industry is concerning. Currently, there is a lack of information about spoilage microorganisms, their possible sources or routes of entry, and their relevant

characteristics. This lack of information hinders our understanding of the associated microbiological quality and food safety risks (Menz et al., 2011). There are no statistics available, but it is estimated that gram-positive bacteria belonging to the lactic acid bacteria (LAB) group cause 70% of contamination in beer and the brewery environment (Garofalo et al., 2015; Rodríguez-Saavedra et al., 2020). LAB have been demonstrated to enter a viable but nonculturable state in response to conditions commonly encountered in beer, including low temperature and oxidative stress (Oldham & Held, 2023). These bacteria are undetectable by traditional detection methods using selective media. Moreover, many have developed resistance mechanisms to antimicrobial compounds derived from hops.

LABs produce an excess of lactic acid and diacetyl, which gives the beer a butter-like flavor and can cause turbidity and deterioration in quality, ultimately resulting in consumer rejection and product waste. It is possible for other microorganisms to delay fermentation by consuming the nutrients needed for fermenting yeast without affecting taste or aroma (Rodhouse & Carbonero, 2019). In craft breweries, the species with the highest incidence of spoilage are *Lactobacillus brevis* and *Pediococcus damnosus*. However, other detrimental species have also been reported, including *L. lindneri*, *L. paracasei*, *L. plantarum*, *L. backii*, and some *Leuconostoc* species. Half of the documented microbiological incidents have been attributed to secondary LAB contamination (Rodríguez-Saavedra et al., 2020; Vaughan et al., 2005). A novel beer-spoilage bacterium, *Lactobacillus acetotolerans*, has been frequently found in craft beer breweries, imparting an undesirable and unpalatable “buttery” taste and oily mouthfeel to beer (Bregow, 2017; Luan et al., 2023). The prevalence of LAB in American craft beer has been estimated at 15.0% (White, 2008), whereas in Australian microbrewed beer, the rate is reported to be 27.5% (Menz et al., 2010). Furthermore, contamination of canned beers by unidentified microbes has been observed to result in gas production (Rodhouse & Carbonero, 2019).

The production of exopolysaccharide (EPS) by LAB is important in brewery settings, as it may serve to anchor bacterial cells on surfaces, facilitating the initiation of potential biofilms. Biofilms can increase the viscosity of beer, alter its texture, cause turbidity, or result in undesirable physical properties such as ropelike characteristics (Fraunhofer et al., 2018). Despite daily cleaning and sanitation, microbial adhesion and biofilm formation occur in breweries (Bittner et al., 2016). This is because biofilms enable microbes to cope with physical and chemical stress factors, such as mechanical cleaning, disinfection, UV radiation, and desiccation (Wagner et al., 2021). In particular, wet sections of the filling area are conducive to the development of bacterial biofilms owing to the presence of moisture and nutrients. Furthermore, colonization of the area by spoilage bacteria increases the risk of contamination of packaged beer.

Among the gram-negative bacteria that deteriorate beer, there are both anaerobic and aerobic species. In high-volume industrial breweries that maintain low dissolved oxygen levels, strict anaerobic bacteria represent a significant challenge. This group includes the genera *Pectinatus*, *Megasphaera*, *Propionispira*, and *Selenomonas* (Matoulková & Kubizniaková, 2014). Although most craft breweries maintain high oxygen levels that make it impossible for them to grow and recur, *Pectinatus* is a harmful bacterium for beer because of the production of noxious off-flavors that cause serious commercial damage to breweries (Rodríguez-Saavedra et al., 2021b). Furthermore, the use of vegetables as ingredients in beer production may result in the introduction of new species of

Pectinatus into the brewery environment, creating new spoilage risks. The control of *Pectinatus* is a recurring topic in beer research; however, there is a paucity of scientific information concerning its occurrence and the factors affecting its presence and survival mechanisms in the brewing process. Regarding facultative aerobic species, these include *Zymomonas*, certain species of *Enterobacteriaceae*, and acetic acid bacteria such as *Acetobacter* and *Gluconobacter* (Ashtavinayak & Elizabeth, 2016). These microbial contaminations in beer are not well documented, yet they are not neglected due to their important health implications (Wagner et al., 2021). Moreover, spore-forming bacteria in beer that possess hop resistance genes have been identified, predominantly in lager and nonalcoholic beer (Munford et al., 2017).

Brewers have used various methodologies to ascertain the susceptibility of bacteria to spoilage. However, those reliant on species identification have encountered challenges because not all strains of a given species can spoil beers. Furthermore, instances of new bacteria being identified as spoilage agents have been reported (García-García et al., 2017). In addition, the growth of LAB in beer is a multifactorial process that relies on a combination of plasmid-encoded genes that dictate spoilage ability (Bergsveinson & Ziola, 2017). The principal hop resistance-related genes were *hpaA*, *hpaC*, *hpaA*, and ORF5. However, the combination of these genes is not yet well understood, and some studies have reported that the absence of these genes does not necessarily indicate that bacteria lack spoilage abilities (Matoulková & Kubizniaková, 2014). Given the uncertainty and associated costs, craft brewers have adopted the practice of bacterial inoculation into beer as a definitive means of gauging true spoilage potential. Rodríguez-Saavedra et al. (2021a, b) proposed implementing predictive tools for craft brewers. These tools were developed using a mathematical model, the growth/no-growth (G/NG)-type binary logistic model. This approach enables the determination of the susceptibility of beer to microbial growth regarding the principal physicochemical data of the brewery's routine control, despite the identification of the potentially contaminating strain. Regarding methods for the detection and identification of microorganisms that spoil craft beer, quantitative polymerase chain reaction (qPCR) is a targeted method of detection and identification that may be useful in most craft breweries. Currently, the cost of sequencing the 16S rRNA gene, FISH-FCM, and MALDI-ToF-MS is prohibitive for most craft breweries (Oldham & Held, 2023).

2.2 Yeasts and fungi beer spoilage

Brettanomyces and *Saccharomyces* are the most common yeasts that spoil beer products. Two notable examples are *S. diastaticus* and *B. bruxellensis* (Suiker & Wösten, 2022). These spoilage yeasts have been identified within biofilms in breweries; however, the source of these yeasts from the environment remains unclear. This knowledge may provide insight into the prevention of spoilage yeast entry at beer production sites. These data indicated that *S. diastaticus* and *B. bruxellensis* exist in mixed biofilms in the presence of *Candida* and/or *Pichia* spp. (Suiker et al., 2021). In particular, *B. bruxellensis* is responsible for the spoilage of beer through the production of acetic acid and highly volatile phenolic compounds, including 4-ethylguaiacol and 4-ethylphenol. *Dekkera* (*Brettanomyces* in its anamorphous form) can be harmful when bottling processes have been conducted and oxygen is conserved (Sohlberg et al., 2022). *Saccharomyces cerevisiae* var. *diastaticus* (*S. cerevisiae* var. *diastaticus*) is the most dangerous wild yeast species of *Saccharomyces* owing to its capacity to cause

significant disruption to beer during production and distribution. These results indicate that *S. diastaticus* is a relatively uncommon variety of *S. cerevisiae* in nature, but it can be readily isolated from biofilms in breweries. This species can secrete extracellular glucoamylase (1,4- α -D-glucan glucohydrolases; EC 3.2.1.3), which is responsible for the conversion of complex carbohydrates. These include dextrin, starch, and simple sugars. A change in the carbohydrate profile of beer could cause continued fermentation at different stages of the production process, including after the beer is packaged. Its superattenuating ability causes increased carbon dioxide concentrations, beer gushing, and potential for bottle explosion, besides changes in flavor, sedimentation, and increased turbidity (Begrow, 2017; Meier-Dörnberg et al., 2018). Moreover, non-*Saccharomyces* spoilage species belong to the genera *Pichia*, *Hanseniaspora*, *Torulaspora*, and *Williopsis* (Stewart, 2016).

The literature on fungal communities in breweries is limited, with most studies using culture-dependent methodologies. Fungal contamination of beer can cause adverse effects, including reduced gas production and the formation of mycotoxins, which can pose a risk to the health of beer consumers (Mastanjević et al., 2018). The mycelia or spores of molds (filamentous fungi) can infect beer raw materials, primarily grains during cultivation and storage, malted cereals, or the final product. Fungi that cause beer contamination originate from the raw material and belong to the genera *Fusarium*, *Aspergillus*, *Alternaria*, and *Penicillium*. These filamentous fungi produce secondary metabolites, namely mycotoxins, besides those produced by *Rhizopus* and *Cladosporium*. Mycotoxins are pervasive in food products, including cereals, fruits, vegetables, and beverages. It is well established that mycotoxins can produce severe toxic effects, including carcinogenic, genotoxic, teratogenic, nephrotoxic, and hepatotoxic effects (section 3). These effects have become a serious issue for human health (Peters et al., 2017; Rodríguez-Carrasco et al., 2015).

3. Chemical and physical contaminants

3.1 Hazards from microbial metabolism and other chemical contaminants

The beer fermentation process involves complex microbiota, including harmful microorganisms. Nitrogen-derived metabolic hazards are caused by extracellular enzymatic reactions, incomplete metabolism, and spontaneous reactions involving nitrogen-containing compounds during beer fermentation (Liao et al., 2023). Nitrogenous hazards, including BA, ethyl carbamate, nitrosamines, and heterocyclic amines, have been reported to potentially pose a risk to dietary safety and human health (Dong et al., 2020; Xian et al., 2019).

3.1.1 Biogenic amines and other nitrogen-derived metabolic hazards

Given concerns regarding food safety, it is notable that beer and other fermented beverages and foods, including wine, cheese, and meat, have a higher probability of accumulating higher concentrations of BA than nonfermented foods (Alvarez & Moreno-Arribas, 2014). The most common sources of BAs in beer are raw materials, which account for over 60% of the total BAs present. They are naturally present in the wort, hops, and mainly in the malt (Lorencová et al., 2020). Furthermore, contaminating microbiota (predominantly LAB) can cause the formation of BA during the brewing process. The most frequently detected BAs produced by decarboxylase-positive microorganisms (LAB) are histamine,

cadaverine, putrescine, 2-phenylethylamine, and tyramine. Spermidine, 2-phenylethylamine, and spermine are naturally found in malts, worts, and hops. Putrescine can originate from both internal and external sources, whereas histamine is considered the principal compound that indicates external contamination. Given the paucity of published research on bacterial populations involved in BA production and determination in craft beers, it has been a significant challenge for food scientists in recent years (Gil et al., 2022; Palomino-Vasco et al., 2019).

In individuals with healthy intestinal function and under normal conditions, dietary BA are primarily metabolized at the intestinal level by amino oxidase enzymes, which serve as the primary enzymatic activities in regulating its passage into the systemic circulation (Comas-Basté et al., 2020). BA intolerance is a disorder resulting from the inability of amino oxidase enzymes to metabolize dietary BA in the intestine. Although BAs are bioactive compounds with relevant physiological activities, the consumption of high concentrations of BA (i.e., histamine) may produce a wide range of nonspecific symptoms, including gastrointestinal effects, headaches, hypotension or hypertension, kidney failure, anaphylactic shock, and even neurological functions (Kalač, 2014; Ruiz-Capillas & Herrero, 2019). It is important to consider all potential sources of BA in the dietary intake, including food and beverages. This is relevant for consumers who may be at risk of impaired metabolic activity, or for patients undergoing treatment with monoamine oxidase inhibitor drugs. In particular, the high worldwide consumption of beer (1950 million hectoliters) has compelled producers to guarantee both quality and food safety given consumer demand for high-quality products (Angulo et al., 2020). Furthermore, the ethanol present in beer may inhibit monoamine oxidase, which is necessary for BA elimination (Poveda et al., 2017).

It has been demonstrated that LAB metabolism represents the primary source of BA in beer (Rodríguez-Saavedra et al., 2020; Romero et al., 2003). Furthermore, prolonged storage of the product can facilitate BA formation in beers. In LAB, the presence of BAs indicates a defense mechanism during fermentation. However, the production of specific microbial BAs, such as histamine, tyramine, putrescine, cadaverine, and phenylethylamine by other microorganisms, including *Escherichia*, *Enterobacter*, *Pseudomonas*, *Salmonella*, *Shigella*, *Clostridium perfringens*, and *Streptococcus*, is commonly associated with poor hygiene and technology (Vinci & Maddaloni, 2020). Despite the controversy surrounding the issue of overexposure levels, no official concentration limits have been set for histamine or tyramine in beers, the most harmful BA due to their potential allergic or immune response in sensitive individuals. The most effective method for mitigating their adverse effects is through the implementation of prevention strategies, including the selection of LAB with low decarboxylation activity or the adoption of GMP that minimizes favorable conditions for the production of BA (Visciano & Schirone, 2022). The aminogenic potential exhibited considerable heterogeneity within the same species, in accordance with the hypothesis that the presence of amino acid decarboxylase genes is a strain-dependent trait in LAB (Álvarez & Moreno-Arribas, 2014). Although some isolates possessed genes associated with BA formation (*tdc*, *hdc*, or *odc*), they did not produce the corresponding BA in beer (tyramine, histamine, and putrescine, respectively). This phenomenon is likely because of the influence of hop bitter acids, absence of essential amino acids, lack of optimal environmental conditions for decarboxylation, and/or presence of silent genes (Álvarez & Moreno-Arribas, 2014). Moreover, the formation of BAs is related to the amino acid content. Consequently, several stress factors, such as intensive nitrogen fertilization, mold infestation, and other parameters

affected by climate or soil, may have a significant influence on BA content (Moreno-Arribas & Polo, 2008).

Ethyl carbamate (EC) and N-nitrosamines (NAs) are nitrogenous contaminants that can form in fermented alcoholic beverages (Xian et al., 2019). These contaminants possess carcinogenic and toxic properties. Beer contamination is also influenced by the raw materials used. The presence of urea and amino acid precursors in malt can cause the production of EC when metabolized by yeast during fermentation. Although microbial contamination by *Staphylococcus danseri*, *Lactobacillus acidophilus*, and *L. brevis* seems to be the primary source of EC in beer, other factors also contribute (Abt et al., 2021; Du et al., 2018). Xian et al. (2019) developed an approach for simultaneous analysis of EC and NAs in beer and yellow rice wine, which detected EC in all samples. However, the concentrations of the 12 NAs were below their respective limit of detection (LOD). In this context, the presence and concentration of potentially toxic carbonyl compounds and furan derivatives (acetaldehyde, acrolein, formaldehyde, and furfuryl alcohol) were detected at all stages of production in ale and lager craft beers, along with EC and furfural concentrations below the detection limits (Hernandes et al., 2020). Contamination of nitrogenous hazards in fermented beer has been reported to be generally low because of the closed fermentation and pure culture of beer (Visciano & Schirone, 2022). Furthermore, prolonged storage of beverages can facilitate the development of EC contamination, with EC levels in other types of alcoholic beverages exhibiting the highest levels, below 20.0 µg/L, owing to their shorter storage periods (Abt et al., 2021). Consequently, monitoring of toxic carbonyl compounds, such as EC, in commercial fermented alcoholic beverages should be intensified.

3.1.2 Mycotoxins

Mycotoxins are secondary fungal metabolites found in food and beverages. They are produced mainly by *Fusarium* species, including trichothecenes of type A (T-2 and HT-2 toxins) and type B (deoxynivalenol: DON; and nivalenol, fumonisins, and zearalenone). Other genera that produce mycotoxins include *Aspergillus* (aflatoxins: AFs; ochratoxin A: OTA), and *Penicillium* (OTA). These toxins are of concern to human and animal health because of their ability to produce severe toxic effects (carcinogenic, genotoxic, teratogenic, nephro-, and hepatotoxic effects) (Awuchi et al., 2022). The most prevalent occurrence of mycotoxins is in cereals (wheat, barley, corn, and rice), fruits, and vegetables. Contamination of barley grains and malt, particularly by *Fusarium* species, can rapidly compromise the safety and quality of malt and beer, as mycotoxins cause excessive foaming of the beer, leading to a brimming impact (Pascari et al., 2018; Powell & Diacetis, 2007). Schabo et al. (2021) examined pertinent aspects of raw material mycotoxins in artisanal beer, as well as the production and fate of mycotoxins during the beer brewing process. It has been reported that fungi contaminating cereals can infect the field or during storage, and that mycotoxins are produced during beer brewing due to malting conditions that favor their production. The malting step (germination and mashing at 45 °C) can transfer mycotoxins from the grist to the wort, resulting in a significant increase in their concentration. In contrast, a slight reduction has been observed during fermentation (Pascari et al., 2018). Consequently, the most efficacious strategies for preventing the contamination of beer with mycotoxins may be to eliminate the growth of toxigenic fungi and implement robust agricultural and postharvest mitigation measures. Several studies have demonstrated the presence of mycotoxins in craft brewers, although the findings are still debated (Awuchi et al., 2022; Bogdanova et al., 2018; Piacentini et al.,

2017). Peters et al. (2017) evaluated the mycotoxin profile of 1000 beer samples from 47 countries, particularly focusing on craft beer. Their findings indicated that some mycotoxins were present at concentrations above the tolerable daily intake. It can be reasonably assumed that malting barley and malt are not the primary sources of aflatoxin exposure for consumers if they are properly dried, and secondary contamination during storage is avoided (Benešová et al., 2012).

The most frequently reported mycotoxins in barley and beer are deoxynivalenol (DON), zearalenone (ZEA), HT-2, and T-2, along with aflatoxin B1 and ochratoxin A (OTA) (Mateo et al., 2007; Rodríguez-Carrasco et al., 2015). Aflatoxins (AFB1, AFB2, AFG1, and AFG2) are classified as carcinogenic to humans (Group 1) and are produced mainly by species of the *Aspergillus* genus, particularly *A. flavus* and *A. parasiticus*. AFB1 and AFB2 are frequently identified as contaminants of cereals (maize, rice, wheat, sorghum, and barley) and could be found in beers, particularly maize beers (Awuchi, 2023; Schabo et al., 2021). Malting conditions may facilitate the production of thermostable mycotoxins, such as OTA and AFs, which are resistant to beer brewing and persist in the final product. A high level of mycotoxin contamination was not present in beers in different studies, and the values found should not affect consumer health (Ciont et al., 2022; González et al., 2023). Mycotoxins may be removed during the brewing process, which involves steeping, kilning, mashing, fermentation, and clarification. These processes may have an inhibitory effect on mycotoxins or result in their destruction after thermal treatment. However, deoxynivalenol (DON) is chemically stable and heat-resistant during the technological process and is present in the final beer. The maximum tolerable levels of mycotoxins in beer vary across countries. The European Commission established a tolerable daily intake, with a temporary value of 1 mg/kg body weight per day for DON and 2–4 ppm for FB1, which is unusual for moderate beer to reach the maximum value of mycotoxins (Rodríguez-Carrasco et al., 2019). Finally, to prevent contamination by mycotoxins during the brewing process, several physical, chemical, and biological treatments have been proposed. Contamination of beer by fungi may be reduced by the addition of a LAB culture, which acts as an antifungal agent. This is because LAB exhibit environmental sensitivity and sustainability and some yeast strains can bind mycotoxins (Pascari et al., 2018; Tian et al., 2022). The paucity of data on the presence of mycotoxins in craft beer and the inability to detect modified mycotoxins using conventional methods, coupled with the current absence of established limits, underestimate the potential risks to human health. In addition, regulatory agencies must strengthen the laws and define limits based on scientific data regarding the transfer of mycotoxins throughout the artisanal beer brewing process.

3.2 Physical and emergent contaminants

3.2.1 Heavy metals

Heavy metals in beer pose a significant health risk because of their toxic effects on the human body. Once these metals enter an organism, they are oxidized to form stable bonds with enzymes or protein molecules (Scutarușu & Trincă, 2023). Bioaccumulation of heavy metals results in a multitude of deleterious effects on tissues and organs, including disruption of cellular processes, such as proliferation, differentiation, damage-repairing mechanisms, and apoptosis. Consequently, regular intake of these metals can contribute to the development of new illnesses or exacerbate existing ones. Some metals, such as iron (Fe), copper (Cu), and manganese (Mn), play nutritional roles and are vital for

biological processes. However, an excessive intake may be toxic. Nonessential metals such as lead (Pb), mercury (Hg), and cadmium (Cd) can be hazardous even at low concentrations, as they can cause a range of physiological and neurological disorders. The expansion of industrial activities, car traffic, and economic globalization over the past century has led to a significant increase in the environmental concentration of heavy metals worldwide. Furthermore, the intensification of agriculture, which involves the use of various fertilizers, herbicides, and pesticides, can also be a significant source of heavy metals in beer raw materials, including water, yeast, and raw materials such as cereals, rice, maize, and hops. These contaminants do not dissolve and accumulate in the soil, plants, and water sources, representing a significant concern because of their potential to cause extensive human exposure (Izah et al., 2017; Scutarașu & Trincă, 2023). Consequently, hazardous metals can be transferred from the source ingredients and surrounding environment to the final beer product and brewing residues. Other sources of metal contamination include exogenous sources, such as heavy metals released from industrial equipment used for brewing owing to the high temperatures and long maturation times of green beer. Inherent elements from the added chemicals or contaminants may also contribute to metal contamination (Voica et al., 2015). Consequently, the concentration of metals in beer is primarily determined by the chemical composition of the raw materials and their capacity to dissolve during the brewing process. As most of these toxic metals remain in the residuals (spent grains, hot trub, and even yeasts) (Passaghe et al., 2015), only a negligible fraction can be detected in beer, which does not represent a health risk (Scutarașu & Trincă, 2023).

The overall composition of beer metals is also an important factor in beer oxidation, which affects the brewing process and beer quality in terms of flavor stability and haze (Mastanjević et al., 2018). They can facilitate the formation of radicals in beer even in the absence of oxygen. Iron and copper exhibit chemical properties that can be described as redox systems; however, they have the potential to cause damage to the final product in terms of flavor stability. Manganese is a significant yeast nutrient that occurs at higher concentrations in the final beer because of its stability. Nevertheless, its influence on beer flavor stability and haze formation is less pronounced than that of other metal ions and peroxides, which can cause the rapid development of a stale aroma (Zufall & Tyrell, 2008). Another concern is the contamination of beverages with aluminum (Al), which are metabolized in the body and can cause numerous major illnesses when consumed in excess. An increase in the number of individuals with neurological disorders has been reported. Most beer Al originates from tin-made processing equipment and beer containers (barrels, cans), imparting a metallic and bitter flavor profile with no discernible changes in aroma compound profiles (Blanco et al., 2010). The significance of these elements in beer staling provides a rationale for a distinct approach to the selection of raw materials. Several chemical processes are available for the removal of toxic metals, including precipitation, electrodialysis, coagulation, and flocculation. In addition, there are adsorption-based methods and bioremediation methods using the brewing cells of *S. cerevisiae* (Soares & Soares, 2012). However, most of these procedures are expensive and difficult to perform (Scutarașu & Trincă, 2023).

3.2.2 Microplastics

Contamination of the environment with micro- and nanoplastics (MNPs) particles and their putative adverse effects on organisms, ecosystems, and human health has received increasing scientific and public attention (Ramsperger et al., 2023). Although there is no official lower size limit for microplastics, particles between 5 mm and 1 μm in diameter are widely accepted as microplastics (MPs),

whereas particles smaller than 1 μm are usually termed nanoplastics (NPs). Microplastics are regarded as emerging food contaminants with evidence of their presence in a wide range of consumed foods, drinking water, and human stools, confirming their oral intake (Jiménez-Arroyo et al., 2023). The presence of microplastics in foods and beverages is associated with their capacity to accumulate, migrate, and become widely distributed in the environment because of their small size, high specific surface area, and stable chemical properties. Microplastics enter the human body through the food chain as they are readily absorbed by plants, ingested by marine and terrestrial animals at various points in the food chain, and eventually enter the human body through the ingestion of food packaging, which is frequently made with plastic (Rubio-Armendáriz et al., 2022). Ramsperger et al. (2023) delineated three principal pathways by which MNPs enter food. First, the contamination of the environment with MNPs determines the contamination of food items. For example, contamination of waterways determines the contamination of blue meat. Second, microplastics can enter food through industrial processing. Third, microplastics can enter food through packaging and atmospheric deposition. Microplastic contamination of drinking water has been shown to increase the ingestion of these particles, which can subsequently contaminate beverages and food (Jin et al., 2021). However, the consequences of human exposure to MNPs remain unclear. Nevertheless, current *in vitro* and *in vivo* studies, as well as observational human studies, have demonstrated that MNPs of various types and sizes, besides the dangerous chemical substances present in all plastic products (plastic additives), have the potential to affect living organisms, posing risks to both human health and ecosystems (van der Laan et al., 2023; Xu et al., 2022).

Beer is considered one of the products most frequently contaminated with microplastics (Díaz-Basantes et al., 2020; Rubio-Armendáriz et al., 2022). Several studies have identified the presence of microplastic fibers and particles in beer, and the number of MPs detected varies considerably depending on the detection techniques employed (Shruti et al., 2020). In a study by Liebezeit & Liebezeit (2014), microplastic particles (MPs) were found in each of the 24 German beer brands analyzed, with an average concentration of fibers 2–79 MP/L. In contrast, the lowest MP concentrations were observed in all 12 beer brands processed with municipal water from the Laurentian Great Lakes in the United States (up to 14.3 particles/L). These findings were also reported by Kosuth et al. (2018). Furthermore, lower levels of contamination were observed in 13 Mexican beers (0–28 MP/L) than those in German beers, as reported by Shruti et al. (2020). Potential sources of microplastics in beer may include ingredients such as cereals and hops, packaging materials, and atmospheric particles that have been observed to float in the air. As water is the most important beer ingredient, it is likely to be the main contaminant of beer microplastics (Díaz-Basantes et al., 2020). Moreover, the craft beer production process can influence the quantity of microplastics present, with higher concentrations observed in nonfiltered craft beers. This is likely due to filtration techniques, which may reduce microplastic contamination. Microbreweries tend to filter their beer more thoroughly to increase their shelf life. In contrast, artisan brewers remove this step, which degrades the sensory properties of their products.

4. Risk assessment and the HACCP system in beer production

The safety of any food and beverage, including beer, is contingent on compliance with GMP. These practices typically necessitate the implementation of a food safety management system (FSMS) and a

Table 9.1 Steps of HACCP implementation and control measures to apply at each HACCP step for an identified hazard.

	Steps of HACCP implementation and control measures
Step 1	Establish a HACCP team
Step 2	Describe the product
Step 3	Identify the product's intended use
Step 4	Draw up flow diagram
Step 5	On-site confirmation of flow diagram
Step 6 (principle 1)	Identify and analysis of hazard(s) (microbiological, chemical, or physical) at every step in the beer/product making process and consider control measures
Step 7 (principle 2)	Determine the critical control points (CCPs) necessary to control the hazards
Step 8 (principle 3)	Establish critical limits (CLs) for each CCPs
Step 9 (principle 4)	Establish procedures to monitor each CCP and their critical limits
Step 10 (principle 5)	Establish corrective actions that need to be taken at any time a deviation from CLs occurs
Step 11 (principle 6)	Establish an effective record keeping system to document the HACCP plan
Step 12 (principle 7)	Establish procedures to verify that the HACCP system is working effectively

sanitation standard operating procedure (SSOP) besides adherence to good hygienic practices (GHP), also called prerequisite operating programs (PRPP). Furthermore, the principles of HACCP must be upheld. The HACCP is a comprehensive approach to risk management designed to identify and predict potential hazards during each stage of food manufacturing, from the initial agricultural production stage to the final consumption stage. The HACCP places greater emphasis on preventive measures than on end-product assessment for control. Besides identifying and preventing hazards, precautions are recommended to control them. Currently, HACCP is a well-known, effective, and preventive method for food safety management that has been adopted by numerous countries and international organizations, including the World Health Organization (WHO) and Food Agriculture Organization (FAO) (Mureşan et al., 2020). This concept is already being used extensively within the food industry, particularly among large food companies, where it may be integrated into quality management programs.

The adoption of HACCP is relatively straightforward for small processors, including those used in the microbrewing industry. As an analytical approach to food safety, HACCP focuses on the critical points or areas of a food process that may present a hazardous situation needing control (Awuchi, 2023; Kourtis & Arvanitoyannis, 2001). The steps for developing an entire HACCP system in accordance with the Codex Alimentarius and seven HACCP principles are summarized in Table 9.1.

Potential hazards may arise during the stages of craft beer production. The presence of contaminating microorganisms, excessive residues, undesirable derivatives of microbial compounds, secondary metabolites, materials, and chemical compounds can compromise the food safety of craft beer. HACCP streamlines hazard analysis by focusing on three primary categories of contaminants: microbiological (LAB, yeast, and fungi), chemical (mycotoxins, NMH, BA, and NAs), and physical (such as metal fragments and MPs). Thus, it is advisable to use the decision tree depicted in Fig. 9.1 to define a critical control point.

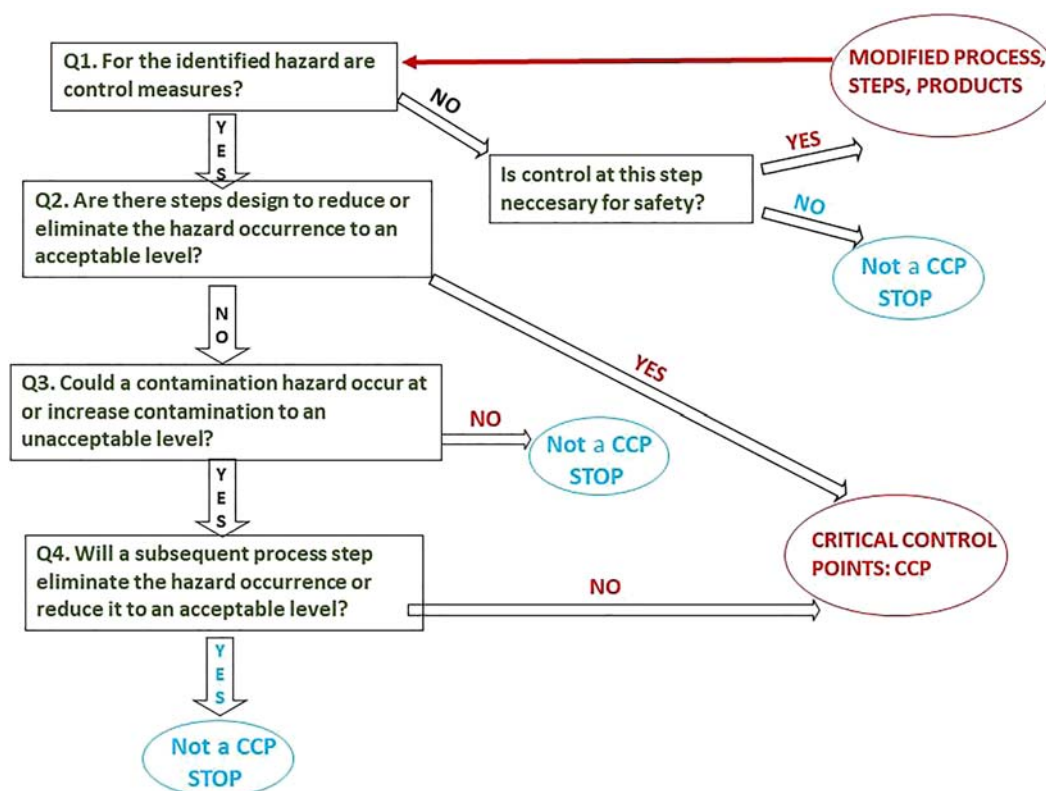


FIGURE 9.1

HACCP decision tree to apply at each step of the process with an identified hazard.

Adapted from Kourtis, L. K., & Arvanitoyannis, I. S. (2001). Implementation of hazard analysis critical control point (HACCP) system to the alcoholic beverages industry. Food Reviews International, 17(1), 1–44. <https://doi.org/10.1081/FRI-100000514>.

Only a few scientific studies have addressed the application of HACCP in the production of brewing products. Singh et al. (2018) comprehensively analyzed the key stages involved in implementing HACCP in the production of beer from barley. A general overview of the technological flow, hazards, and CCP of beer production was recently reported by Ciont et al. (2022). Fig. 9.2 summarizes the main hazards and CCP associated with craft beer production. By implementing the HACCP approach at each stage of the beer production process, the risk of contamination can be reduced and beer safety can be ensured through prevention. The implementation of specific HACCP approaches to beer raw materials and brewery production, packaging, storage, and delivery is necessary to ensure the safety of finished products, thus preventing the outbreak of foodborne diseases.

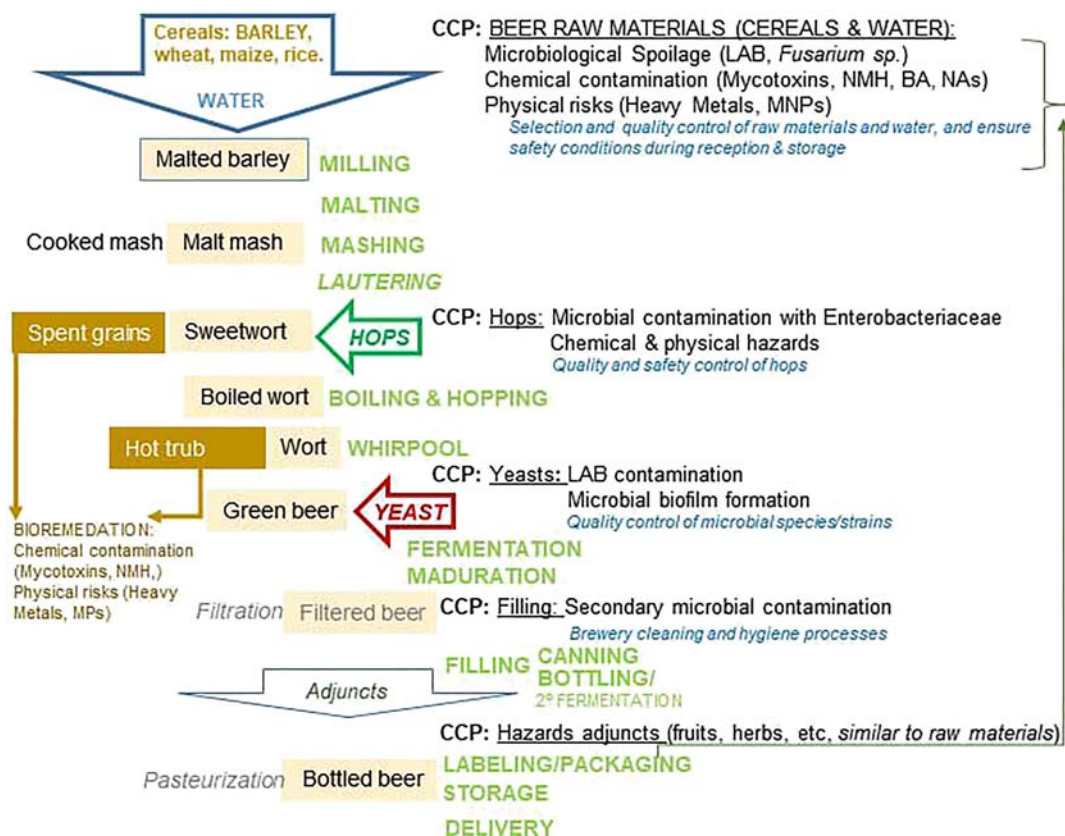


FIGURE 9.2

Main hazards and critical control points (CCPs) of craft beer production.

5. Conclusions and future perspectives

The craft beer market encompasses both innovative and traditional beers that have gained consumer appreciation because of their distinctive and unexpected sensorial characteristics, which have attracted a growing audience. However, because of its nature, craft beer is more susceptible to spoilage and chemical hazards because it is rarely pasteurized or sterile filtered, along with microbiota and contaminants associated with auxiliary materials and unconventional ingredients added to achieve many beer styles and flavors. Consequently, it is imperative that manufacturers are aware of the techniques employed to limit the formation of contaminants and remove them. The implementation of a HACCP plan with practical preventive procedures for manufacturing beer, particularly craft beer, can ensure food safety. A multitude of potential hazards in terms of biological, chemical, and physical factors have been documented during the processing of beer (from the stage of raw materials until distribution). It is of paramount importance to monitor raw materials as they are more likely to be

contaminated. Regardless of the state of the raw materials at the outset of the beer production process, the final product may be susceptible to biological, chemical, or physical risk. Current limitations include a lack of knowledge about the molecular mechanisms of some spoilage microorganisms that find their most suitable niche in craft beer, as well as a lack of tools and methodologies for their detection in the brewery environment. Regarding microbial metabolites, climate change may affect the production of higher concentrations of BA or other nitrogen derivatives by microbiota that are not well controlled in beer. The final concentration of mycotoxins in craft beer is challenging to predict, as it is contingent upon the initial concentration present in the raw material and brewing conditions. The current lack of mycotoxin limits in craft beer underestimates its potential risks to human health. Similarly, the development of robust, standardized, and validated methodologies to analyze heavy metals and MNPs in craft breweries and identify their potential main sources, exposure, and toxicity represents a significant challenge. Although there are still very few studies aimed at a comprehensive HACCP program in the craft beer sector, HACCP records and documentation can facilitate traceability of the source of contamination, ensure the safety and quality of beer, effectively manage control points and CCPs, and increase microbrewery sales and standing.

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Functional traits of craft beers

10

Emanuele Zannini^{1,2}

¹*Università di Roma La Sapienza, Department of Environmental Biology, Rome, Italy;* ²*University College Cork, School of Food and Nutritional Sciences, Cork, Ireland*

1. Introduction

Over the last few decades, the increasing scientific evidence linking nutrition and health/well-being has pushed the modern consumer's choices toward healthier and natural food alternatives. This trend forces food industries to work continuously on innovative formulations or adapt older ones to new standards. Increasing the “functionality” is a widespread and commercially appealing way to boost the potential health benefits of food products.

Malted-based beverages have been tested as functional and probiotic systems because of their nutritious and health-promoting properties, for example, soluble fibers and phytoestrogens. To address the sustainability trends, naturalness, and healthiness, brewery companies have started employing transformation practices and choosing low-processing and low-energy methods. The brewing industry has devoted much research effort to developing new technologies and innovations to expand the assortment of craft beer in response to increased consumer demand. Craft beer, also known as speciality beers, typically originates from small breweries employing traditional methods in beer production, omitting pasteurization at the end of the process (Artesana, 2024), prioritizing quality rather than quantity.

Generally, these breweries are managed by a master brewer, who tries to manufacture a unique taste and quality product. Currently, it has become fashionable to produce beer in little independent craft breweries, where unique beers (gluten-free, functional or probiotic beers, etc.) are looked for, knowing that consumers are now searching for health-related and tasty foods and beverages (Carbone & Quici, 2020). The category of speciality beers is the catch-all for beer styles which only fit in the market. Five types of speciality beers are of particular interest—low-calorie beer, low-alcohol or nonalcohol beer, novel-flavored beer, gluten-free beer, and functional beer. These novel and *avant-garde* brews usually leave long-lasting and memorable impressions on drinkers and are getting popular among consumers pursuing unique drinking experiences. While speciality beers are generally brewed to a classic style with some new flavor added, they can also be the end-products of brewing processes starting with unconventional yeast and/or Lactic Acid Bacteria (LAB). The addition of fruits, herbs and spices, miscellaneous flavorings (e.g., liquorice, smoke, and hot pepper), and alternative fermentable substrates (e.g., honey, maple syrup, and molasses) provide the beer's distinct flavors (Brewers

Association Beer Style Guidelines, 2023) turning, thus, an ordinary beer into a speciality beer (Nachel & Ettlinger, 2012). In addition to the different raw materials used in its production, industrial beer receives a second filtration (microfiltration) and the beers bottled are pasteurized to eliminate microorganisms before packaging (Pascari et al., 2018), while craft beer is usually a nonfiltered/clarified and unpasteurized product without additional nitrogen or carbon dioxide pressure (Berg et al., 1993). Being an unpasteurized and unfiltered craft beer could potentially represent a new vehicle for delivering health effects.

Regarding the potential superiority of the health profile of craft beer when compared to the industrial one, it is largely agreed that craft beer, generally unfiltered and unpasteurized, could secure a significant retention of health-promoting compounds. In a study assessing the occurrence of healthy compounds in beers produced on small and large, Bertuzzi et al. (2020) found significantly higher values of antioxidant activity and total phenolics due to the absence of steps, such as pasteurization and filtration (known to decrease the total phenolic contents), nitrogen, and folate, in small-scale beers.

2. Potential health benefits of probiotic-enriched beer

In the development of probiotic-enriched craft beers, the inclusion of probiotics imposes the implementation of approaches to adapt, endure, or resist several compounds that generate a challenging environment within the beer environment. Some of these conditions include substances with antimicrobial properties, such as ethanol and hops, high carbon dioxide levels, low pH, limited oxygen availability, and readily available nutrients.

In general, probiotic strains are introduced during wort fermentation, allowing these beneficial bacteria to thrive alongside the yeast responsible for alcohol production. The result is a beer that delivers the complex flavors and aromas typical of craft brews and offers potential digestive health benefits. While the exact health advantages are still being researched, probiotic craft beer appeals to health-conscious consumers seeking innovative ways to incorporate probiotics into their diets without sacrificing enjoyment. In addition, the diversity of flavors and styles within the craft beer movement provides ample opportunities for brewers to experiment with different probiotic strains, offering drinkers a wide range of options to explore while potentially supporting their well-being.

3. Functional yeasts

Several investigations have recently been conducted to assess the functionality of different unconventional microbial starter (probiotic) cultures in craft beer-making.

Among them, in the work conducted by Mulero-Cerezo et al. (2019), the probiotic *Saccharomyces cerevisiae* var. *boulardii* and a commercial *Saccharomyces cerevisiae* strain, commonly employed in the brewing industry, were compared as single starters. Craft beer produced with probiotic yeast had lower alcohol content, higher antioxidant activity, similar sensory attributes, much higher yeast viability after 45 days, and more acidification, allowing for reduced contamination risks at large-scale production. Furthermore, the probiotic yeast showed faster growth and larger cell volumes than the commercial yeast, increasing the probiotic bulk of the final craft beer. The same species of probiotic yeast, *S. cerevisiae* var. *boulardii*, was also tested in mixed cultures during wort fermentation with five different *S. cerevisiae* strains to produce craft beers with increased health benefits (Capece et al.,

2018). In almost all the fermented samples, at the end of the process, the probiotic yeast was predominant in the *S. cerevisiae* strain, with a high number of viable cell counts (ranging between 8×10^6 and 7.0×10^7 /mL). The inclusion of probiotic yeast strain in mixed starter did not negatively affect the beer aroma. It promoted an increase in the antioxidant activity and polyphenols content compared with beers produced from single starter yeast, indicating the positive influence of probiotic yeast strain on these parameters. In a similar study, the probiotic *S. cerevisiae* var. *boulardii* was also used as an adjunct starter in wort samples fermented with low-fermentation commercial yeast with the aim of challenging its probiotics traits (Reitenbach et al., 2021). The *S. boulardii* strain preserved its probiotic viability during the 28 days of storage while securing a sensory profile compared with the commercial beer.

The suitability of *S. cerevisiae* var. *boulardii* as a yeast candidate for crafting functional alcoholic beer remains remarkable and widely investigated (Díaz et al., 2023). The patenting of a specific variant *S. boulardii* T1 has highlighted its concrete applicability. This patented beer closely mimics *S. cerevisiae* beer characteristics, with a 7.83 °Brix, pH of 4.19, and cell concentration of 7.41×10^5 CFU/mL after 21 days of bottle conditioning (Iniesta et al., 2015). Besides its capability to resist the brewery environment without impairing the beer sensory profile, different strains of *S. boulardii* (CNCM 1–745, 3, 146, 6, 7, 8, 10c, 2 PV, AKP1, CCMA 1746) have shown antiinflammatory properties, resistance to stomach acids, and the ability to affect signaling pathways in the host's intestinal cells (Pais et al., 2020; Pereira de Paula et al., 2021). Mixed fermentation of *S. boulardii* CODEX with other *S. cerevisiae* strains enhanced antioxidant activity and polyphenol content, suggesting potential medical applications for conditions like antibiotic-induced diarrhea and gut dysbiosis (Capece et al., 2018; Kaźmierczak-Siedlecka et al., 2020).

Several other yeast genera are known to thrive in beer. A step further in the development of new functional craft beer has been taken by Canonico et al. (2021) where 43 wild yeast strains belonging to the genera *Rhodospiridiobolus*, *Candida*, *Lachancea*, *Rhodotorula*, *Torulaspora*, *Kazachstania*, *Brettanomyces*, *Pichia*, *Kluyveromyces*, *Metschnikowia*, *Hanseniaspora*, and *Saccharomyces*, previously validated for their probiotic traits, have been tested for craft beer production. The fermentative performance of the different yeast strains was assessed on pils wort (PW) alone or enriched with hydrolyzed chickpea wort (PCW) or lentil wort (PLW (20% of wort replacement)). The presence of lentils (PLW) confers compelling fermentative characters and pleasant aromatic notes to the final beers while The PCW results in beer with an unpleasant aromatic taste. The selected strains *Kazachstania unispora*, *Lachancea thermotolerans*, and *Saccharomyces cerevisiae* determined a significant increase in ethyl acetate, isoamyl acetate, and higher alcohols as the main aromatic compounds in PLW. The sensory profile indicated that emphasized aromatic attributes characterized the beers. The selected yeasts could represent potential microbial candidates for obtaining premium craft beer with highly nutritional and functional characteristics with a distinctive aromatic character. In Table 10.1, few examples of probiotic yeasts utilized in experimental brewery process are reported.

4. Functional LAB

Beneficial live microorganisms for human health, primarily lactobacilli and bifidobacteria, are known as probiotics. They play a key role in enhancing immunity and balancing gut microflora. Despite their recognized benefits, limited information is available regarding beer as a carrier for probiotics

Table 10.1 Probiotic and functional properties of some representative yeast strains found in beer.

Yeast strains	ABT	Tt	GIT	AA	CA	AI	AS	AmA	AfA	AdA	AOx	References
<i>Brettanomyces custersianus</i> VIT-MN05	✓	✓	✓	✓	—	—	✓	✓	—	—	—	(Ragavan & Das, 2017; Ragavan & Das, 2017)
<i>Candida orthopsilosis</i> CCMA 1748	✓	✓	✓	✓	✓	✓	—	✓	—	—	—	Ragavan & Das (2020)
<i>Candida tropicalis</i> CCMA 1751	✓	✓	✓	✓	✓	✓	—	✓	—	—	—	Simões et al. (2021)
<i>Debaryomyces hansenii</i> CCMA 1761	✓	✓	✓	✓	✓	✓	—	✓	—	—	—	Simões et al. (2021)
<i>Kluyveromyces lactis</i> VIT-MN02	✓	✓	✓	✓	—	—	✓	✓	—	✓	✓	(Ragavan & Das, 2017, 2020)
<i>Lachancea thermotolerans</i> B13	—	✓	✓	—	—	—	—	—	—	—	✓	Agarbati et al. (2020)
<i>Pichia guilliermondii</i> CCMA 1753	✓	✓	✓	✓	✓	✓	—	✓	—	—	—	Simões et al. (2021)
<i>Saccharomyces cerevisiae</i> 10C, 8, 6, 2 PV, 7	—	✓	✓	—	—	—	—	—	—	—	✓	Agarbati et al. (2020)
<i>Saccharomycopsis fibuligera</i> VIT-MN04	✓	✓	✓	✓	—	—	✓	✓	—	—	—	(Ragavan & Das, 2017, 2020)
<i>Torulaspora delbrueckii</i> 1.1 t2, 7.3 t2, 35, C 7.4, J401, Td vcs ff	—	✓	✓	—	—	—	—	—	—	—	✓	Agarbati et al. (2020)

Probiotic and functional properties: Acid and bile salt tolerance (ABT), thermotolerance 25–45°C (Tt), gastrointestinal tolerance (GIT), auto-aggregation (AA), coaggregation against pathogens (CAs), adhesion to intestinal cells (AIs), antibiotic sensitivity (AS), antimicrobial activity (AmA), antifungal activity (AfA), antidiabetic activity (AdA), antioxidant activity (AOX).

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Table 10.2 Probiotic and functional properties of some representative yeasts and LAB strains able to growth in malted-beverages.

Strains	Functional properties	References
<i>Brettanomyces custersianus</i> VIT-MN05	Inhibition of pathogenic <i>Streptococcus</i>	Mortazavi and Akhlaghi (2012)
<i>Brettanomyces custersianus</i> VIT-MN05	Inhibition of angiotensin-converting enzyme	Donkor et al. (2005)
<i>L. paracasei</i> F19 and 431	Improve of immune functions	Herkenhoff et al. (2023)
<i>L. paracasei</i> strain Lpc-37	Reduce stress-induced blood pressure and psychological markers of stress in human	Loh et al. (2021)
<i>L. paracasei</i> DTA-81	Reduce anxious behaviors	Silva et al. (2021)
Kefir biomass (<i>Leuconostoc</i> , <i>Lactobacillus lactic cremoris</i>)	Antiinflammatory and antiulcerogenic activities	Rodrigues et al. (2016)

(Table 10.2). This gap presents an opportunity to develop an innovative probiotic alcoholic beverage. This specialized brewing process enhances the complexity of the beer's flavors and aromas and potentially boosts its digestive health benefits. Probiotic Lactic acid bacteria (LAB) craft beer appeals to health-conscious consumers seeking flavorful beverages that support gut health, offering a refreshing alternative to traditional beers while promoting a holistic approach to well-being. As interest in probiotics continues to grow, LAB craft beer represents an exciting frontier in the evolving landscape of craft brewing. LAB, known for their probiotic properties, are naturally present in fermented foods like yoghurt and *kimchi*, and their introduction into craft beer offers a novel way to incorporate these beneficial microorganisms into the diet. Among all beer types, unpasteurized and unfiltered beer is potentially a new vehicle for delivering probiotics and has recently received research attention. Pasteurized and filtered beers are not suitable because of the nature of the processes involved in production unless probiotics are added postpasteurization or postfiltration, as probiotics may be killed by heat treatment or removed by filtration.

Lactic acid bacteria have long been used as starter cultures in many animal and plant-based foods, as well as being part of their indigenous microflora (Holzapfel, 1997), and many of them have been granted with the GRAS (Generally Recognized as Safe)/QPS (Qualified Presumption of Safety) status. The genera *Enterococcus*, *Lactocaseibacillus*, *Lactococcus*, *Leuconostoc*, *Pediococcus*, *Streptococcus*, and *Weissella* are naturally found on the surface of grains and in the surrounding environment (Guyot, 2012). Hence, they frequently constitute the innate starter culture, alongside fungal strains, in fermented cereal porridges regularly enjoyed in numerous rural communities globally (Nout, 2009). Research has demonstrated that the fermentation of cereal-based drinks, along with various food substrates, by LAB, enhances protein digestibility (Holzapfel, 1997; Taylor & Taylor, 2002), strengthens (micro/macro) nutrient bioavailability (Greffeuille et al., 2011), prolongs shelf life (Gupta et al., 2010), and finally enhances organoleptic qualities (Peyer et al., 2015).

In malting, LAB has been applied to control the endogenous microbial population that naturally colonizes the cereal surface. Due to their production of antimicrobial metabolites and their ability to outcompete nutrients, LAB effectively enhance malt safety by inhibiting spoilage from aerobic heterotrophic bacteria and filamentous fungi. In addition, the mild acidification resulting from lactic acid

has a positive impact on malt quality and processing efficiency. This includes benefits such as higher soluble extract content and reduced wort viscosity, leading to improved wort filtration. LAB has also been recognized as a necessary agent to acidify mash and wort during brewing operations naturally and to produce sour beers, for example, *Berliner Weisse* or *Lambics*. LAB can also serve as an agent for enhancing functionality and ameliorating the low sensory qualities of raw cereals. Starter cultures can produce desired flavors or changes in the food architecture and have been effectively used to enhance cereal-based beverages' palatability and reduce the need for additives, that is, flavor enhancers, processing aids, enzymes, or thickeners.

LAB fermentation generates a wide range of volatile and nonvolatile compounds, for example, organic acids, alcohols, aldehydes, ketones, and carbonyl compounds (McFeeters, 2004). Depending on the metabolic features of the starter culture, the cereal substrate, and the manufacturing processes involved, different strains of LAB may generate characteristic and contrasting end products (Blandino et al., 2003). The knowledge of which organoleptic-active compounds are being released during wort fermentation can help in the tailored formulation of the novel, speciality "sour beers" inspired by European styles such as the *Belgian Lambic* or the *German Berliner Weisse* or from traditional fermented beverages (Waters et al., 2015). The malting process of cereals improves the nutrient conditions for sustaining LAB fermentation due to the higher amount of assimilable nutrients (Charalampopoulos et al., 2002) contributing to the aroma profile of the substrate (Kunze, 2004).

The potential for producing probiotic beer was initially demonstrated when probiotic strains *Bifidobacterium lactis* BB-12 and *Lactobacillus acidophilus* LA-5 were successfully maintained in nonalcoholic beer at levels of approximately 10^6 CFU per mL following 20 days of refrigerated storage. This achievement was realized by introducing the probiotics after heat inactivation of residual yeast (Sohrabvandi et al., 2010). Subsequent investigations have confirmed similar opportunities and obstacles regarding the incorporation of probiotics into beer, both during and after the brewing process. Chan et al. (2021) comprehensively reviewed these aspects.

Incorporating probiotics in the wort would lead to acidification due to the production of lactic/acetic acid, which has a protective effect on microbial stability and organoleptic impact on the final products. Moreover, because viability is crucial for the efficacy of probiotics, attention must be paid to the sensitivity of probiotics to hop-bitter acids, which can inhibit the survival of Gram-positive lactic acid bacteria (Vriesekoop et al., 2012). In such regards, the work conducted by Silva et al. (2021) showed how the survivability of lacticaseibacillus spp. in beer can be enhanced through semiseparated cocultivation systems, underscoring the significance of cultivating lacticaseibacillus spp. in the wort prior to yeast pitching. Moreover, kettle hopping was recommended for hop addition to produce functional sour beer, as the dry-hopping method may hinder iso-alpha formation in the wort, although with a tendency to sediment that could adversely impact *L. paracasei* DTA 81 viability. Despite stress factors encountered in the matrices and during gastrointestinal transit potential probiotic *L. paracasei* DTA 81 endured at sufficient doses to elicit antidepressant effects in the mice groups treated with PWB or PSB, respectively. Recently, Herkenhoff (2024) showed that the probiotic strains of *Lacticaseibacillus paracasei* subsp. *paracasei* F19 (F19) and 431 (L431), used to develop probiotic-containing functional sour beers were well-suited to produce sour beers with elevated hop levels, demonstrating excellent stability.

Additional investigations have been conducted to assess the feasibility of producing probiotic craft beer by using selected probiotic LAB strains. In general, wort represents a suitable substrate to successfully support the growth of probiotic LAB with the brewery yeast acting as a mutualistic factor. In the work conducted by [Chan et al. \(2019\)](#), the growth and survival of the probiotic *Lactobacillus paracasei* L26, were assessed during a 10-day cofermentation period with a brewer's yeast, *S. cerevisiae* S-04, in unhopped wort. Most fermentable sugars were attenuated by *S. cerevisiae* S-04, with concomitant production of alcohols and esters, while *L. paracasei* L26 produced significant amounts of lactic acid. The viability of the probiotic LAB was significantly reduced when hop was added to the wort due to its antimicrobial effect. However, during storage, the maximal probiotic viability-enhancing effects were observed in the presence of live *S. cerevisiae* S-04, in combination with refrigeration. The resulting beer showed elevated levels of ethyl lactate, contributing to the fruity and buttery notes that were noticeably lost during storage.

In a similar vein, the application of the coculture method involving *S. pastorianus* SafLager W-34/70 and *L. plantarum* MTCC 1407 resulted in the creation of a sour beer characterized by pH levels ranging from 3.89 to 4.56 and lactic acid concentrations spanning from 0.61 to 3.79 mg/mL. Despite these variations, this beer garnered a higher overall acceptance rating while maintaining comparable scores in appearance and carbonation levels. Nevertheless, it displayed weaker performance in terms of aroma, bitterness, and alcohol content ([Mahanta et al., 2022](#)). Likewise, [Herkenhoff et al. \(2023\)](#) explored the volatile compounds present in different beer formulations containing potential probiotic LAB. These beers shared common waxy and aldehydic aromas, along with soapy flavors. Notably, *Streptococcus thermophilus* TH-4 samples exhibited additional fruity, floral, aldehydic, and phenolic aromas, accompanied by waxy, nutty, and tropical flavor profiles. Conversely, cultures of *L. paracasei* F19 and *L. paracasei* L.431 featured green, minty, floral, and woody scents paired with woody, medicinal, citrus, and floral flavor notes. When analyzed individually, *L. paracasei* F19 axenic beer displayed fruity, fatty, and waxy aromas, complemented by fruity, green, and tropical flavors. Meanwhile, the L.431 strain showcased naphthyl, aldehydic, citrus, green, and herbal aromas, with accompanying flavors of naphthyl, aldehydic, citrus, and hints of cocoa. It's worth noting that sensory panel assessments are essential for more precise comprehension of consumer acceptance and sensory perceptions.

5. Functional AAB

Acetic acid bacteria (AAB) have found applications in food technology, including beverages and vinegar. The study conducted by Alcine [Chan et al. \(2019\)](#) and [Neffe-Skocińska et al. \(2022\)](#) investigated the suitability of the AAB strain *Gluconobacter oxydans* H32 (with in vitro probiotic properties—surviving at a minimum of 1×10^6 CFU/mL in the human GIT) as a starter culture to produce functional light and dark beers enriched with fruit juices (cherry and plum). The craft beer samples resulted in a significant concentration of probiotic bacteria, ranging from 1.58×10^4 to 1.55×10^7 CFU/mL, and with ABV of 2.53 %–3.79%, a content of acetic acid (9.60–46.93 mg/100 mL), gluconic acid (8.60–69.07 mg/100 mL), and Vitamin C (1.03–2.83 mg/100 mL), with positive sensory profiles detected in the samples. The conclusion drawn from the study suggests that pasteurized light beer serves as a more suitable growth medium for *G. oxydans* H32.

Effective probiotic integration requires precise postfermentation addition stages to ensure viability, taste, and aroma. The sensory impact and consumer response of probiotic-enriched beers demand further research as well as the implementation of strategies that could enhance a sustained live cell count throughout beer shelf-life. Probiotics must survive storage conditions to qualify as functional beer. Moreover, additional effort is needed to refine strain selection and production methods to support microbial growth and longevity during the intended shelf life. However, the results clearly show that craft beer can serve as a medium for live yeasts, bacteria, and probiotic delivery even though the optimal mixed-species communities are not fully explored in beer.

While research into probiotic-enriched beer is ongoing, existing evidence remains quite narrow, with a paucity of standard human clinical validation demonstrating their properties or health benefits. Encouraging the health benefits of probiotic-enriched beer should be careful due to potential misperception, as it might nullify the negative effects of alcohol (Zugravu et al., 2023). Scientific literature indicates higher alcohol content in beer impairs gut health and overall well-being, especially with excessive consumption (Marques et al., 2022). Therefore, the incorporation of probiotics into beer may suggest an innovative approach to improve flavor and health features as long as beer consumption remains within moderate limits.

6. Functional constituents in craft beers

The nutrient and bioactive compound content, primarily polyphenolic compounds (PCs), in beer, is influenced by various ingredients and their geographic origin, including malted grains like barley, wheat, oats, and rice, hops, adjuncts such as fruits and spices, and microorganisms such as *Saccharomyces* yeasts or cofermenting bacteria like *Lactobacillus* (De Simone et al., 2021). The PCs identified in different craft beers are shown in Table 10.3. Key PCs in beer originate from malt and hops (Salañă et al., 2020). Among these, prominent PCs include xanthohumol (XH), isoxanthohumol (IXH), and other prenylflavonoids like 6-prenylnaringenin (6-PN), and 8-prenylnaringenin (8-PN), derived from hops (Osorio-Paz et al., 2020).

Beer polyphenols actively stimulate the growth of gut microbiota, leading to local anti-inflammatory and antioxidant effects. In addition, their breakdown products exhibit prebiotic properties, potentially addressing intestinal dysbiosis (Tirado-Kulieva et al., 2023). Beer's composition includes dietary fiber like β -glucans, arabinoxylans, mannose, and fructose polymers, which interact with gut bacteria through fermentation, providing a nutritional source. Melanoidins, compounds with minimal digestibility similar to fiber, provide antioxidant and antibacterial properties to beer. Despite limited research, beer emerges as a promising functional food candidate, with multiple pathways to positively influence human gut microbiota. Technological challenges remain but developing nonalcoholic or low-alcohol functional beers fortified with fiber, antioxidants, and probiotics could mitigate negative perceptions and alcohol-related health risks while enhancing gut health (Zugravu et al., 2023).

Diversification within craft brewing aims to create beverages with distinct aromas, colors, and flavors while also imparting health benefits similar to other polyphenol-rich foods. For instance, craft beers' antioxidant activity (AA) varies significantly depending on the *Saccharomyces cerevisiae* strains used for fermentation (Postigo et al., 2021). In a recent work conducted by Viana et al. (2021),

References	Type of beer	Technique	Compounds identified
Anderson et al. (2021)	From the United States	LC-QTOF-MS	Catechin, benzoic acid, chlorogenic acid, caffeic acid, cinnamic acid, esculin, myricetin, quercetin, vanillin, naringin, and naringin
Hinojosa-Avila et al. (2024)	From Portugal	HPLC	Gallic acid, ferulic acid, caffeic acid, p-coumaric acid, trans-cinnamic acid, luteolin, gallo catechin, epicatechin, quercetin, catechin and kaempferol-3-O-glucoside
Cortese et al. (2020)	Ego, Alter, Triplo Malto, Ubi, Fiat Lux” and Maior	HPLC-ESI-MS/MS	Gallic acid, 3-caffeoylquinic acid, 4-caffeoylquinic acid, epicatechin, caffeic acid, trans-ferulic acid, vanillic acid, syringic acid, trans-p-coumaric acid, sinapic acid, 4-hydroxybenzoic acid, quercetin, isoxanthohumol, 8-prenylnaringenin, catechin, xanthohumol, cohumulone, humulone, colupulone, and lupulone
dos Santos et al. (2020)	Made with brown, red and black rice.	HPLC–ESI-MS	Peonidin, cyanidin, cyanidin-3-O-rutinoside, cyanidin-3-O-glucoside, cyanidin-3,5-O-diglucoside and peonidin-3-O-glucoside
Marques et al. (2017)	Brown porter, classic American Pilsner, Irish red ale and American pale ale	HPLC-DAD	Gallic acid, p-coumaric acid, ferulic acid and caffeic acid
Nardini & Garaguso (2020)	Made with fruit and fruit by-products	HPLC-DAD	Chlorogenic acid, sinapic acid, p-coumaric acid, caffeic acid, vanillic acid, syringic acid, and ferulic acid

Continued

Table 10.3 Identification of PCs in craft beers from different countries.—cont'd

References	Type of beer	Technique	Compounds identified
Petrucci et al. (2020)	American pale ale, India pale ale and Amber ale	HPLC-PDA-ESI-MS/MS	Caffeic acid, ferulic acid, sinapic acid, p-coumaric acid, 5-caffeoylquinic acid, gallic acid, 3,4-dihydroxybenzoic acid, syringic acid, vanillic acid, p-hydroxybenzoic acid, m-hydroxybenzoic acid, quercetin, rutin, and kaempferol
Silva et al. (2021)	From Brazil	HPLC-DAD	Caffeic acid, epicatechin, trans-ferulic acid, p-coumaric acid, formononetin, kaempferol, quercetin, catechin and hydrated rutin
Zapata et al. (2019)	Made with quince of different varieties	HPLC-DAD-ESI-MS	3-O-caffeoylquinic acid, 3-O-coumaroylquinic acid, 5-O-caffeoylquinic acid, 5-O-coumaroylquinic acid, and 3,5-di-O-caffeoylquinic acid

DAD, diode array detector; ESI, electrospray ionization; HPLC, high-performance liquid chromatography; LC, liquid chromatography; MS, mass spectroscopy; PDA, photodiode array detector; QTOF, quadrupole time-of-flight.
 Reprinted from Tirado-Kulieva, V. A., Hernández-Martínez, E., Minchán-Velayarce, H. H., Pasapera-Campos, S. E., & Luque-Vilca, O. M. (2023). A comprehensive review of the benefits of drinking craft beer: Role of phenolic content in health and possible potential of the alcoholic fraction. *Current Research in Food Science*, 6. <https://doi.org/10.1016/j.crfs.2023.100477>. Copyright (2024), with permission from Elsevier.

the impact of four *S. cerevisiae* strains on polyphenolic compounds (PCs) and AA in Pale Ale-style craft beers was examined. They were observed that strain US-05 yielded a superior polyphenolic profile and higher AA. Postigo et al. (2021) experimented with over 100 *Saccharomyces* strains, comparing them with beers fermented using the commercial yeast *S. cerevisiae* Safale S-04. Notably, strain G 520 demonstrated the highest melatonin content (a hormone with antioxidant properties) and exhibited increased AA. Furthermore, the inclusion of the probiotic yeast *S. cerevisiae* *bouardii* in craft beers brewed with traditional *S. cerevisiae* led to elevated total polyphenol content (TPC) and AA (Capece et al., 2018). In exploring alternative ingredients, Nardini & Garaguso (2020) experimented with crafting beers using eight different fruits or fruit by-products. They found that compared with commercial beers, these brews exhibited a 32.3% increase in total polyphenol content (TPC), a 51.33% increase in total flavonoid content, and a 42.73% increase in antioxidant activity (AA). Similarly, Nunes Filho et al. (2021) investigated the impact of adding spices to green (15 days postbottling) and

aged (196 days postbottling) red ale craft beers on TPC and AA. Their findings revealed that the addition of a blend comprising turmeric extract (25% and 50%), black pepper extract (37.5% and 20%), and aromatic hops (37.5% and 30%) resulted in the highest TPC (295 and 221 mg gallic acid equivalents (GAE)/L, respectively) and optimal AA as measured by DPPH (0.65 and 0.56 mmol Trolox equivalent (ET)/L, respectively) and FRAP assay (408 and 350 mmol ET/L, respectively) (Piva et al., 2021). In addition, Piva et al. (2021) observed that incorporating leaves (0.1% and 0.5%) or extract (0.05% and 0.1%) of *Ocimum selloi* before or after fermentation in Pilsner craft beer production led to an increase in TPC from 291.2 to 360.6–371.9 μg GAE/mL and AA from 45.1% to 50.3%–82.5%.

Polyphenolic compounds are extensively researched for their capacity to neutralize reactive oxygen species and free radicals, mitigating oxidative damage and cellular stress (Vazquez-Cervantes et al., 2021). In a study by Caon et al. (2021), the impact of moderate commercial and craft beer consumption over 30 days on the redox status and liver integrity of Wistar rats exposed to carbon tetrachloride was investigated. Craft beers exhibited higher total polyphenol content (TPC), including catechin, rutin, epicatechin, caffeic acid, and isoxanthohumol, and greater antioxidant activity than traditional beers. Neither type of beer induced adverse effects on the rats' organisms, nor did they confer hepatoprotective benefits. However, consumption of imperial red craft beer notably bolstered the enzymatic and nonenzymatic redox balance, enhancing the activity of catalase (CAT), superoxide dismutase 2 (SOD2), and SOD3 enzymes (Caon et al., 2021).

The phenolic compounds identified in craft beer also exhibited potential antiviral properties. Research investigating the impact of American pale ale and stout craft beers on Tamiflu-resistant H1N1, a strain of influenza virus, demonstrated their ability to inhibit virus-induced cytotoxicity. These findings suggest that phenolics sourced from barley malt and hops in craft beers contribute to their capacity to mitigate or prevent complications associated with influenza infection, complementing other dietary phenolics (Elrod et al., 2014). In addition, aging craft beers in barrels previously used for red wine storage can substantially elevate total phenolic content, reaching up to 2.5 times higher levels compared with unaged craft beers and 3.6 times higher levels compared with industrial beer (Sanna & Pretti, 2015).

To enhance the antioxidant capacity of craft beer, Taiti et al. (2023) assess the impact of incorporating spirulina during the brewing process. Specifically, the levels of polyphenols and tannins were assessed, along with the cytoprotective effects against oxidative damage induced by tert-butyl hydroperoxide (tBOOH) in human H69 cholangiocytes.

Both variants of beer (craft beer with and without spirulina addition) exhibited comparable levels of total polyphenols and tannins, with slightly elevated levels observed in the sample containing spirulina at 0.25% w/v. Furthermore, both beers demonstrated radical scavenging capabilities against DPPH and ABTS radicals, although with a minor enhancement attributed to spirulina. However, yeast cells treated with spirulina had higher riboflavin content.

Conversely, the incorporation of spirulina at a concentration of 0.25% w/v appeared to enhance the beer's cytoprotective properties against tBOOH-induced oxidative damage in H69 cells, reducing intracellular oxidative stress. Incorporating legumes like chickpea and lentils into the brewing process results in craft beers characterized by an important mineral profile, abundant phenolic content, and the presence of isoflavones such as genistin and daidzin. Isoflavones, known for their potent antioxidant properties, also demonstrate potential in preventing various diseases, including inflammatory conditions, cardiovascular ailments, and postmenopausal issues (Luneia et al., 2018). Craft beer also serves

as a source of folate, vitamin B6, and vitamin B12, all crucial in homocysteine metabolic pathways, a risk factor for cardiovascular disease. Notably, research comparing industrial and craft beer consumption has revealed distinct effects: industrial beer consumption lowers homocysteine levels while elevating folic acid, whereas craft beer intake increases γ -glutamyl transpeptidase and decreases vitamin B6 levels (Rossi et al., 2021).

The current trend in crafted beer involves the production of “isotonic” beers, claiming to offer low alcohol content (<0.5% v/v) combined with isotonic properties, thus creating new beverages similar to sports drinks. These beers are rich in antioxidant compounds and possess appropriate osmolality. As per the European Food Safety Authority (EFSA), isotonic beverages typically exhibit osmolality within the 270–330 mOsmol/kg range. They are formulated to contain water and nutrients in proportions similar to human blood, facilitating rapid replenishment of fluids lost through sweating and providing glucose. This emerging beer category targets a specific market niche and can find its niche within craft beers. De Fusco et al. (2019) developed low-alcohol isotonic beer through interrupted fermentation, and some alcohol-free isotonic craft beers are already commercially available. Examples include Nirvana Helles Lager, Mikkeller Energibajer Alcohol-Free Wheat Beer, and NL Dark Low Alcohol Organic Craft Beer, among others.

Even though several in vitro studies highlighted the potential beneficial effect of PCs and AA compounds in beer, human intervention studies trials or in vivo studies are rare due to their complexity, typically conducted after extensive in vitro and in vivo investigations. Several criteria must be met in these studies regarding participants, interventions, and pre- and during-intervention conditions, including geographic location, race/ethnicity, gender, age, and dosage frequency. Few studies on cardiovascular diseases with incomplete designs were identified, such as that by Bassus et al. (2004), involving only 12 male subjects and a brief 3-hour intervention period without specifying study type, standardization, or monitoring of lifestyle or diet. Martínez Alvarez et al. (2009) solely included female participants, lacked information on age range or lifestyle/diet standardization before the intervention, and utilized only nonalcoholic beers as samples. Duration across studies ranged from 3 h to 45 days, and although lifestyle or diet during the intervention was standardized, study nature limited control over environmental conditions, potentially yielding inconclusive results. Another notable aspect affecting study robustness is illustrated by Padro et al. (2018), who monitored adherence to adaptation periods, interventions, and lifestyle/diet during the intervention via telephone contact and postintervention interviews. Despite meticulous attention to detail, the validity of results hinges on participant responsibility. In summary, cautious interpretation is warranted due to insufficiently detailed study designs and inherent shortcomings, while findings may suggest beneficial effects of craft beer.

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Sustainability in the craft beer brewing industry

11

Nancy Hoalst-Pullen

Kennesaw State University, Professor of Geography and Geospatial Sciences, Department of Geography and Anthropology, Kennesaw, GA, United States

1. Introduction

Broadly speaking, sustainability refers to the balanced relationship of behavioral conditions that impact economics, society, and the environment in a manner that maintains an indefinitely viable present and future. However, [Hoalst-Pullen et al. \(2014\)](#) argue that sustainability in the brewing industry is harder to define, as it commonly represents different things to different brewers located in different geographic locales. [Jones \(2018\)](#) notes this as well, asking “what exactly does sustainability mean, in theory and practice, within the craft brewing industry?”

Unfortunately, there is no one entity to answer Jones’ question. Quite commonly, (craft) brewers view sustainability in terms of *economics*—equipment upgrades and related “green” investments that may be costly in the short term but usually achieve high-efficiency gains and cost optimization in the long term. For others, sustainability relates more directly to the *environment*, usually with a focus on resource management—water conservation, waste reduction, carbon footprint reduction, and other environmental stewardship practices. Still others focus on *social* sustainability, emphasizing quality of life, livelihood security, and social cohesion, among other factors. Many brewers¹ understand the importance and interconnectedness of all three pillars of sustainability—economic, environmental, and social—and thereby mandate their breweries to incorporate some form of the triple bottom line of sustainability, informally known as the “Three Es” (Economics, Environment, and Equity) or “Three Ps” (Profits, Planet, and People).

Considering the brewing industry is a global multibillion-dollar enterprise with a now-maturing craft beer market ([Rhodes, 2023](#)) that brings hundreds of styles and varieties to the masses, the need for sustainability practices is apparent. Sustainability is becoming increasingly important in the brewery industry in part because of the sheer number of breweries found globally. For example, the 89

¹I define breweries as all production sites that make beer, including contract breweries, brewpubs, microbreweries, taprooms, regional craft breweries, and (large, noncraft) macrobreweries. I align craft breweries by the most recent [Brewers Association \(2023\)](#) definition of craft—a small (<6 million barrels of beer) independent brewery own and controlled (>75%) by a craft brewer that holds a Brewer’s Notice (or equivalent) or controls the intellectual property for one or more brands of beer, and has as its primary business purpose in the resale of the brand(s) it controls. That said, I also recognize the definition of “craft” can be as varied as the definition of “sustainability” for the brewery industry

breweries operating in 1978 in the United States (US) burgeoned into nearly 10,000² breweries as of 2023 (Brewers Association, 2024a), with over 80% of Americans now living within 10 miles of one (Patterson et al., 2022).³ Similar explosions in the number of breweries have been found in Europe, increasing by almost 6000 breweries just within the decade between 2008–18 (Arthur, 2018). In China, over 13,000 enterprises are now involved in craft beer-related businesses as of 2023, with the vast majority of these (~8400) registered since 2020 (Xing, 2023). Even with the craft beer industry adjusting to an increase in closures (Patterson et al., 2024; Staples et al., 2024) and strategic acquisitions (Howard, 2014), the global beer industry is still expected to show robust global growth (King, 2024). Approximately 23 million jobs—equivalent to 1 in every 110 worldwide—are linked directly, indirectly, or via related channels to the beer sector (Lindenberger, 2022). With these numbers, it is no surprise that beer remains the most consumed alcohol in the world (Hoalst-Pullen & Patterson, 2023).

With these high numbers come an industry that is traditionally at odds with environmentalism (Fish, 2015). Although beer is one of the oldest beverages in the world (BBC, 2018), having been invented even before written language, it is also one of the most resource intensive. Brewing beer uses significant amounts of water and energy, has a massive carbon footprint, and produces (in addition to beer) substantial air, water, and solid wastes (Hoalst-Pullen et al., 2014). The environmental impacts only further increase as one examines the entire life cycle of beer—from sourcing (e.g., cultivation of raw ingredients, raw material origins) to production (e.g., brewing and packaging) to the distribution and disposal of beer packaging. Life cycle assessments (LCA)⁴ evaluate and highlight “hotspots” where potential environmental-based improvements can occur before, during, and after the brewing process (Fig. 11.1). LCA studies on beer involve thorough inventories of energy, resources, and materials required to make beer, and calculate emissions in addition to other potential environmental impacts⁵ such as total waste, toxicity, fossil resource depletion, global warming potential, acidification, terrestrial, freshwater, and/or marine eutrophication, and photochemical ozone formation, to name a few (Cimini & Moresi, 2017, 2018; Morgan et al., 2021; Salazar Tijerino et al., 2023). While many of these LCA studies are done with large industrial macrobreweries associated with multinational corporations (Koroneos et al., 2005), studies that have included smaller (craft) breweries that have shown them to be less environmentally sustainable than their industrial counterparts (Cimini & Moresi, 2018; Salazar Tijerino et al., 2023).

Brewers with sustainability strategies (Patterson et al., 2016) are likely to lean on sustainability initiatives developed by groups and coalitions in the brewing industry. For example, The Brewers Association, BIER, and other agencies create instructive manuals and research reports that cover such topics as energy consumption, greenhouse gas (GHG) reduction, carbon footprint reduction, and water conservation, among other sustainable topics. Brewers can even take advantage of instructive tools that help measure and improve their sustainability practices. For example, the Sustainability

²According to the Brewers Association (2024a), there were 9906 breweries in 2023, with the most identified as either taprooms (3910) or brewpubs (3902).

³Although it should be noted that there are regional distinctions, with the West, Mid-Atlantic and Northeastern regions being higher than 80%, while the South and North having a sizable portion (33+ percent) living outside the 10-mile radius. See Patterson et al. (2022) for more details on the geographic demographics of breweries in the US

⁴The origin of LCAs in the USA is an interesting one, starting in 1969 when Coca-Cola assessed the environmental impact of switching from glass to plastic bottles. As Hunt et al. (1996) note, the primary emphasis of the early LCAs was solid wastes, with “energy and other environmental information ... just a ‘bonus.’”

⁵Deemed “environmental burdens” by Morgan et al. (2021), which I quite like and hope that the literature makes a point to use this language when referring to such environmental impact indicators

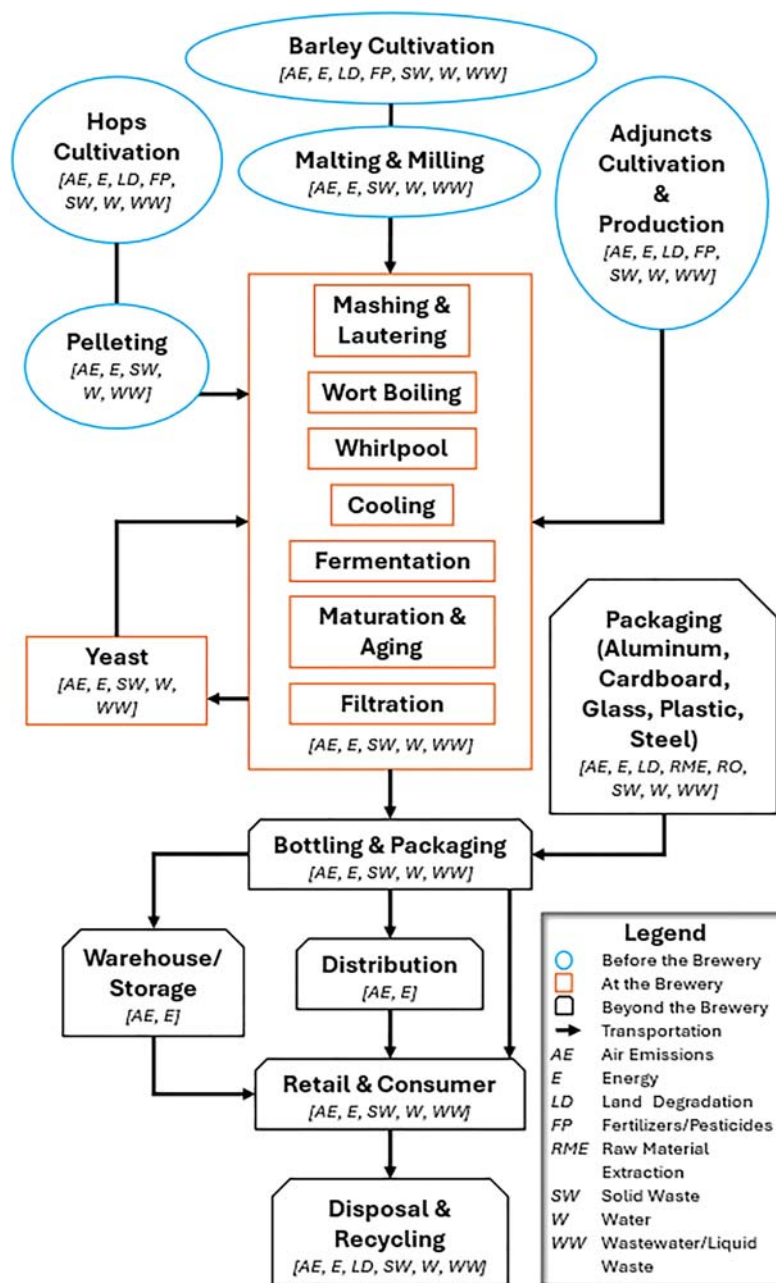


FIGURE 11.1

Life cycle assessment for a brewery.

Benchmarking Tool developed by the [Brewers Association \(2024b\)](#) is an easy-to-use spreadsheet-based template designed to track and decrease the use of natural resources by US craft breweries. Likewise, [Bowler et al. \(2024\)](#) created a freely available carbon calculator to benchmark and identify opportunities to reduce the carbon footprint of UK craft breweries. Breweries responding to sustainability challenges may generate sustainability reports ([Ness, 2018](#)) and participation in benchmark assessments ([Brewers Association, 2022](#)) that gauge the environmental performance and best practices. Brewers and/or brewery owners that do not know how or where to start when implementing sustainability strategies ([Tokos et al., 2012](#)), or lack the scale to do so, may opt to work directly with outside organizations to assist them with the heavy lifting of measuring, reporting, reducing, and/or certifying given aspects of sustainability (e.g., carbon neutrality, decreased water consumption, and organic foods) for their beers and/or breweries.

Breweries that incorporate sustainability strategies and practices usually lead to the generation of sustainability reports ([Ness, 2018](#)) and/or sustainable/green certifications ([Rosburg & Grebitus, 2020](#)). In some cases, sustainability reporting brought economic value to breweries ([Amahalu et al., 2018](#)) by promoting innovations and management policies that in turn encourage resource and energy savings. On the contrary, certifications highlight environmental measures promoting an aspect of sustainability (economic, environmental, or social), as related before, during, or after the brewing process or the brewery itself. [Table 11.1](#) highlights a selection of sustainability-related certifications that sustainably driven breweries have adopted. Of course, the more sustainability strategies and practices a given brewery adopts, the more sustainability becomes part of its brand ([Gatrell & Steiger, 2018](#)).

For many breweries, problems arise when craft brewers (and their breweries) have underlying goals that are at odds with sustainability. While more sustainable decisions and practices in sourcing, production, distribution, and disposal can lead economic savings (e.g., lower energy costs), environmental benefits (e.g., decreased greenhouse gas emissions), and social gains (e.g., improved work conditions), the adoption of sustainable practices is not always an (easy) option. The inherent characteristics that make craft beer “craft” purposefully includes:

1. small(er) production scales ([Brewers Association, 2023](#)) and traditional practices that are inherently less energy efficient;
2. creative and innovative beers ([Alonso et al., 2017](#)) that frequently use more raw ingredients, more types of raw ingredients (e.g., multigrain malt bill, fruit adjuncts, and spices), and more raw ingredients from nonlocal sources; and
3. local identity ([Schnell & Reese, 2014](#)) by way of neolocal practices that use beer and/or the brewery to foster a sense of place and community in locations that may increase transportation, energy, and other resource costs.

Indeed, the characteristics that define craft beer may make it difficult for some brewers—especially small brewers who do not benefit from economies of scale⁶—to be simultaneously innovative, financially solvent, and maintain a triple bottom line, all while staying “authentic” to their craft.

⁶Economies of scale are cost advantages gained by larger companies due to increased efficiency because of size

Table 11.1 Selected sustainability certifications adopted by breweries.

Certification	Purpose	Pillar ^a	Scale	Breweries (#) ^b	Website
<i>1% for the Planet</i>	Members contribute at least 1% of annual revenue to environmental causes.	ENV	International	N/A	https://www.onepercentfortheplanet.org/
<i>A Greener World</i>	Supports sustainable practices, particularly in agriculture. Breweries can use farmers that are AGW certified.	ENV	International	N/A	https://agreenerworld.org/
<i>B Corp Certified</i>	Certifies companies with high social and environmental performance, a commitment in governance documents, and transparency in performance.	ENV SOC	International	35	https://www.bcorporation.net/
<i>Good Bines</i>	Best practices in hop management by hop growers of America. Breweries can use farmers with good bines logos.	ECON ENV SOC	National (USA)	N/A	https://www.usahops.org/brewers/good-bines/
<i>Brewers Association Independent Seal</i>	A certified mark to easily identify independent US craft breweries and their beers.	SOC	National (USA)	4000+	https://www.brewersassociation.org/
<i>Brewshed Alliance</i>	Nationwide effort to protect wild waterways by recognizing the role breweries can play in water conservation.	ENV	National (USA)	200+	Only state-level websites exist
<i>Cicerone</i>	Certification demonstrates professional knowledge and tasting skills related to beer.	SOC	International	158,870 (individuals certified)	https://www.cicerone.org/
<i>Craft Malt Certified Seal</i>	The craft maltsters guild certifies brewers that purchase and/or produce beer with 10% or more craft malt.	ENV SOC	National (USA)	150 (breweries and distilleries)	https://craftmalting.com/
<i>Iowa Green Brewery Certification</i>	Environmental auditing and implementation of sustainable practices for the craft beer industry in Iowa.	ECON ENV	State (Iowa)	33	https://iwrc.uni.edu/food-beverage/green-brewery

Continued

Table 11.1 Selected sustainability certifications adopted by breweries.—cont'd

Certification	Purpose	Pillar ^a	Scale	Breweries (#) ^b	Website
<i>LEED Certification</i>	Recognizes sustainable practices in building design and construction.	ECON ENV SOC	International	12	https://www.usgbc.org/leed
<i>Living Wage Network</i>	Living wage certification recognizes employers who pay workers fairly.	SOC	National (USA)	N/A	https://www.livingwagenetwork.org/
<i>NetZero Brewery Initiative</i>	Breweries follow a dedicated roadmap for achieving net zero carbon emissions.	ECON ENV	National (UK)	N/A	https://netzeronow.org
<i>Salmon Safe</i>	Certification and accreditation programs linking land management practices with climate resilient agricultural and urban watersheds.	ENV	Regional	N/A	https://salmonsafe.org/
<i>TRUE Certification for Zero Waste</i>	Assessor-based program that rates how well facilities perform in minimizing their solid wastes and maximizing their efficiency.	ECON ENV SOC	International	N/A	https://true.gbci.org/
<i>USDA Certified Organic</i>	Sole organic certification in US. Incorporates standards of cycling resources, conserving biodiversity, and preserving ecological balance.	ENV	National (USA)	N/A	https://www.ams.usda.gov/services/organic-certification/

^aECON, economic sustainability; ENV, environmental sustainability; SOC, social sustainability.
^bAs of July 2024.

In the following chapter, the roles of sustainability will be assessed in the brewing industry, with a focus on craft brewers and craft breweries in the United States.⁷ This will be done using the framework for craft beer supply chain that consists of three overarching life-cycle stages: before the brewery (e.g., ingredient cultivation, processing, and procurement), at the brewery (e.g., beer production), and beyond the brewery (e.g., packaging, transportation and distribution, and packaging waste) (Cimini &

⁷While the focus is on modern craft brewers in the USA, the underlying need for LCAs for any brewer/brewery is warranted. For example, Sawadogo-Lingani et al. (2021) looked at sorghum-based traditional beers in West Africa, detailing how traditional beers are made and sustainability strategies that could/should be adopted for raw material acquisition, the use of cook stoves and starter cultures, waste valorization, and appropriate packaging and stabilization

Moresi, 2018; Morgan et al., 2021). While all three pillars of sustainability (i.e., profit, planet, and people) are important during all three stages, there may be an emphasis made on any given pillar anywhere in the supply chain.

2. Before the brewery

Food and beverages are key contributors to intensive water and energy use and play an outsized role in the creation of anthropogenic greenhouse gases (GHG) (Ivanova et al., 2015) which in turn contribute to climate change and environmental degradation. For the brewing industry, sustainability and sustainable practices begin long before brewing and fermentation occur. This is particularly true of water, of which beer uses massive amounts. Mekonnen & Hoekstra (2010) found that beer had a global water footprint of 300 m³/ton,⁸ while SABMiller & WWF (2009) reported a total net water footprint for industrial beer at about 191 L of water per liter of beer (155/L if excluding gray water).⁹ Due to economies of scale, increased costs, and ingredients used, craft breweries are likely to have higher water footprints on average than their macrobrewery counterparts. However, one thing remains universal—the total water used to produce beer overwhelmingly comes from the farm (in cultivating the ingredients) rather than in the brewery (in producing the beer) (Elliot & McKinney, 2022).

Like water, the emissions for craft breweries are highest before and after the brewery via fertilizer, fuel, transport, packaging, cooling, and so on. Unfortunately, as noted by Hansen et al. (2022), most of these emissions will be underreported. This in and of itself is an issue, as Bowler et al. (2024) note how “most craft breweries often lack the resources, finances, and technical knowledge to evaluate the carbon footprint of their entire value chain and typically focus their sustainability efforts on reducing their energy consumption.” Even though beer is a culmination of four main ingredients—grains/malt, hops, yeast, and water (see Table 11.2)—the following will focus primarily on barley (grain and malt) and hops, as they are the most common agricultural products used in craft beer.

2.1 Grains

While the beer brewing process can use a variety of ingredients depending on the beer style, some of the most popular cultivated grains include barley (*Hordeum vulgare* L.), wheat (*Triticum aestivum*), oats (*Avena sativa*), and rye (*Secale cereale*), in addition to maize/corn (*Zea mays*) and rice (*Oryza sativa*). For most craft brewers, malted barley is overwhelmingly preferred, and the only fermentable grain not considered an adjunct. Davison et al. (2012) estimate that 25% of all barley grain in the US is used for beer and other malted beverages. Unfortunately, most cereal grains used in brewing beer have a large carbon footprint and contribute to environmental degradation by way of intensive water requirements/irrigation, fertilizers, and chemical inputs (e.g., pesticides) used in cultivation, as well as energy costs and emissions released during planting, harvesting, processing, malting, and trans-

⁸which to note, is appreciably smaller than wine, at 870 m³/ton

⁹A commonly cited Water Footprint Network (WFN) number attributes as much as 300 L of water needed to create 1 liter of beer; however, I could not find the original source to validate this finding

Table 11.2 Brewery ingredients and sources.			
Ingredient	Purpose	Examples	Source
Water	Primary ingredient. Solvent for wort.	Rainwater, surface freshwater (lakes, rivers, streams), groundwater (wells, springs, aquifers), municipality water supplies	Local
Grain/Malt	Wort extracted for fermentation. Bestows color and flavor.	Barley (<i>Hordeum vulgare</i> L.), wheat (<i>Triticum aestivum</i>), oats (<i>Avena sativa</i>), rye (<i>Secale cereale</i>), sorghum (<i>Sorghum bicolor</i>), maize/corn (<i>Zea mays</i>), rice (<i>Oryza sativa</i>), fonio (<i>Digitaria exilis</i>), kernza (<i>Thinopyrum intermedium</i>)	Local to global
Hops	Provides bittering, flavor, and aroma via alpha acids, beta acids and essential oils. Acts a preservative.	<i>Humulus lupulus</i> , with nearly 300 varieties include bittering hops such as columbus, galena, and nugget; aromatic hops such as cascade, motueka, and saaz; and dual-purpose hops like centennial, mosaic, and nelson sauin	Local to global
Yeast	Converts dissolved sugars into alcohol and carbon dioxide. Impacts flavor from esters.	<i>Saccharomyces pastorianus</i> (lager yeast), <i>Saccharomyces cerevisiae</i> (ale yeast), <i>Brettanomyces</i> , kveik (Norwegian)	Local to global (can be collected locally, but usually obtained from commercial yeast labs)
Modified from Gebhardt, S. L. (2018). <i>Sustainable Beer: An overview of sustainable practices in Southern California craft breweries</i> . Thesis, California State Polytechnic University: Pomona.			

portation. Rajaniemi et al. (2011) found barley grown in Finland to have a greenhouse gas emission akin to wheat and oats of 570–590 g (about 1.3 lb.) CO₂-eq. per kg,¹⁰ which was comparatively smaller than rye, at 870 g (about 1.92 lb.) CO₂-eq. per kg.

So, what does a brewer do to be more sustainable with its ingredients? The first apparent option for brewers is to decrease the use of high GHG grains, like rye, and introduce grains that are more climate friendly—like fonio¹¹ (*Digitaria exilis*), an ancient cereal crop from West Africa that is drought-resistant and requires no fertilizer, pesticides, or chemical inputs, or kernza (*Thinopyrum intermedium*), a newly domesticated form of wheatgrass that requires less water, less fertilizer, and reduced tilling (Marcus & Fox, 2021). Adoption of these more sustainable grains would decrease the overall carbon footprint of many craft ales (Tse, 2024). Likewise, buying heritage and drought-resistant grain varieties will decrease the water footprint and potentially build stronger connections between the brewers and their ingredients. Sourcing such grains may also help with another problem inherent to most craft brewers—the amount of grain used in making beer. Compared with their industrial

¹⁰For perspective, each kilogram (2.2 pounds) produced an equivalent of driving 2.4 km (1.5 miles) in a gas-powered vehicle

¹¹also known as acha, fundi, findi, iburu, Asian millet, fonio millet, pearl millet, and hungry rice

counterparts, craft brewers use up to four times more barley (per barrel) grain when creating their brews. [Lee-Weitz \(2022\)](#) estimates that somewhere between 1.8 billion lbs. (816.5 kilotons) to 20 billion lbs. (907.2 kilotons) of grains (as base malt and specialty malt) were used to produce beer in the US in 2021, while [Watson \(2022\)](#) attributes 1.56 billion pounds (about 707.6 kilotons) of barley malt were used solely by US craft brewers.

Although beer styles¹² were once historically tied to location ([Mittag, 2014](#)), beer styles today can theoretically be made without geographic limitations. Indeed, one no longer has to live where the ingredients originate. As a result, craft brewers replicate certain styles using certain recipes requiring specific ingredients, many of which may not be local. This may be further amplified by some craft brewers who focus on quality and/or innovation. Indeed, craft brewers commonly purchase multiple types of grain from various locations all over the world. Such investments may increase innovation and creativity in the brewery, but they also increase both production costs and the beer's overall carbon footprint. This is compounded with the fact that the economies of scale allow for macrobreweries to pay a lower price for, say, barley, when compared to craft brewers, due to the high amount purchased and limited varieties needed. In addition, macrobrewers (and even some craft brewers) are located adjacent to railroad tracks which allowed them to have their grain shipped in railcars ([Bahl et al., 2021](#)), furthering the cost-savings. Of course, craft brewers are unlikely to relocate near railroads for the cost-savings potential. However, craft brewers could create new recipes that source high-quality and/or organic grains and adjuncts from local farms to reduce transportation costs and emissions while supporting local agricultural communities. Moreover, brewers could purchase grains and adjuncts from farmers who use regenerative agriculture practices (e.g., no till) that enhance soil health and further sequester carbon ([Burrows, 2023](#)).¹³

2.2 Malts

Malting is a process that modifies barley until it is suitable to produce beer. More specifically, malting converts starch and enzymes in barley into fermentative sugars that yeast then converts into ethanol and carbon dioxide ([Bettenhausen et al., 2018](#)). Malt also provides color (melanoidins) and flavor (e.g., esters and aldehydes) to beer ([Fox & Bettenhausen, 2023](#)). The process of malting comprises of three steps—steeping, germination, and kilning ([Dugulin et al., 2021](#); [Mussatto et al., 2006](#)). The level of heat applied during the kilning stage determines the color and flavor, with lower temperatures resulting in pale malts, and higher temperatures producing darker, more roasted malts. Of the three steps, steeping is the most water intensive, as this is when the grains are submerged to increase their moisture content, while kilning (drying the malt) creates the largest carbon footprint, and accounts for up to 80% of the energy used in the malting process ([Dugulin et al., 2021](#)). Any increase in the number of breweries or amount of craft beer produced equates to increases in the water and carbon footprints related to malt. In 2018, when craft beer was only 14% of the market share, about 40% of the total malt was consumed by U.S. craft brewers ([Brewers Association, 2019](#)). By 2023, when craft beer

¹²the term “beer style” is a more recent phenomenon, and one attributed to Michael Jackson with his book *The World Guide to Beer* (1977). For more on the concept and categorization of beer styles, see [Hoalst-Pullen & Patterson \(2023\)](#).

¹³In fact, Patagonia Provisions and Aslan Brewing Company released the world's first regenerative organic certified IPA in April 2024, made with Regenerative Organic Certified Kernza, Regenerative Organic Certified Pilsner Malt from Breathe Deep Farm in New York's Hudson Valley, and organic Chinook Hops and Strata Hops from Washington's Roy Farms ([Brewbound, 2024](#)).

encompassed 24.7% of the market share (Brewers Association, 2024a), the total amount of malt also increased, as did its associated water and carbon footprints.

Craft malting has shifted consumer preferences to craft ingredients, much like craft brewing did for beer. In response, the Craft Maltsters Guild was created in 2013 to identify, promote, and support craft malt, which is defined as “small (5 metric tons (5.5 US tons) to 10,000 metric tons (11,000 US tons) of malthouse production per year), local (50% of grains sourced within 500 miles of the malthouse) and independent (malthouse independently owned by 76% majority)” (Craft Maltsters Guild, 2024). Reminiscence of the craft beer scene in the 1990s up until the mid-2000s, these small-batch, independent maltsters (numbering 67 as of 2024) may make up a small percentage of the total volume of malt produced but still accounting for an estimated 24.7 million lbs. (about 11.2 kilotons) in 2022, a 60% increase from 3 years prior (Holl, 2022).

Moreover, case studies have noted potential ways to decrease the overall carbon footprint (specifically GHG) up to 90% by adopting anaerobic digestion and biomass heating as alternative energy sources (Davies, 2020). Brewers have also decreased their water and carbon footprints by (partially) replacing barley with (usually less expensive) adjuncts, including unmalted cereals (e.g., barley, corn, wheat, and oats), green malts (freshly germinated but not kilned grains), and liquid syrups and sugars (Dugulin et al., 2021). While adjuncts are not always adopted for sustainability reasons if at all (e.g., style recipes, high gravity worts, tradition, etc.), adjuncts can be sustainable as they save on costs during production (e.g., water and energy use) and postproduction (e.g., via alcohol tax) with no additional investments required by the brewery (Bogdan & Kordialike-Bogacka, 2017).

2.3 Hops

Hops (*Humulus lupulus* L.) is an herbaceous climbing bine with strobile (flowers) that give beer—particularly IPAs and other hop-forward styles—its distinctive flavor and aroma. Hops are usually the most expensive part of the brewing process, and one of the most water intensive. Hops cost around 5–6 USD per pound to buy and take 251–450 gallons (about 950–1700 L) of water per pound to grow (Craig, 2021; O’Hare et al., 2013). For context, 1 kg of dry whole cone hops equates to approximately 100 pints of beer (Elliot & McKinney, 2022), although this ratio is quite dependent on the beer style.¹⁴

Hops is a global commodity, with 118,415 metric tons produced in 2023 (Crowell, 2024). Where high-quality hops can grow, however, is restricted to relatively small regions across the globe (at about the 48 parallel), with the US and Germany accounting for 71% of the world hop acreage. Due to its limited geography and niche growing conditions,¹⁵ hop growers have long been aware of the potential impact of climate change on hop farming (Fish, 2015). For example, hops in major beer-producing European countries are ripening earlier (by 20 days), producing less (0.13–0.27 t/ha/yr) and losing their alpha bitter acids content (–0.6%) (Mozny et al., 2023). With rising temperatures and more frequent and severe droughts, hop quantity and/or quality may decline. This is compounded with the fact that it takes nearly 70% more water to grow aroma hop pellets than alpha hop pellets. Fortunately,

¹⁴This does indeed look to be the case, as imperial IPA has 3.8 lbs of per keg (1984 oz), barley wine has 1.57 lbs/keg, IPA has 1 lb/keg, pale ale has 0.56 lb/keg, ESB and 0.47 lb/keg, pilsner has 0.31 lb/keg, and hefeweizen has 0.19 lb/keg (from [usahops.org](https://www.usahops.org)).

¹⁵including full sun, long summer days, cool temperatures with four frost-free months in the summer and one or two cold (<40°F/<4°C) months in the winter, a moist climate, soil that is fertile, well drained, and well aerated, and with a pH of 6.0–7.0, and a trellis system for the bines

some hops do better in drier, hotter climates (e.g., Chinook) while others grow better in warm moist conditions (e.g., Golding), allowing farmers to replant with varieties that are less climate sensitive. In addition, using hop varieties with lower carbon footprints is another environmental sustainability option for many brewers (see Fig. 11.2). Moreover, demand for local hops has resulted in its cultivation of hops outside of core regions (Reid et al., 2020). While beer is not always made where hops are grown, sourcing hops from local farms can also reduce transportation costs and associated environmental impacts.

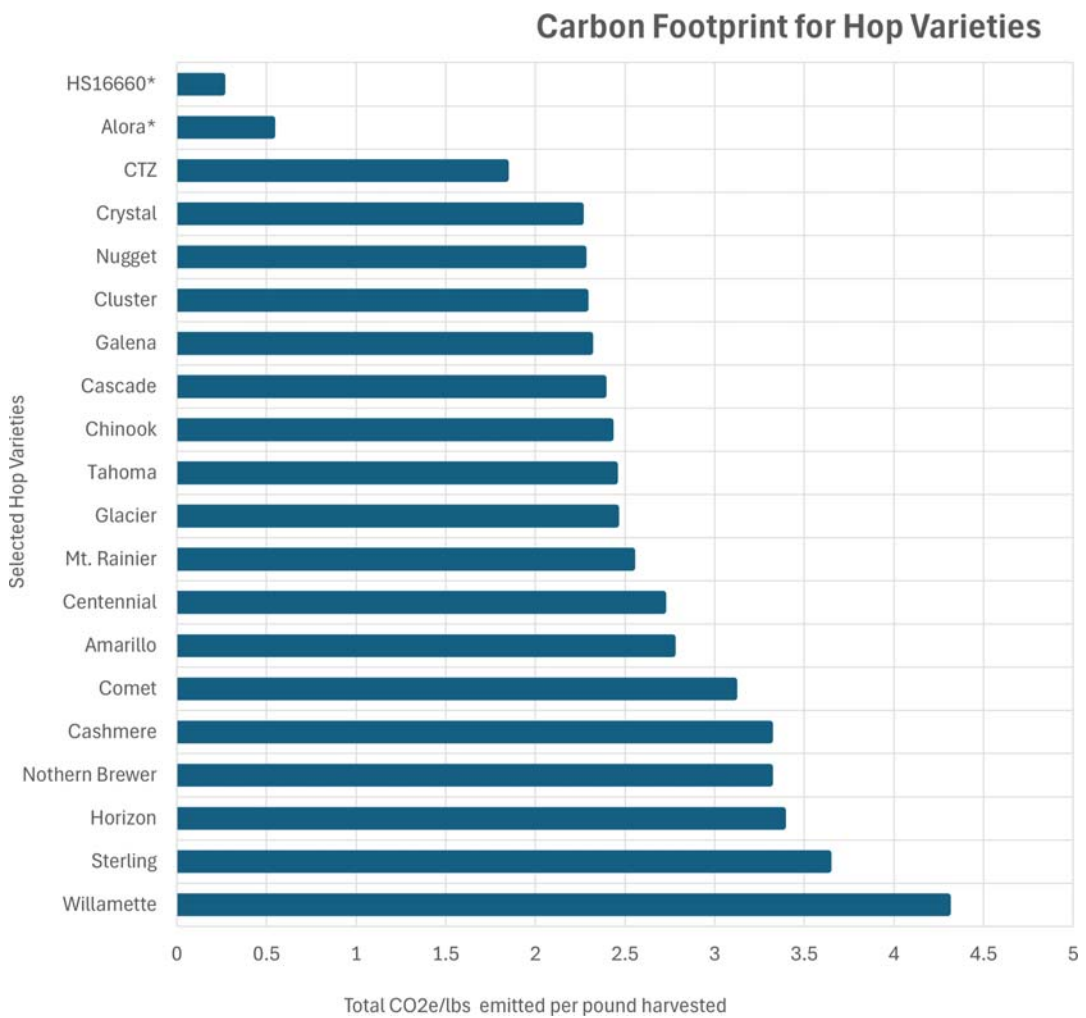


FIGURE 11.2 Carbon footprint for select hop varieties grown on Hopsteiner farms.

Total CO₂ equivalents are based on averages from mature hops varieties cultivated across 6 crop years except for Helios (4 crop years), Alora (2 crop years), and HS16660 (1 crop year).

Data from Hopsteiner. (2024). 2023 year in review with Hopsteiner. Retrieved July 2, 2024 <https://www.hopsteiner.com/blog/2023-year-in-review-with-hopsteiner/>.

Arguably, published studies on cradle-to-gate life cycle assessments for hop production are few (Bristol, 2022). In an LCA study (starting with seedlings and ending with hop pellets), Bristol (2022) determined that 1 kg of aroma hop pellets resulted in the global warming potential (GWP) of 4.42 kg CO₂-eq. and a eutrophication potential (EP) of 12.17 gNe/kg. In another study, Elliot & McKinney (2022) found lower GHG emissions for dry whole hops, with 3.1 kg CO₂-eq for alpha whole hops, and 3.7 kg CO₂-eq. for aromatic hops.¹⁶ Akin to barley and malt, hop farming was the largest contributor toward its overall carbon footprint, while the kilning process was the most significant single carbon contributor. Using hop producers that use alternative energy sources for kilning is one way for brewers decrease their overall carbon footprint. Additional goals across the hop production chain—farming, processing, packaging, and shipping—can include increased energy efficiency, use of renewable energy, implementation of closed-loop¹⁷ technologies for carbon emissions, reduction in water consumption, and reduction in overall wastes (Bahl et al., 2021; Bristol, 2022; Elliot & McKinney, 2022; Gebhardt, 2018).

While decreasing the use of barley and malt may be an option for some brewers, decreasing the use of hops is a less likely alternative, particularly for brewers who specialize in IPA styles (especially hazy IPA) that remain central to the craft beer sector. Yet, hop alternatives do exist, including industrial yeasts that impart hop flavors deemed hoppier (via a double-blind taste test) than traditionally hopped beers (Denby et al., 2018).

3. At the brewery

Compared with other stages, the brewing stage might be a relatively minor contributor to the overall carbon footprint of beer (Bahl et al., 2021, Fig. 11.3). Yet modern brewing facilities remain both water-intensive (Olajire, 2020) and energy intensive (Cimini & Moresi, 2018), especially in the steps of malt mashing, wort fermentation, and rough beer maturation and filtration (Olajire, 2020). Sustainable brewing practices at the brewery involve establishing reliable water sources, reducing water usage, minimizing waste production, conserving energy, utilizing renewable energy sources, fabricating resource-efficient facilities, and adopting socially sustainable practices.

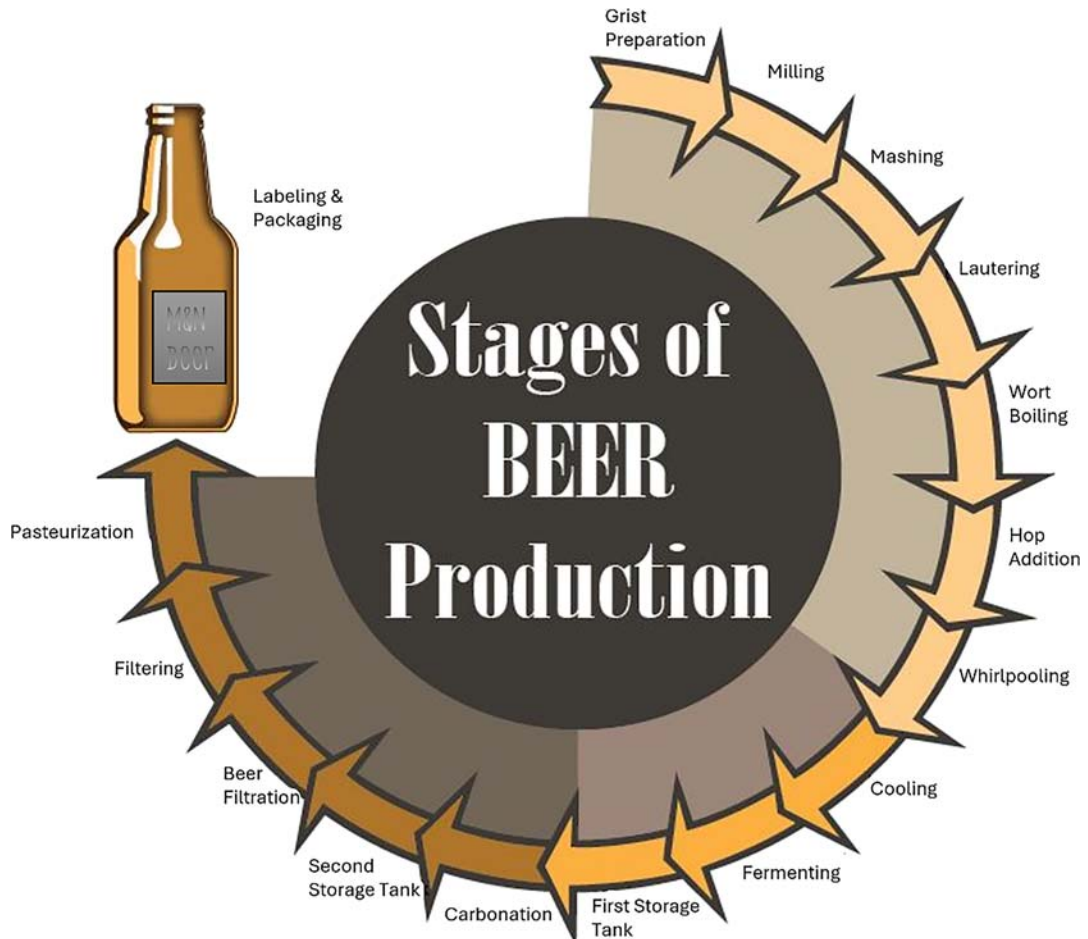
3.1 Water sources

The beer industry may be global, but 100 years ago (before modern refrigeration in particular) it was a local commodity, with breweries centered near local, reliable water sources that both facilitated the brewing process and created a high-quality beer (Gatrell et al., 2014). Beer can be brewed using rainwater,⁴ surface water (e.g., lakes, rivers, and streams), groundwater (e.g., wells, springs, and aquifers), and/or municipal water supplies.¹⁸ Reliance on “free” water sources such as groundwater does decrease the overall costs of the brewery; however, some sources may not be ideal where water is limited or regulated, especially where water conservation efforts exist.

¹⁶Akin to Bristol (2022), Elliot & McKinney (2022) found higher GHG emissions with hop pellets, with 3.5 kg CO₂-eq. for alpha hop pellets, and 4.1 kg CO₂-eq. for aroma hop pellets

¹⁷Closed-loop systems enable breweries to capture and reuse waste materials such as spent grains and wastewater

¹⁸Technically, any water can work. As examples, Antares (Argentina) brewed a beer made using seawater, while Devil’s Canyon Brewery (California, USA) made a Kolsch-style beer using highly purified recycled greywater

**FIGURE 11.3**

Stages of beer production.

Brewers know that the quality of water can impact the quality of beer. Due to variations in seasons (for surface water) and geology (for groundwater), the local water profile may be as distinctive as its location from where it is found. It therefore makes sense that many beer styles were historically linked to certain locations, due in part to the geographic nuances of water composition ([Gatrell et al., 2014](#)). For example, the mineral-laden waters in and around the town of Burton-on-Trent, Staffordshire, England, are exceptionally rich in calcium, magnesium, and sulfate yet low in sodium and bicarbonate. By contrast, the “soft” water of Pilsen, with its low mineral composition, was ideally suited for the clean, crisp Bohemian Pilsner style. Today, brewers pay meticulous attention to what is found in the water profile by testing and modifying the mineral content to produce the ideal characteristics of the given beer style ([Mosher & Trantham, 2017](#)). Surface water sources are typically low(er) in

dissolved minerals and high(er) in organic matter when compared to groundwater sources. Municipality water sources—which originate from surface water, groundwater, and/or in some cases, reverse osmosis from oceans—are usually treated and filtered and therefore lack any inherent mineral composition. Disinfectants used by municipalities to clean water supplies are usually persistent and can be found in measurable quantities in the brewery water (Mosher & Trantham, 2017).

3.2 Water reduction

Brewing beer is water intensive. Water is not just used for the creation of beer itself but plays an oversized role in virtually every stage of beer production. Water in the brewery can be categorized as product water (used in making beer), process water (for cleaning and sanitation), or service water (for heat or energy in the brewery) (Lewis & Young, 2001). Of the three categories, process water and service water represent the primary demand for water in a brewery. Water is added to the grain bill (or malt extract in some cases) to create the mash, as well as during the processes of underletting, sparging, and blending. Water is used to transfer the resulting wort between vessels. Water washes, rinses, and sterilizes equipment after each batch of beer, most of which ends up as discharge to drains. It is also lost to spent grain and to waste hop material known as trub. Water is also used in wort cooling, added during yeast pitching, and lost after fermentation with the removal of yeast and trub. Sometimes yeast is further removed via centrifugal separation, which also uses water. If beer is filtered, more water is used.¹⁹ If carbonated, the carbonated water is added back, and the beer is stored in bright tanks. More water is used for cleaning—flushing lines, bottle washing, washing/rinsing equipment, pasteurizing, and transferring beer. Packaging also uses water. Any water that does not become beer, effluent, or waste is lost to evaporation.

Reduction in water usage in a brewery, particularly for process water and service water, is crucial for economic, environmental, and social reasons. Economies of scale play a role in the water footprint. Large breweries can fully automate the brewing process, establish onsite wastewater treatment, integrate equipment upgrades, require efficient cleaning procedures, and when applicable, implement cross-flow or membrane filtration systems, among other optimized processes and protocols. In addition to funneling significant investments into meeting global sustainability standards, larger breweries can also staff individuals in sustainability-related positions (Olajire, 2020). Indeed, Patterson et al. (2016) found larger regional craft breweries were more apt to reduce, reuse, and recycle water as well as review water practices when compared to microbreweries and brewpubs. It is therefore not surprising that many larger breweries and those that are sustainably focused can reduce water-to-beer ratios at or below 4:1 while smaller breweries can use up to 20 or more barrels of water to make one barrel of beer.²⁰

While not every brewer can automate or fully upgrade their brewery, any brewery regardless of size can improve their water efficiency (Uligian & Hoxmeier, 2020). All craft breweries can (and should)

¹⁹Filtering and pasteurization are less common with craft beer, as such processes generally reduce the flavor and aroma (Cimini & Moresi, 2017).

²⁰This number varies remarkably in the literature. In a study by the WWF/SAB Miller, 60–180 L of water is used to create 1 liter of beer. In another study, Van der Merwe & Friend (2002) found that 5 cubic meters of water was used to produce 1 cubic meter of beer. Online, the “300 L to one pint” is often cited but lacks a reliable source. Much of this variability may be due to the style of the beer, the data used (e.g., averages and max/min), the water reduction practices being used, and of course, the overall water footprint of the supply chain

incorporate into reducing their water footprint include checking and repairing water leaks, changing to low-flow spray valves and faucets, and identifying overflow points to employees. Other actions, like reusing surplus wort from previous batches or adopting a dry floor policy (e.g., using rakes and brushes rather than water to wash effluent into drains) can save breweries money, reduce the water footprint of the brewery, and improve the health and safety of the work environment for their employees. Even the installation of meters to track water use and incorporation of hot water recovery systems to redistribute water for other brewhouse purposes (e.g., tank cleaning) can decrease the water footprint of any size brewery. The [Brewers Association \(2014a\)](#) provides a detailed water manual and associated checklist of water efficiency opportunities related to cooling and heating, process use, cleaning and sanitation, and landscape and irrigation. In just 2 years (2014–16), microbreweries that responded to the Brewers Association annual sustainability benchmarking survey reduced their average water use ratio by more than half ([Brewers Association, 2022](#)). By implementing these strategies, breweries can significantly reduce their water consumption and lessen the strain on local water resources.

3.3 Waste production

Breweries generally produce four types of solid waste: 1. waste from the brewing process; 2. waste from packaging; 3. waste from food service; and 4. waste from special events ([Fish, 2015](#)). Waste from the brewery processes includes wastewater and solid wastes. With up to 70% of water leaving the brewery as effluent ([Binnie & Partners, 1986](#)), the processing and disposal of brewery wastewater is one of the most observable environmental issues facing most breweries. Wastewater from brewing can be high in organic residues (e.g., sugars, starch, ethanol, and volatile fatty acids),²¹ high in temperature, low in pH from cleaning and sanitizing chemicals, with the potential to cause environmental pollution if not properly disposed ([Ashraf et al., 2021](#)).

Wastewater treatment in breweries is designed to not only remove contaminants but reduce the environmental impact of the water before it is discharged or reused. This can be done with simple screening to remove large solids, sedimentation to allow solids to settle out, dewatering hoppers and screw presses to remove liquid from solid waste, biological treatments that use bacteria to consume organic matter, and advanced treatments like reverse osmosis or electrodialysis reversal to purify water for reuse ([Ashraf et al., 2021](#)). High-temperature lift stations can also be implemented to handle hot temperatures and corrosive wastewater, while pH balancing systems can help meet the environmental regulatory requirements (i.e., the Clean Water Act) of discharge set between 5.0 and 11.0. The goal is always to minimize the impact on the environment and comply with water quality standards.

Spent grains, spent hops, hot trub, and residual brewer's yeast are the major byproducts of the brewing process, and can weigh up to 50 pounds per barrel of beer produced ([Gribbins, 2013](#)). Proper management of these wastes may bring economic, environmental, and social benefits ([Rachwał et al., 2020](#)). About 85% of the solid waste generated originates in the mashing stage as spent grains ([Bachmann et al., 2022](#)). Until recently, spent grains were not considered a tradable commodity; however, hauling away such copious amounts of waste usually comes with huge costs and repercussions, both economic (e.g., costs to haul) and environmental (e.g., increased carbon footprint

²¹ with elevated levels of biological oxygen demand (BOD) and total suspended solids (TSS). To note, BOD is a measure of the amount of dissolved oxygen required to remove waste organic matter by anaerobic bacteria. TSS is the dry weight of particles not dissolved in water (like soil particles, organic matter, and microscopic organisms) that exceed 2 microns in size

from transportation). As such, brewers have looked to a variety of alternative, artisanal, energy-producing, and sustainable uses for spent grains, including animal feed, dog snacks, upcycled food products for human consumption,²² biochar used in soil remediation, as plant food (due to N, P, and C content), in anaerobic digestion, as cationic and anionic species adsorption with metals and dyes, and as a component of biofuel (Bachmann et al., 2022; Mainardis et al., 2024; Mussatto et al., 2006). Innovative solutions to return wastes and by-products (like spent grain, see Zeko-Pivač et al., 2022) to the production cycle promote a circular bioeconomy that replaces nonrenewable, fossil-based products with sustainable alternatives. Even spent hops from dry hopping can be reused for bittering beer, with a study by Alfeo et al. (2023) highlighting how sensory evaluations of beers with spent hops do not always correlate with the evaluated differences in hop characteristics (e.g., hop oil, polyphenol content, etc.).

3.4 Energy conservation and alternative energy sources

U.S. Breweries spend hundreds of millions in energy costs annually. Galitsky et al. (2003) note that as much as 8% of a brewery's overhead costs can come from energy expenditures. Multiple unit operations can occur at the brewhouse, and may include malt milling, mashing, lautering, boiling, whirlpooling, and cooling, which convert raw materials into chilled wort for fermentation and maturation (Salazar Tijerino et al., 2023). Overall, refrigeration creates the largest electrical load, while brewing consumes the largest amount of natural gas (used for heat) (Brewers Association, 2014b). Energy uses in breweries also vary widely and depend on several factors. Like water consumption, the size of the brewery matters. In essence, the larger the brewery, the more energy efficient it can be. As an example, Cimini & Moresi (2017) show the overall brewing energy needs for brewing operations increased from 128 to 613 MJ hL⁻¹ as the brewery size decreased from 10 million to 500 hL/yr. The location of the brewery—in terms of the building as well as its geographic location—can also influence the cost and amount of energy used. In the US, locations like Hawaii, and New England (e.g., Rhode Island, Connecticut) have some of the most expensive energy rates in the USA, while other locales (e.g., Alaska, southwestern US) have high energy consumption due to extreme climate conditions. Finally, the brewing techniques and the scale of production can impact energy use. Small batches, experimental beers, and styles that require additional steps, extra ingredients, prolonged fermentation times (e.g., lager yeast, secondary fermentation) or extended temperature controls can increase the energy requirements. By comparing lagers to ales, Cimini & Moresi (2018) note that lagers take more energy than ales due to lagering, a process that benefits from economies of scale. In fact, Salazar Tijerino et al. (2023) found that the lower temperatures and longer time necessary for lager fermentation and maturation (due to yeast strain requirements) resulted in a 24% higher total electricity use for lager production than ale production at a craft (micro) brewery scale (i.e., 21.5 bbl brewhouse).

Breweries can reduce their economic costs and overall environmental impact by improving brewhouse energy efficiency and lowering energy demands. While some energy-efficient practices are easy to incorporate (e.g., energy-efficient lighting, turning off equipment when not in use, vinyl door flaps on cooler doors), many brewery upgrades can be capital intensive and create a fiscal barrier for adoption despite high-efficiency gains and long(er) term cost reductions. The size of the brewery

²²See Rachwał et al. (2020) for a detailed list of developed/brand food products that use spent grains and/or spent yeast

seems to matter, as some of the best sustainable practices and energy-efficient upgrades are more ubiquitous among larger craft breweries (Bahl et al., 2021) and industrial macrobreweries rather than smaller microbreweries, taprooms, and especially brewpubs (Patterson et al., 2016). Economies of scales dictates this, as increase production equates to energy costs as a smaller portion of the total expense (Uligian & Hoxmeier, 2020). Such upgrades include the installation of (or replacement with) modern, energy-efficient brewing systems and using automated control systems and smart brewing technologies (Brewers Association, 2014b). Large breweries are also more apt to develop on-site renewable energy generation, such as recovering carbon dioxide from fermentation, repurposing waste heat for mashing or pasteurization via (retrofitted) heat transfer area or capturing methane from on-site treated wastewater. Even if such large investments cannot be made, any form of energy utilization should be a primary focus of every brewery to keep overall energy costs down. Thus, measuring and benchmarking current utility data should be the first step in improving efficiencies and decreasing costs of brewhouse operations (Cimini & Moresi, 2018). Apart from benchmarking tools (e.g., Brewers Association, 2024b), brewers may have access to local programs and free services that will audit the energy efficiency of their building, operations, and/or equipment (Uligian & Hoxmeier, 2020). With such data, informed decisions can be made on what (and where) upgrades, replacements, and/or more sustainable practices should be made to decrease energy costs and thus, the brewery's overall carbon footprint. Moreover, shifting from fossil to renewable energies by purchasing (or creating!) energy from renewable sources and/or purchasing carbon offsets for energy can and should be attained by all brewers. As many breweries are in sunny regions, installation of solar collectors would be one option for cheap solar energy (Olijare 2020).

4. Beyond the brewery

Sustainability does not end once the beer is made. Unless consumed on premise, beer still must leave its origin and travel to the consumer. However, the craft brewing industry faces challenges to be both efficient and sustainable in terms of packaging and distribution. Due to a plethora of alcohol rules and regulations at different geographic scales (e.g., city, county, and state), there is no “one size fits all” in terms of how to best meet these challenges. Solutions require a more individualistic approach (Morgan et al., 2022) to decrease the weight, distances, and end-of-life strategies. The following focuses on economic and environmental costs and impacts with packaging, disposal, and transportation.

4.1 Packaging

While packaging can be considered as the final step of the brewing process, it is considered in this section (rather than the former) as it directly relates to the consumer. Moreover, packaging can be its own manufacturing labyrinth—with operations that can include bottling, canning, kegging, labeling, sorting, accumulating, boxing, palletizing, and shipping. As noted by Morgan et al. (2022), packaging “plays a crucial role in protecting food and drink, extending the shelf life, and ensuring food safety standards can be maintained from the postproduction stage through to consumption.” But LCA studies show packaging as a hotspot for beer's carbon footprint, and according to Amienyo & Azapagic (2016), accounts for 35–55% of the GWP in the life cycle of beer. This leads to the question: is there an ideal, sustainable packaging for beer?

The short answer is no, and not all packaging is the same. Indeed, multiple LCA studies have found the carbon footprint of beer varies significantly with package use (Cimini & Moresi, 2016; Marrucci et al., 2024; Morgan et al., 2022). Although comparing LCA findings is challenging, as they usually contain incomplete data and use different methods, findings have shown that beer in single-use PET kegs can be more sustainable than steel kegs (Marrucci et al., 2024), steel kegs are more sustainable than aluminum cans, and aluminum cans are more sustainable than glass (Cimini & Moresi, 2016).²³ In fact, the only thing going for glass is its low toxicity potential (Amienyo & Azapagic, 2016).

Many craft (and other) breweries have switched their single-use packaging from glass to aluminum, and multiuse from returnable steel kegs to single-use plastic ones. In terms of single-use packaging, aluminum cans now account for over 60% of the craft beer market packaging (MCC.com). And it is not just the carbon footprint that makes aluminum more sustainable—it costs less, is lighter weight for shipping (also a cost), increases shelf life of the beer, and is easy to recycle. Unfortunately, the surge in demand—along with supply chain issues, aluminum tariffs, and even declines in aluminum recycling—created shortage of aluminum cans that was exacerbated by COVID-19 lockdowns (Ahamed, 2024). New innovations in packaging promise to further decrease the carbon footprint of beer, including the Carlsberg Group’s Green Fiber Bottle, deemed the world’s first “paper bottle” for beer (Carlsberg Group, 2019).

Steel kegs, once the dominant packaging in the draught beer market, are increasingly being replaced by single-use plastic kegs, or PET (Martin et al., 2022) due in part to their lower investment cost, lighter weight, and overall ease of use. This may be surprising, especially considering the increasing importance of circular economies and the cultural and policy-related deterrence from single-use plastic consumption. While single-use PET kegs may be more sustainable than single-use glass bottles or aluminum cans, they still have a higher carbon footprint than steel kegs, primarily because its higher GHG emissions from manufacturing and single-use cannot be offset from being reused like steel kegs. And the more steel kegs are used (>40 times, or 7 years²⁴), the lower the GHG emissions and life cycle costs. The introduction of a closed-loop recycling scheme for plastic kegs would improve the environmental performance of PET kegs (Martin et al., 2022).

Packaging does not just include what the beer is in but can also include cardboard or plastic can holders. For cardboard, almost all forms can be recycled, leading to the potential reduction in the carbon footprint of beer. In contrast, typical plastic can carriers and plastic rings common for carrying four and six packs of canned beer are not recyclable, despite consumers thinking otherwise (Yoder & Grist, 2024). Although popular because of their functionality, convenience, and cheap price tag, craft brewers have mimicked macrobreweries in adopting more sustainable alternatives, including plant based, compostable rings, cardboard rings, and 100% recycled plastic rings. Some macrobreweries are creating and adopting no packaging solutions. For example, Carlsberg is using recyclable glue to hold cans together, while Corona has invented stackable cans that screw into one another that require no packaging at all (Kell, 2019). In the spirit of closed-loop systems, Corona has also created 100% biodegradable packaging from surplus barley straw and reported to use one third of the water required for “traditional technologies” and prevent approximately 1400 kg (about 3086.47 lb) CO₂ emissions/ton from being produced (PR Newswire, 2022).

²³with the exception of reusable glass. Amienyo & Azapagic (2016) found that at 85% reuse, the contribution of returnable bottles was larger than that of nonreturnable for all the environmental impacts

²⁴Kegs can last up to 30 years (Morgan et al., 2022).

4.2 Transportation

Transportation does play an outsized role in the brewing industry. After the beer is brewed and packaged, the distribution of beer occurs (direct to consumers, to retail, or to distribution centers).²⁵ Some breweries contract out the bottling and canning to other facilities and back. Transportation can also include other aspects of the brewhouse, including the transfer of wastewater to treatment plant as well as spent grains and other solid waste to become energy (anaerobic digestion), end up on a farm for feed or compost/fertilizer, or sent directly to the landfill.

Buying locally is not only popular with consumers ([Atallah et al., 2021](#)), but it shortens the supply chains. Reduced distances decrease the costs and carbon footprint for transporting beer in almost any geographical locale. However, little exists in the literature on the types of transportation. Should one use gas or diesel? Large trucks or small vans? Specific packaging to take? For example, [Morgan et al. \(2022\)](#) found the optimal combination of packaging and transportation was reusable glass bottles delivered by lorry, which reduced carbon footprints by between 45% and 55%. They also highlighted how the dependence on small (gasoline-powered) vans for local deliveries represents a major environmental hotspot for craft beer. While no studies look directly at the life cycle impacts and costs of using electronic vehicles (EV) for beer distribution, adoption of EVs²⁶ would significantly decrease (by over 40%) the transportation emissions. Operating emissions would solely be from charging the batteries from the local power grid (which would be lowered further if the energy were from renewable sources like wind and solar). Even the adoption of hybrid and plug-in hybrid electronic vehicles (PHEV) emit less than standard gasoline or diesel vehicles, decreasing the total carbon footprint associated with distribution.

5. Final thoughts

For craft brewers, sustainability in theory might be easier than sustainability in practice, due in part to the modern-day disconnects between the brewery locations and the raw ingredients (thus leading to de facto increases in the carbon footprints), the innate need for large amounts of water and energy, and the production of liquid or solid wastes at nearly every stage of beer production. This is exacerbated by economies of scale, with many small breweries unable to ever produce or package beer at the same levels of sustainability—in particular, economic, and environmental sustainability—as their corporate industrial counterparts ([Patterson et al., 2016](#)).

Unfortunately, [Patterson et al. \(2016\)](#) showed that smaller breweries (especially brewpubs) were less likely to engage in environmental, economic, and social sustainability practices, including a systematic review of operations to improve sustainability. Many of these founders and brewers expressed how prioritization of sustainability would come once they reached a larger production size or more financial success. Ironically, sustainability actions that reduce carbon emissions, minimize waste, and decrease water use (as well as lower expenses) do not need stem from a deep environmental commitment but as enlightened self-interest or even smart long-term investing ([Jones, 2018](#)).

²⁵One could argue that the antiquated three tier system, established in the USA after Prohibition in which brewers sell to wholesale distributors who then sell to retailers who then sell to consumers, does not promote sustainability, especially given the extra transportation required (and hence, an expanded carbon footprint) to move beer between the tiers

²⁶when replacements are needed. The carbon footprint for an electric vehicle is high during its production, but over time decreases to have (most likely) a smaller carbon footprint than a fossil fuel vehicle

Consumer attitudes impact their purchase intentions regarding craft beer noted that sustainability measures like water conservation, organic and local sourcing, and carbon footprint reduction can hold considerable influence on consumer purchasing in the craft beer market (Lourenco & Piotto, 2024). Locally brewed craft beer, which Gatrell & Steiger (2018) argue is a proxy for authenticity, is also a common preference for consumers, and one reason for craft beer's continued popularity. Of course, gaps do occur between a consumer's perception of what measures are sustainable and the reality of what actually is. For example, in a Danish study by Boesen et al. (2019), consumers assessed the environmental sustainability of packaging based on material type and means of disposal, rather than impacts of production and transportation. As a result, glass is seen as "good" (which it is not) while plastic is seen as "bad" (which in some cases, is the best option).

Given consumer preference for sustainability and authenticity, it is common for a brewery to promote sustainable beliefs and efforts by way of websites, social media campaigns, news articles, and other brand-awareness tactics (Patterson et al., 2016). While the planet reaps the benefits of sustainable decisions, the brewer may just reap a bounty too, as studies have found that beer drinkers are willing to pay more for more sustainably produced beer (Carley & Yahng, 2018). And while the economies of scale work in the favor of the macrobreweries, it is evident that many craft brewers are willing to implement economic, environmental, and social sustainability practices while continuing to balance the following dichotomies: (1) small-batch innovations and experimentation versus financial and business strategies for increased production and growth; (2) ingredients with large(r) carbon footprints versus more locally sourced ingredients or those with smaller carbon footprints; (3) local "authenticity" versus geographic expansion; and (4) independent brewery ownership versus corporate partnerships (Erhardt et al., 2022).

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CRAFT BEER

New Frontiers in Brewing Science

Edited by

PATRIZIA ROMANO, MAURIZIO CIANI
and **ANGELA CAPECE**

This first and only scientific book covers the craft beer market, brewing process, and the qualities and styles of craft beer

Key features

- Covers the latest developments in craft beer production
- Summarizes the main factors affecting craft beer quality
- Reports the qualities responsible for the differentiation between craft and industrial beer
- Explores modern approaches to craft brewing

Recently, both researchers and consumers have shown a strong and growing interest in craft beer, which is characterized by its small batches, local breweries, and unique varieties. *Craft Beer: New Frontiers in Brewing Science* provides a comprehensive picture of the craft beer market, history and legal definition, brewing processes, and the qualities and styles of the end product. It emphasizes the traditional and innovative microbial resources involved in the fermentation process, as well as the safety, nutrition, and emerging problems, such as the sustainability of the production process.

Craft Beer: New Frontiers in Brewing Science is the first and only scientific text to comprehensively summarize this exciting field. Researchers and students interested in brewing science, as well as industry specialists from beer companies, will all benefit from this unique resource.

About editors

Patrizia Romano

Professor, Department of Economy, Universitas Mercatorum, Rome, Italy

Maurizio Ciani

Professor, Department of Life and Environmental Sciences (DiSVA), Polytechnic University of Marche, Ancona, Italy

Angela Capece

Associate Professor; Department of Agricultural, Forest, Food, and Environmental Sciences (DAFE); University of Basilicata; Potenza; Italy



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An imprint of Elsevier

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ISBN 978-0-443-16015-8



9 780443 160158